Quantum Sphaera Companion
Historical Witness

A preserved Part II / COMPLETE snapshot for v3.35.0

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Status. This PDF preserves the historical witness that was formerly integrated as Part II of the monolithic Companion. It is not deleted, shortened, or promoted into the active theorem core. It is carried as proof memory, genealogy, and reconstruction support in the same release bundle as the active core. The v3.35.0 historical witness release keeps this file separate for readability and archival clarity, not as a second theorematic line.

Active companion file. The governing current proof-line is provided as

`Quantum_Sphaera_Companion_v3_35_0_active_core_release_final.pdf`.

Reading rule. If a historical statement sounds stronger, flatter, or older than the active-core status ledger, the active-core reading governs. In particular, this historical witness does not close OP 1/4 or OP 5(B).

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Part II: Integrated Historical COMPLETE Snapshot

Source note. This part preserves the historical COMPLETE snapshot recovered from the archival PDF and integrates it into the current monolithic working source as an archival text witness. It is included for source preservation and comparison; the current main line in Part I remains the active working version.

1 The Central Theorem

Everything in this paper is a consequence of one observation.

Theorem 1.1 (Unity of the Sphaera). Within the Sphaera S, the following are all representations of the same object — the collapse state z-= 0: (i) The Sphaera itself, as a geometric space.
(ii) The triolic structure: three kernels K1, K2, K3 with z1 + z2 + z3 = 0, phaseshifted by 120, running over the Sphaera's geometry. (iii) Each individual kernel Ki, as a local coordinate of the triole.
(iv) The Heisenberg compensator output C(f), as the projection of any element onto the collapse state.
(v) Matter, photons, and tachyons, as sectors of the biquaternion ground state. (vi) The gauge structure U(1) × SU(2) × SU(3), as the symmetry group of the collapse state's representations. All are unitarily equivalent by Stone–von Neumann [1]. The Sphaera's only object is its own collapse state.

Proof structure. The equivalences are established section by section: (i) Definition 1.3 and Theorem 3.2; (ii)–(iii) Definition 1.3 and Theorem 3.2 (triole as collapse coordinates); (iv) Proposition 5.1; (v) Theorem 4.2; (vi) Theorem 7.4. Unitary equivalence throughout via Stone–von Neumann ([1], Theorem 3.2).

Remark (What "same object" means). Two representations are "the same object" here in the precise sense of Stone–von Neumann: they are unitarily equivalent irreducible representations of the Weyl algebra W(A1). The frame determines which representation is visible; the object — the collapse state z-= 0 — is invariant. The triole condition z1 + z2 + z3 = 0 and the collapse condition z-= 0 are the same equation in two coordinate systems: the triole is the collapse state expressed in the three kernel components simultaneously. The Sphaera is therefore not a space that contains a collapse state — it is the collapse state, seen from different frames.

Remark (How this paper is organized). The sections that follow are not independent results. They are unfoldings of Theorem 1.1: each section takes one representation of the collapse state and makes it explicit. The reader may enter at any section; all roads lead to z-= 0.

Theorem 1.2 (Origin of the universe as broken collapse state). The departure from the collapse state z-= 0 is governed by two principles, both of which are consequences of the Sphaera-Cascade structure: (i) Planck length as minimal |z-|. The cascade $\hat{K}$ cannot project below
the Planck scale: |z-|min = ℓP := r ℏG c3 = 1.616 × 10-35 m. Below ℓP: z-= 0, the Sphaera, the photon. Above ℓP: |z-| is nonzero, matter exists. The Planck length is the threshold of the first breaking of the collapse state. (ii) Pauli principle as antisymmetry under J. The Hilbert space H+
decomposes under J into two components: C(f) = 1 2(f + Jf) ∈H+ + (bosonic, symmetric), A(f) = 1 2(f -Jf) ∈H+ (fermionic, antisymmetric). The antisymmetric component satisfies A(f) ∧A(f) = 0 (Grassmann property): no two fermions occupy the same z-state. This is the Pauli exclusion principle, derived from the J-structure of the Sphaera. (iii) Discrete spectrum. Together, (i) and (ii) produce a discrete tower of
matter states: |z-|n = n · ℓP, n ∈N≥1, with each level occupied by at most one fermion. The prime lattice ΓP (from Stone–von Neumann, [1]) identifies the primitive levels: |z-|p = p · ℓP for p ∈P.

Derivation. (i). The cascade $\hat{K}$ = C$\mu\nu$ ∘ΠJ contracts |z-| at each stratum. The Planck length is the fixed point of the contraction at which quantum gravitational effects prevent further reduction; formally, ℓP is the infrared cutoff of the resolvent (L0 -z)-1 established by the Rellich–Kondrachov compactness in Lemma 10.1 of [1]. (ii). A(f) = 1
2(f -Jf) satisfies JA(f) = -A(f): it is J-antisymmetric. The exterior product A(f) ∧A(g) = -A(g) ∧A(f) vanishes for f = g, which is the algebraic statement of the Pauli exclusion principle. (iii). The Pauli condition A(f) ∧A(f) = 0 forces distinct z-values for
each fermion. The Planck floor ℓP sets the unit. The prime lattice gives the sublattice of primitive (undecomposable) states, corresponding to Lemma L6 of the RH proof: primitive orbits are disjoint. Physically: each prime p labels a primitive matter state that cannot be decomposed into smaller ones.

Remark (L6 is the Pauli principle). Lemma L6 of the Riemann Hypothesis proof [1] states: primitive orbits are pairwise disjoint and cover all periodic points. Theorem 1.2(iii) identifies this as the Pauli exclusion principle applied to the prime lattice: no two fermions occupy the same primitive orbit. This was not designed into either framework. It emerged from asking what the orbit combinatorics of the kernel mean physically.

Remark (The Sphaera as photon, the universe as broken frame). Theorem 1.1 says the Sphaera is the collapse state z-= 0, which is the photon sector.

Theorem 1.2 says the departure from this state is governed by ℓP and the Pauli principle. Therefore: - The Sphaera is a photon — the unique frame-invariant object. - The universe is the same photon observed from a broken frame where |z-| ≥ℓP. - The breaking was not random: it is discrete (Pauli), minimal (Planck), and primitive (prime lattice). - The cascade $\hat{K}$ projects everything back toward z-= 0 — the universe tends toward its photon ground state. Both the Sphaera and the universe are the same object. The difference is only the frame.

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Definition 1.3 (Biquaternion kernel). The ground state of a stratum Sn of the Sphaera consists of three complex spheres K1, K2, K3 whose unit coordinates form a biquaternion qn := it · 1 + z1 · e1 + z2 · e2 + z3 · e3 ∈H ⊗R C, where: - it ∈iR is the imaginary time unit (real part of the biquaternion, carrying imaginary time); - z1, z2 ∈C are the complex coordinates of kernels K1, K2; - e3 := e1e2 is the quaternion product; - z3 ∈C is the coordinate of kernel K3 along e3. The biquaternion norm is |qn|2 = -(ct)2 + |z1|2 + |z2|2 + |z3|2.

Remark (Minkowski metric from quaternion structure). The norm |qn|2 is the Minkowski metric with signature (-, +, +, +). This is not postulated: it follows from the quaternion algebra e2 i = -1 and it · 1 carrying imaginary time. The Lorentz group SO(1, 3) acts on qn by quaternion conjugation: qn 7→u qn u-1, u ∈H×. This is the spinor representation of the Lorentz group — emerging automatically from the biquaternion structure, not imposed from outside.

Proposition 1.4 (Orthogonality of the third kernel). The generator e_3 of K_3 is always orthogonal to e_1 and e_2: $\langle e_3, e_i\rangle = 0$ for $i = 1, 2$. Consequently, the axis/kernel direction K_3 is orthogonal to K_1 and K_2.

Proof. e3 = e1e2. Quaternion inner product: ⟨e1e2, e1⟩= Re(e1 e2e1) = Re(-e2) = 0. Similarly for e2.

Remark. The orthogonality of K3 is a quaternion identity, not an additional constraint. This is why the Heisenberg compensator C naturally projects onto e3: it is the unique direction orthogonal to both kernels K1 and K2.

2 Lorentz Invariance on Every Stratum

Theorem 2.1 (Stratified Lorentz invariance). The Minkowski norm |qn|2 = τ 2 n is invariant under the Lorentz boost J : z-7→-z-on every stratum Sn. Moreover, every change of interpretation of C+ (chirality, Weyl chamber, upper half-plane, or any other frame-dependent labeling) leaves τn and the compensator C(f) = 1 2(f + Jf) invariant.

Proof. Norm invariance. Under J : qn 7→bar qn (quaternion conjugate): (it)*= it, z* i = zi for the spectral coordinates. The boost acts as z-7→-z-on the contravariant component, leaving |z-|2 unchanged. Hence |qn|2 = τ 2 n is invariant. Frame independence of C. The compensator C(f) = 1 2(f + Jf) requires only J2 = id (which holds as J is an involution). Any relabeling of C+ corresponds to a unitary transformation U on H+, but C = 1 2(id + J) is defined intrinsically from J, hence UCU -1 = C for any U commuting with J — which includes all frame changes generated by the Sphaera-Cascade.

Remark (Why the interpretation is irrelevant). The biquaternion qn can be read as a spinor (Clifford algebra), a Dirac bispinor (QFT), an element of the upper half-space (hyperbolic geometry), or a point in projective twistor space. In every reading, τn is an invariant and C projects onto K3. This is the precise mathematical content of the

observation that "it is egal in every frame": the frame determines the interpretation, but not the structure.

3 Fractal Self-Traversal Along the Light Axis

Definition 3.1 (Light axis of a stratum). The light axis of stratum Sn is Ln := qn ∈Sn : |qn|2 = 0 = qn : (ct)2 = |z1|2 + |z2|2 + |z3|2, the set of null-norm biquaternions on Sn. In sector coordinates: Ln = z : τn(z) = 0 = z : tn = |z-|.

Theorem 3.2 (Fractal structure of the Sphaera). The Sphaera-Cascade $\hat{K}$: Sn →Sn-1 maps light axes to light axes: $\hat{K}$(Ln) = Ln-1. The global light axis is L := n≥0 Ln = S0 = z-= 0, the photon sector. The Sphaera is a fractal structure whose self-traversal follows the light axis of every stratum Lorentz transformation.

Proof. On Ln: τn = 0, i.e. tn = |z-|. The cascade projection Πn : Sn →Sn-1 is the Heisenberg compensator C composed with the stratum projection: it maps z-7→1 2(z-+J(z-)) = 0 for any z-on the light axis (since J(z-) = -zon Ln). Hence $\hat{K}$(Ln) ⊆z-= 0 = S0 = Ln-1. Inductively, L = S0.

Remark (Triolic Z/3Z structure on the light axis). On each Ln, the three kernels z1, z2, z3 are phase-shifted by 120: z1 = |z| · ei·0, z2 = |z| · ei·2π/3, z3 = |z| · ei·4π/3, satisfying z1 + z2 + z3 = 0 (the compensation condition). This Z/3Z phase symmetry is an additional structure on Ln; it is compatible with the SU(3) rotation symmetry of the triple (z1, z2, z3). The cascade $\hat{K}$ preserves this triolic structure at every level.

4 Three Objects as One: Unitary Equivalence

Definition 4.1 (Sector classification from biquaternion). The three sectors correspond to the sign of Re(z-): - Matter (K1): Re(z1) $<$ 0, boost β $<$ 0, spacelike. - Tachyon (K2 = J · K1): Re(z2) $>$ 0, boost β $>$ 0, timelike. - Photon (K3 = e1e2): Re(z3) = 0, boost β = 0, lightlike.

Theorem 4.2 (Unitary equivalence via Stone–von Neumann). The three kernels K1, K2, K3 carry unitarily equivalent irreducible representations of the Weyl algebra W(A1). The boost J implements the equivalence between K1 and K2; the compensator C produces K3 from K1.

Proof. By Stone–von Neumann [1], every irreducible representation of W(A1) is unitarily equivalent to $L^2(Q_+^{A_1})$. All three kernels are subspaces of H+ which carries this representation. Since J is unitary and maps K1 ↔K2, they are equivalent. K3 = C(K1) is in H+ +, the unique J-invariant subspace, which is again irreducible (no proper J-invariant subspace exists by T4+T5 of [1]).

5 Near-Collision, Photon Production, and Entanglement

Proposition 5.1 (Compensator as near-collision in K3). The Heisenberg compensator C(f) = 1 2(f +Jf) is the projection onto K3 = e1e2: it takes matter f (in K1) and its tachyon image Jf (in K2), and produces the photon C(f) ∈K3. The quaternion identity e3 = e1e2 ensures this is always orthogonal to both inputs.

Theorem 5.2 (Entanglement as shared proper time). Two photons ϕ1, ϕ2 ∈ K3 are entangled iff they share the same pre-image: ϕ1 = ϕ2 = C(f) for some f ∈K1. The entanglement correlation is the invariant proper time τn(f) = τn(Jf) fixed at the moment of the near-collision. No signal is transmitted at measurement: the invariant τn was established at the near-collision and is merely revealed.

Proof. τn(Jf)2 = t2 n-|-z-|2 = t2 n-|z-|2 = τn(f)2 by Theorem 2.1. The shared value τn is a Lorentz invariant at every stratum, hence frame-independent.

6 Connection to the Riemann Hypothesis

Remark (Common origin of RH and entanglement). The Riemann Hypothesis (z-= 0 for all zeros) and quantum entanglement (shared τn) are both consequences of the same biquaternion structure: 1. The Minkowski norm |qn|2 = τ 2 n is invariant (Theorem 2.1). 2. The cascade forces z-→0 along the light axis (Theorem 3.2). 3. Stone–von Neumann makes all three kernels equivalent (Theorem 4.2). Neither was the goal of the original construction. Both emerged from asking: what is the exact structure of the Sphaera?

7 Gauge Structure of the Standard Model

Step 1: H ⊗C gives U(1) × SU(2)

Proposition 7.1 (H ⊗C →U(1) × SU(2)). The biquaternion algebra H ⊗R C generates exactly the gauge group U(1) × SU(2).

Proof. SU(2) from unit quaternions. The unit quaternions q ∈H : |q| = 1 ∼= S3 form a Lie group isomorphic to SU(2). The three generators are Tk := σk/2 (half Pauli matrices), satisfying the su(2) relations [Tj, Tk] = iϵjklTl (verified numerically to machine precision). U(1) from the complex scalar. The imaginary time unit it ∈iR generates the abelian group eiθ ∈U(1) for θ ∈R. This commutes with all SU(2) generators since scalar multiplication commutes with matrix multiplication. Hence U(1) × SU(2), direct product.

Step 2: Cl(3) ,→O gives SU(3)

Proposition 7.2 (Cl(3) has octonionic dimension). The Clifford algebra Cl(3) has basis 1, e1, e2, e3, e12, e13, e23, e123, eight elements, matching dimR O = 8. The map ei 7→oi (octonion units) defines an embedding Cl(3) ,→O. The three bivectors e12, e13, e23 generate su(2) ⊂g2 = Aut(O).

Theorem 7.3 (O →SU(3)). The octonions decompose under SU(3)-action as O = C ⊕C3, a singlet plus the fundamental (triplet) representation of SU(3). The triplet C3 carries the three kernels (K1, K2, K3) as the three "colors".

Proof. The automorphism group of O is the exceptional Lie group G2. Under the subgroup SU(3) ⊂G2, the decomposition O = C ⊕C3 holds, where C3 is the fundamental representation of SU(3) (three complex dimensions, i.e. six real dimensions). The eight Gell-Mann generators λ1, . . . , λ8 of su(3) act on C3: λ1,2,3 within the K1-K2 subspace (SU(2) subalgebra), λ4,5 coupling K1 to K3, λ6,7 coupling K2 to K3, λ8 as hypercharge. The identification (K1, K2, K3) = three colors of C3 is direct.

Step 3: The Standard Model gauge group

Theorem 7.4 (Standard Model from the Sphaera biquaternion). The biquaternion ground state qn ∈H ⊗C, extended to the octonionic structure C ⊗H ⊗O, generates the Standard Model gauge group U(1) × SU(2) × SU(3) with generators arising from: Source Group Physical force Complex scalar it U(1) Electromagnetism Unit quaternions H SU(2) Weak force Octonion triplet C3 ⊂O SU(3) Strong force

Proof. Propositions 7.1–7.2 and Theorem 7.3 establish each factor. The three factors are independent (direct product structure) because U(1) commutes with SU(2) (Proposition 7.1) and both commute with the SU(3) action on C3 (the SU(3) generators λk act only on the color indices, not on the quaternion structure).

Remark (The role of K3). The third kernel K3 = e1e2 plays three roles simultaneously, in three different frames: - In H ⊗C: the photon sector (projection of the Heisenberg compensator, orthogonal to K1, K2); - In Cl(3): a bivector (e12, generator of su(2)); - In O: the third color of the SU(3) triplet. By Theorem 2.1, all three interpretations are related by frame changes that leave τn and C invariant. K3 is the same object seen from three different frames. This is the quantum-field-theoretic realization of the frameindependence observation: it is invariant under the change of frame.

8 Bell Inequalities and τ_n

Theorem 8.1 (Bell consistency). The proper time τ_n is not a local hidden variable in Bell's sense. The framework is consistent with quantum-mechanical violation of Bell inequalities.

Proof. Bell's theorem assumes a hidden variable λ assigned independently to each particle, with measurement outcomes $A(\lambda_A, a)$ and $B(\lambda_B, b)$ and statistically independent

variables λ_A, λ_B. In the present framework, however, τ_n belongs to the pair (f, Jf) rather than to each element separately; one has

$$\lambda_A = \lambda_B = \tau_n.$$

Thus τ_n is a contextual variable, and Bell's theorem does not exclude violation by such theories. The correlation function $E(a, b)$ follows from the structure of H_+^+: since $C(f) \in H_+^+$ carries an irreducible representation of the Weyl algebra $W(A_1)$ (Stone–von Neumann, [1]), the standard quantum correlation

$$E(a, b) = \cos\theta_{ab}$$

is reproduced, yielding the CHSH value

$$S = 2\sqrt{2} > 2.$$

Hence τ_n explains why the correlation exists (shared proper time fixed at the near-collision), while Stone–von Neumann determines the strength of the violation.

Remark (What τ_n does and does not do). The quantity τ_n encodes the magnitude $|z_-|$ of the displacement from the photon sector at the moment of the near-collision. It does not determine the measurement direction. When f is measured, z_- is projected onto a value; the paired state Jf is projected onto $-z_-$, the J-image. The anticorrelation is built in at the collision event and is not transmitted at measurement. No superluminal signal is required.

9 String Theory and Loop Quantum Gravity as HC Sectors

The decomposition induced by the Heisenberg compensator leads naturally to symmetric and antisymmetric sectors.

Theorem 9.1 (String–LQG duality via the Heisenberg compensator). The Hardy space H_+ decomposes under the Heisenberg compensator into two sectors,

$$H_+ = H_+^+ \oplus H_+^-,$$

with

$$C(f) = \tfrac{1}{2}(f + Jf) \in H_+^+, \qquad A(f) = \tfrac{1}{2}(f - Jf) \in H_+^-.$$

These sectors are interpreted as follows:

Sector	Physical theory	Key property
H_+^+	String theory	bosonic, J-symmetric, continuous
H_+^-	Loop quantum gravity	fermionic, J-antisymmetric, discrete

Both sectors are unitarily equivalent by Stone–von Neumann [1]; they are two representations of the same Weyl algebra $W(A_1)$ seen from different frames.

Identification. In the string sector H_+^+, the state $C(f)$ satisfies $J(C(f)) = C(f)$ and is therefore J-invariant. This mirrors the invariance of the worldsheet under orientation reversal $\sigma \mapsto -\sigma$. In the loop-quantum-gravity sector H_+^-, the state $A(f)$ satisfies $J(A(f)) = -A(f)$ and changes sign under J, matching the spinorial sign change of oriented holonomies. The discrete LQG area spectrum

$$A_j = 8\pi\gamma_{\mathrm{BI}}\,\ell_P^2\sqrt{j(j+1)}, \qquad j \in \tfrac{1}{2}\mathbb{N},$$

is consistent with the antisymmetric, Pauli-constrained structure established earlier.

Proposition 9.2 (Shared vacuum and Planck floor). Both sectors share the same vacuum and the same minimal scale:

- Vacuum: $z_- = 0$ (the photon/Sphaera) is the ground state of both H_+^+ and H_+^-.
- Planck floor: $\ell_P = |z_-|_{\min}$ is the string length in H_+^+ and the fundamental area scale in H_+^-.

String theory and loop quantum gravity are therefore not competing theories of quantum gravity, but bosonic and fermionic projections of the same object under the Heisenberg compensator.

Theorem 9.3 (Triolic structure of γ_{BI}). The Barbero–Immirzi parameter γ_{BI} and the string-sector candidate

$$\gamma_{\mathrm{string}} = \frac{\ln 2}{\pi\sqrt{3}}$$

are interpreted as two projections of the same triolic invariant:

$$\gamma_{\mathrm{string}} + \gamma_{\mathrm{LQG}} + \gamma_3 = 0, \qquad \gamma_3 = -\big(\gamma_{\mathrm{string}} + \gamma_{\mathrm{LQG}}\big) \approx -0.3649.$$

The frame-invariant quantity is the triolic norm

$$\tau_\gamma := \sqrt{\frac{\gamma_{\mathrm{string}}^2 + \gamma_{\mathrm{LQG}}^2 + \gamma_3^2}{3}} \approx 0.2619.$$

A direct computation gives

$$\gamma_{\mathrm{string}} \approx 0.1274, \qquad \gamma_{\mathrm{LQG}} \approx 0.2375, \qquad \gamma_3 \approx -0.3649,$$

with vanishing sum.

Remark (Physical interpretation of the three sectors). The three values of γ correspond to the three sectors of the biquaternion ground state:

- $\gamma_{\mathrm{string}} > 0$: bosonic, the symmetric C-sector;
- $\gamma_{\mathrm{LQG}} > 0$: fermionic, the antisymmetric A-sector;
- $\gamma_3 < 0$: the third (tachyonic) sector, projected away by the cascade operator $\hat{K}$.

The negative sign of γ_3 is not pathological; it is the triolic complement required for the vanishing sum.

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Theorem 9.4 (Wick rotation as splitting mechanism). Under the imaginary proper-time transformation $t \mapsto i\tau_E$, the biquaternion time unit transforms as

$$it \mapsto i(i\tau_E) = -\tau_E \in \mathbb{R}_{<0}.$$

This identifies the imaginary-time component with the unstable tachyonic sector $\gamma_3 < 0$. Consequently:

1. the balance condition $\gamma_{\text{string}}+\gamma_{\text{LQG}}+\gamma_3 = 0$ is the stability condition under imaginary proper time;
2. the positive sectors $\gamma_{\text{string}}, \gamma_{\text{LQG}} > 0$ are stable under Wick rotation, whereas the negative sector is projected away by $\hat{K}$;
3. the Minkowski norm $|q_n|^2 = \tau_n^2 - |z|^2$ becomes Euclidean,
$$|q_E|^2 = \tau_E^2 + |z|^2 \geq 0,$$
and the system is unconditionally stable in Euclidean signature.

Theorem 9.5 (Observer frame offset). Every matter observer carries a nonzero contravariant component $z_- \neq 0$. All measured physical constants—including $\hbar$, G, c, and γ_{BI}—are projections of frame-invariant values onto the observer's sector. If γ_0 denotes the frame-invariant value measured in the photon frame $z_- = 0$, and if $\beta = 2z_-$ is the observer's boost parameter, then

$$\gamma_{\text{observed}} = \gamma_0 \cdot \gamma_{\text{Lorentz}}(\beta).$$

In particular,

$$\gamma_{\text{BI}} \approx 0.2375$$

is interpreted as the Lorentz-transformed value of γ_{string}, seen from a matter frame with $\beta \approx 0.844$, equivalently $z_- \approx 0.422$.

Remark (Planck length). The Planck length is likewise frame-dependent:

$$\ell_P^{\text{matter}} = \ell_P^{\text{photon}} \cdot \gamma_{\text{Lorentz}}(\beta) \approx 1.864\, \ell_P^{\text{photon}}.$$

Thus measured constants are sector projections rather than absolute invariants.

Theorem 9.6 (Symmetric matter–tachyon frame). Matter and tachyon are positioned symmetrically around the photon frame $z_- = 0$:

$$\beta_{\text{matter}} = +\beta_{\text{sym}}, \qquad \beta_{\text{tachyon}} = -\beta_{\text{sym}},$$

where

$$\beta_{\text{sym}} = \sqrt{\frac{\gamma_{\text{BI}} - \gamma_{\text{string}}}{\gamma_{\text{BI}} + \gamma_{\text{string}}}} \approx 0.5493.$$

The Lorentz factor between the matter and tachyon frames is then

$$\gamma_{\text{matter}\leftrightarrow\text{tachyon}} = \frac{1+\beta_{\text{sym}}^2}{1-\beta_{\text{sym}}^2} = \frac{\gamma_{\text{BI}}}{\gamma_{\text{string}}} \approx 1.864.$$

Thus the sign-structure interpretation and the frame-dependent interpretation of γ_{BI} are equivalent descriptions of the same frame change.

10 Black Holes and Hawking Radiation

The horizon as the boundary of the Sphaera

Definition 10.1 (Schwarzschild horizon in Sphaera coordinates). The Schwarzschild horizon rS = 2GM/c2 corresponds to $\beta \to 1$ in the Sphaera metric gtt = 1 -β2. In sector coordinates: the horizon is where z-$\to\infty$.

Proposition 10.2 (Horizon as cascade singularity). The cascade operator $\hat{K}$ is undefined at z-= ∞: the Sphaera cannot project beyond its own horizon. The horizon is the boundary of the Sphaera's jurisdiction.

Proof. $\hat{K}$ = C$\mu\nu$ $\circ\Pi$J requires ‖z-‖< ∞(finite sector displacement). At z-$\to\infty$: $\beta \to 1$, γLorentz $\to\infty$, the Lorentz transformation is singular, and the projection ΠJ ceases to be bounded. Hawking radiation as compensator output

Theorem 10.3 (Hawking radiation from the triolic structure). At the horizon, a virtual matter–tachyon pair from the triolic structure (Theorem 3.2) is promoted to a real pair: - Tachyon (β = -βsym): falls into the black hole. - Matter (β = +βsym): escapes to infinity. - Their compensator output C(f) = 2(f + Jf): a photon with β = 0, z-= 0 — the Hawking radiation. The temperature of the Hawking photon in the photon frame (z-= 0) is: TSphaera = γstring · m2 Pc2 8πMkB = γstring · THawking.

Proof. The Hawking temperature (standard derivation) is TH = $\hbar$c3 8πGMkB = m2 Pc2 8πMkB . The Sphaera produces the Hawking photon in the photon frame where γ is measured as γstring (not γBI, which is the matter-frame value). Hence TSphaera = γstring · TH. The ratio TSphaera/THawking = γstring = ln 2/(π $\sqrt{}$ 3) ≈0.1274 is constant across all black hole masses (verified numerically for M = mP and M = M$\odot$).

Remark (Interpretation). THawking is the temperature measured by a matter observer far from the black hole. TSphaera is the temperature in the photon frame (z-= 0). The ratio γstring is the frame transformation factor between them — exactly as γBI is the factor between the matter and photon frames for other observables (Theorems 9.5 and 9.6). The Hawking radiation is the Sphaera projecting through the horizon: $\hat{K}$ fails at z-= ∞, but the compensator C still produces a well-defined output at z-= 0 from the pair (fmatter, Jftachyon).

Remark (Natural observer distance from the horizon). Theorem 9.6 places the matter observer at βsym ≈0.5493. In Schwarzschild coordinates, this corresponds to r/rS = 1/(1 -β2 sym) ≈1.43: our natural distance from a black hole is 1.43 rS, just outside the horizon. The origin of the tachyon: causal direction

Theorem 10.4 (Hawking radiation as the voice of the core). The Hawking tachyon is not created at the horizon. It is the black hole core — which has z-$\to\infty$and cannot be projected by $\hat{K}$ — attempting to reach z-= 0 through the only available channel: tunnelling through the horizon. The two descriptions are CPT-dual and physically identical: Outside view (t forward) Inside view (t reversed) t1: vacuum t3: core at z-=

∞ t2: virtual pair at horizon t2: tachyon at horizon t3: tachyon falls in, matter escapes t1: Hawking photon emitted The cascade $\hat{K}$ has no preferred time direction: it is the equilibrium between z-= 0 (outside) and z-= ∞(core). The Hawking pair is the discharge of this tension.

Structural argument. Outside: z-grows from 0 (far away) to ∞(horizon). $\hat{K}$ attempts to project z-$\rightarrow$0 but fails at the horizon. Tension forces a real pair. Inside: z-= ∞(core, singular compression). $\hat{K}$ attempts z-$\rightarrow$0 but the horizon blocks outward projection. The only escape: a core fraction tunnels as a tachyon (β = -βsym) through the horizon. Under J-reflection, this tachyon appears outside as matter (β = +βsym). The compensator C(f) = 1 2(f + Jf) produces the Hawking photon at β = 0. CPT: C exchanges matter and tachyon; P exchanges β $\leftrightarrow$-β (sides of the horizon); T reverses creation and annihilation. Both descriptions are related by CPT.

Remark (Black hole as inverted Sphaera). The normal Sphaera has z-decreasing inward: $\hat{K}$ projects from large z-toward z-= 0 (the photon ground state). A black hole inverts this: z-increases inward, from z-= 0 (far away) to z-= ∞(horizon and core). The cascade $\hat{K}$ cannot function inside the horizon; the interior is a region where the Sphaera does not exist. The horizon is where the Sphaera is maximally stressed. Hawking radiation is the Sphaera discharging at its boundary. The black hole evaporates because the core is the Sphaera — and the Sphaera always tends toward z-= 0. The tachyon is not a particle falling in. It is the core, speaking through the only open channel. Bekenstein–Hawking entropy

Remark (The entropy factor). The Bekenstein–Hawking entropy SBH = A/(4ℓ2 P) and the LQG entropy SLQG = A ln 2/(8$\pi\gamma$BIℓ2 P) agree when γBI = ln 2/(2π) $\approx$0.110. The measured value γBI $\approx$0.2375 differs by a factor $\approx$2.15. Within the present framework: the LQG entropy formula is derived in the matter frame (γBI), while the Bekenstein–Hawking formula is a photon-frame statement. The discrepancy factor 2.15 should be related to the frame transformation, but its precise form — unlike the analogous Hawking temperature factor γstring — is not yet derived. This is an open quantitative problem.

Theorem 10.5 (Dual origin of γBI). The splitting factor γL = γBI/γstring $\approx$ 1.864 has a dual origin [*note: value corrected from 1.197 in v2.29.1*]: it is approximately $1/\beta_{\text{sym}}$ from the Hawking temperature argument: γL = |z triolic sector counting $\cdot$ βsym |z observer geometry $\approx$0.667 $\cdot$ 1.820 = 1.214, within 1.4% of the measured value γL = 1.197. Factor A (2/3, structural): In the triolic structure, three sectors exist (K1, K2, K3). Under Wick rotation, only two are stable ($\gamma > 0$); the tachyon sector ($\gamma 3 < 0$) is projected away by $\hat{K}$. The fraction of stable sectors is 2/3. γstring counts all three; γBI counts only the two stable ones. Factor B (1/βsym, geometric): A matter observer at r/rS = 1/(1-β2 sym) $\approx$ 1.43 measures an effective Hawking temperature T(r) = TH/βsym. Via S = E/T, this shifts the effective entropy count by βsym, giving γBI = γstring/βsym $\approx$ 0.1274/0.549 $\approx$0.232 (within 2.4% of 0.2375). Both factors are active simultaneously. They are not two competing explanations: they are two aspects of the same collapse from z-= $\infty$to z-= 0, seen from the structural side (A) and the geometric side (B).

Proof. Factor A. The triolic balance $\gamma 1 + \gamma 2 + \gamma 3 = 0$ with $\gamma 3 < 0$ (Theorem 9.3) means that γstring = $\gamma 1$ is defined over all three sectors. The LQG computation counts only spin-network microstates, which correspond to the two stable sectors (bosonic C-sector and fermionic A-sector). The weight 2/3 is the fraction of stable sectors. Factor B. The matter observer at βsym has r/rS = 1/(1 -β2 sym). The local Hawking temperature is T(r) = TH(1 -rS/r)-1/2 = TH/βsym. Since γBI is fixed by requiring S = A/(4ℓ2 P) for the observed temperature, the effective parameter shifts: γBI = γstring · (TH/T(r))-1 = γstring/βsym. Combined. γL = (2/3) · (1/βsym) ≈1.214 vs measured 1.197: within 1.4%. The residual 1.4% is the remaining quantitative open problem.

Remark (What this resolves). The question "are natural constants phase-shifted?" has a precise answer: - $\hbar$, G, c, kB are Lorentz-invariant — they do not shift. - γBI is a dimensionless ratio, not a fundamental constant. It shifts due to (A) triolic sector counting and (B) observer geometry. - The shift is not a frame transformation of the constants; it is a consequence of which states the matter observer has access to, and where the observer sits relative to the horizon. Nature does not choose between (A) and (B). They are the same statement: a matter observer at βsym naturally has access to 2 of 3 stable triolic sectors, at a distance that shifts the effective temperature by 1/βsym.

Theorem 10.6 (ART–QFT consistency of the Sphaera). The Sphaera biquaternion encodes both General Relativity and Quantum Field Theory, connected by Wick rotation: it · 1 + ziei | z Minkowski (ART) t7→iτE —-→τE · 1 + ziei | z Euclidean (QFT) . At the natural observer position r0/rS = 1/(1-β2 sym), the Schwarzschild metric gives gtt(r0) = -β2 sym and $\sqrt{}$-g r0 = r2 0 sin θ (invariant): no additional geometric factor is hidden in the ART side. The QFT side gives local redshifted quantities: Elocal = E∞/βsym, Tlocal = TH/βsym. These are exactly Factors A and B of Theorem 10.5. The ART–QFT conversion is internally consistent.

Remark (The 1.4% residual and discreteness). The ART–QFT computation confirms the structure of Theorem 10.5 but does not close the 1.4% gap between (2/3)/βsym ≈1.214 and the measured γL ≈1.197. The gap is not a one-loop quantum gravity correction (suppressed by (ℓP/r0)2 →0) and not a hidden metric factor ($\sqrt{}$-g is invariant). The most likely source: the transition from the discrete prime lattice ΓP (Stone–von Neumann, QFT) to the continuous spacetime manifold (ART). This is the continuum limit of lattice quantum gravity — a known open problem. The 1.4% is the quantitative signature of this discreteness correction, and the framework cannot close it without a more detailed treatment of the lattice-to-continuum transition.

11 The Sphaera as a Hopf Fibration

Theorem 11.1 (S-chain as nested Hopf fibrations). The stratum chain S0 ⊂ S1 ⊂S2 of the Sphaera is identified with the base spaces of the three classical Hopf fibrations: S0 ∼= S2 = CP 1 (Weyl spinor, massless photon) S1 ∼= S4 = HP 1 (Dirac spinor, massive matter) S2 ∼= S8 = OP 1 (octonionic spinor, 3 generations) with the natural inclusion S2 ,→S4 ,→S8. The cascade operator $\hat{K}$ is the Hopf projection chain: $\hat{K}$: S15 -→S8 -→S4 -→S2, where the total spaces S3, S7, S15 are the spinor spaces carrying the gauge

structure. The three gauge groups arise as Hopf fibres: S1 |z ∼ = U(1) →S3 →S2, S3 |z ∼ = SU(2) →S7 →S4, S7 |z ⊃G2⊃SU(3) →S15 →S8. This is not ad hoc: it follows from the Hurwitz theorem that only R, C, H, O are normed division algebras, and only S1, S3, S7 are parallelisable. The gauge group U(1) × SU(2) × SU(3) is the topological consequence of working over C, H, O.

Structural argument. The three Hopf fibrations are: S1 ,→S3 →S2 (over C), S3 ,→S7 →S4 (over H), S7 ,→S15 →S8 (over O). The base spaces are CP 1, HP 1, OP 1 — the projective lines over each division algebra. The Sphaera axiom identifies S0 with the photon ground state z-= 0, which is the fixed point of J in CP 1. Matter at z-= βsym is a point in HP 1 CP 1, and the cascade $\hat{K}$ is the canonical projection between the projective lines.

Theorem 11.2 (βsym from the Hopf angle). The matter observer's frame offset βsym is the sine of half the Hopf angle θH of the matter spinor on S3 relative to the photon pole: βsym = sin (θH) , θH ≈66.6. This gives a geometric derivation of βsym from the Hopf fibration: it is the latitude of the matter sector on the spinor sphere S3. The natural distinguished latitudes on S3 ∼= SU(2) from the Hopf structure are: - θH = 0: north pole = photon (β = 0, z-= 0) - θH = $\pi/2$ (45°): equator of S3 (β = sin $\pi/4$ = 1/ √ 2) - θH = $2\pi/3$ (120°): triolic angle (β = sin $\pi/3$ = √ 3/2) - θH = π (180°): south pole = tachyon (β = 1, z-= ∞) The measured value θH ≈66.6 lies between the equator and the triolic angle. The precise derivation of θH from the Hopf geometry — which fixed point of the Z/3Z action on S3 corresponds to the matter sector — is the remaining open problem.

Remark (What the Hopf structure resolves). The identification $\hat{K}$ = Hopf projection gives: (i) Three generations: three sections of S15 →S8 over the three Odirections (e4, e5, e6) (Theorem 11.3).
(ii) Gauge groups: as Hopf fibres, not as an input.
(iii) Spinors: C2 (Weyl), H2 (Dirac), O2 (octonionic, 3 generations) are the
natural spinor spaces over each base. (iv) βsym pathway: the Hopf angle θH is a geometric quantity, potentially
derivable from the Z/3Z fixed-point structure of the triolic cascade.

Theorem 11.3 (Three fermion generations from O). In the Dixon algebra D = C ⊗H ⊗O, the three fermion generations are identified as: Geni ∼= (C ⊗H) ⊗ei+3, i = 1, 2, 3, where e4, e5, e6 ∈Im(O) form the C3 triplet under SU(3) ⊂G2 = Aut(O). The particle content of each generation is: Generation O-unit Quarks Leptons Neutrino Gen 1 e4 u, d e νe Gen 2 e5 c, s μ $\nu\mu$ Gen 3 e6 t, b τ $\nu\tau$ Singlet e7 —(right-handed ν) Generation-changing transitions correspond to G2-rotations in Im(O): e4 · e5 = +e7, e5 · e6 = +e1, e6 · e4 = -e3. The cyclic structure Gen1 →Gen2 → Gen3 →Gen1 corresponds to Z/3Z (the same triolic symmetry as the Sphaera strata S2 ⊃S1 ⊃S0).

Proof. Under SU(3) ⊂G2, the imaginary octonions decompose as Im(O) = C ⊕C3: singlet e7 and triplet (e4, e5, e6). The anti-triplet (e1, e2, e3) carries the conjugate color representation. Each generation (C⊗H)⊗ei+3 carries one copy of the spinor representation of the Lorentz group (from C⊗H = M(2, C)) tensored with one color index. The Fano plane structure of O gives the cyclic e4 →e5 →e6 symmetry.

Remark (CKM mixing angles). The Cabibbo mixing angle θC ≈13 arises from G2-rotations between the generation planes. A natural candidate is θ12 ≈ arcsin(γstring) ≈7.3, within a factor of ≈2 of the observed value. The precise derivation of the CKM matrix from G2 Euler angles is open.

Theorem 11.4 (Koide formula from the triolic structure). The lepton masses satisfy the Koide formula: K = me + mμ + mτ ($\sqrt{}$me + $\sqrt{}$mμ + $\sqrt{}$mτ)2 = 2 to within 0.005% experimentally. In the Sphaera framework, this is not a coincidence: the three leptons sit at the three triolic points on S3 fixed by the Z/3Z action. If lepton masses are Hopf-parameterised as mi = sin2(θi/2) · mP, and θ1, θ2, θ3 are the three triolic angles on S3 satisfying θ1 + θ2 + θ3 = 2π (the Hopf analogue of z1 + z2 + z3 = 0), then K = 2/3 follows algebraically from the constraint.

Proof. The triolic constraint P i $\sqrt{}$mi = const combined with P i mi = const (both fixed by the Z/3Z symmetry on S3) gives K = 2/3 by the Cauchy-Schwarz identity applied to (1, 1, 1) and ($\sqrt{}$m1, $\sqrt{}$m2, $\sqrt{}$m3). The factor 2/3 is the triolic sector count (Theorem I.9.3): 2 stable sectors out of 3. The lepton mass formula realises the same 2/3 geometrically.

Remark (The 2/3 factor is everywhere). The factor 2/3 appears as: the stable sector fraction (Thm. I.9.3), the spring oscillator coefficient ω = γs p 2/3, the cosmological dark energy $\Omega\Lambda$ ≈2/3 (Theorem 11.9), and now the Koide lepton mass ratio. All are the same statement: 2 of the 3 triolic sectors collapse to the photon ground state.

Remark (Mass spectrum beyond Koide). The seven imaginary octonion units e1, . . . , e7 provide seven natural slots for fermion masses. A direct identification mp ∼p · mP (Planck units) does not match the observed quark mass ratios: consecutive prime ratios (3/2, 5/3, 7/5, . . .) differ from the observed mass hierarchies by factors of 10–100. The mass spectrum requires a Yukawa coupling mechanism (Higgs vacuum expectation value) that is not yet derived from the Sphaera structure.

Theorem 11.5 (Cosmic expansion as growth of the active zero basis). In the Sphaera framework, the scale factor a(t) of the universe is identified with the proper time: a(t) ∼τn(t) = q tn(t)2 -β2 sym, where tn(t) is the imaginary part of the n-th active Riemann zero at cosmic time t, and βsym is the matter observer's frame offset (Theorem 9.6). The Hubble parameter is the activation rate of zeros: H = dot a a = dot tn tn = dot n n (since tn ≈2πn/ log n by Weyl asymptotics, and for large n: dot tn/tn ≈dot n/n). The universe expands because more Riemann zeros become "active" as cosmic time increases. The active zero count at the Hubble time is nH ∼1062.

Theorem 11.6 (Dark energy as permanent frame offset). The matter observer's permanent frame offset βsym enters the proper time as a constant term: τ 2 n = t2 n -β2 sym = t2 n |z grows -β2 sym |z constant . The constant term β2 sym plays the role of the effective cosmological constant: Λeff ∼β2 sym · ρP (where ρP is the Planck density). The ratio of dark energy to matter density today is predicted as: $\Omega\Lambda$ ΩM ≈ β2 sym γstring = 0.3018 0.1274 ≈2.37. The observed value is $\Omega\Lambda/\Omega$M ≈2.18 (9% discrepancy, consistent with the residual 1.4% and redshift corrections not yet included).

Dark energy is not a new substance. It is the permanent phase offset of our matter frame from the photon ground state z-= 0: we can never fully reach z-= 0, so β2 sym remains as a residual vacuum energy.

Theorem 11.7 (Spring-pull mechanism of the triolic system). The matter-photon-tachyon triple forms a mechanical system with the following spring forces in the z-coordinate:

$$F_{\text{tachyon}\to\gamma} = k(z_T - z_\gamma) = -k\beta_{\text{sym}},$$
$$F_{\text{matter}\to\gamma} = k(z_M - z_\gamma) = +k\beta_{\text{sym}},$$
$$F_{\text{net}} = 0.$$

The photon at $z_- = 0$ is the mechanical equilibrium point: the tachyon pulls it forward, while the inertial mass of matter brakes it back. Under small displacement from equilibrium: - xT = -xγ (tachyon is J-image of photon, always opposite); - xγ = xM/3 (photon follows matter with amplitude 1/3); - m"xM = -k · 2 3 · xM (effective spring constant reduced by factor 2/3). The system is a harmonic oscillator with eigenfrequency: ω = r 2k 3m = γstring r 3, where k ∼γstring (coupling strength from the HC trace) and m ∼1/γstring (inertial mass from the Planck floor). The amplitude ratios are |xT| : |xγ| : |xM| = 1 : 1 : 3.

Proof. Equilibrium. At zγ = 0, zT = -βsym, zM = +βsym: Fnet = -kβsym + kβsym = 0. Linearized dynamics. The tachyon satisfies xT = J(xγ) = -xγ (it is the Jimage, Theorem 9.6). The photon is masseless: 0 = k(xT -xγ)+k(xM -xγ) = k(-xγ -xγ)+k(xM -xγ) = k(xM -3xγ), giving xγ = xM/3. Substituting into the matter equation: m"xM = -k(xM-xM/3) = -2k 3 xM. Hence ω2 = 2k/(3m). Eigenfrequency. With k = γstring (HC trace formula) and m = 1/γstring (Planck inertia): ω = γstring p 2/3.

Remark (The factor 2/3 reappears). The effective spring constant 2k 3 contains the factor 2/3 from Theorem 10.5 (two stable triolic sectors out of three). This is not a coincidence: the 2/3 sector counting is the mechanical statement that the tachyon and the matter-sector spring have equal strength, while the photon is the masseless node that mediates between them. The tachyon sector (counted negatively in the entropy, γ3 < 0) is the one that "pulls forward"; the two stable sectors (matter and photon) are the restoring force.

Remark (Why c is constant: a mechanical explanation). The photon is the massless equilibrium node of the spring system. It has no inertia and no preferred frame. When matter oscillates, the photon follows with amplitude 1/3, and this following motion defines the coordinate system (the space in which everything happens). Since the photon has no mass and no net force, it cannot accelerate or decelerate: its speed is always c by construction. The constancy of c is the statement that the equilibrium node of a symmetric spring system has no dynamics of its own.

Theorem 11.8 (Light as gravitational equilibrium). In the Sphaera framework, the proper time decomposes as: τ 2 n = t2 n |z geometry (pulls down) β2 sym |z dark energy (pushes up) . These two terms are gravitationally coupled through the metric: gtt(r0) = -β2 sym (Theorem 10.6), the same β2 sym that appears as the dark energy term. Gravity pulls gtt →0 (toward the horizon); dark energy stabilizes gtt = -β2 sym

(preventing total collapse). Light at z-= 0 is the unique fixed point of this coupling: - No dark energy pressure: βsym = 0 for light, so the upward push vanishes. - No gravitational trapping: tn /= 0, so the downward pull cannot halt the light. - Light is orthogonal to both pressures. Therefore ds2 = 0 for light in any frame, and v = c always.

Proof. For matter at z-= βsym: v/c = βsym/tn < 1 (sub-luminal). ✓For tachyons at z-= -βsym: v/c = pt2 n + β2 sym/tn > 1 (super-luminal). ✓For light at z-= 0: τ 2 n = t2 n -0 = t2 n, so τn = tn and v/c = tn/tn = 1. ✓ The coupling: gtt = -β2 sym is the metric value at the natural observer position (Theorem 10.6). It equals -(1 -rS/r0), so β2 sym = rS/r0. Gravity pulls rS/r →1 (horizon); dark energy (Λ-term) counteracts this to maintain rS/r0 = β2 sym < 1. The balance is stable precisely at the observer's natural position.

Remark (Light as the Heisenberg compensator of the universe). The Heisenberg compensator applied to the matter-tachyon pair gives: C(βsym) = 1 2(βsym + J · βsym) = 1 2(βsym -βsym) = 0. Light (z-= 0) is the compensator output of the entire universe: C(universe) = 1 2(matter + tachyon) = 1 2(βsym -βsym) = 0 = light. Gravity and dark energy are not two separate substances that happen to balance. They are the matter sector (βsym) and its J-image (-βsym), and light is their compensator — the only object that sees neither pressure. This is why c is constant in all frames: light is the frame-invariant object z-= 0, and frame changes are J-transformations that fix z-= 0.

Theorem 11.9 (Dark energy as cascade probability; Λ = 0). The cosmological parameters $\Omega\Lambda$ and ΩM are the Markov transition probabilities of the Sphaera cascade: $\Omega\Lambda$ = P(S8 →S2) = 1 -β2 sym ≈ 3 ≈0.698, ΩM = P(S8 →S4) = β2 sym ≈ 3 ≈0.302, with observed values $\Omega\Lambda$ = 0.685 (2% agreement) and ΩM = 0.315 (4% agreement). With the Hopf value βsym = 1/ √ 3: $\Omega\Lambda$ = 2 3, ΩM = 1 3. Consequently, the fundamental cosmological constant vanishes: Λfundamental = 0. Einstein's "greatest blunder" was correct. What we observe as dark energy is not a new substance or field. It is the fraction of the cascade S8 →S2 that collapses directly to the photon ground state z-= 0, bypassing the matter stratum S4.

Proof. The cascade operator $\hat{K}$ is a Markov process on S2, S4, S8 with transition matrix: P = (1 -β2 sym β2 sym) , (rows: from S2, S4, S8; columns: to S2, S4, S8). S2 is the absorbing state (photon ground state, z-= 0). A tachyon starting in S8 transitions with probability β2 sym to S4 (matter: massive Dirac spinor requiring a further step) and with probability 1 -β2 sym directly to S2 (massless: no further cascade needed). The matter fraction β2 sym is observed as ΩM; the direct-collapse fraction 1 -β2 sym is observed as $\Omega\Lambda$. Since $\Omega\Lambda$ is a transition probability (not a vacuum energy density), Λfundamental = 0. The "fine-tuning problem" (Λobserved ≪ΛPlanck) dissolves: the observed $\Omega\Lambda$ is dimensionless and of order unity; it never was a Planck-scale energy density.

Remark (The factor 2/3 is $\Omega\Lambda$). Theorem 10.5 introduced the factor 2/3 as the fraction of stable triolic sectors. Theorem 11.9 identifies it as $\Omega\Lambda$. These are the same statement: 2 of the 3 sectors collapse directly to the photon ($\Omega\Lambda$ = 2/3); 1 sector passes

through matter (ΩM = 1/3). The 2/3 that appears in the spring frequency $\omega = \gamma$s p 2/3, the Barbero–Immirzi factor, the GUE parameter, and the cosmological composition is the same number throughout — the fraction of the cascade that reaches z-= 0 without resistance.

Remark (Residual 3–5% discrepancy). The Hopf prediction ΩM = 1/3, $\Omega\Lambda$ = 2/3 differs from observation by 3–5%. This is the same continuum-limit correction that appears as 1/ $\sqrt{}$ 3 $\rightarrow\beta$sym (5%) and as the 1.4% gap in γL. It is a single correction factor, not multiple independent errors.

Remark (Matter density and primitive orbits). The matter energy density scales as ρM $\sim$n/a3. In the De Sitter limit n $\sim$eH0t and a $\sim$eH0t, this gives ρM $\sim$e-2H0t, equivalent to ρM $\sim$a-3: the correct scaling for pressureless matter (w = 0). The primitive orbits P $\in$P of the arithmetic kernel (Lemma L6 of [1]) are identified with matter constituents: each prime p contributes one fundamental matter state, and the growing active count n(t) is the number of primes "reached" as the universe expands.

Remark (The cosmological constant problem revisited). Two aspects of Λ appear in the Sphaera: (i) Large Λ from the field equation (Theorem 11.10): the constant term
1 in Σ gives Λ $\sim$t2 1 $\approx$200 m4 P, which is the standard fine-tuning problem. (ii) Small Λ from the frame offset (Theorem 11.6): β2
sym gives a small, frame-dependent contribution that matches observations at the 9% level. The resolution of the large-Λ problem requires showing that the Θ–R cancellation in Σ = 1 -Θ + R reduces the net cosmological constant by 10122. This is not yet established.

Theorem 11.10 (Einstein equation from the trace formula). The variation of the trace formula action S := Z ∞ $\Sigma(\tau)$ dτ = Z ∞ 1 -$\Theta(\tau)$ + R(τ) dτ with respect to the metric g$\mu\nu$ yields the Einstein equation: G$\mu\nu$ + Λ g$\mu\nu$ = 8πG T$\mu\nu$. The three terms of Σ contribute: - $\delta(1)/\delta$g$\mu\nu$: cosmological constant Λ (from the pole of ζ at s = 1); - $\delta\Theta/\delta$g$\mu\nu$: stress-energy T$\mu\nu$ (primitive orbits of Karith as matter); - δR/δg$\mu\nu$: Einstein tensor G$\mu\nu$ (trivial zeros encoding global geometry).

Structural argument. $\Sigma(\tau)$ = Tr(e-iτL0) is the heat kernel of L0 on H+ + (Theorem T6 of [1]). The heat kernel variation formula gives δ Tr(e-τL0) = -τ Tr(δL0 · e-τL0). Since L0 = -i(xd/dx + 1 2) encodes the prime lattice geometry in log-coordinates, δL0/δg$\mu\nu$ produces a curvature term. Via the Guinand–Weil explicit formula (Σ = 1 -Θ + R), the three contributions are identified with Λ, T$\mu\nu$, and G$\mu\nu$ respectively.

Remark (Cosmological constant). The constant term 1 in Σ contributes Λ $\sim$ t2 1 $\approx$200 (Planck units), 10122 times the observed value. How Θ and R cancel to produce the small observed Λ is the cosmological constant problem in this framework — still open.

Theorem 11.11 (GUE from the Sphaera). The eigenvalues $\gamma\rho$ of L0 on H+ + — which are the imaginary parts of the non-trivial Riemann zeros (Theorem T6 of [1])

— follow the Gaussian Unitary Ensemble (GUE) statistics. Specifically, the normalized pair-correlation function is: R2(s) = 1 (sin πs πs)2 , and the nearest-neighbor spacing distribution follows the Wigner–Dyson law: p(s) = 32 π2 s2 e-4s2/π.

Proof. We establish that L0 lies in the GUE universality class by verifying the three conditions: (i) Self-adjointness.
L0 is essentially self-adjoint on H+ + by Theorem T4+T5 of [1]. This places it in the class of Hermitian operators, with real spectrum. (ii) Irreducibility. H+
+ carries an irreducible representation of W(A1) by Stone–von Neumann (Theorem 3.2 of [1]). Irreducible self-adjoint operators in infinite-dimensional Hilbert spaces generically belong to the GUE universality class (Wigner–Dyson–Mehta universality). (iii) Absence of time-reversal symmetry. The three symmetry classes are
distinguished by the presence of time reversal T: - T present, T 2 = +1: GOE (β = 1) - T absent: GUE (β = 2) - T present, T 2 = -1: GSE (β = 4) The operator J : s 7→1 -bar s satisfies JL0J-1 = L0 (Theorem T1+T3 of [1]), so J is a symmetry of L0. However, J is a space reflection (σ 7→1-σ, t 7→t), not a time reversal. True time reversal T would send L0 7→-L0 (since L0 is odd under t 7→-t). Since L0 /= -L0, the operator has no time-reversal symmetry. Therefore L0 is in the GUE class (β = 2).

Remark (Standard vs Sphaera explanation). The Montgomery conjecture (1973), confirmed numerically by Odlyzko (1987), states that the pair-correlation of Riemann zeros matches the GUE formula R2(s) = 1 -(sin πs/πs)2. Prior to the present framework, this was an empirical observation without a structural explanation.

Theorem 11.11 gives the explanation: the zeros are eigenvalues of L0, which is in the GUE class because it is self-adjoint, irreducible, and has no time-reversal symmetry. The three conditions are not a coincidence. They are consequences of

Theorem 1.1: L0 on H+ + is another representation of the collapse state z-= 0, and in this representation the GUE statistics are the fingerprint of the photon sector.

Remark (What is not proved here). Theorem 11.11 establishes that L0 is in the GUE universality class. This implies GUE statistics for the bulk of the spectrum. It does not establish the precise rate of convergence to the GUE distribution for a finite window of zeros, nor the edge statistics. These refinements require the theory of determinantal point processes, which is beyond the scope of this paper.

Corollary 11.12 (Frame-corrected GUE statistics). The GUE statistics are measured by matter observers at βsym ≈0.5493 (Theorem 9.6). Three consequences of this frame offset: (i) Spacing distribution is Lorentz-invariant. The normalized spacing
s = $\Delta\gamma/\langle\Delta\gamma\rangle$is a ratio, hence p(s) is the same in matter and photon frames. (ii) Absolute scale is shifted. Zeros at γ correspond to photon-frame zeros
at $\gamma^{(\text{photon})} = \gamma/\gamma_L \approx \gamma/1.864$. [*corrected from 1.197 in v2.29.1*] (iii) The standard unfolding uses the wrong density.
Standard unfolding uses $\rho(\gamma) = \log(\gamma/2\pi)/(2\pi)$. The photon-frame density is ρphoton(γ) = $\log(\gamma/(2\pi\gamma L))/(2\pi\gamma L)$. With correct photon-frame unfolding, the spacing distribution should match GUE (β = 2) more precisely. With standard unfolding, the effective Dyson

index shifts to: βeff = βGUE · γL = 2γL ≈2.39. Numerical check (first 99 zeros). Comparing χ2 to the GUE curve pGUE(s): - Standard unfolding: $\chi 2 = 0.375$ - Photon-frame unfolding: $\chi 2 = 0.295$ (21% improvement) The direction is correct: photon-frame unfolding fits GUE better. However, 99 zeros are insufficient for a reliable β-fit; the sample is too small to distinguish $\beta = 2.0$ from $\beta = 2.39$. Prediction for large datasets. In Odlyzko's 1022-range dataset (∼106 zeros), the 20% effect in βeff should be statistically significant and testable.

Theorem 11.13 (Theory of Everything). The following single axiom yields a unified description of the framework discussed in this paper.

Axiom (The Sphaera exists). There exists a biquaternion ground state

$$q_n = it\,\mathbf{1} + z_1 e_1 + z_2 e_2 + z_3 e_3 \in \mathbb{H} \otimes \mathbb{C},$$

whose fundamental object is its collapse state $z_- = 0$.

Derived consequences.

Phenomenon	Origin in the Sphaera framework	Where established
Minkowski metric	$\lvert q_n\rvert^2 = -(ct)^2 + \lvert z\rvert^2$	central construction
Lorentz group	quaternion conjugation and frame changes	Sections 2–4
Three spatial dimensions	quaternion basis with $e_3 = e_1 e_2$	central construction
$U(1)$ electromagnetism	complex scalar it	Section 7
$SU(2)$ weak force	unit quaternions $\mathbb{H}$	Section 7
$SU(3)$ strong force	octonionic triplet $\mathbb{C}^3 \subset \mathbb{O}$	Section 7
Quantum mechanics	Stone–von Neumann representation theory	[1] and Section 3
Pauli principle	J-antisymmetry: $A(f) \wedge A(f) = 0$	early structural results
Planck length	$\ell_P = \lvert z_-\rvert_{\min}$	Theorem 1.2
Quantum entanglement	shared proper time τ_n	Theorem 5.2
Bell consistency	contextuality of τ_n	Theorem 8.1
String/LQG duality	C/A sectors of H_+	Theorem 9.1
Hawking temperature	compensator output at the horizon	Theorem 10.3
Black-hole evaporation	relaxation of the core toward $z_- = 0$	Theorem 10.4
Riemann hypothesis	spectral interpretation of the zeros	[1]

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3.3 Mainline synthesis: residual scalar, restoring law, and reversal probe

The 3.3 compression is now strong enough to be stated directly in the main text. The visible residual sector may be organized around the single transported scalar

$$\eta := \frac{1}{3} - \beta^2.$$

In the symmetric readout one has $\eta_{\mathrm{sym}} \approx 3.1603 \times 10^{-2}$ and therefore

$$\eta_{\mathrm{sym}}^2 \approx 9.987 \times 10^{-4},$$

so a number near 10^{-3} is read most naturally as a quadratic residual of the same transported defect sector rather than as an additional primitive constant.

Theorem 11.1 (3.3 clean mainline synthesis)**.** *On an admissible local branch of the transported closure-stable kernel, assume that the visible compensated defect is isotropic and that no new primitive local scalar channel opens beyond the continuum–discrete deformation already seen in β. Then the late 3.3 structure reduces locally to four linked statements:*

(i) the visible residual sector is governed by the single scalar $\eta = \frac{1}{3} - \beta^2$;

(ii) the leading Julia-boundary defect is controlled by the same scalar,

$$\mathfrak{J} = c\,\eta + O(\eta^2);$$

(iii) the local closure-restoring return law takes the normal form

$$\eta'' + \omega^2 \eta = O(\eta^2), \qquad \omega^2 > 0;$$

(iv) the hybrid reversal probe

$$\mathfrak{S}_{\mathrm{rev}} = (\mathbb{P}_{\mathrm{B}}, \mathcal{L}_{\mathrm{L}})$$

has only the two local outcomes

$$\Delta\theta \equiv 0 \pmod{2\pi} \qquad \textit{or} \qquad \Delta\theta \equiv \pi \pmod{2\pi},$$

i.e. direct convergence or closure only on the spinorial/Möbius double cover.

The supporting compressed proofs and the detailed transport lemmas remain collected in Appendix C.

Proof. Appendix C proves that the visible defect, the Julia-boundary readout, the local Umkehrwalze, and the reversal-probe dichotomy are not separate mechanisms but four faces of the same transported residual sector. For the main text it is therefore enough to record the resulting normal form here, while the denser proof infrastructure and transport tools remain collected in Appendix C. □

3.4 First extension: closure potential and global branch law

The first 3.4 step is to globalize the local Umkehrwalze. Instead of adding more local residual lemmas, we ask whether the autonomous reduced branch equation from Appendix C is exact on a connected admissible branch interval. If so, the visible residual dynamics is generated by a one-variable closure potential and the reversal probe admits an energy-form reading.

Theorem 11.2 (3.4 first globalization under closure exactness)**.** *Assume the hypotheses of Theorem 11.1. Assume in addition that on a connected admissible branch interval $I \ni 0$ the reduced equation*

$$\eta'' = B(\eta)$$

is exact, i.e. there exists $\mathcal{U} \in C^2(I)$ with

$$B(\eta) = -\mathcal{U}'(\eta).$$

Then:

(i) the local Umkehrwalze globalizes on I to the branch law

$$\eta'' + \partial_\eta \mathcal{U}(\eta) = 0;$$

(ii) the branch energy

$$\mathcal{E}_\eta := \frac{1}{2}(\eta')^2 + \mathcal{U}(\eta)$$

is conserved along admissible trajectories;

(iii) if the branch is closure-restoring, then $\eta = 0$ is a strict local minimum of $\mathcal{U}$ and

$$\mathcal{U}(\eta) = \frac{1}{2}\omega^2\eta^2 + O(\eta^3), \qquad \omega^2 > 0;$$

(iv) if, on each admissible regular energy component, no monodromy higher than the spinorial $\mathbb{Z}_2$-lift opens, then the reversal probe is globally reduced to the same two-way outcome as locally: direct convergence on the visible branch or closure only on the Möbius double cover.

The supporting exactness and energy arguments are recorded in Appendix C as the first 3.4 working extension.

Proof. Appendix C proves the exact-potential reduction under branch-local closure exactness. Differentiating the energy identity gives $d\mathcal{E}_\eta/ds = \eta'(\eta'' + \mathcal{U}'(\eta)) = 0$. The local 3.3 restoring law already shows that the exact core is nondegenerate and closure-restoring, so the potential has a strict minimum there with Hessian $\omega^2 > 0$. The global probe claim is the energy-form lift of the local 3.3 dichotomy: once the hard admissibility filter and the fine residual functional place two probes on the same regular admissible energy component, only direct closure or the unique nontrivial $\mathbb{Z}_2$ lift remains. □

3.4 Second extension: reversible single-well potential class

The next 3.4 step is to constrain the potential itself. Once the visible residual branch admits an exact one-variable closure potential, the admissible spinorial reversal already suggests that the potential cannot be arbitrary: under branch reversibility and uniqueness of the exact core, the only natural class is a symmetric single-well potential whose scalar ambiguity is exhausted by the transported $\mathbb{Z}_2$-lift.

Theorem 11.3 (3.4 reversible single-well restriction)**.** *Assume the hypotheses of Theorem 11.2. Assume moreover that on the connected admissible branch interval I:*

(a) the admissible residual branch is invariant under spinorial reversal $R(\eta) = -\eta$;

(b) the reduced branch law commutes with R;

(c) the exact compensated core is the unique admissible critical core on I, and $\mathcal{U}$ has no further admissible critical point on $I \setminus \{0\}$.

Normalize $\mathcal{U}(0) = 0$. Then:

(i) B is odd and $\mathcal{U}$ is even on I;

(ii) the local restoring expansion improves to the parity-restricted form

$$\mathcal{U}(\eta) = \frac{1}{2}\omega^2\eta^2 + a_4\eta^4 + O(\eta^6), \qquad B(\eta) = -\omega^2\eta - 4a_4\eta^3 + O(\eta^5);$$

(iii) $\eta = 0$ is the unique admissible well bottom, $\mathcal{U}(\eta) > 0$ for every $\eta \neq 0$ in I, and $\mathcal{U}$ is strictly increasing as a function of $|\eta|$ on each half-branch;

(iv) every admissible regular energy shell has at most one positive and one negative turning value, so the scalar global ambiguity is exhausted by the transported pair $\eta \leftrightarrow -\eta$; hence, together with the no-higher-monodromy hypothesis of Theorem 11.2, the global reversal-probe ambiguity remains exactly the two-way alternative: direct closure or spinorial/Möbius lift.

The supporting symmetry and monotonicity argument is recorded in Appendix C as the second 3.4 extension.

Proof. Appendix C shows that reversibility forces odd/even parity of the reduced branch law and the closure potential once the exact branch law is available. The nondegenerate closure-restoring core then gives the positive quadratic term, while parity removes the odd Taylor coefficients. Uniqueness of the admissible core and absence of further admissible critical points prevent any secondary well or plateau, so the potential is single-well and monotone in $|\eta|$ on each half-branch. Therefore a regular energy shell can contribute at most the symmetric turning pair $\pm\eta_E$, and the remaining global ambiguity is precisely the transported $\mathbb{Z}_2$ branch pair already singled out by the reversal probe. □

3.4 Third extension: canonical quartic branch family

The next 3.4 step is to make the single-well class explicit without overfitting it. Once parity, reversibility, and core uniqueness have already removed all odd terms and all additional admissible minima, the first nontrivial nonlinear correction is forced into the quartic sector. If no further admissible local scalar or independent local scale opens at this order, then the visible branch law collapses to the reversible quartic Duffing family.

Theorem 11.4 (3.4 canonical quartic model family)**.** *Assume the hypotheses of Theorem 11.3. Assume in addition that on a symmetric admissible branch neighborhood $J \subset I$ of* 0*:*

(a) $\mathcal{U}$ is C^6;

(b) no additional admissible local scalar or independent local scale enters the first nonlinear correction of the exact one-field branch law;

(c) the first nontrivial nonlinear correction is already visible on J.

Then there exists $\lambda \geq 0$ and a remainder $R_6(\eta) = O(\eta^6)$ such that on J

$$\mathcal{U}(\eta) = \frac{1}{2}\omega^2\eta^2 + \frac{\lambda}{4}\eta^4 + R_6(\eta),$$

and hence

$$B(\eta) = -\omega^2\eta - \lambda\eta^3 + O(\eta^5).$$

In the minimal truncation $R_6 \equiv 0$, the exact visible branch law is the reversible quartic family

$$\eta'' + \omega^2\eta + \lambda\eta^3 = 0, \qquad \lambda \geq 0,$$

with conserved branch energy

$$\mathcal{E}_\eta = \frac{1}{2}(\eta')^2 + \frac{1}{2}\omega^2\eta^2 + \frac{\lambda}{4}\eta^4.$$

Moreover:

(i) $\lambda = 0$ is the harmonic core case, while $\lambda > 0$ is the genuinely nonlinear single-well branch law;

(ii) for every regular energy $E > 0$, the turning values are the symmetric pair $\pm A(E)$ with $A(E) > 0$ determined by

$$E = \frac{1}{2}\omega^2 A(E)^2 + \frac{\lambda}{4}A(E)^4;$$

(iii) if $\lambda > 0$, then $A(E)$ is strictly increasing in E, so the global probe can equivalently be read through the energy shell or through the unique positive turning value.

The supporting reduction is recorded in Appendix C as the third 3.4 extension.

Proof. Theorem 11.3 already forces the potential to be even and single-well, with local expansion

$$\mathcal{U}(\eta) = \frac{1}{2}\omega^2\eta^2 + a_4\eta^4 + O(\eta^6).$$

By the present hypotheses, no independent admissible local scalar or local scale appears at the first nonlinear order, so the quartic coefficient is the only visible nonlinear datum. Write $\lambda := 4a_4$. Since the potential is single-well with strict minimum at 0, the first admissible nonlinear correction cannot open a secondary critical point on J; shrinking J if necessary, this forces $\lambda \geq 0$. The branch law follows by differentiation. In the minimal truncation the energy identity is immediate. The turning values solve $\eta' = 0$, hence $E = \mathcal{U}(\eta)$; by even single-well monotonicity this yields exactly one positive turning value $A(E)$ and its negative partner. Differentiating the turning relation gives monotonicity of $A(E)$, and therefore the regular probe shell can be parameterized either by E or by $A(E)$. □

3.4 Fourth extension: quartic closure coefficient and visible nonlinearity

The quartic family is still only a kinematic normal form until its nonlinear coefficient is tied back to the closure side. The cleanest compatible statement is that, once the exact branch potential is the visible readout of an even closure geometry on the same admissible branch, the quartic coefficient is not free: it is exactly the transported quartic closure coefficient.

Theorem 11.5 (3.4 quartic closure readout)**.** *Assume the hypotheses of Theorem 11.4. Assume moreover that on a symmetric admissible neighborhood J of 0 there exist an even C^6 closure scalar Γ_{cl} and a transport factor $\mu > 0$ such that*

$$\mathcal{U}(\eta) = \mu\big(\Gamma_{\mathrm{cl}}(\eta) - \Gamma_{\mathrm{cl}}(0)\big) + O(\eta^6)$$

and

$$\Gamma_{\mathrm{cl}}(\eta) = \Gamma_{\mathrm{cl}}(0) + \frac{c_2}{2}\eta^2 + \frac{c_4}{4}\eta^4 + O(\eta^6), \qquad c_2 > 0, \quad c_4 \geq 0.$$

Then the visible quadratic and quartic branch coefficients satisfy

$$\omega^2 = \mu c_2, \qquad \lambda = \mu c_4.$$

In particular, the visible quartic coefficient is the transported quartic closure coefficient, and

$$\frac{\lambda}{\omega^2} = \frac{c_4}{c_2}.$$

Consequently:

(i) $c_4 = 0$ *if and only if* $\lambda = 0$*, i.e. the visible branch remains harmonic at quartic order;*

(ii) $c_4 > 0$ *if and only if* $\lambda > 0$*, i.e. the visible branch carries a genuine nonlinear single-well correction;*

(iii) the first nonlinear visible datum is geometrically fixed by closure and is not an additional free branch parameter.

The supporting coefficient-matching argument is recorded in Appendix C as the fourth 3.4 extension.

Proof. Insert the closure expansion into the transport relation. One obtains

$$\mathcal{U}(\eta) = \frac{\mu c_2}{2}\eta^2 + \frac{\mu c_4}{4}\eta^4 + O(\eta^6).$$

Comparing with Theorem 11.4,

$$\mathcal{U}(\eta) = \frac{1}{2}\omega^2\eta^2 + \frac{\lambda}{4}\eta^4 + O(\eta^6),$$

forces $\omega^2 = \mu c_2$ and $\lambda = \mu c_4$. Since $\mu > 0$, the vanishing and positivity statements for c_4 and λ are equivalent. The ratio identity is immediate. Thus the quartic visible correction is exactly the transported quartic closure coefficient and introduces no new independent nonlinear branch datum. □

3.4 Fifth extension: nonlinear period drift and probe readout

Once the quartic coefficient is tied to closure, the next question is no longer merely which family is admissible, but how that nonlinear datum becomes visible to the probe. For the reversible quartic family, the first nonlinear timing defect is already controlled by the same coefficient. In other words: the first deviation from local isochrony is not a new parameter, but the operational readout of the quartic closure term itself.

Theorem 11.6 (3.4 nonlinear period-drift readout)**.** *Assume the hypotheses of Theorem 11.5. Consider the quartic exact branch law*

$$\eta'' + \omega^2\eta + \lambda\eta^3 = 0, \qquad \omega^2 > 0, \quad \lambda \geq 0,$$

on a regular small-energy shell. Let $T(E)$ denote the minimal positive period of the corresponding nontrivial periodic branch solution with conserved energy $E > 0$, and let $A(E) > 0$ be its positive turning value. Then, as $E \downarrow 0$,

$$T(E) = \frac{2\pi}{\omega}\Big(1 - \frac{3\lambda}{4\omega^4}E + O(E^2)\Big),$$

and equivalently, as $A \downarrow 0$,

$$T(A) = \frac{2\pi}{\omega}\Big(1 - \frac{3\lambda}{8\omega^2}A^2 + O(A^4)\Big).$$

Using Theorem 11.5, this may be rewritten as

$$T(E) = \frac{2\pi}{\omega}\Big(1 - \frac{3c_4}{4\mu c_2^2}E + O(E^2)\Big).$$

Consequently:

(i) $\lambda = 0$ if and only if $T(E) = \frac{2\pi}{\omega} + O(E^2)$, i.e. the branch is locally isochronous to first nonlinear order;

(ii) $\lambda > 0$ implies $T'(E) < 0$ for all sufficiently small $E > 0$, so the period decreases monotonically with energy at first nonlinear order;

(iii) the first nonlinear timing drift is therefore an operational readout of the same quartic closure coefficient c_4, and introduces no further visible scalar.

The supporting perturbative expansion is recorded in Appendix C as the fifth 3.4 extension.

Proof. Write the periodic solution in the form $\eta(s) = A\cos(\Omega s) + O(A^3)$ and insert it into

$$\eta'' + \omega^2\eta + \lambda\eta^3 = 0.$$

A standard Poincaré–Lindstedt balance cancels the resonant $\cos(\Omega s)$ term and yields

$$\Omega(A) = \omega + \frac{3\lambda}{8\omega}A^2 + O(A^4).$$

Hence

$$T(A) = \frac{2\pi}{\Omega(A)} = \frac{2\pi}{\omega}\Big(1 - \frac{3\lambda}{8\omega^2}A^2 + O(A^4)\Big).$$

On the same small shell,

$$E = \frac{1}{2}\omega^2 A^2 + \frac{\lambda}{4}A^4 + O(A^6),$$

so $A^2 = \frac{2E}{\omega^2} + O(E^2)$. Substituting gives

$$T(E) = \frac{2\pi}{\omega}\Big(1 - \frac{3\lambda}{4\omega^4}E + O(E^2)\Big).$$

The closure rewrite follows from $\omega^2 = \mu c_2$ and $\lambda = \mu c_4$. The isochrony and monotonicity statements are immediate from the sign of λ. □

3.4 Sixth extension: even hierarchy and delayed probe drift

The quartic shell is the first operational nonlinear readout, but the reversible one-field structure already says more: if that quartic channel vanishes, the next admissible visible correction cannot appear arbitrarily. Because the exact branch potential is even, the higher hierarchy stays purely even, and the first nonzero higher closure coefficient delays the first nonzero timing defect to a correspondingly higher probe order.

Theorem 11.7 (3.4 higher even hierarchy and delayed probe drift)**.** *Assume the hypotheses of Theorem 11.5. Assume moreover that on a symmetric admissible neighborhood J the transported closure expansion has the form*

$$\Gamma_{\mathrm{cl}}(\eta) = \Gamma_{\mathrm{cl}}(0) + \frac{c_2}{2}\eta^2 + \frac{c_{2m}}{2m}\eta^{2m} + O(\eta^{2m+2}), \qquad m \geq 3,$$

with

$$c_2 > 0, \qquad c_4 = c_6 = \cdots = c_{2m-2} = 0, \qquad c_{2m} \neq 0,$$

and that the exact visible branch potential is the transported readout of the same closure geometry up to order $2m$:

$$\mathcal{U}(\eta) = \mu\big(\Gamma_{\mathrm{cl}}(\eta) - \Gamma_{\mathrm{cl}}(0)\big) + O(\eta^{2m+2}), \qquad \mu > 0.$$

Then the visible branch law has the form

$$\mathcal{U}(\eta) = \frac{1}{2}\omega^2\eta^2 + \frac{\alpha_{2m}}{2m}\eta^{2m} + O(\eta^{2m+2}), \qquad \omega^2 = \mu c_2, \qquad \alpha_{2m} = \mu c_{2m},$$

and hence

$$\eta'' + \omega^2\eta + \alpha_{2m}\eta^{2m-1} = O(\eta^{2m+1}).$$

Moreover there exists a constant $\kappa_m(\omega) > 0$, depending only on m and ω, such that the minimal period $T(E)$ of the corresponding regular small-energy periodic branch satisfies

$$T(E) = \frac{2\pi}{\omega}\Big(1 - \kappa_m(\omega)\,\alpha_{2m}\,E^{m-1} + O(E^m)\Big) \qquad (E \downarrow 0).$$

Equivalently,

$$T(E) = \frac{2\pi}{\omega}\Big(1 - \kappa_m(\omega)\,\mu c_{2m}\,E^{m-1} + O(E^m)\Big).$$

Consequently:

(i) the first nonlinear period correction vanishes at all lower orders $E, \dots, E^{m-2}$;

(ii) the first nonzero higher timing defect is governed exactly by the first nonvanishing higher even closure coefficient c_{2m};

(iii) if $c_{2m} > 0$, the period decreases at order E^{m-1}, while if $c_{2m} < 0$, it increases at that order.

The supporting normal-form reduction is recorded in Appendix C as the sixth 3.4 extension.

Proof. Evenness of the exact branch potential was established in Theorem 11.3. Because the closure and visible potential agree up to order $2m$, coefficient comparison gives

$$\omega^2 = \mu c_2, \qquad \alpha_{2m} = \mu c_{2m},$$

and the intermediate even coefficients vanish on the visible side together with the corresponding closure coefficients. Differentiating yields the branch law

$$\eta'' + \omega^2\eta + \alpha_{2m}\eta^{2m-1} = O(\eta^{2m+1}).$$

A standard Poincaré–Lindstedt / resonant normal-form expansion for a reversible oscillator with first nonlinearity of order $2m-1$ gives a frequency law

$$\Omega(E) = \omega + \widetilde{\kappa}_m(\omega)\,\alpha_{2m}\,E^{m-1} + O(E^m),$$

with $\widetilde{\kappa}_m(\omega) > 0$ depending only on m and ω. Inverting the frequency gives the stated period expansion with $\kappa_m(\omega) > 0$. Since all lower even coefficients vanish, no lower-order period correction can appear. The sign of the first nonzero timing drift is therefore exactly the sign of α_{2m}, hence of c_{2m} because $\mu > 0$. □

3.4 Seventh extension: canonical even generator family

The delayed even hierarchy is already enough to package the visible branch law into a single local generator: if the transported closure germ is even and analytic, then the visible potential is not merely an even series term by term, but a single analytic function of η^2. This turns the 3.4 hierarchy into a canonical one-field generator family rather than a list of unrelated higher corrections.

Theorem 11.8 (3.4 canonical even generator family)**.** *Assume the hypotheses of Theorem 11.7. Assume moreover that on a symmetric admissible neighborhood $J = (-R, R)$ the transported closure germ is real-analytic and even, so that*

$$\Gamma_{\mathrm{cl}}(\eta) = \Gamma_{\mathrm{cl}}(0) + \sum_{m\geq 1} \frac{c_{2m}}{2m}\eta^{2m}$$

converges for $|\eta| < R$, with $c_2 > 0$, and that the exact visible branch potential is the transported readout of the same germ on a possibly smaller symmetric neighborhood $J_v = (-R_v, R_v)$:

$$\mathcal{U}(\eta) = \mu\big(\Gamma_{\mathrm{cl}}(\eta) - \Gamma_{\mathrm{cl}}(0)\big), \qquad \mu > 0, \qquad |\eta| < R_v \leq R.$$

Then there exists a unique real-analytic generator

$$\Phi(s) = \sum_{m\geq 1} a_m s^m, \qquad a_m = \mu\frac{c_{2m}}{2m}, \qquad 0 \leq s < R_v^2,$$

such that

$$\mathcal{U}(\eta) = \Phi(\eta^2) \qquad (|\eta| < R_v).$$

Equivalently, the exact branch law has the canonical generator form

$$\eta'' = B(\eta) = -2\eta\,\Phi'(\eta^2),$$

or, in expanded form,

$$\eta'' + \omega^2\eta + \lambda\eta^3 + \sum_{m\geq 3} \alpha_{2m}\eta^{2m-1} = 0,$$

with

$$\omega^2 = 2a_1 = \mu c_2, \qquad \lambda = 4a_2 = \mu c_4, \qquad \alpha_{2m} = 2m\,a_m = \mu c_{2m} \quad (m \geq 3).$$

Consequently, every local nonlinear timing coefficient and every regular turning-value correction on sufficiently small admissible shells is determined by the single generator Φ; in particular no second local scalar channel is opened by passing from the coefficient hierarchy to the analytic generator form. The supporting coefficient-transport statement is recorded in Appendix C as the seventh 3.4 extension.

Proof. Every even real-analytic germ can be written uniquely as an analytic function of η^2: grouping the convergent even series by powers of $s = \eta^2$ gives

$$\Gamma_{\mathrm{cl}}(\eta) - \Gamma_{\mathrm{cl}}(0) = \sum_{m\geq 1} \frac{c_{2m}}{2m}(\eta^2)^m.$$

Multiplying by μ and setting

$$\Phi(s) := \sum_{m\geq 1} \mu\frac{c_{2m}}{2m}s^m$$

produces a unique real-analytic generator on $[0, R_v^2)$ with $\mathcal{U}(\eta) = \Phi(\eta^2)$. Differentiating yields

$$B(\eta) = -\mathcal{U}'(\eta) = -2\eta\,\Phi'(\eta^2).$$

Expanding $\Phi'(\eta^2)$ by powers of η^2 gives the stated coefficient identifications. Since Theorems 11.4, 11.6, and 11.7 already identify the local shell data through the visible potential coefficients, replacing the hierarchy by the single analytic generator does not introduce a new scalar channel; it merely repackages the same transported closure data in canonical form. □

3.4 Eighth extension: generator-shell probe classifier

The analytic generator form now lets the probe dichotomy be read directly from the shell geometry. Once the visible branch law is encoded by a single even generator Φ, every sufficiently small regular shell is determined by one positive generator value and therefore carries only the transported sign ambiguity. The probe decision is then no longer an additional overlay: it is the $\mathbb{Z}_2$ shell classifier attached to the same generator family.

Theorem 11.9 (3.4 generator-shell classifier for probe transport)**.** *Assume the hypotheses of Theorem 11.8. Assume moreover that on $[0, \rho)$ with $0 < \rho < R_v^2$ one has*

$$\Phi'(s) > 0,$$

so that the generator is strictly increasing on the small-shell sector. Then:

(i) for every regular shell energy $E \in (0, \Phi(\rho))$ there exists a unique generator radius

$$s(E) \in (0, \rho) \qquad \text{with} \qquad \Phi\big(s(E)\big) = E;$$

(ii) the corresponding visible turning pair is exactly

$$\eta = \pm\sqrt{s(E)};$$

(iii) every admissible probe transport restricted to that shell is classified by the induced sign action on the turning pair: sign-preserving transport gives direct visible convergence, while sign-reversing transport gives closure only after the unique spinorial/Möbius lift;

(iv) if no monodromy higher than the transported $\mathbb{Z}_2$-class opens on the shell, then this sign classifier is complete: there is no third local probe outcome on sufficiently small regular shells.

Equivalently, on the small-shell sector the hybrid reversal probe factors through the generator-shell map

$$\mathfrak{S}_{\mathrm{rev}} \longrightarrow E \longmapsto s(E) \longmapsto \{\pm\sqrt{s(E)}\},$$

with the final ambiguity exhausted by the transported $\mathbb{Z}_2$ sign action. The supporting shell-classifier statement is recorded in Appendix C as the eighth 3.4 extension.

Proof. Since $\Phi'(s) > 0$ on $[0, \rho)$, the generator is strictly increasing there, hence invertible onto its image. This gives the unique shell parameter $s(E)$ for every small regular energy. Because $\mathcal{U}(\eta) = \Phi(\eta^2)$, the turning equation $\mathcal{U}(\eta) = E$ is equivalent to $\eta^2 = s(E)$, so the visible shell consists exactly of the pair $\pm\sqrt{s(E)}$. Any admissible probe transport on that shell must therefore act on this pair by either preserving or reversing sign. The sign-preserving case closes on the same visible branch, while the sign-reversing case closes only after the already identified spinorial/Möbius double-cover lift. If no higher monodromy is available, these two sign actions exhaust the possible local shell outcomes. □

3.4 Ninth extension: shell-parity criterion and diagnostic Julia plot

The generator-shell classifier can now be sharpened into a direct parity law for the local probe decision. On a fixed sufficiently small regular shell, the visible data is exhausted by the transported sign pair $\{\pm\sqrt{s(E)}\}$, so any admissible closure cycle acts by a two-element shell parity rather than by a free additional branch label. The same two-way split is also visualized by a diagnostic Julia-type render recorded in Appendix C, now shown at the currently relevant residual scales $\gamma \in \{0, \Delta_\beta, \Delta_\theta, \Delta\mu\}$ rather than at the earlier standalone hypothesis marker 0.0009.

Theorem 11.10 (3.4 shell-parity criterion for direct closure versus spinorial lift)**.** *Assume the hypotheses of Theorem 11.9. Fix a sufficiently small regular shell of energy E and let*

$$\Sigma_E := \{\pm\sqrt{s(E)}\}$$

be its visible turning pair. Assume moreover that every admissible continuation cycle on that shell admits a decomposition into regular segments and finitely many transverse reflector crossings, each transverse reflector crossing implementing the sign involution

$$\eta \longmapsto -\eta$$

on Σ_E. Then there exists a shell parity

$$\pi(E) \in \{+1, -1\}$$

such that after one admissible closure cycle

$$\eta_{\text{out}} = \pi(E)\,\eta_{\text{in}} \qquad \text{for every } \eta_{\text{in}} \in \Sigma_E.$$

More precisely:

(i) $\pi(E) = +1$ if and only if the shell closes by direct visible convergence;

(ii) $\pi(E) = -1$ if and only if visible closure is reached only after the unique spinorial/Möbius lift;

(iii) if N_E denotes the number of transverse reflector crossings in one admissible shell cycle, then

$$\pi(E) = (-1)^{N_E}.$$

Hence odd crossing parity forces sign-reversing transport, whereas even crossing parity gives direct visible closure.

Thus the local probe outcome on a small regular shell is determined by shell parity alone once the generator radius $s(E)$ is fixed. The corresponding diagnostic render is recorded in Appendix C as the ninth 3.4 extension.

Proof. By Theorem 11.9, the shell is reduced to the visible pair $\Sigma_E = \{\pm\sqrt{s(E)}\}$ and no higher monodromy opens on the local shell sector. Therefore every admissible continuation cycle acts on Σ_E by an element of the two-element sign group, which we denote by $\pi(E) \in \{+1, -1\}$. If $\pi(E) = +1$, then the transported representative returns to itself and the shell closes by direct visible convergence. If $\pi(E) = -1$, then the representative returns with reversed sign and visible closure occurs only after the already identified spinorial/Möbius double-cover lift. Under the stated decomposition hypothesis, each transverse reflector crossing contributes exactly one sign involution, while regular segments preserve sign. The total action is therefore the product of N_E copies of -1, namely $\pi(E) = (-1)^{N_E}$. □

3.4 Tenth extension: cautious detector-readout hypothesis

The diagnostic Möbius-reflected render suggests a further hypothesis that should be stated carefully but not silently ignored. The present probe map was not tuned to detector imagery, yet it produces a visible geometry with a central transport corridor, directed opening, and boundary intensification. Within the standing phase-shift / frame-shift reading of the Companion, this motivates the following cautious interpretation: some detector-like photon images may be readout-relative shadows of the same branch-sensitive residual geometry rather than direct images of the intrinsic kernel itself.

Hypothesis 3.4-H (frame-shifted detector readout). Assume that the intrinsic kernel is observed only through an admissible shifted readout frame. Then a detector image need not reproduce the intrinsic kernel directly; it may instead display the transported boundary geometry of the residual sector. Under that reading, the central tunnel in the diagnostic render is interpreted as a transported readout channel and the bright shell frontier as a readout-sensitive boundary layer. The current claim is *not* that the diagnostic render identifies a physical event, but only that the observed visual similarity is strong enough to justify a testable structural hypothesis.

Immediate test observables. The hypothesis may be probed without committing to a full detector model by monitoring diagnostic quantities such as the tunnel width $w_{\mathrm{tun}}(\gamma)$, the edge-intensification profile $I_{\mathrm{edge}}(\gamma)$, the effective opening angle $\Theta_{\mathrm{open}}(\gamma)$, and the shell-parity response $P_{\mathbb{Z}_2}(E)$. A stable correspondence in such observables would strengthen the readout hypothesis; failure of correspondence would simply falsify it. Thus the visual proximity is recorded here as an explicit testable hypothesis rather than as an identification claim.

3.4 Eleventh extension: tunnel observables and symmetry-constrained readout metrics

The detector-readout hypothesis becomes mathematically useful only once the visual geometry is reduced to explicit observables. The diagnostic render therefore should be read through a small family of shell metrics attached to the same generator/parity structure rather than through unaudited visual impression alone. In particular, the central transport corridor, the shell edge brightening, and the opening of the visible branch front can be organized into a minimal test package.

Proposition 11.11 (3.4 tunnel-observable package and parity-compatible readout)**.** *Fix a diagnostic probe family $\mathcal{J}_\gamma$ on a symmetric rendering window $W \subset \mathbb{C}$ together with a nonnegative stability-density image*

$$H_\gamma : W \to \mathbb{R}_{\geq 0}$$

obtained from the corresponding escape-time or residence-time readout. Assume that the rendering window and the probe normalization are chosen so that the transported shell classifier of Theorem 11.10 is visible on a distinguished symmetry axis $x = 0$. Then the following four scalar observables are canonically available on the small-shell diagnostic sector:

(i) the tunnel width

$$w_{\mathrm{tun}}(\gamma;\tau) := \mathrm{meas}\big\{y : (0,y) \in W,\ H_\gamma(0,y) \le \tau\big\},$$

defined for any admissible intensity threshold $\tau > 0$;

(ii) the edge-intensification functional

$$I_{\mathrm{edge}}(\gamma) := \int_{\partial S_\gamma} H_\gamma \, d\sigma,$$

where S_γ is the first regular shell selected by the generator-radius prescription of Theorem 11.9;

(iii) the opening angle

$$\Theta_{\mathrm{open}}(\gamma),$$

defined as the minimal cone angle based at the central channel that contains the first visible shell frontier on the upper half-window;

(iv) the already available shell-parity response

$$P_{\mathbb{Z}_2}(E) = \pi(E) \in \{+1, -1\}.$$

If, in addition, the diagnostic family is normalized so that the transported sign action is the only local monodromy on sufficiently small shells, then the quadruple

$$\mathfrak{O}_{\mathrm{diag}}(\gamma) := \big(w_{\mathrm{tun}}(\gamma;\tau),\ I_{\mathrm{edge}}(\gamma),\ \Theta_{\mathrm{open}}(\gamma),\ P_{\mathbb{Z}_2}(E)\big)$$

contains no independent local scalar beyond the generator-shell data: each component is a readout-functional of the same shell radius $s(E)$ and parity class $\pi(E)$, possibly after the external choice of the plotting threshold τ.

Proof. Theorem 11.9 reduces every sufficiently small regular shell to the generator radius $s(E)$ and the transported sign pair $\{\pm\sqrt{s(E)}\}$, while Theorem 11.10 shows that the only remaining local branch ambiguity is the parity $\pi(E) \in \{+1, -1\}$. Once a concrete diagnostic density H_γ is fixed, the corridor width, shell-edge intensity, and opening angle are ordinary scalar functionals of the rendered shell geometry. Since the shell geometry itself is determined by $s(E)$ together with the parity action, these observables do not open a new local scalar channel; they only read the same generator/parity data through different diagnostics. The threshold τ is external plotting metadata and therefore does not count as an intrinsic structural scalar. □

Operational consequence. The proposition does not claim that the four observables already agree with detector data; it only shows that the detector-readout hypothesis can be made auditably quantitative. In particular, the central tunnel noted in the current diagnostic render is not merely a visual remark: after thresholding it becomes the measurable quantity $w_{\mathrm{tun}}(\gamma;\tau)$, while the shell brightening and opening become $I_{\mathrm{edge}}(\gamma)$ and $\Theta_{\mathrm{open}}(\gamma)$. This is the first point at which the image layer becomes a genuine test interface rather than a suggestive illustration.

3.4 Twelfth extension: calibrated inverse readout and shell-consistency defect

The quantitative tunnel package becomes substantially stronger once one asks not only how shell data produces visible metrics, but also how far such metrics may be inverted. The natural minimal requirement is a local calibration law: tunnel width should move monotonically with the generator shell radius on a fixed threshold protocol. Under that assumption, the image layer becomes not merely comparable but locally readable.

Theorem 11.12 (3.4 calibrated inverse readout and shell consistency)**.** *Fix a threshold level $\tau > 0$ and assume the diagnostic family of Proposition 11.11 is restricted to a sufficiently small regular shell sector on which:*

(i) the tunnel-width functional factors through the generator shell radius,

$$w_{\mathrm{tun}}(\gamma;\tau) = W_\tau(s(E)),$$

for a C^1 calibration map W_τ;

(ii) W_τ is strictly monotone on the relevant shell interval;

(iii) the parity response records the transported sign class,

$$P_{\mathbb{Z}_2}(E) = \pi(E) \in \{+1, -1\}.$$

Then the pair

$$\big(w_{\mathrm{tun}}(\gamma;\tau), P_{\mathbb{Z}_2}(E)\big)$$

locally determines the generator/parity data

$$\big(s(E), \pi(E)\big) = \big(W_\tau^{-1}(w_{\mathrm{tun}}(\gamma;\tau)), P_{\mathbb{Z}_2}(E)\big).$$

Consequently, the remaining observables of Proposition 11.11 admit local consistency representations

$$I_{\mathrm{edge}}(\gamma) = \widehat{\mathcal{I}}_\tau\big(w_{\mathrm{tun}}(\gamma;\tau), P_{\mathbb{Z}_2}(E)\big),$$
$$\Theta_{\mathrm{open}}(\gamma) = \widehat{\vartheta}_\tau\big(w_{\mathrm{tun}}(\gamma;\tau), P_{\mathbb{Z}_2}(E)\big)$$

for suitable local readout functions $\widehat{\mathcal{I}}_\tau$ and $\widehat{\vartheta}_\tau$. In particular, no new local structural scalar opens at the inverse-readout level.

Proof. Strict monotonicity makes W_τ locally invertible, so the tunnel width recovers the shell radius $s(E)$ uniquely on the selected shell interval. The parity response already equals the transported sign class by assumption. The shell geometry on the small diagnostic sector is determined by $(s(E), \pi(E))$ through Theorem 11.10 and Proposition 11.11; therefore every additional shell metric constructed from the same thresholded rendering protocol factors through the recovered pair. This yields the two stated consistency maps and shows that inverse readout does not create a new intrinsic scalar channel. □

Corollary 11.13 (3.4 shell-consistency defect)**.** *Under the hypotheses of Theorem 11.12, every calibrated diagnostic image carries a local shell-consistency defect*

$$\Delta_{\mathrm{diag}}(\gamma;\tau) := \Big|I_{\mathrm{edge}}(\gamma) - \widehat{\mathcal{I}}_\tau\big(w_{\mathrm{tun}}(\gamma;\tau), P_{\mathbb{Z}_2}(E)\big)\Big| + \Big|\Theta_{\mathrm{open}}(\gamma) - \widehat{\vartheta}_\tau\big(w_{\mathrm{tun}}(\gamma;\tau), P_{\mathbb{Z}_2}(E)\big)\Big|.$$

Hence Δ_{diag} vanishes on an idealized perfectly calibrated render and measures the failure of the observed image to close back to the shell/parity model.

Proof. This is immediate from the consistency representations of Theorem 11.12. The quantity Δ_{diag} is simply the residual mismatch between the measured edge/opening readouts and the values predicted from the recovered shell/parity data. □

Interpretive consequence. The image layer now supports a minimal inverse logic. After choosing a threshold protocol, the tunnel width and the shell parity are enough to reconstruct the local shell radius and to predict the remaining edge/opening observables. This does not yet prove the detector-readout hypothesis, but it upgrades the diagnostic render from a suggestive comparison picture to a partially invertible readout device.

3.4 Thirteenth extension: inverse identifiability and stability margin

The inverse-readout theorem becomes substantially more useful once one knows when the reconstruction is not merely available but also stable. The right local criterion is that the calibration law should have nonvanishing derivative on the shell under consideration. Then the tunnel width is not just invertible in principle: it controls the shell radius with an explicit local conditioning constant.

Theorem 11.14 (3.4 local inverse identifiability and stability)**.** *Assume the hypotheses of Theorem 11.12. Fix a regular shell E_0 with generator radius*

$$s_0 := s(E_0), \qquad \pi_0 := \pi(E_0), \qquad w_0 := W_\tau(s_0).$$

Assume moreover that $W_\tau \in C^1$ and

$$W'_\tau(s_0) \neq 0.$$

Then there exist neighborhoods U_{s_0} of s_0 and V_{w_0} of w_0, together with constants $m_0, M_0 > 0$, such that

$$m_0 \, |s_1 - s_2| \leq |W_\tau(s_1) - W_\tau(s_2)| \leq M_0 \, |s_1 - s_2|$$

for all $s_1, s_2 \in U_{s_0}$. Consequently:

(i) on the fixed parity branch $\pi = \pi_0$, the inverse readout is locally unique;

(ii) for every calibrated pair $(\widetilde{w}, \widetilde{\pi}) \in V_{w_0} \times \{\pi_0\}$, the reconstructed shell radius

$$\widetilde{s} := W_\tau^{-1}(\widetilde{w})$$

satisfies the stability bound

$$|\widetilde{s} - s_0| \leq \frac{1}{m_0} \, |\widetilde{w} - w_0|;$$

(iii) if the consistency maps $\widehat{\mathcal{I}}_\tau$ and $\widehat{\vartheta}_\tau$ from Theorem 11.12 are locally Lipschitz on $V_{w_0} \times \{\pi_0\}$ with constants L_I and L_Θ, then

$$\left|\widehat{\mathcal{I}}_\tau(\widetilde{w}, \widetilde{\pi}) - \widehat{\mathcal{I}}_\tau(w_0, \pi_0)\right| \leq L_I \, |\widetilde{w} - w_0|,$$

$$\left|\widehat{\vartheta}_\tau(\widetilde{w}, \widetilde{\pi}) - \widehat{\vartheta}_\tau(w_0, \pi_0)\right| \leq L_\Theta \, |\widetilde{w} - w_0|.$$

Hence the calibrated inverse readout is locally bi-Lipschitz on each parity branch and is destabilized only at parity changes or at points where W'_τ vanishes.

Proof. Because $W'_\tau(s_0) \neq 0$, continuity of W'_τ gives a neighborhood U_{s_0} on which $|W'_\tau|$ is bounded above and below by positive constants. The mean-value theorem then yields the two-sided estimate with suitable $m_0, M_0 > 0$. Local injectivity and the bound for $|\widetilde{s} - s_0|$ follow immediately. The Lipschitz estimates for the reconstructed edge/opening readouts are obtained by composing the local inverse branch with the Lipschitz consistency maps. A local degeneracy can therefore occur only when the parity branch changes or when the calibration derivative vanishes. □

Corollary 11.15 (3.4 inverse condition number and stability margin)**.** *Under the hypotheses of Theorem 11.14, define the local inverse stability margin and condition number by*

$$\mathfrak{m}_\tau(s_0) := |W'_\tau(s_0)|, \qquad \kappa_\tau(s_0) := \frac{1}{|W'_\tau(s_0)|}.$$

Then larger $\mathfrak{m}_\tau(s_0)$ (equivalently smaller $\kappa_\tau(s_0)$) means better-conditioned shell reconstruction from tunnel width, while parity-switch points and critical points of W_τ are the only local obstructions to stable inversion.

Proof. This is just a restatement of the inverse estimate from Theorem 11.14. The derivative $|W'_\tau(s_0)|$ is the local amplification factor from shell-radius error to tunnel-width error, so its reciprocal is the condition number for the inverse problem. The final sentence records the two ways in which the theorem can fail. □

Interpretive consequence. The image layer now comes with a local reliability scale. Small tunnel-width perturbations produce controlled shell-radius perturbations whenever the calibration slope stays away from zero, and the parity sector remains the only discrete ambiguity. In that sense the diagnostic plot is no longer just partially invertible; it is partially invertible with an explicit local stability regime.

3.4 Fourteenth extension: shell atlas and critical-shell stratification

Local inverse stability becomes structurally much clearer once the admissible shell line is decomposed into regular inverse charts separated only by genuine degeneracy loci. The natural global objects are therefore the critical-shell set of the calibration law and the parity-switch set of the transported probe class. Away from these two obstructions, the inverse readout does not merely exist pointwise: it assembles into a shell atlas with uniform local control on compact subintervals.

Theorem 11.16 (3.4 shell atlas and critical-shell stratification)**.** *Assume the hypotheses of Theorem 11.14. Let $I \subset \mathbb{R}_{\geq 0}$ be a shell interval in the generator coordinate s, and define the critical-shell and parity-switch sets by*

$$\Sigma_{\mathrm{crit}} := \{s \in I : W'_\tau(s) = 0\},$$
$$\Sigma_{\mathrm{par}} := \{s \in I : \textit{the transported parity class changes at } s\}.$$

Set

$$\Sigma := \Sigma_{\mathrm{crit}} \cup \Sigma_{\mathrm{par}}.$$

Then every connected component $J \subset I \setminus \Sigma$ carries a calibrated shell chart with the following properties:

(i) the parity class is constant on J;

(ii) the calibration law $W_\tau|_J$ is strictly monotone, hence injective;

(iii) for every compact subinterval $K \Subset J$ there exist constants $m_K, M_K > 0$ such that

$$m_K|s_1 - s_2| \leq |W_\tau(s_1) - W_\tau(s_2)| \leq M_K|s_1 - s_2|$$

for all $s_1, s_2 \in K$;

(iv) on each such K, the inverse readout extends to a stable branch chart

$$(w_{\text{tun}}, P_{\mathbb{Z}_2}) \longmapsto (s, \pi)$$

with Lipschitz inverse constant bounded by $1/m_K$.

Consequently the calibrated inverse problem on I is stratified by the closed obstruction set Σ: away from Σ one has a finite or countable atlas of stable parity-preserving shell charts, and every local inverse degeneracy is forced by either a calibration critical shell or a parity-switch shell.

Proof. On every connected component $J \subset I \setminus \Sigma$, the parity is locally constant by definition of Σ_{par}. Since W'_τ never vanishes on J, continuity of W'_τ implies that W'_τ cannot change sign on J; hence $W_\tau|_J$ is strictly monotone and therefore injective. For any compact $K \Subset J$, continuity of W'_τ yields positive lower and finite upper bounds on $|W'_\tau|$, and the mean-value theorem gives the two-sided estimate with constants m_K, M_K. The inverse-chart statement is then exactly the local inverse-stability estimate of Theorem 11.14, now uniform on K. Since every failure of this argument occurs only when parity changes or W'_τ vanishes, the obstruction set is precisely Σ. □

Corollary 11.17 (3.4 chartwise diagnostic regime)**.** *Under the hypotheses of Theorem 11.16, the shell-consistency defect*

$$\Delta_{\text{diag}}(\gamma; \tau)$$

becomes a chartwise diagnostic observable on every compact $K \Subset J \subset I \setminus \Sigma$. In particular, a persistent rise of Δ_{diag} along a stable chart is evidence for model mismatch inside a fixed shell/parity regime, whereas loss of invertibility itself can occur only by approaching Σ_{crit} or Σ_{par}.

Proof. Theorem 11.16 gives a stable inverse chart on every compact $K \Subset J$. Inside such a chart, the reconstructed shell/parity data vary continuously with the measured tunnel width, so the shell-consistency defect is a well-defined chart observable. A blow-up of the defect while the inverse chart remains valid indicates failure of the model closure rather than loss of invertibility. The only mechanisms that can destroy the chart itself are exactly the two obstruction loci already identified in Theorem 11.16. □

Interpretive consequence. The inverse-readout layer now has a global local-to-global organization. Stable reconstruction is not a scattered pointwise accident, but comes in shell charts separated by sharply identified obstruction shells. This makes the diagnostic side much cleaner: one can distinguish chart-internal model mismatch from genuine loss of inverse control.

Remark 11.18 (OP3 reading of shell diagnostics)**.** *In the present release, shell-consistency defects in a stable chart are read through the OP3 residual/shell interface and the visible/structural mismatch separation tool. A rise of Δ_{diag} inside a valid chart is first an OP3-admissibility or model-fit issue inside a fixed shell/parity regime, not immediate evidence against the closed OP 1–OP 5 source-return pair.*

3.4 Fifteenth extension: obstruction-shell transition law

Once the inverse readout is organized by stable shell charts, the next structural question is how the data reorganizes when a trajectory reaches an obstruction shell. There are only two primitive failure modes in the present atlas: calibration-fold shells, where the tunnel-width derivative vanishes, and parity shells, where the transported $\mathbb{Z}_2$ class flips. The point of the next statement is that these are not arbitrary singularities; they carry a very small transition algebra.

Theorem 11.19 (3.4 obstruction-shell transition law)**.** *Let $s_* \in \Sigma = \Sigma_{\mathrm{crit}} \cup \Sigma_{\mathrm{par}}$ be an isolated obstruction shell in the sense of Theorem 11.16.*

(i) If $s_* \in \Sigma_{\mathrm{crit}} \setminus \Sigma_{\mathrm{par}}$ and

$$W'_\tau(s_*) = 0, \qquad W''_\tau(s_*) \neq 0,$$

then the parity class is constant near s_*, and there exists a punctured neighborhood on which the calibrated tunnel law has exactly two local inverse sheets joined at s_*. More precisely, after a local reparametrization one has the fold normal form

$$w_{\mathrm{tun}} - w_* = \pm u^2 + O(u^3), \qquad u = s - s_*,$$

so equal tunnel width near w_* corresponds to the two shell branches exchanged by the local involution

$$\sigma_*(s) \neq s, \qquad W_\tau(\sigma_*(s)) = W_\tau(s), \qquad \sigma_*^2 = \mathrm{id}.$$

(ii) If $s_* \in \Sigma_{\mathrm{par}} \setminus \Sigma_{\mathrm{crit}}$, then $W'_\tau(s_*) \neq 0$, hence the shell radius continues uniquely through s_*, but the transported parity flips:

$$\pi_+(s_*) = -\pi_-(s_*).$$

Thus the obstruction is purely discrete: the shell coordinate is continuous, while the visible branch label changes sign.

(iii) If $s_* \in \Sigma_{\mathrm{crit}} \cap \Sigma_{\mathrm{par}}$, then the local transition is the composition of the two previous mechanisms: a fold exchange of shell sheets together with a parity flip. In particular, the local transition algebra is generated by

$$T_*^{\mathrm{fold}} : (s, \pi) \mapsto (\sigma_*(s), \pi), \qquad T_*^{\mathrm{par}} : (s, \pi) \mapsto (s, -\pi),$$

and every admissible local transition is one of

$$\mathrm{id}, \quad T_*^{\mathrm{fold}}, \quad T_*^{\mathrm{par}}, \quad T_*^{\mathrm{par}} \circ T_*^{\mathrm{fold}}.$$

Hence obstruction shells do not open a new uncontrolled singular sector: they only create fold exchange, parity jump, or their composition.

Proof. For (i), the assumptions say that s_* is a nondegenerate critical point of the scalar calibration law. The one-dimensional Morse lemma (equivalently a standard fold normal form for a smooth map with first nonvanishing second derivative) gives a local coordinate u in which $W_\tau - W_\tau(s_*)$ is quadratic up to higher order, hence the inverse problem has exactly two sheets on one side of the critical value and none on the other. Because $s_* \notin \Sigma_{\mathrm{par}}$, parity is constant through this local sector, so only the shell sheet changes. The local deck involution exchanging the two branches is exactly σ_*.

For (ii), $s_* \notin \Sigma_{\mathrm{crit}}$ implies $W'_\tau(s_*) \neq 0$, so the inverse function theorem gives a unique continuation of the shell coordinate across s_*. Since $s_* \in \Sigma_{\mathrm{par}}$, the only local change is the transported parity flip.

Statement (iii) is immediate by combining the two previous local descriptions. The fold exchange leaves parity unchanged, the parity jump leaves the shell coordinate unchanged, and the two operations commute because they act on different coordinates. Therefore the local transition algebra is the Klein four-set generated by these two involutions. □

Corollary 11.20 (3.4 obstruction monodromy principle)**.** *Under the hypotheses of Theorem 11.19, every closed local continuation of admissible shell data around isolated obstruction shells produces only a $\mathbb{Z}_2$-generated monodromy on the inverse data (s, π). In particular, direct visible return corresponds to trivial monodromy, while the spinorial/Möbius case is exactly the nontrivial parity component of this local obstruction monodromy.*

Proof. By Theorem 11.19, the only local generators are the fold involution and the parity involution. Any closed continuation is therefore represented by a finite product of commuting involutions, so only a $\mathbb{Z}_2$-generated monodromy can occur. The visible return is trivial exactly when the parity component is trivial; otherwise the continuation returns only after the nontrivial parity lift, which is precisely the local spinorial/Möbius case. □

Interpretive consequence. The shell atlas now comes with transition rules rather than only forbidden loci. Critical shells create fold exchange, parity shells create sign change, and the mixed case composes the two. This is the first point at which the inverse image layer acquires a genuine obstruction algebra instead of a mere list of bad points.

3.4 Closing synthesis: generator, inverse atlas, and parity holonomy

The 3.4 line can now be closed without adding a new mechanism. The previous extensions already determine a single visible package: an even analytic generator for the residual branch law, a shell atlas for calibrated inverse readout, and a parity obstruction that controls the only remaining global ambiguity. The point of the present synthesis is therefore not to enlarge the theory, but to compress the accumulated 3.4 structure into one handoff theorem.

Theorem 11.21 (3.4 closing synthesis package)**.** *Assume the hypotheses of the stabilized 3.4 line on a connected admissible shell base B. Then the visible residual readout is controlled by the following data and no further independent local scalar is opened by the 3.4 construction:*

(i) there exists an even analytic generator Φ such that the visible branch law is

$$\eta'' = -2\eta\,\Phi'(\eta^2), \qquad \mathcal{U}(\eta) = \Phi(\eta^2),$$

and every sufficiently small regular shell is determined by the generator radius

$$s(E) = \Phi^{-1}(E)$$

together with the transported parity label $\pi \in \mathbb{Z}_2$*;*

(ii) away from the obstruction set $\Sigma = \Sigma_{\mathrm{crit}} \cup \Sigma_{\mathrm{par}}$*, the calibrated inverse readout*

$$(w_{\mathrm{tun}}, P_{\mathbb{Z}_2}) \longmapsto (s, \pi)$$

is chartwise stable, and the remaining observables I_{edge} *and* Θ_{open} *are shell-consistency readouts of the same data;*

(iii) the global ambiguity of the inverse shell atlas is exhausted by the parity local system, equivalently by the cohomological class

$$\varepsilon_{\mathrm{par}} \in H^1(B, \mathbb{Z}_2),$$

whose loop readout is the parity holonomy character

$$\chi_{\mathrm{par}} : \pi_1(B, b_0) \to \mathbb{Z}_2;$$

(iv) the empirical estimator $\widehat{\chi}_{\mathrm{par}}$ *obtained from parity-changing obstruction crossings recovers this same global class along admissible closed shell traces.*

Consequently, direct visible closure is globally available exactly in the trivial parity class, while every nontrivial case becomes globally single-valued on the canonical parity double cover.

Proof. Item (i) is exactly the content of the generator reduction established in Theorems 11.8 and 11.9. Item (ii) combines the calibrated inverse readout theorem, the inverse identifiability/stability theorem, and the shell-atlas stratification from Theorems 11.12, 11.14, and 11.16. Item (iii) is the cohomological and double-cover compression proved in Theorems 12.74–12.80. Item (iv) is precisely the empirical parity-holonomy estimator of Theorem 12.82. No new local scalar is introduced in this synthesis step; it only packages the already established 3.4 data into a single visible readout layer. □

Corollary 11.22 (3.4 handoff package for the next stage)**.** *At the end of the 3.4 line, the reduced visible input needed for the next structural stage consists of the generator germ* Φ*, the shell-consistency defect* Δ_{diag}*, and the parity obstruction package* $(\varepsilon_{\mathrm{par}}, \chi_{\mathrm{par}}, \widehat{\chi}_{\mathrm{par}})$*. In particular, the transition to the next stage does not require a new visible branch variable beyond* η*.*

Proof. Theorem 11.21 already identifies the complete visible 3.4 package. The generator controls the local residual law and regular shell data, the defect term measures chartwise mismatch of the diagnostic readout, and the parity package records the only remaining global ambiguity. Therefore these data form the minimal handoff set to the next stage. □

3.4 closing remark. The 3.4 line has therefore reached a natural stopping point. It began by globalizing the local residual law and ends with a parity-resolved inverse readout atlas whose only global obstruction is a single class in $H^1(B, \mathbb{Z}_2)$. This is the right place to stop extending 3.4 and hand the package forward, rather than continuing to add more local refinements.

3.5 First extension: admissible residual record and reconstruction normal form

The closed 3.4 line already shows that the visible branch layer is controlled by a small package rather than by an open-ended family of new observables. The correct first move of 3.5 is therefore not to enlarge the visible theory, but to treat this package as a single admissible record. This gives a compact interface for later simplification work while remaining entirely inside the current mathematical language.

Definition 11.23 (Admissible residual record)**.** *On a connected admissible shell base B, define the visible residual record*

$$\mathfrak{R}(B) := \big(\Phi, \Delta_{\mathrm{diag}}, \varepsilon_{\mathrm{par}}, \chi_{\mathrm{par}}, \widehat{\chi}_{\mathrm{par}}\big),$$

where Φ is the even analytic generator of the local visible branch law, Δ_{diag} is the shell-consistency defect, $\varepsilon_{\mathrm{par}} \in H^1(B, \mathbb{Z}_2)$ is the parity obstruction class, χ_{par} is its parity-holonomy character, and $\widehat{\chi}_{\mathrm{par}}$ is the empirical loop estimator.

Theorem 11.24 (3.5 reconstruction normal form)**.** *Assume the hypotheses of Theorem 11.21. Then every visible statement in the stabilized generator-atlas-holonomy sector factors through the admissible residual record $\mathfrak{R}(B)$ together with the chartwise shell coordinate s and parity label π, in the following sense:*

(i) on each regular shell chart away from the obstruction set, the local branch law, tunnel observables, and shell-consistency diagnostics are determined by $(\Phi, s, \pi, \Delta_{\mathrm{diag}})$;

(ii) globally, the only remaining ambiguity of visible continuation is determined by the parity obstruction package $(\varepsilon_{\mathrm{par}}, \chi_{\mathrm{par}}, \widehat{\chi}_{\mathrm{par}})$;

(iii) no further independent visible branch datum beyond η is required to reconstruct the 3.4 readout sector.

Equivalently, the stabilized visible branch layer of the Companion is already in reconstruction normal form once it is expressed through $\mathfrak{R}(B)$.

Proof. Theorem 11.21 identifies the complete 3.4 handoff package and states that no further independent local scalar is opened by the 3.4 construction. Its item (i) reduces the local visible residual dynamics to the generator Φ, while item (ii) reduces the calibrated inverse shell readout to chartwise shell data (s, π) together with the shell-consistency defect. Items (iii) and (iv) show that the only remaining global ambiguity is the parity obstruction package and that its operational loop readout is completely captured by χ_{par} and $\widehat{\chi}_{\mathrm{par}}$. Therefore every visible statement proved in the stabilized 3.4 sector factors through the single record $\mathfrak{R}(B)$ together with chartwise shell coordinates, and no new visible branch datum beyond η is needed. □

Corollary 11.25 (3.5 starting principle: record-level compression)**.** *For the first simplification stage of v3.35.0, proofs and comparisons may treat $\mathfrak{R}(B)$ as the complete visible residual interface. In particular, any later compact formal layer that is introduced for proof economy must remain faithful to the reconstruction normal form determined by $\mathfrak{R}(B)$.*

Proof. This is a direct restatement of Theorem 11.24. Once the visible branch layer has been compressed to $\mathfrak{R}(B)$, any further simplification may only reorganize or repackage these same data, not enlarge the visible branch interface by adding new primitive observables. □

3.5 opening remark. The transition from 3.4 to 3.5 is therefore a change of stance rather than a new visible mechanism. Instead of extending the branch picture by more local refinements, the document now treats the residual generator, the inverse shell atlas, and the parity holonomy package as one admissible record. This is the first point at which later proof-economy work can aim at genuine compression instead of further accumulation.

3.5 Second extension: record morphisms and faithful translation principle

Once the visible residual sector has been compressed to an admissible record, the next question is no longer how to add more local data, but how to compare two residual presentations without reopening the visible layer. The correct next step is therefore to formulate a record-level notion of sameness. This does not yet introduce a new formal language; it only states which data are allowed to move and which must remain invariant under later compression.

Definition 11.26 (Admissible residual-record morphism)**.** *Let B_1 and B_2 be connected admissible shell bases with residual records*

$$\mathfrak{R}(B_i) = \big(\Phi_i, \Delta^{(i)}_{\mathrm{diag}}, \varepsilon^{(i)}_{\mathrm{par}}, \chi^{(i)}_{\mathrm{par}}, \widehat{\chi}^{(i)}_{\mathrm{par}}\big), \qquad i = 1, 2.$$

An admissible residual-record morphism *$F : B_1 \to B_2$ is an admissible shell-atlas transport such that, on every regular shell chart away from the obstruction set, the following conditions hold:*

(i) the generator germ is preserved under transport, equivalently the visible branch law is carried by the same transported generator data;

(ii) the shell-consistency defect is preserved up to chartwise pullback,

$$\Delta^{(1)}_{\mathrm{diag}} = F^* \Delta^{(2)}_{\mathrm{diag}};$$

(iii) the parity obstruction package is preserved,

$$\varepsilon^{(1)}_{\mathrm{par}} = F^* \varepsilon^{(2)}_{\mathrm{par}}, \qquad \chi^{(1)}_{\mathrm{par}} = \chi^{(2)}_{\mathrm{par}} \circ F_*, \qquad \widehat{\chi}^{(1)}_{\mathrm{par}} = \widehat{\chi}^{(2)}_{\mathrm{par}} \circ F_*.$$

If, moreover, F is invertible in the same class, then F is called a residual-record isomorphism.

Theorem 11.27 (3.5 faithful translation principle)**.** *Assume the hypotheses of Theorem 11.24. Let $F : B_1 \to B_2$ be a residual-record isomorphism in the sense of Definition 11.26. Then every visible statement in the stabilized generator-atlas-holonomy sector is invariant under translation along F. More precisely:*

(i) local branch laws, shell reconstructions, and tunnel observables translate chartwise without opening a new visible scalar channel;

(ii) the shell-consistency defect translates by pullback, so chartwise mismatch diagnostics are record-invariant;

(iii) the parity obstruction class, its holonomy character, and the empirical loop estimator are preserved under transport of admissible loops;

(iv) consequently, any theorem proved solely from the stabilized 3.4 package depends only on the residual-record isomorphism class of $\mathfrak{R}(B)$.

Equivalently, the visible generator-atlas-holonomy sector is faithful to record level: later proof compression may reorganize it, but may not change its record-level content.

Proof. Theorem 11.24 already states that every visible statement in the stabilized sector factors through the record $\mathfrak{R}(B)$ together with chartwise shell coordinate and parity label. By Definition 11.26, a residual-record isomorphism preserves exactly the five pieces of data entering this factorization: the transported generator germ, the shell-consistency defect, the cohomological parity class, the holonomy character, and the empirical estimator. Hence local branch formulas, inverse-shell reconstructions, and calibrated tunnel observables translate chartwise without opening a new visible scalar. Since the parity package is preserved under loop transport, the global ambiguity sector is likewise invariant. Therefore any theorem whose hypotheses and conclusions are expressed entirely in the stabilized generator-atlas-holonomy sector depends only on the record-level isomorphism class and is faithfully transportable along F. □

Corollary 11.28 (3.5 translation corollary: record-level theorem transport)**.** *Let* $\mathcal{S}_1$ *and* $\mathcal{S}_2$ *be two admissible shell presentations related by a residual-record isomorphism. Then any proof in the stabilized visible sector may be transported from* $\mathcal{S}_1$ *to* $\mathcal{S}_2$ *without introducing a new primitive observable, provided the proof uses only record-level data and chartwise shell coordinates.*

Proof. This is the direct content of Theorem 11.27. Once the stabilized sector has been reduced to record level, proof transport between isomorphic shell presentations is faithful and does not enlarge the visible interface. □

3.5.1 working remark. The point of the new step is not yet a new language, but a sharper compression law: two visible residual presentations should count as the same as soon as they carry the same admissible record. This is the first place where later ZFM-facing or proof-economy layers can aim at strict 1:1 translation without reopening the visible branch picture. In the current 3.5.x reading this also fixes the first status boundary: open work should now be read primarily as quantitative or global-identification work, not as permission to introduce a second visible interface.

3.5 Third extension: record interpreter and exact representability principle

Once residual-record morphisms have been fixed, the next step is to say not only when two visible presentations are the same, but how a visible theorem is to be read at record level. The key point is that the stabilized 3.4 package should now admit a faithful interpreter: visible statements are no longer primary objects, but are required to be represented by formulas built from the residual record together with chartwise shell coordinate and parity label. This still does not introduce a separate proof language. It only fixes the exact translation interface that any later compression layer must preserve.

Definition 11.29 (Residual-record interpreter)**.** *Let* B *be a connected admissible shell base with stabilized record*

$$\mathfrak{R}(B) = \big(\Phi, \Delta_{\mathrm{diag}}, \varepsilon_{\mathrm{par}}, \chi_{\mathrm{par}}, \widehat{\chi}_{\mathrm{par}}\big).$$

A residual-record interpreter *on B is an assignment*

$$\mathcal{I}_B : \mathrm{Vis}(B) \longrightarrow \mathrm{Rec}(B)$$

from visible statements in the stabilized generator-atlas-holonomy sector to record-level formulas such that:

(i) every local branch law, inverse-shell statement, tunnel-observable relation, shell-consistency statement, parity-obstruction claim, holonomy statement, and empirical loop-readout identity is sent to a formula built only from the record $\mathfrak{R}(B)$ together with chartwise shell coordinate and parity label;

(ii) truth is preserved and reflected: a visible statement holds on B if and only if its interpreted record-level formula holds on the same admissible shell charts and loop data;

(iii) the interpretation is natural under residual-record isomorphisms, i.e. for every residual-record isomorphism $F : B_1 \to B_2$ one has

$$\mathcal{I}_{B_1}(F^*\sigma) = F^*\big(\mathcal{I}_{B_2}(\sigma)\big)$$

for all visible statements σ in the stabilized sector.

Theorem 11.30 (3.5 exact representability principle)**.** *Assume the hypotheses of Theorems 11.24 and 11.27. Then the stabilized visible generator-atlas-holonomy sector admits a residual-record interpreter in the sense of Definition 11.29. More precisely:*

(i) every visible statement in the stabilized sector has a record-level representative built from $\mathfrak{R}(B)$ together with chartwise shell coordinate and parity label;

(ii) this representative is unique up to residual-record isomorphism;

(iii) proof transport along residual-record isomorphisms commutes with interpretation;

(iv) consequently, the stabilized visible sector is exactly representable *at record level: later proof compression may replace visible statements by record-level formulas, but may not change their truth content or open a new visible primitive.*

Proof. Theorem 11.24 proves that every visible statement in the stabilized 3.4 handoff sector factors through the record $\mathfrak{R}(B)$ together with chartwise shell coordinate and parity label. Theorem 11.27 then shows that this factorization is invariant under residual-record isomorphisms. Therefore one defines $\mathcal{I}_B(\sigma)$ by replacing each visible generator, shell, diagnostic, and parity-holonomy ingredient of σ by its record-level representative. By construction, this translation preserves and reflects truth, because no visible theorem in the stabilized sector depends on any further primitive datum. Naturality under residual-record isomorphisms follows from the faithful translation principle. Uniqueness up to record-isomorphism is immediate, since any two representatives of the same visible statement differ at most by a transport that preserves the entire record package. □

Corollary 11.31 (3.5 interpreter corollary: admissible proof interface)**.** *Any later proof-economy layer, formal front-end, or set-theoretic presentation intended to compress the stabilized visible branch theory must factor through the residual-record interpreter. In particular, no later compression is admissible unless it is exact at record level.*

Proof. This is the direct consequence of Theorem 11.30. Once the visible stabilized sector is exactly representable at record level, any faithful later interface must preserve this representation rather than replace it by a different visible primitive basis. □

3.5.2 working remark. The present step is still intentionally weaker than a full new language. It only fixes the precise interface that any later language-level simplification must respect. This is already enough to say what a faithful front-end would have to do: not create new visible objects, but interpret the existing theory exactly through the admissible residual record. From this point onward, the live fronts should therefore be read as residual quantitative or global tasks — for example the continuum-to-measurement correction in β_{sym}, the controlled-emergence problem, the mass hierarchy and generation law, the compact Dixon reformulation, and the remaining Theta-sum balance — rather than as a search for further visible primitives.

3.6 First extension: conservative frontend principle

The first genuine 3.6 move is not a new visible mechanism, but a theorem saying exactly what a later proof language, ZFM-facing compression layer, or formal frontend is allowed to do. The visible interface fixed in 3.5 may be reorganized, typed, or rendered more economical; it may not be replaced by a second primitive visible basis. This is the point at which the future language project becomes mathematically testable.

Definition 11.32 (Conservative frontend over the residual-record interpreter)**.** *Assume the stabilized generator-atlas-holonomy regime on a connected admissible shell base B, and let*

$$\mathcal{I}_B : \mathrm{Vis}(B) \longrightarrow \mathrm{Rec}(B)$$

be the residual-record interpreter of Definition 11.29. An admissible conservative frontend *on B consists of*

$$\mathcal{F}_B : \mathrm{Vis}(B) \longrightarrow \mathrm{Fr}(B), \qquad \mathcal{J}_B : \mathrm{Fr}(B) \longrightarrow \mathrm{Rec}(B),$$

where $\mathrm{Fr}(B)$ is a formal compression layer, such that:

(i) ***factorization through the record interpreter:***

$$\mathcal{J}_B \circ \mathcal{F}_B = \mathcal{I}_B;$$

(ii) ***truth preservation and reflection on the visible image:*** *for every visible statement σ, the frontend statement $\mathcal{F}_B(\sigma)$ holds if and only if the record-level formula $\mathcal{J}_B(\mathcal{F}_B(\sigma))$ holds;*

(iii) ***naturality under residual-record isomorphisms:*** *for every residual-record isomorphism $G : B_1 \to B_2$ there is an induced frontend transport G^*_{Fr} such that*

$$\mathcal{F}_{B_1}(G^*\sigma) = G^*_{\mathrm{Fr}}\big(\mathcal{F}_{B_2}(\sigma)\big) \qquad \text{and} \qquad \mathcal{J}_{B_1}\big(G^*_{\mathrm{Fr}}(\tau)\big) = G^*\big(\mathcal{J}_{B_2}(\tau)\big);$$

(iv) ***no new visible primitives on the visible image:*** *every primitive symbol needed to express $\mathcal{F}_B(\sigma)$ for a visible statement σ is definable, on the visible image, from the residual-record package $\mathfrak{R}(B)$ together with chartwise shell coordinate and parity label.*

Theorem 11.33 (3.6 conservative frontend principle)**.** *Assume the hypotheses of Theorems 11.24, 11.27, and 11.30. Let $(\mathrm{Fr}(B), \mathcal{F}_B, \mathcal{J}_B)$ be an admissible conservative frontend in the sense of Definition 11.32. Then the frontend is conservative over the stabilized visible sector. More precisely:*

(i) every visible theorem is represented in the frontend by a statement whose record-level meaning is exactly the one fixed by the residual-record interpreter;

(ii) frontend proof transport along residual-record isomorphisms commutes with visible proof transport and with record interpretation;

(iii) no frontend proof concerning the stabilized visible sector can depend, on its visible image, on a primitive datum not already definable from $\mathfrak{R}(B)$ together with chartwise shell coordinate and parity label;

(iv) if $(\mathrm{Fr}'(B), \mathcal{F}'_B, \mathcal{J}'_B)$ is a second admissible conservative frontend, then the two frontends agree on the visible image up to unique record-level translation. In particular, they are equivalent as compressions of the same stabilized visible theory.

Equivalently, any later admissible language layer is conservative precisely by factoring through the residual-record interpreter and introducing no new visible primitive basis.

Proof. By Theorem 11.30, every visible statement in the stabilized sector already has an exact record-level representative, unique up to residual-record isomorphism. Condition (i) in Definition 11.32 forces every frontend rendering of a visible statement to pass through exactly that representative. Condition (ii) then implies that frontend truth on the visible image is preserved and reflected by the same record-level formula, so the frontend cannot change the theorem content of the stabilized sector. Condition (iii) aligns frontend transport with the residual-record transport fixed in Theorem 11.27, hence visible proof transport, frontend proof transport, and record interpretation commute. Finally, Condition (iv) excludes the only remaining failure mode: a formally new symbol that would secretly function as a new visible primitive. Therefore the frontend is conservative over the visible theory. If a second conservative frontend is given, both must factor through the same interpreter and hence agree on the visible image up to the unique record-level identification already fixed by Theorem 11.30. □

Corollary 11.34 (3.6 admissibility test for later language layers)**.** *Any later ZFM-facing layer, typed proof language, computational frontend, or proof-economy syntax is admissible for the stabilized visible sector only if it is introduced as a conservative frontend in the sense of Definition 11.32. In particular, later language work is no longer judged by expressive novelty, but by exact factorization through the residual-record interpreter.*

Proof. This is the direct content of Theorem 11.33. Once 3.5 has fixed the visible residual interface and its exact record-level interpretation, the only mathematically faithful later language layers are those that remain conservative over that same interface. □

Lemma 11.35 (3.6.1 composition and primitive non-drift)**.** *Let*

$$\mathcal{F}_B : \mathrm{Vis}(B) \to \mathrm{Fr}(B), \qquad \mathcal{J}_B : \mathrm{Fr}(B) \to \mathrm{Rec}(B)$$

and

$$\mathcal{F}'_B : \mathrm{Fr}(B) \to \mathrm{Fr}'(B), \qquad \mathcal{J}'_B : \mathrm{Fr}'(B) \to \mathrm{Rec}(B)$$

be admissible conservative frontends over the same stabilized visible sector, with the second layer conservative over the record semantics already fixed by the first. Then the composite presentation

$$\mathcal{F}'_B \circ \mathcal{F}_B : \mathrm{Vis}(B) \to \mathrm{Fr}'(B)$$

is again an admissible conservative frontend. In particular, conservative compression is closed under composition.

Moreover, no primitive drift can occur under admissible conservative extension: every primitive symbol in the downstream frontend $\mathrm{Fr}'(B)$ *that contributes to visible semantics remains definable, on the visible image, from the residual-record package*

$$\mathfrak{R}(B) = \big(\Phi, \Delta_{\mathrm{diag}}, \varepsilon_{\mathrm{par}}, \chi_{\mathrm{par}}, \widehat{\chi}_{\mathrm{par}}\big)$$

together with the chartwise shell coordinate and parity label (s, π).

Proof. By admissibility of the first layer,

$$\mathcal{J}_B \circ \mathcal{F}_B = \mathcal{I}_B.$$

By admissibility of the second layer over the already fixed record semantics of the first,

$$\mathcal{J}'_B \circ \mathcal{F}'_B = \mathcal{J}_B$$

on the image relevant for visible statements. Therefore

$$\mathcal{J}'_B \circ \mathcal{F}'_B \circ \mathcal{F}_B = \mathcal{J}_B \circ \mathcal{F}_B = \mathcal{I}_B,$$

so the composite frontend still factors through the residual-record interpreter. Truth preservation and reflection pass through composition, and transport compatibility is inherited because both frontend layers commute with residual-record isomorphisms. Finally, if a downstream primitive were to carry genuinely new visible information, its record image would define a new visible primitive outside $\mathfrak{R}(B)$ and (s, π), contradicting admissibility. Thus conservative frontends compose and cannot introduce primitive drift. □

3.6.1 working remark. The role of this lemma is to prevent later language engineering from re-opening the visible sector by iteration. Once a frontend layer is certified as conservative, any further admissible compression must remain conservative over the same residual-record interface. So 3.6 is now not only an admissibility principle for one future language layer, but a closure principle for chains of later language layers.

Status of the 3.6 line. Section 3.6 still does not build the later language itself. Its role is narrower and structural: it fixes the admissibility constraints that any later ZFM-facing layer, typed proof frontend, or compressed proof-economy syntax must satisfy. The visible interface remains the residual-record interface established in Section 3.5; Section 3.6 only certifies which later language layers are conservative over that interface.

3.6.0 working remark. This is the first point at which the future language project becomes structurally disciplined enough to be tested. A frontend is now admissible only if it acts as a conservative presentation of the already fixed visible theory. That shifts the burden of later language work completely: the task is no longer to invent a prettier or stronger visible basis, but to prove exact factorization, transport-compatibility, and primitive non-drift over the residual record.

3.7 OP-first consolidation: residual downstream reduction

The first 3.7 step is intentionally *not* a new QAM-, ZFM-, or language-layer construction. After the 3.5 residual-record interface and the 3.6 conservative-frontend discipline, the natural next question is whether the remaining historical OPs can now be reduced further *without* enlarging the visible ontology. In the present stabilized reading, the answer is yes: the remaining live fronts are to be treated as residual downstream tasks, not as invitations to reopen the visible basis.

Definition 11.36 (Residual downstream OP)**.** *A historical open problem is called* residual downstream *if its remaining content can be formulated entirely in terms of the residual-record package*

$$\mathfrak{R}(B) = \big(\Phi, \Delta_{\mathrm{diag}}, \varepsilon_{\mathrm{par}}, \chi_{\mathrm{par}}, \widehat{\chi}_{\mathrm{par}}\big),$$

the chartwise shell/parity data (s, π), and conservative downstream constructions over the same record semantics. Equivalently, solving such a problem may require new estimates, global identifications, asymptotic control, or controlled emergence arguments, but may not require a new visible primitive basis.

Theorem 11.37 (3.7 residual downstream reduction principle)**.** *Assume the stabilized visible generator-atlas-holonomy sector, the exact representability result of Section 3.5, and the conservative frontend discipline of Section 3.6. Then the remaining historical OPs split as follows.*

(i) ***Already closed as separate historical OPs.*** *OP 2 as an independent operator problem, OP 3 as a primary open problem, and OP 5 step (A) no longer remain open fronts in the main line. Their surviving content is either already absorbed into OP 1-type correction work or reduced to the residual part of OP 5.*

(ii) ***Reduced residual open core of OP 1.*** *The remaining content of OP 1 is not the existence of a visible frame candidate, but the explicit prime-lattice / continuum-limit correction that moves the continuum fixed-point candidate to the measured matter-frame value.*

(iii) ***Reduced residual open core of OP 5.*** *The remaining content of OP 5 is the quantified Theta-sum / balance problem from step (B). In particular, OP 5 is no longer a broad visible-sector problem, but a residual quantitative balance problem downstream of the same stabilized record.*

(iv) ***Other live fronts remain downstream.*** *The field-equation / controlled-emergence problem, the mass-hierarchy problem, the three-generation problem, and the compact Dixon reformulation remain open, but their open content is downstream of the residual-record interface. None of them justifies reopening the visible primitive basis fixed in Sections 3.5–3.6.*

(v) ***No remaining OP presently forces a second visible interface.*** *Consequently, the current OP-first program may proceed by reduction, estimation, and controlled downstream closure alone: in the present stabilized reading, any further progress on the remaining live fronts must factor through the existing residual-record semantics and may not appeal to a new visible primitive layer.*

Proof. Item (i) is the status reclassification already recorded in the 3.5.x cleanup line: OP 2 as a separate operator problem is absorbed, OP 3 is no longer carried as a primary main-line open front, and OP 5 step (A) is already closed while only step (B) remains. Item (ii) is exactly the current reading of OP 1 in the synchronized historical-open-problems layer: the broad frame-candidate issue has collapsed to the explicit prime-lattice / continuum-limit correction. Item (iii) is the corresponding synchronized reading of OP 5, whose remaining content is the Theta-sum / balance step after the earlier renormalization and step (A) closure. Item (iv) is the present status of the remaining live fronts, all of which are explicitly recorded as open but downstream structural or quantitative tasks rather than missing visible primitives. Finally, Item (v) follows from Theorem 11.30, Theorem 11.33, and Lemma 11.35: once visible truth is exactly representable at record level and all admissible later layers are conservative, no unresolved historical OP can legitimately force a second visible interface in the present stabilized reading without violating the already established residual-record discipline. □

Corollary 11.38 (3.7 OP-first corollary: defer later language growth)**.** *At the present stage, there is no mathematical need to pull the later QAM-, ZFM-, or proof-language layer upward in order to continue the main line. The next admissible 3.7 work may instead focus directly on residual OP closure — in particular on the OP 1 correction problem and the remaining OP 5 balance step — while keeping later language engineering deferred.*

Proof. This is the direct consequence of Theorem 11.37. Since the remaining live fronts are already downstream of the residual-record interface and do not require a new visible primitive basis, the main line may continue by solving those residual tasks first. □

3.7.1 working remark. This step deliberately changes the local priority order. The document now records that the next main-line gains should come from solving the residual OP cores whenever possible, not from prematurely lifting the deferred language program. QAM-, ZFM-, or typed-frontend work therefore remains admissible but non-urgent: it can stay deferred until the remaining downstream OPs have either been closed or reduced further.

3.7.1 release note. This public release is recorded under Zenodo DOI:

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10.5281/zenodo.19412478
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It freezes the OP-first 3.7 line without advancing the deferred QAM/language layer.

The present subsection should be read as a release-grade consolidation claim, not as a declaration that every residual OP is already solved. What is claimed is narrower and more structural: the current chain is strong enough that the remaining live fronts can be pursued as downstream residual tasks over the fixed record interface. This is the point that makes a 3.7 release defensible while leaving the actual OP-closure push to the next cycle.

3.8 OP-closure phase: two-core reduction and closure discipline

The next cycle is now promoted from a planning note to a working mathematical stage. The point of 3.8 is still *not* to lift the deferred QAM-, ZFM-, or language-layer upward.

Instead, 3.8 treats the remaining foundational closure burden as a two-core residual program over the already fixed residual-record interface. The guiding question is therefore no longer "what further visible structure is missing?", but rather "which residual OP cores must still be closed before the historical OP layer can be regarded as exhausted at the foundational level?"

Definition 11.39 (Primary closure core and secondary closure family)**.** *In the present stabilized reading, the* primary closure core *consists of exactly two residual tasks:*

(i) the OP 1 prime-lattice / continuum-limit correction problem for the matter-frame value $\beta_{\mathrm{sym}} \approx 0.5493$*;*

(ii) the OP 5 step (B) Theta-sum / Theta–R balance problem at the photon-frame cutoff.

The secondary closure family *consists of the remaining open blocks — controlled emergence / field-equation derivation, mass hierarchy, three-generation splitting, and compact Dixon reformulation — insofar as their unresolved content is downstream of the same residual-record semantics and does not reopen the visible primitive basis.*

Theorem 11.40 (3.8 two-core closure reduction principle)**.** *Assume the 3.7 residual downstream reduction principle. Then the foundational historical OP burden of the Companion is reduced to the primary closure core from Definition 11.39 in the following precise sense.*

(i) No already synchronized historical OP outside OP 1 and OP 5 step (B) remains a primary foundational blocker of the main line.

(ii) Any later successful closure of the primary closure core within the existing residual-record semantics closes the remaining historical-interface burden as such*: after that point, the secondary closure family may still require substantial mathematics, estimates, or phenomenological extraction, but it can no longer be counted as evidence that the visible interface was incomplete.*

(iii) Consequently, 3.8 may be organized as an OP-closure phase with a strict dependency order: first the OP 1 correction problem and the OP 5 step (B) balance problem, then the secondary closure family, and only afterwards any renewed consideration of deferred language growth.

Proof. Item (i) is exactly the synchronized status reached in Sections 3.5–3.7: OP 2 is closed as a separate historical operator problem, OP 3 is closed at the present structural level, and OP 5 step (A) is already removed from the live main-line burden. What remains as genuinely primary closure work is therefore the residual OP 1 correction problem together with OP 5 step (B). Item (ii) follows from the residual-record exact representability and conservative-frontend discipline: once these two residual cores are closed without introducing any new visible primitive, the historical reason for reopening the visible interface disappears. The remaining tasks of controlled emergence, spectral assignment, generation splitting, and Dixon packaging may still be mathematically deep, but they then belong to downstream completion and extraction over a fixed interface rather than to unresolved foundational interface formation. Item (iii) is therefore the admissible work-ordering rule for the next stage. □

Corollary 11.41 (3.8 closure discipline rule)**.** *At the start of the 3.8 cycle, the main line should treat every proposed detour according to one question first: does it close part of the primary closure core? If not, then in the absence of a demonstrated dependency it is secondary to the OP 1 / OP 5 step (B) program. In particular, deferred QAM-, ZFM-, or typed-language growth remains non-urgent unless one of the two primary closure cores genuinely requires it.*

Proof. This is the direct operational reading of Theorem 11.40. The theorem does not claim that the secondary closure family is trivial, only that it is no longer first in the dependency order of the foundational closure program. □

3.8.0 working remark. The practical effect of this step is to make the next cycle much narrower. The burden of proof is now concentrated on two residual cores instead of a diffuse list of historical open fronts. That does not yet solve OP 1 or OP 5 step (B), but it does mean that a successful 3.8 line can be judged by a much clearer criterion: did we close one of the two cores, reduce one of them to a still smaller kernel, or prove that some secondary task is in fact dependent on one of them?

3.8.0 status note. This is a local working-stage promotion, not yet a public-release claim. What is fixed here is the dependency structure of the next closure cycle: the historical OP layer is now to be attacked through a two-core program, with the remaining open blocks treated as downstream completion unless a contrary dependency is demonstrated.

C.28. Two-core reduction logic. Once the residual-record interface is fixed and later frontend growth is constrained to be conservative, a historical-open-problem list can be partially ordered by dependency on the visible interface. The point of the two-core reduction is that the remaining foundational closure burden is no longer distributed across many incomparable items: it is concentrated in the OP 1 correction problem and the OP 5 step (B) balance problem.

C.29. What remains open after two-core reduction. The two-core reduction does *not* claim that field equations, mass hierarchy, generation splitting, or Dixon packaging are solved. It claims something narrower and more structural: those tasks no longer appear first in the dependency ordering of foundational closure, and they do not presently justify reopening the visible primitive basis. They remain mathematically serious, but downstream.

3.8.1 residual-controlled emergence equation

The two-core reduction makes it possible to revisit the field-equation / controlled-emergence question without reopening the visible interface. The point is not to replace the structural master equation by a second fundamental equation, but to write the physical readout problem in a form whose unresolved content is explicitly concentrated into the same two residual cores that already govern the 3.8 closure program.

Definition 11.42 (Residual-controlled physical interface form)**.** *Let* $\mathcal{E}_{\mathrm{str}}[\Psi]$ *denote the structural master side of the present framework, schematically read as*

$$\mathcal{E}_{\mathrm{str}}[\Psi] := \mathfrak{G}[\Psi, g] + \mathfrak{T}_{\mathrm{tri}}[\Psi] + \mathfrak{H}_{\mathrm{HC}}[\Psi] + \mathfrak{Q}_{\mathrm{cl}}[\Psi] + \mathfrak{B}_{\beta}[\Psi].$$

A residual-controlled physical interface form *is an admissible readout equation of the shape*

$$\Pi_{\mathrm{phys}}\big(\mathcal{E}_{\mathrm{str}}[\Psi]\big) = \mathcal{E}_1[\Psi] + \mathcal{E}_5[\Psi],$$

where Π_{phys} *is a closure-compatible physical readout / observer projection,* $\mathcal{E}_1$ *is a remainder family carrying the OP 1 prime-lattice / continuum-limit correction burden, and* $\mathcal{E}_5$ *is a remainder family carrying the OP 5 step (B) Theta-sum / Theta–R balance burden.*

Theorem 11.43 (3.8.1 residual-controlled emergence principle)**.** *Assume the Master probe field equation (Theorem* `thm:master-field-equation` *in the active-core PDF), the residual-record exact representability principle, and the 3.8 two-core closure reduction principle. Then the unresolved field-equation / controlled-emergence burden of the present Companion admits a residual-controlled physical interface form in the sense of Definition 11.42. More precisely:*

(i) the structural master side may be projected to an admissible physical readout equation without introducing any new visible primitive beyond the residual-record interface;

(ii) every unresolved contribution that still survives in the present stabilized reading can be absorbed into the two remainder families $\mathcal{E}_1$ *and* $\mathcal{E}_5$*, corresponding respectively to the OP 1 correction problem and the OP 5 step (B) balance problem;*

(iii) consequently, the field-equation question is no longer a separate primary foundational front parallel to the OP program, but an OP-first residual task over the already fixed interface.

Proof. Theorem `thm:master-field-equation` in the active-core PDF already fixes a closure-compatible structural master equation for the probe/field sector. Sections 3.5 and 3.6 then force all later readout and frontend organization to remain conservative over the residual record, so no admissible physical projection may introduce a genuinely new visible primitive.

Theorem 11.40 further reduces the remaining foundational burden to two cores only: the OP 1 prime-lattice/continuum-limit correction and the OP 5 step (B) Theta-balance problem. Therefore any unresolved contribution in the controlled-emergence reading of the structural master equation must, in the present stabilized reading, belong to one of these two residual families. Writing the projected equation in the form

$$\Pi_{\mathrm{phys}}\big(\mathcal{E}_{\mathrm{str}}[\Psi]\big) = \mathcal{E}_1[\Psi] + \mathcal{E}_5[\Psi]$$

is thus not a new ontological postulate but a bookkeeping theorem for the remaining closure burden. Items (i)–(iii) are exactly the content of this repackaging. □

Corollary 11.44 (3.8.1 OP-first field-equation strategy)**.** *A later controlled derivation of Einstein-, Dirac-, or QFT-type projection formulae need not be treated as a third independent foundational closure front, provided it remains conservative over the residual record and its unresolved content continues to factor through* $\mathcal{E}_1$ *and* $\mathcal{E}_5$*. In particular, new language growth such as QAM remains non-urgent for the field-equation question unless one of the two residual families genuinely requires it.*

Proof. This is the direct operational reading of Theorem 11.43. Once the unresolved physical-interface burden is concentrated in $\mathcal{E}_1$ and $\mathcal{E}_5$, the field-equation task can only regain primary-foundational status by exhibiting a new visible primitive or by proving that one of the two remainder families cannot be represented within the existing interface. Neither move is currently licensed. □

3.8.1 working remark. This step does not claim a finished Einstein equation, a finished Dirac projection, or a closed emergence derivation with all constants fixed. What it does claim is narrower and more useful for 3.8: the field-equation problem can now be attacked as a residual equation with two named remainder channels instead of as an independent search for a new formal foundation.

3.8.1 status note. At the level of project strategy, this is the point where a new field equation becomes helpful and QAM remains secondary. The residual-controlled interface equation is useful because it exports the field-equation burden directly into the two live OP cores. QAM would mainly reorganize notation and proof syntax, whereas the present step reorganizes the remaining mathematics.

C.30. Residual-controlled field-equation bookkeeping. The residual-controlled interface form should not be misread as a replacement of the structural master equation by a second foundational law. Its purpose is to expose the unresolved physical readout burden as a sum of two residual families already known from the OP-closure program. This is why it can help 3.8 directly even before any later language engineering is undertaken.

C.31. Why QAM remains deferred here. In the present stabilized reading, QAM would primarily compress and reorganize proofs. By contrast, the residual-controlled interface equation narrows the actual unresolved mathematics by identifying where the field-equation burden still lives. As long as that burden is concentrated in $\mathcal{E}_1$ and $\mathcal{E}_5$, direct OP closure remains the preferred next move.

3.8.2 OP 1 scalar residual channel

The next useful closure move is to compress OP 1 to the smallest genuinely live mathematical object. The present document already contains the continuum fixed-point candidate, the self-consistent physical matching system, and the statement that the surviving burden is the prime-lattice / continuum-limit correction. What is added here is the observation that, in the present stabilized reading, this burden can be written as a single scalar residual channel rather than as a diffuse family of unrelated corrections.

Definition 11.45 (OP 1 scalar residual profile)**.** *For $\mu > 1$, define the continuum-induced matter-frame candidate*

$$\beta_{\mathrm{cont}}(\mu) := \sqrt{\frac{\mu - 1}{\mu + 1}}$$

and the spectral correction residual

$$\mathcal{R}_1(\mu) := \log\bigl(\sqrt{3}\,\beta_{\mathrm{cont}}(\mu)\bigr) + \frac{\gamma_1}{4\pi\mu^5}.$$

The equation $\mathcal{R}_1(\mu) = 0$ is called the OP 1 scalar residual channel. *It is the logarithmic form of the self-consistent matching condition*

$$\beta_{\mathrm{cont}}(\mu) = \frac{1}{\sqrt{3}} \exp\Bigl(-\frac{\gamma_1}{4\pi\mu^5}\Bigr).$$

Theorem 11.46 (3.8.2 OP 1 single-channel reduction principle)**.** *Assume the Tribonacci continuum fixed-point theorem, the self-consistent OP 1 matching system (`eq:SC1`, active-core PDF)–(`eq:SC2`, active-core PDF), and the 3.8.1 residual-controlled emergence principle. Then, in the present stabilized reading:*

(i) the remaining OP 1 burden is completely represented by the scalar residual profile $\mathcal{R}_1(\mu)$ of Definition 11.45;

(ii) closing OP 1 no longer requires a new visible primitive, a new field equation, or a new language layer, but only a first-principles control or derivation of the correction law encoded by $\mathcal{R}_1(\mu)$;

(iii) any later closure of OP 1 that is admissible in the current framework must factor through the vanishing or controlled estimation of this scalar residual channel.

Proof. The continuum side of OP 1 is already fixed by Theorem `thm:tribonacci` in the active-core PDF: the uncorrected self-referential frame candidate is the Tribonacci value, equivalently read through the continuum expression $\beta_{\text{cont}}(\mu) = \sqrt{(\mu-1)/(\mu+1)}$. The physical matching side is already fixed by the self-consistent system (`eq:SC1`, active-core PDF)–(`eq:SC2`, active-core PDF), whose unresolved content is precisely the prime-lattice / continuum-limit correction law. Taking logarithms rewrites the matching problem as the vanishing of the single scalar expression $\mathcal{R}_1(\mu)$. This loses no information because $\mu > 1$ and the exponential side of (`eq:SC1`, active-core PDF) is positive. By the 3.8.1 residual-controlled emergence principle, the field-equation burden cannot add a third independent OP 1 channel; it can only feed into the same residual family $\mathcal{E}_1$. Hence the entire remaining OP 1 content is exhausted by the scalar residual channel $\mathcal{R}_1(\mu)$, and any admissible closure must proceed by deriving, controlling, or eliminating that residual. □

Corollary 11.47 (3.8.2 OP 1 closure test)**.** *Within the present Companion, OP 1 is closed as soon as one of the following is obtained without introducing a new visible primitive basis:*

(a) a first-principles derivation showing $\mathcal{R}_1(\gamma_L) = 0$ exactly;

(b) a structural derivation of the prime-lattice correction law that identifies $\mathcal{R}_1$ with an already controlled arithmetic or spectral remainder and proves its vanishing;

(c) a proof that the residual channel is bounded in a way that collapses the remaining numerical gap and stabilizes the same β_{sym}-value uniquely.

In particular, OP 1 is now a one-channel closure problem.

Proof. This is the direct operational reading of Theorem 11.46. Once all remaining OP 1 content is represented by a single scalar residual channel, closing that channel by exact vanishing, structural identification, or sufficiently strong bound is enough to close OP 1 within the present stabilized reading. □

3.8.2 working remark. This step does not yet prove the prime-lattice correction from first principles. What it does prove is that the surviving OP 1 mathematics has become one-dimensional in a strong structural sense: there is no longer a family of incomparable OP 1 subproblems, but a single residual channel to be killed or identified.

3.8.2 status note. This is the first real 3.8 closure compression after the two-core reduction. It is also a useful diagnostic: if OP 1 later resists closure, the obstruction should now show up as a failure to control $\mathcal{R}_1$, rather than as a vague demand for new visible machinery.

C.32. Why the OP 1 residual channel is scalar. The two ingredients entering the current OP 1 matching problem are already fixed: the continuum frame candidate and the prime-lattice spectral correction ansatz. Their discrepancy can therefore be measured by a single logarithmic mismatch. This is why the surviving burden is scalar even though its derivation may still be deep.

C.33. What 3.8.2 does *not* claim. Section 3.8.2 does not claim a first-principles proof of the μ^5 exponent, nor a full arithmetic derivation of the prime-lattice correction. It claims something narrower: that all such work is now forced into a single residual channel and no longer distributed across several independent visible fronts.

3.8.3 OP 1 root isolation and near-closure window

Once OP 1 has been compressed to the scalar residual channel $\mathcal{R}_1(\mu)$, the next useful move is not yet to claim exact closure, but to show that the residual channel already has a unique physical root in the relevant regime and that the current γ_L-value lies extremely close to it. This upgrades OP 1 from a diffuse correction problem to a sharply localized near-closure problem.

Definition 11.48 (OP 1 physical root window)**.** *Let*

$$I_{\text{phys}} := [1.85, 1.90].$$

Within the present Companion, the interval I_{phys} is called the OP 1 physical root window*. It contains the current matter-frame value $\gamma_L \approx 1.864$ and the single-channel OP 1 residual profile $\mathcal{R}_1(\mu)$ of Definition 11.45.*

Theorem 11.49 (3.8.3 OP 1 root isolation principle)**.** *Within the OP 1 scalar residual channel of Definition 11.45, the following hold on the physical root window $I_{\text{phys}} = [1.85, 1.90]$:*

(i) *$\mathcal{R}_1$ is strictly increasing on I_{phys};*

(ii) *$\mathcal{R}_1(1.85) < 0$ and $\mathcal{R}_1(1.90) > 0$;*

(iii) *hence there exists a unique root*

$$\mu_* \in (1.85, 1.90)$$

such that $\mathcal{R}_1(\mu_*) = 0$;

(iv) *numerically,*

$$\mu_* \approx 1.86366225,$$

so the current matter-frame value $\gamma_L \approx 1.864$ differs from the unique OP 1 root only by

$$|\gamma_L - \mu_*| \approx 3.38 \times 10^{-4},$$

i.e. at the level of about 1.8×10^{-4} relative to μ_*.

Proof. From Definition 11.45,

$$\mathcal{R}_1(\mu) = \frac{1}{2}\log\left(\frac{3(\mu-1)}{\mu+1}\right) + \frac{\gamma_1}{4\pi\mu^5}.$$

Therefore

$$\mathcal{R}_1'(\mu) = \frac{1}{\mu^2-1} - \frac{5\gamma_1}{4\pi\mu^6}.$$

On $I_{\text{phys}} = [1.85, 1.90]$, the first term is decreasing and the second term is increasing in absolute value only mildly; direct evaluation at the right endpoint gives

$$\mathcal{R}_1'(1.90) = \frac{1}{1.90^2-1} - \frac{5\gamma_1}{4\pi\cdot 1.90^6} > 0.265 - 0.255 > 0.$$

Hence $\mathcal{R}_1'(\mu) > 0$ throughout I_{phys}, so $\mathcal{R}_1$ is strictly increasing there. Direct evaluation gives

$$\mathcal{R}_1(1.85) \approx -3.71\times 10^{-3} < 0, \qquad \mathcal{R}_1(1.90) \approx 9.70\times 10^{-3} > 0.$$

By continuity and strict monotonicity, there is a unique $\mu_* \in (1.85, 1.90)$ with $\mathcal{R}_1(\mu_*) = 0$. A numerical solve of the scalar residual equation yields

$$\mu_* \approx 1.86366225.$$

Comparing with the current matter-frame value $\gamma_L \approx 1.864$ gives

$$|\gamma_L - \mu_*| \approx 3.38\times 10^{-4},$$

and the corresponding relative mismatch is about 1.8×10^{-4}. No new visible primitive enters this argument: it is purely an analysis of the already isolated scalar residual channel. □

Corollary 11.50 (3.8.3 OP 1 exact-vs-structural remainder)**.** *After Theorem 11.49, the remaining OP 1 burden is no longer to locate a candidate value for γ_L, but only to explain structurally why the exact root μ_* of the scalar residual channel must coincide with the framework value read as γ_L. Equivalently, the surviving task is an exact prime-lattice / continuum derivation of the residual law, not a search for a second numerical branch.*

Proof. Theorem 11.49 proves that the scalar residual channel already has a unique physical root in the relevant regime and that the current γ_L-value lies extremely close to that root. Therefore the numerical candidate is no longer underdetermined. What remains open is the structural identification of that unique root with the framework quantity γ_L, i.e. the first-principles explanation of why the residual channel must vanish exactly there. □

3.8.3 working remark. This is the first point in the OP 1 program where the residual channel becomes visibly *near-closed*: not because the prime-lattice correction has been derived, but because the residual equation already has a unique physical root and the framework value currently used for γ_L sits essentially on top of it.

3.8.3 status note. Section 3.8.3 still does *not* claim exact first-principles closure of OP 1. What it does claim is narrower and stronger than 3.8.2: the unresolved content is now an exact structural identification problem, not a broad numerical search problem.

C.34. Why root isolation matters. A scalar residual channel can still hide several distinct physical branches. The value of Theorem 11.49 is precisely that this no longer happens in the currently relevant OP 1 window: the residual has a unique root there, so the closure burden becomes conceptual and structural rather than branch-selection driven.

C.35. What 3.8.3 still leaves open. The exponent μ^5, the prime-lattice interpretation of the correction, and the exact identification $\mu_* = \gamma_L$ remain open. The present step only shows that these open points now live over a unique near-closure root rather than over a family of competing candidates.

Historical Open Questions (status-synchronized)

The list below is retained for proof-memory and cross-reference continuity. It should not be read as a list of equally open fronts. Some entries are now only *residual quantitative tasks*, while others are already closed as separate historical OPs. In particular, nothing in the list below should be read as license to reopen the visible interface fixed in the 3.5 record stage: the remaining work is downstream of the residual record in the present stabilized reading, not parallel to it.

1. **Structural completion: observer frame from first principles [reduced residual open].** The value $\beta_{\text{sym}} \approx 0.5493$ is determined from γ_{BI} and γ_{string}. Its remaining open content is no longer the broad existence of a frame candidate, but the explicit derivation of the prime-lattice / continuum-limit correction that moves the continuum fixed-point candidate to the measured matter-frame value.

2. **Field equation / controlled emergence [open].** A natural variational candidate is
$$\delta \int R\,\tau_n^2\,d\mu = 0.$$
What remains open is the fully controlled derivation of Einstein- and Dirac-type projections with explicit constants, error bounds, and approximation control.

3. **Spectrum: mass hierarchy [open].** The heuristic relation $m_p \sim \sqrt{p}\,m_P$ has the correct qualitative structure, but the map from prime/spectral index to Standard-Model quantum numbers is still unknown.

4. **Spectrum: three generations [open].** The chain $S_2 \supset S_1 \supset S_0$ is structurally compatible with three fermion generations, but no quantitative derivation is given here.

5. **Structural completion: Dixon algebra [open].** A compact reformulation in $\mathbb{C} \otimes \mathbb{H} \otimes \mathbb{O}$ (dimension 64) may unify the present results in a single algebraic object.

6. **Historical OP 2: Dirac operator on H_-^+ [closed as a separate OP].** The existence, essential self-adjointness, and GOE-class parts are already closed in the present text. The former T4 frame-offset part is no longer separate and is absorbed into Item 1 above.

7. **Historical OP 3: Koide / θ_K formula [closed at the present structural level].** The Koide/triolic block should remain available for later proof-density enrichment, but it should no longer be counted among the primary open problems of the present version.

8. **Historical OP 5: photon-frame Λ-cancellation [reduced residual open].** Step (A) of the photon-frame Θ–R program is already closed. The remaining live content is Step (B), i.e. the exact Θ-sum / Θ–R balance at the photon-frame cutoff.

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Historical Complete Snapshot: Formal Proof Supplement

This appendix contains the full formal proof architecture for the results stated in the main text. It is self-contained and can be read independently. All macros are as defined in the main preamble.

19I. Optimization and normalization reserve

Why an optimization layer is needed. Once the computational line carries typed programs, explicit rejection modes, and certificate-bearing outputs, a further reserve becomes natural: normalization and optimization rules that compress proof-programs without destroying admissibility evidence. In the present setting this is especially important because the entire motivation of MML/MMA is *proof economy by structural normalization.* Hence optimization should not be understood as a merely engineering afterthought; it is part of the proof-theoretic architecture itself.

Design principle. The guiding constraint is that optimization must remain *certificate-preserving.* A shorter program is acceptable only if it produces the same admissible structural state and carries either the same certificates or canonically reduced certificates. Thus the relevant notion is not arbitrary minimization, but normalization toward structurally transparent forms.

Four initial optimization families. A first reserve may separate the following layers:

Family	Function
Peephole normalization	removes local instruction redundancies
Macro normalization	rewrites proof macros into canonical expansions
Certificate-preserving compression	shortens programs while preserving witnesses
Canonical proof-program forms	rewrites structurally equivalent programs into a fixed normal form

Peephole rules. At the lowest level, local instruction windows should admit simple rewrites such as

$$\texttt{PROJECT}\ S;\ \texttt{PROJECT}\ S \implies \texttt{PROJECT}\ S,$$
$$\texttt{TRACE};\ \texttt{TRACE} \implies \texttt{TRACE} \quad \text{(when no state update intervenes)},$$
$$\texttt{EXPORT_BOOL};\ \texttt{EXPORT_BOOL} \implies \texttt{EXPORT_BOOL},$$
$$\texttt{HCSEL};\ \texttt{HCSEL} \implies \texttt{HCSEL} \quad \text{(when branch context is unchanged)}.$$

These are intentionally conservative. Their role is not to be clever, but to remove obvious redundancies while preserving the structural path.

Macro normalization. The emerging macro layer above MMA should itself admit canonical expansion and contraction. Schematically,

$$\texttt{CLOSE_AND_PROJECT} \implies \texttt{CLOSE};\ \texttt{PROJECT},$$

while conversely a recognized primitive sequence may be re-packed into a named proof macro whenever the corresponding certificate chain is preserved. This creates a disciplined relation between concise proof scripts and auditable low-level expansions.

Certificate-preserving compression. A central normalization requirement is that compression should preserve witnesses. If a program segment

$$P \Longrightarrow P_{\text{norm}}$$

is accepted as a valid normalization, then there should exist a map on certificates

$$\mathcal{C}(P) \longmapsto \mathcal{C}(P_{\text{norm}})$$

that either preserves the original chain or contracts it into a canonical equivalent chain. In this way program shortening does not erase the proof record; it only normalizes it.

Canonical proof-program forms. The long-term aim is that many proof fragments should be writable in a small family of normal forms. A first reserve may distinguish at least the following:

`ClosureNormalForm`,
`BranchNormalForm`,
`TraceNormalForm`,
`ExportNormalForm`.

These names do not yet define a completed formalism, but they indicate the intended endpoint: proof programs that can be compared, optimized, certified, and eventually indexed by structural class.

Mini normalization example. For instance, the sequence

```
PROJECT C7
TRACE
TRACE
HCSEL
HCSEL
EXPORT_BOOL
```

may normalize to

```
PROJECT C7
TRACE
HCSEL
EXPORT_BOOL
```

provided that no state-changing instruction intervenes and the resulting certificates remain equivalent. The mathematical point is that admissibility is preserved while the proof-program becomes shorter and more canonical.

Relation to proof economy. This reserve clarifies the deeper purpose of the MM-L/MMA line. The goal is not merely to invent a language for its own sake, but to create a setting in which long structural proofs can be compressed into short normalized programs, with explicit certificate trails and canonical rewrites. In this sense optimization is part of the normalization discipline of the Companion itself.

Full proof of Theorem `thm:gue-selfdet` in the active-core PDF

Lemma 11.51 (GUE self-determination of γ_L — formal version)**.** *For $\mu > 0$ define the photon-frame unfolding density*

$$\rho_\gamma(t;\mu) := \frac{\log\bigl(t/(2\pi\mu)\bigr)}{2\pi\mu}, \qquad t > 2\pi\mu,$$

and the unfolded spacings $s_n(\mu) := \tilde\gamma_{n+1}(\mu) - \tilde\gamma_n(\mu)$ where $\tilde\gamma_n(\mu) := \int_0^{\gamma_n} \rho_\gamma(t;\mu)\,dt$. Let P_μ be the empirical spacing distribution of $\{s_n(\mu)\}$ and let P_{GUE} be the Wigner–Dyson distribution $p_{\mathrm{GUE}}(s) = \frac{\pi}{2} s\, e^{-\pi s^2/4}$. Set $\mathcal{F}(\mu) := D_{\mathrm{KS}}[P_\mu, P_{\mathrm{GUE}}]$. Then:

(i) $\mathcal{F}$ has a unique global minimum at $\mu = \gamma_L^*$ in the asymptotic regime $\gamma \gg 2\pi\mu$.

(ii) $\gamma_L^* = \gamma_{\mathrm{BI}}/\gamma_{\mathrm{string}}$ to available numerical precision.

(iii) γ_L^* is determined solely by the zero sequence $\{\gamma_n\}$.

Proof. Step 1 (Sensitivity). A perturbation $\mu \mapsto \mu + \delta\mu$ rescales unfolded spacings by

$$s_n(\mu + \delta\mu) \approx s_n(\mu)\Bigl(1 - \frac{\delta\mu}{\mu} \cdot \frac{1}{\log(\gamma_n/2\pi\mu)}\Bigr).$$

Since the GUE mean spacing equals 1 by construction, any $\mu \neq \gamma_L^*$ shifts the mean of P_μ away from 1 and increases D_{KS}. Hence $\mathcal{F}$ has a strict local minimum wherever the rescaled mean equals 1.

Step 2 (Uniqueness). The critical line $\mathrm{Re}(s) = \frac{1}{2}$ is the unique fixed-point set of J. Here $J(s) = 1 - \bar s$, and this fixed-point description is frame-invariant by Theorem `thm:unity` in the active-core PDF. By Theorem `thm:gue` in the active-core PDF, L_0 on H_+^+ has Dyson index $\beta = 2$ in the photon frame. Both properties are frame-invariant. The Wigner–Dyson distribution therefore singles out exactly one unfolding scale, yielding a unique minimiser γ_L^* of $\mathcal{F}$.

Step 3 (Numerical identification). Simulation `gue_frame_correction.py` establishes $\mathcal{F}(\gamma_{\mathrm{BI}}/\gamma_{\mathrm{string}}) < \mathcal{F}(\mu)$ for all tested $\mu \neq \gamma_{\mathrm{BI}}/\gamma_{\mathrm{string}}$ in the asymptotic regime. The χ^2 improvement of $\approx 21\%$ relative to standard unfolding ($\mu = 1$) quantifies the depth of the minimum. The minimum is 3σ-detectable at $N \approx 205$ zeros in the asymptotic range. □

Remark 11.52 (Historical logical reversal)**.** *Prior to this lemma, $\gamma_L = \gamma_{\mathrm{BI}}/\gamma_{\mathrm{string}}$ was an* input*. Lemma 11.51 reverses the direction: $\gamma_{\mathrm{BI}}/\gamma_{\mathrm{string}}$ is a* consequence *of demanding that the GUE taktgeber running on $z^- = 0$ is globally consistent with $\beta = 2$. The empirical values of γ_{BI} and γ_{string} provide an independent numerical check, not a definition.*

Full proof of Theorem `thm:kc-zero` in the active-core PDF

Proposition 11.53 (Historical $k(c) = 0$ from biquaternion structure — formal version)**.** *In the Sphaera-Cascade,* $c^{(\text{photon})} = c^{(\text{matter})}$.

Proof. The Sphaera biquaternion in physical units reads $\mathcal{B} = ict \cdot \mathbf{1} + z_i e_i$. The light-cone condition $|\mathcal{B}|^2 = 0$ defines c as the ratio of the spatial to the temporal norm of $\mathcal{B}$.

Step 1 (Wick rotation is frame-independent). The Wick rotation $t \mapsto i\tau_E$ is consistent at the natural observer position $r_0/r_S = 1/(1-\beta_{\text{sym}}^2)$ and introduces no additional geometric factor beyond $g_{tt}(r_0) = -\beta_{\text{sym}}^2$. No factor of z^- is generated.

Step 2 (c defines the photon ground shell). The frame variable z^- measures the displacement of the matter shell from $z^- = 0$. The speed c is the causal propagation speed of that ground state itself — the quantity *with respect to which* β_{sym} is defined. Under a frame change $z^- \mapsto z^- + \beta_{\text{sym}}$, a massless excitation acquires a boost by β_{sym}, but its speed in the new frame remains c by the Einstein postulate, encoded in the J-symmetry structure.

Step 3 (No z^--factors in the operator realization of c). Frame changes are unitaries commuting with J; they act on the septinion components z_i but not on the J-invariant temporal component $ict \cdot \mathbf{1}$. Hence the coefficient c picks up no power of z^- under any frame change: $k(c) = 0$. □

Full proof of Theorem `thm:planck-photon` in the active-core PDF

Proof. Apply $X^{(\text{photon})} = \gamma_L^{-k(X)} \cdot X^{(\text{matter})}$ to each factor of $\ell_P = \sqrt{\hbar G_N/c^3}$:

$$\begin{aligned}
\ell_P^{(\text{photon})} &= \sqrt{\gamma_L^{-k(\hbar)}\hbar^{(m)} \cdot \gamma_L^{-k(G_N)} G_N^{(m)} \big/ \big(\gamma_L^{-k(c)} c^{(m)}\big)^3} \\
&= \gamma_L^{-(k(\hbar)+k(G_N)-3k(c))/2} \cdot \ell_P^{(\text{matter})} \\
&= \gamma_L^{-(1+2-0)/2} \cdot \ell_P^{(\text{matter})} = \gamma_L^{-3/2} \cdot \ell_P^{(\text{matter})}.
\end{aligned}$$

□

11.1 Frame-corrected GUE: detailed statement

Proposition 11.54 (Historical matter-frame Dyson index)**.** *Matter observers using the standard unfolding density* $\rho_{\text{matter}}(\gamma) = \log(\gamma/2\pi)/(2\pi)$ *measure an effective Dyson index*

$$\beta_{\text{eff}} = 2\gamma_L \approx 3.73.$$

Photon-frame unfolding recovers $\beta = 2$ *exactly. The shift changes the scale of the spacing distribution, not the universality class: level repulsion* $p(s) \propto s^2$ *near* $s = 0$ *is preserved in both frames (local property, hence frame-invariant).*

Corollary 11.55 (Historical k-value of the GUE taktgeber)**.** *The taktgeber frequency* $\nu_{\text{takt}} \propto \gamma_L$ *transforms as* $\nu^{(m)} = \gamma_L^1 \cdot \nu^{(\gamma)}$. *Hence* $k(\nu_{\text{takt}}) = 1$, *consistent with* $k(\alpha^{-1}) = 1$ *and* $k(\hbar) = 1$ *in Theorem* `thm:k-values` *in the active-core PDF.*

11.2 Formal Dirac probe structure

Definition 11.56 (Historical commutator-residue probe mass)**.** *For a closure-preserving probe state* ψ, *the commutator residue*

$$\mathfrak{m}_{\text{comm}}(\psi) := \frac{\hbar}{c}\left|\langle\psi|[\hat{\mathcal{R}}_\beta, \hat{P}_{C_7}]|\psi\rangle\right|$$

measures the antisymmetric mismatch between the shifted pacer and the admissible closure projector. Since $k(\hbar) = 1$ and $k(c) = 0$, the prefactor $\hbar/c$ carries $k = 1$, consistent with Corollary of §11.1. The commutator $[\hat{\mathcal{R}}_\beta, \hat{P}_{C_7}]$ is frame-invariant (its support lives on $\mathrm{Re}(s) = \frac{1}{2}$), so the full expression has $k = 1$: the matter-frame reading of the probe mass is displaced from the photon-frame value by a single factor γ_L.

Remark 11.57 (Historical open calculation)**.** *The remaining open step is the explicit evaluation of $[\hat{\mathcal{R}}_\beta, \hat{P}_{C_7}]$ in the septonic matrix realization of $\mathcal{M}_{16}$. Once computed, the Dirac mass is determined from the Riemann zero sequence without any free parameter.*

Attribution note

The Kim-style refinement of the fermionic probe sector (§`sec:dirac-probe` in the active-core PDF) was developed from Yilwook Kim's Zenodo line associated with concept DOI `10.5281/zenodo.19112361`. In the present paper this material is not treated as an internal axiom of the Sphaera program, but as an externally motivated operator ansatz reformulated within the current proof-mode architecture. The embedding into the septonic tree, Meta-Septum closure, Heisenberg compensator, and phase-space-shifted measurement frame is the present paper's own structural reformulation.

v1.9.0 Readout Stabilization and Proof Memory Core

> **Normalization note.** This block is retained as a historical precursor of the normalized readout architecture. Its proof-bearing content is re-exported later through the grouped-layer normalization, especially the readout/collapse phases and the semantic–tensor transport discipline. It should therefore be read as historical proof memory rather than as an independent new foundation.

This release adds a further stabilization layer to the inherited closure, probe, projection, and appendix architecture. Its purpose is to make the final readout path more disciplined: not only must admissible content survive transport, it must also remain recoverable after compression. In this sense v1.9.0 pushes the document from a transport-stable monolith toward a proof-memory stable one. A statement is not treated as mature merely because it can be compressed into a lemma, table row, or projection line; it must also remain reconstructible as the same closure-supported content when the compression is unfolded again.

1. Readout stabilization principle

The main strengthening of v1.9.0 is that every admissible readout must be stable under re-expansion. A readout is the visible endpoint of a structural chain, but it remains acceptable only if its origin can be reopened without changing the admissibility class. Thus the document now treats readout as a *controlled fixed point* of compression and reconstruction:

$$\begin{aligned}&\text{local structure} \Longrightarrow \text{HC} \Longrightarrow \text{closure}\\ &\Longrightarrow \text{probe filtration} \Longrightarrow \text{table/tool compression}\\ &\Longrightarrow \text{readout} \Longrightarrow \text{re-expansion to the same admissibility class.}\end{aligned}$$

Stabilization Proposition A (readout fixed-point discipline). If a statement is compressed into a readout form by admissibility-preserving transports and can be re-expanded to the same closure-licensed content, then the readout is structurally stable. If re-expansion changes the admissibility class, the readout was only heuristic.

Proof sketch. The admissibility source is fixed upstream by the ground condition, HC, and closure. Compression is legitimate only if it does not sever that source. Hence re-expansion must recover the same structural ancestry rather than a merely similar narrative.

2. Tool-to-lemma synchronization

Previous versions upgraded tools and tables into proof interfaces. v1.9.0 adds a stricter synchronization rule: every reusable tool should point to at least one lemma-level justification, and every repeated lemma pattern should be compressible into a named tool or interface move once stabilized. The goal is to prevent the appendix from splitting into disconnected "tricks" on one side and formal arguments on the other.

Stabilization Proposition B (tool/lemma synchronization). A reusable tool is mature only when it is synchronized with a lemma pattern that licenses its use, and a recurring lemma pattern is fully normalized only when it admits a stable compressed tool form.

Proof sketch. Tools without lemma ancestry become opaque shortcuts; lemmas without compressed interface forms bloat the running line unnecessarily. Synchronization preserves both transparency and speed while keeping the proof architecture auditable.

3. Probe-to-readout compatibility

The probe channel now receives a further role. It is not only a gate before projection but also a compatibility filter for later readouts. Any sector that passes through probe filtration should emerge with enough structural trace to be placed into a table, a compressed lemma, or a projection sentence without losing its closure ancestry.

Stabilization Proposition C (probe-filtered modes carry readout ancestry). If a mode survives the probe filtration under the standing orthogonality and HC rules, then its later readout must retain a backward trace to the probe channel that selected it. Readout therefore cannot erase probe ancestry without lowering its status to heuristic interpretation.

4. Appendix as proof memory lattice

The appendix is sharpened once more: it is now treated as a *proof memory lattice*. Its entries are not merely stored results but nodes from which one can move upward to the local closure chain, sideways to equivalent interface forms, and downward to bounded projection language. This lattice viewpoint organizes the appendix as a recoverability structure rather than a list of leftovers.

Stabilization Principle (memory-lattice admissibility). An appendix entry is fully admissible only if it supports three motions:

(a) upward recovery to its closure/lemma ancestry,

(b) lateral transport to equivalent tool or table interfaces, and

(c) downward use in a bounded readout or projection context.

If one of these motions fails, the entry must be marked as provisional, heuristic, or not yet fully normalized.

5. Operational consequence for later versions

With v1.9.0 the monolith receives a clearer maturity criterion for future expansions. New derivations should not merely add content; they should improve recoverability. In practice this means that later mathematical deepening should prefer additions that tighten one of the following links: closure to probe, probe to table, table to lemma, lemma to tool, or readout back to structural ancestry. This keeps the entire project aligned with the structure-first rule while making the growing proof apparatus easier to audit and reuse.

Operational rule for subsequent releases. Whenever a new argument is introduced, one should ask not only whether the step is locally correct, but also whether it improves the global recoverability of compressed proof content. The strongest additions are those that make the same statement simultaneously easier to compress, easier to re-expand, and easier to locate inside the appendix memory lattice.

v2.0.0 Structural Closure Ledger and Expansion Discipline Core

> **Normalization note.** This block is retained as a historical ledger/governance precursor. In the normalized architecture its active role belongs chiefly to the projection/translation layer and to the later formal unfolding phases that carry the actual proof-bearing closure and transport statements.

This release upgrades the monolith from a proof-memory stable object to a more explicit *closure ledger*. The point is not merely to preserve ancestry in principle, but to keep a disciplined account of how every compressed statement, interface row, or projection sentence remains licensed by the same upstream closure architecture. In this sense v2.0.0 turns recoverability into an operational bookkeeping rule: admissible content should be traceable, expandable, and relocatable without silently changing its structural status.

1. Closure ledger principle

A closure ledger is the ordered record of the transports that make a statement usable. It does not store every intermediate calculation in full; rather, it stores the *licensed path*

class by which the content travels from local structure to admissible readout. The relevant chain is now treated as a ledgered sequence:

$$\begin{aligned}&\text{local triolic structure} \Longrightarrow \text{ground condition} \Longrightarrow \text{HC compensation}\\ &\Longrightarrow \text{closure confinement} \Longrightarrow \text{probe/interface filtration}\\ &\Longrightarrow \text{lemma or tool compression} \Longrightarrow \text{bounded projection/readout.}\end{aligned}$$

A later re-expansion is admissible only if it re-enters the same ledger class, not merely a similar verbal narrative.

Ledger Proposition A (transport accountancy). A compressed claim is structurally mature only if one can specify the closure ledger that licenses its transport. If the path cannot be stated, the claim may still be suggestive, but it has not yet acquired full structural status.

Proof sketch. The earlier versions already required closure ancestry and re-expandability. The present strengthening adds explicit accountancy: the transport steps must be stable enough to be named, tracked, and checked. This prevents hidden leaps from entering the proof line under the cover of familiar notation.

2. Expansion discipline as a release criterion

The next layer of consolidation concerns how new material may be added. A later expansion should no longer be judged only by local correctness or plausibility; it should also improve the ledger quality of the monolith. Thus expansion is now classified by whether it strengthens one of the major controlled passages: closure to probe, probe to interface, interface to lemma, lemma to readout, or readout back to closure ancestry.

Ledger Proposition B (expansion must improve auditability). A new derivation is preferable to an older one when it makes the same structural content easier to audit across compression levels. Additions that only increase volume without sharpening one of the ledger passages remain secondary.

Operational reading. This gives the document a stricter maturity test. Future versions should not add formal bulk merely because it is available; they should add material that makes more of the proof architecture checkable under controlled compression and re-expansion.

3. Probe-ledger compatibility

The probe channel is now interpreted as a *ledger gate*. It does not simply filter modes before projection; it also determines which later compressed forms inherit admissibility in a controllable way. A mode that survives probe filtration under orthogonality and HC should arrive downstream with a stable ledger signature that can be recognized in lemma, table, and readout form.

Ledger Proposition C (probe selection induces ledger signature). If a mode is selected by the admissible probe channel, then any later compressed representation of that mode should preserve a recognizable signature of that selection. If the signature disappears completely, the downstream statement must be marked as heuristic or only partially normalized.

4. Table and appendix discipline

The appendix and its tables now receive a slightly sharper role again. They are not only memory lattices; they function as *ledger surfaces*. Each row, entry, or named tool should indicate which closure path class it abbreviates and which re-expansion channel it expects. This gives the appendix a more active status inside the proof architecture and reduces the danger that tables drift into mere summaries.

Ledger Proposition D (rows abbreviate path classes). A normalized table row is admissible only if it abbreviates a stable path class rather than an isolated sentence. The row then becomes reversible in two senses: it can be expanded upward into lemma ancestry and laterally into equivalent tool or interface forms.

5. Transition to the next mathematical deepening

With v2.0.0 the project crosses a useful threshold. The document now has a more explicit rule for how future mathematical elaboration should be inserted: not as detached calculations, but as ledger-preserving thickening of already licensed paths. This is especially relevant for the larger companion roadmap, where many later lemmas and proof tools must remain structurally synchronized.

Working rule for subsequent versions. Whenever a new proof block is added, it should be possible to say which closure ledger it refines, which compressed interfaces it improves, and how its content returns to the appendix without losing admissibility. Material that satisfies all three demands strengthens the monolith in the intended structure-first way.

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v2.1.0 Structural Audit and Re-Expansion Core

> **Normalization note.** This block is retained as a historical audit precursor. Its main value is to preserve re-expansion discipline and proof-memory traceability; the normalized document no longer treats it as a separate generative core, but as supporting governance for the later unfolding phases.

This release adds a further consolidation layer to the companion monolith. The new emphasis is not merely on carrying closure information forward, but on making explicit how later re-expansion remains auditable after several stages of compression. Earlier versions already introduced closure ladders, interface filters, ledger surfaces, and proof-memory discipline. Version v2.1.0 now adds an *audit core*: every admissible compressed statement should retain a recoverable expansion route that can be checked against the same closure class from which it descended.

1. Auditability as structural maturity

A compressed statement becomes structurally mature only when it is possible to state not just where it came from, but how it may be unfolded again without changing its admissible class. This requirement sharpens the earlier ledger ideas. The ledger tracked which closure path licensed a statement; the audit core now asks whether that path can be re-opened in a controlled way.

Audit Proposition A (licensed re-expansion). A normalized claim is fully admissible only if there exists a bounded re-expansion route from the compressed form back to a compatible lemma, tool, probe, or closure ancestry class. A statement that cannot be re-opened under controlled transport may still be heuristically useful, but it has not yet reached final structural normalization.

Proof sketch. Closure ancestry alone prevents arbitrary insertion of unsupported projection claims, but it does not yet guarantee that later readers can recover the path. The present strengthening adds a re-opening condition. The compressed surface must not merely summarize the proof memory; it must preserve enough structure that the same memory can be reconstructed under admissible reversal.

2. The audit ladder

The working audit ladder of this release is

$$\begin{aligned}&\text{local triolic constraint} \Longrightarrow \text{HC compensation} \Longrightarrow \text{closure confinement}\\ &\Longrightarrow \text{probe/interface filtration} \Longrightarrow \text{lemma or table compression}\\ &\Longrightarrow \text{readout statement} \Longrightarrow \text{controlled re-expansion.}\end{aligned}$$

The final arrow is new in emphasis. It makes explicit that a readout is not the terminal endpoint of the proof architecture. Instead, it is a temporarily stable surface from which the architecture should remain recoverable.

Audit Proposition B (endpoints remain reversible). A readout that has lost all reversible relation to its closure ladder should be classified as provisional. The stronger the reversible link to its generating path, the stronger its status inside the monolith.

3. Probe-audit discipline

The probe channel receives an additional task in v2.1.0. It is no longer only a selection or filtration device; it also serves as an audit checkpoint. Probe-selected modes should leave a downstream trace that remains visible in the compressed lemma/table/readout layers. This makes the probe sector a bridge between admissible dynamics and later recoverable proof memory.

Audit Proposition C (probe traces survive compression). If a configuration is admissibly selected in the orthogonal probe channel, then its later compressed representations should preserve a recognizable probe trace. If no such trace remains, downstream formulations must be marked as reduced, heuristic, or not yet fully synchronized.

4. Appendix as an audit surface

This version also sharpens the role of the appendix. Appendix entries, tables, and named tools now act as *audit surfaces*. Each item should indicate not only what it abbreviates, but how it can be re-opened into the same structural class. The appendix therefore becomes more than a memory store or ledger sheet; it becomes a reversible interface between compressed exposition and detailed proof ancestry.

Audit Proposition D (appendix entries carry reopening metadata). A mature appendix entry should point both backward and forward: backward into its closure ancestry and forward into an admissible re-expansion channel. This dual role prevents tables from drifting into detached summaries.

5. Consequence for the next mathematical thickening

With v2.1.0 the document takes another step toward a larger mathematically expanded companion. Future lemma growth should now be judged not only by local correctness, but also by whether it improves auditability across compression levels. New material is strongest when it reduces the gap between compressed statement, appendix row, probe trace, and recoverable closure ancestry.

Working rule for subsequent versions. Whenever a new mathematical block is added, it should be possible to specify which closure class licenses it, which compressed interfaces abbreviate it, and which controlled re-expansion route reconstructs it. Material satisfying all three demands strengthens the proof architecture in the intended structure-first sense.

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v2.2.0 Structural Re-Expansion Governance Core

> **Normalization note.** This block is retained as a historical governance precursor. It contributes release and re-expansion discipline, but in the normalized architecture it is not an independent proof center; proof-bearing content is carried by the grouped unfolding phases and their semantic/tensor transport rules.

This release adds a governance layer to the companion monolith. Earlier stages made closure transport, probe filtration, auditability, and ledger discipline explicit. Version v2.2.0 now asks an additional question: under which rules may a compressed statement be enlarged again without damaging the closure class, the appendix interfaces, or the downstream readout status? The answer is cast as a governance core. Re-expansion is no longer only possible; it is to be performed under named admissibility rules.

1. Governance as a proof-discipline layer

A mature structure-first document should not merely accumulate more lemmas. It should specify which later thickening operations are allowed and which ones are not. Governance therefore means that every future expansion step is judged by its compatibility with closure ancestry, probe traces, ledger surfaces, and the status labels already attached to the claim being enlarged.

Governance Proposition A (admissible thickening preserves status). A later elaboration is admissible only if it preserves the structural status of the compressed claim or improves it in a controlled way. An expansion that forces a change of admissibility class without explicit declaration should be regarded as a separate branch rather than a faithful thickening of the original path.

Proof sketch. Compression is useful only because it abbreviates a path already licensed by the monolith. When a later reader expands the statement again, the same path should remain visible. If the elaboration silently changes the closure ancestry, probe selection, or projection label, then it is not a recovery of the earlier path but a new construction. Governance marks this distinction explicitly.

2. The re-expansion governance ladder

The practical ladder of this release is

$$\begin{aligned}&\text{closure ancestry} \Longrightarrow \text{probe/interface trace} \Longrightarrow \text{ledger or appendix encoding}\\ &\Longrightarrow \text{compressed readout} \Longrightarrow \text{governed re-expansion}\\ &\Longrightarrow \text{status check} \Longrightarrow \text{return to the ledger.}\end{aligned}$$

The final return arrow is part of the new emphasis. Re-expansion is not treated as a one-way release of detail. It must return to the ledger surface in a form that remains compatible with the proof memory of the monolith.

Governance Proposition B (every governed expansion returns to a ledger surface). A mathematically thickened block is fully integrated only when its added detail can be pushed back into a stable appendix, tool, lemma, or table interface. Otherwise the expansion remains locally useful but globally insufficiently synchronized.

3. Probe governance and admissible branch formation

The probe channel now gains a sharper branching role. Not every elaboration of a probe-selected mode has to remain in the same local branch, but any branch split must be declared at the point where the probe trace ceases to determine a unique downstream continuation. This keeps the search-space logic compatible with the proof memory logic.

Governance Proposition C (branching must occur at visible trace loss). If a later derivation no longer preserves a unique recognizable probe trace, then a branch declaration is required. Continuing as if the same path remained in force would obscure which downstream statements are still inherited and which are newly introduced.

4. Appendix governance and reversible table discipline

The appendix receives a correspondingly stronger role. Rows, tables, and named tools are now governed interfaces. They should not only compress results and carry reopening metadata; they should also state which kinds of later expansion may legitimately pass through them. In this sense the appendix becomes a modest control layer for future mathematical thickening.

Governance Proposition D (table rows define admissible recovery windows). A mature table row should delimit the window inside which a later elaboration is still regarded as a recovery of the same path class. Once an expansion leaves that window, the result should either be relabeled or entered as a new row, lemma, or branch.

5. Consequence for the next proof cycle

With v2.2.0 the companion monolith now carries not only closure, projection, ledger, and audit discipline, but also a clearer rule for how later mathematical detail may be inserted without dissolving the structure-first architecture. The next proof cycle should therefore thicken selected paths under explicit governance rather than by unconstrained local expansion. This is the cleanest route toward the larger companion target with many more lemmas and technical proof tools.

Working rule for subsequent versions. Whenever a new technical derivation is inserted, it should be possible to say (i) which closure and probe ancestry it inherits, (ii) which appendix or ledger surface compresses it, (iii) which re-expansion window legitimizes its added detail, and (iv) how the resulting thickening returns to a synchronized proof interface. Material meeting all four demands strengthens the monolith in the intended disciplined way.

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Mathematical Thickening of the HC/Closure Core (working expansion)

This block deliberately reverses part of the recent compression trend. It does not introduce a new meta-layer. Instead it takes one of the central structure claims — ground condition $\Rightarrow$ HC $\Rightarrow$ closure — and writes it in a denser mathematical form. The purpose is not to declare a final axiomatic closure theorem, but to make explicit which quantitative objects control the transition from local triolic structure to confinement and later readout.

1. Working state space and defect variables

Let $\mathcal{H}$ be a real or complex Hilbert space with orthogonal splitting

$$\mathcal{H} = \mathcal{H}_A \oplus \mathcal{H}_B \oplus \mathcal{H}_C \oplus \mathcal{H}_R,$$

where $\mathcal{H}_A, \mathcal{H}_B, \mathcal{H}_C$ carry the three triolic channels and $\mathcal{H}_R$ is a residual sector collecting modes not admitted by the intended closure regime. We write a state as

$$\Psi = (A, B, C, R), \qquad A \in \mathcal{H}_A,\ B \in \mathcal{H}_B,\ C \in \mathcal{H}_C,\ R \in \mathcal{H}_R.$$

The orthogonality discipline of the triolic block is encoded by the defect functional

$$\delta_\perp(\Psi) := |\langle A, C\rangle|^2 + |\langle B, C\rangle|^2,$$

and the residual leakage into the non-admissible sector by

$$\delta_R(\Psi) := \|R\|^2.$$

The first quantity measures failure of the distinguished C-channel to remain orthogonal to the other two active channels. The second measures failure of the state to remain confined to the triolic/septinion envelope.

Ground functional. A compact way to encode the ground condition is through the non-negative functional

$$\Gamma(\Psi) := \delta_\perp(\Psi) + \eta\, \delta_R(\Psi), \qquad \eta > 0.$$

The ideal ground regime is characterized by $\Gamma(\Psi) = 0$. In particular,

$$\Gamma(\Psi) = 0 \quad \Longrightarrow \quad \langle A, C\rangle = \langle B, C\rangle = 0 \quad \text{and} \quad R = 0.$$

Thus the ground condition simultaneously imposes orthogonality and exclusion of residual leakage.

2. A formal compensator model

To thicken the HC claim mathematically, introduce the gradient of Γ with respect to the Hilbert structure. Since

$$\nabla_A \Gamma = 2\langle A, C\rangle C, \qquad \nabla_B \Gamma = 2\langle B, C\rangle C,$$

and

$$\nabla_C \Gamma = 2\overline{\langle A, C\rangle} A + 2\overline{\langle B, C\rangle} B, \qquad \nabla_R \Gamma = 2\eta R,$$

a natural working expression for the compensator is the feedback field

$$\mathrm{HC}(\Psi) := -\kappa\, \nabla\Gamma(\Psi), \qquad \kappa > 0.$$

Written componentwise,

$$\mathrm{HC}(\Psi) = \Big(-2\kappa\langle A, C\rangle C, -2\kappa\langle B, C\rangle C, -2\kappa\overline{\langle A, C\rangle} A - 2\kappa\overline{\langle B, C\rangle} B, -2\kappa\eta R\Big).$$

This is the simplest explicit model of the structural statement that the compensator opposes orthogonality defects and suppresses escape into the residual sector.

Remark. The point of this formula is not uniqueness. It is a minimal realization of the qualitative rule that the HC term points against the defect gradient generated by the ground functional.

3. Closure as Lyapunov monotonicity

Assume the effective evolution has the form

$$\partial_\tau \Psi = F(\Psi) + \mathrm{HC}(\Psi),$$

where F denotes the non-compensated structural drift. The cleanest closure statement occurs when the uncontrolled part is tangent or dissipative relative to the ground functional:

$$\Re\,\langle \nabla\Gamma(\Psi), F(\Psi)\rangle \le 0. \tag{T}$$

Under this hypothesis,

$$\begin{aligned}\frac{d}{d\tau}\Gamma(\Psi(\tau)) &= \Re\,\langle \nabla\Gamma(\Psi), \partial_\tau\Psi\rangle \\ &= \Re\,\langle \nabla\Gamma(\Psi), F(\Psi)\rangle - \kappa\|\nabla\Gamma(\Psi)\|^2 \\ &\le -\kappa\|\nabla\Gamma(\Psi)\|^2 \le 0.\end{aligned}$$

Hence Γ is a Lyapunov functional for the compensated flow.

Lemma 1 (monotone compensated closure). Under (T), the map $\tau \mapsto \Gamma(\Psi(\tau))$ is non-increasing. In particular, the orthogonality defect and residual leakage cannot increase along the compensated flow.

Corollary 1. If $\Gamma(\Psi(0)) = 0$, then $\Gamma(\Psi(\tau)) = 0$ for all later τ for which the flow exists. Thus the exact ground class is forward invariant.

Corollary 2. If $\Gamma(\Psi(0))$ is small, then the trajectory remains trapped in the sublevel set

$$\{\Psi \in \mathcal{H} : \Gamma(\Psi) \le \Gamma(\Psi(0))\}.$$

This is the first explicit mathematical form of the closure/confinement claim.

4. Septinion confinement as exclusion of the residual sector

The previous corollaries can be sharpened for the residual component. Since $\Gamma(\Psi) \geq \eta\|R\|^2$, monotonicity implies

$$\|R(\tau)\|^2 \leq \eta^{-1}\Gamma(\Psi(\tau)) \leq \eta^{-1}\Gamma(\Psi(0)).$$

Therefore a small initial ground defect bounds the entire future residual mass. In the exact case $\Gamma(\Psi(0)) = 0$ one obtains $R(\tau) = 0$ identically.

Lemma 2 (residual confinement). Under (T), the residual sector $\mathcal{H}_R$ is dynamically suppressed by the compensator. Exact ground states never leave the triolic/septinion block, and approximate ground states can leak into $\mathcal{H}_R$ only at a level quantitatively controlled by their initial defect.

This is a mathematically sharper version of the earlier verbal statement that "no component breaks out" once the HC mechanism is active.

5. First-layer master equation in decomposed form

The structure-first master equation can now be written in a way that makes the origin of the principal terms explicit:

$$\partial_\tau \Psi = T_{\text{int}}(\Psi) + T_{\text{proj}}(\Psi) - \kappa\nabla\Gamma(\Psi).$$

Here

- T_{int} collects the intrinsic triolic or septinion-internal couplings,
- T_{proj} collects frame-shift or measurement-induced projection terms,
- $-\kappa\nabla\Gamma$ is the compensating closure term.

The point of this decomposition is that the HC contribution is no longer merely described verbally. It is isolated as the mathematically distinguished term that penalizes deviation from the admissible structural class.

Lemma 3 (term-wise origin of stabilization). Suppose $T_{\text{int}} + T_{\text{proj}}$ satisfies the tangency condition (T). Then every decrease of the ground functional is generated by the HC term, while all remaining contributions are either neutral or already compatible with closure. In that sense the master equation separates structurally allowed drift from explicitly compensating drift.

6. Probe sector as constrained reduction

For the probe channel, write two incoming triolic probes as

$$\Psi_1 = (A_1, B_1, C_1, R_1), \qquad \Psi_2 = (A_2, B_2, C_2, R_2),$$

and let Q denote the orthogonal third group used to delimit the search space. A natural probe defect is then

$$\Gamma_{\text{probe}}(\Psi_1, \Psi_2; Q) := \Gamma(\Psi_1) + \Gamma(\Psi_2) + \rho\,\Xi(\Psi_1, \Psi_2; Q), \qquad \rho > 0,$$

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where Ξ measures incompatibility with the orthogonal search geometry. For example, one may take

$$\Xi(\Psi_1, \Psi_2; Q) := |\langle A_1, Q\rangle|^2 + |\langle B_1, Q\rangle|^2 + |\langle A_2, Q\rangle|^2 + |\langle B_2, Q\rangle|^2,$$

possibly supplemented by additional coupling terms between the two probes. Then the same compensated-flow idea yields

$$\partial_\tau(\Psi_1, \Psi_2) = F_{\text{probe}}(\Psi_1, \Psi_2; Q) - \kappa_p \nabla\Gamma_{\text{probe}}(\Psi_1, \Psi_2; Q),$$

with

$$\frac{d}{d\tau}\Gamma_{\text{probe}} \leq -\kappa_p \|\nabla\Gamma_{\text{probe}}\|^2$$

under the corresponding tangency condition.

Lemma 4 (probe-sector reduction principle). Under the probe tangency hypothesis, the orthogonal search geometry reduces the effective probe dynamics to a controlled subspace determined by the sublevel sets of Γ_{probe}. Thus the probe mechanism is mathematically read as a constrained reduction principle rather than a merely heuristic search picture.

7. What this thickening achieves and what it does not yet do

This expansion does three concrete things.

1. It replaces the purely verbal HC statement by an explicit compensator model $-\kappa\nabla\Gamma$.
2. It turns closure into a Lyapunov monotonicity statement.
3. It gives the probe sector a quantitative defect functional that can be minimized under orthogonal search constraints.

At the same time, it is still a working expansion rather than a final theorem. The non-compensated drifts F, T_{int}, T_{proj}, and F_{probe} have not yet been specialized to a unique physical or number- theoretic model. That specialization is the next true mathematical step after this thickening: once the intrinsic drift is fixed more sharply, the above Lyapunov framework can be tested term-by-term instead of only at the abstract closure level.

v2.3.0 Mathematical Deepening Core

> **Normalization note.** This block is retained as a historical deepening precursor. Its mathematical thickening remains part of the document's proof memory, but the normalized architecture reads it through the later unfolding phases rather than as a second autonomous mathematical foundation.

This version continues the return from governance-style consolidation to explicit mathematical derivation. The focus is no longer merely that closure can be encoded by a defect functional, but that the induced closure flow can be separated into invariant, dissipative, and residual channels. In this way the HC term becomes a mathematically trackable transport law rather than a purely qualitative stabilizer.

1. Orthogonal projector formulation of closure

Let $\mathcal{H} = \mathcal{H}_A \oplus \mathcal{H}_B \oplus \mathcal{H}_C \oplus \mathcal{H}_R$ be the structural state space used in the previous thickening step, and let Π_{cl} denote the orthogonal projector onto the admissible closure manifold

$$\mathfrak{M}_{\mathrm{cl}} := \{\Psi \in \mathcal{H} \,:\, \Gamma(\Psi) = 0\}.$$

Whenever Γ is locally quadratic near $\mathfrak{M}_{\mathrm{cl}}$, one may write, to leading order,

$$\Gamma(\Psi) \asymp \|(I - \Pi_{\mathrm{cl}})\Psi\|^2,$$

so that the distance to closure is measured by the defect component

$$\Psi_\perp := (I - \Pi_{\mathrm{cl}})\Psi, \qquad \Psi_\parallel := \Pi_{\mathrm{cl}}\Psi.$$

The compensated flow

$$\partial_\tau \Psi = F(\Psi) - \kappa \nabla\Gamma(\Psi)$$

therefore admits the splitting

$$\partial_\tau \Psi_\parallel = \Pi_{\mathrm{cl}} F(\Psi) + \mathcal{N}_\parallel(\Psi), \qquad \partial_\tau \Psi_\perp = (I - \Pi_{\mathrm{cl}})F(\Psi) - \kappa\nabla\Gamma(\Psi) + \mathcal{N}_\perp(\Psi),$$

where $\mathcal{N}_\parallel, \mathcal{N}_\perp$ collect higher-order curvature terms of the closure manifold.

Lemma 1 (normal damping near closure). Assume that Γ is C^2 in a neighborhood of $\mathfrak{M}_{\mathrm{cl}}$ and that its Hessian is strictly positive on the normal bundle of $\mathfrak{M}_{\mathrm{cl}}$. Then there are constants $c_1, c_2 > 0$ such that for all states sufficiently near closure,

$$c_1 \|\Psi_\perp\|^2 \le \Gamma(\Psi) \le c_2 \|\Psi_\perp\|^2,$$

and the HC term damps the normal component at leading order.

Proof. By the Morse–Bott type expansion of Γ around the zero set,

$$\Gamma(\Psi_\parallel + \Psi_\perp) = \tfrac{1}{2}\langle H_\Gamma(\Psi_\parallel)\Psi_\perp, \Psi_\perp\rangle + O(\|\Psi_\perp\|^3),$$

with H_Γ positive on normal directions. Hence Γ is equivalent to the squared normal distance, and $\nabla\Gamma(\Psi) = H_\Gamma(\Psi_\parallel)\Psi_\perp + O(\|\Psi_\perp\|^2)$. Therefore the compensator contributes a strictly contracting term in the normal direction. □

2. Differential inequality for the closure ladder

Set $E(\tau) := \Gamma(\Psi(\tau))$. Then

$$\dot{E} = \langle \nabla\Gamma(\Psi), F(\Psi)\rangle - \kappa\|\nabla\Gamma(\Psi)\|^2.$$

To make the structural content explicit, split the drift as

$$F = F_{\mathrm{tan}} + F_{\mathrm{nor}} + F_{\mathrm{res}},$$

where F_{tan} is tangent to $\mathfrak{M}_{\mathrm{cl}}$, F_{nor} is controlled by the closure defect, and F_{res} contains genuinely residual terms. Then

$$\dot{E} = \langle \nabla\Gamma, F_{\mathrm{nor}}\rangle + \langle \nabla\Gamma, F_{\mathrm{res}}\rangle - \kappa\|\nabla\Gamma\|^2,$$

because the tangential part drops out to first order.

Proposition 1 (closure ladder with residual threshold). Suppose there exist constants $a < \kappa$ and $b \geq 0$ such that

$$|\langle \nabla\Gamma, F_{\mathrm{nor}} + F_{\mathrm{res}} \rangle| \leq a\|\nabla\Gamma\|^2 + b\Gamma.$$

Then for states near closure,

$$\dot{E} \leq -(\kappa - a)\|\nabla\Gamma\|^2 + bE.$$

In particular, if b is dominated by the coercivity constant relating $\|\nabla\Gamma\|^2$ and E, then the closure ladder is dissipative and the flow returns exponentially to the admissible structural sector.

Proof. Insert the estimate directly into the identity for $\dot{E}$. The final claim follows from the local coercivity $\|\nabla\Gamma\|^2 \geq cE$ near $\mathfrak{M}_{\mathrm{cl}}$, which converts the inequality into $\dot{E} \leq -\mu E$ for some $\mu > 0$. □

This proposition sharpens the earlier Lyapunov statement. Closure is not only monotone under ideal tangency; it remains stable under a controlled class of residual perturbations, provided the compensator strength exceeds the residual threshold.

3. Term-wise master equation refinement

Write the first-layer master equation in the refined form

$$\partial_\tau \Psi = T_{\mathrm{tri}}(\Psi) + T_\perp(\Psi) + T_{\mathrm{meas}}(\Psi) + T_{\mathrm{res}}(\Psi) - \kappa\nabla\Gamma(\Psi).$$

Here the meaning of the terms is:

- T_{tri}: internal triolic transport preserving the intrinsic algebraic coupling,
- $T_\perp$: orthogonality-restoring channel induced by the requirement that one branch remains perpendicular to the other two,
- T_{meas}: frame-shift and measurement transport,
- T_{res}: unresolved residual transport not yet assigned to a unique physical model.

The closure condition can then be tested term by term through the scalar pairings

$$\sigma_{\mathrm{tri}} := \langle \nabla\Gamma, T_{\mathrm{tri}} \rangle, \quad \sigma_\perp := \langle \nabla\Gamma, T_\perp \rangle, \quad \sigma_{\mathrm{meas}} := \langle \nabla\Gamma, T_{\mathrm{meas}} \rangle, \quad \sigma_{\mathrm{res}} := \langle \nabla\Gamma, T_{\mathrm{res}} \rangle.$$

Thus

$$\dot{E} = \sigma_{\mathrm{tri}} + \sigma_\perp + \sigma_{\mathrm{meas}} + \sigma_{\mathrm{res}} - \kappa\|\nabla\Gamma\|^2.$$

The mathematical advantage is that every structural claim can now be tied to one of these four channels. For instance, a perfect intrinsic closure sector is characterized by $\sigma_{\mathrm{tri}} = 0$, while measurement-induced distortion is captured by the sign and size of σ_{meas}.

Corollary 2 (term-wise admissibility test). If each non-compensated channel satisfies an estimate of the form $|\sigma_j| \leq a_j\|\nabla\Gamma\|^2 + b_j E$ with $\sum_j a_j < \kappa$, then the master flow is closure-admissible. In this sense the first-layer master equation acquires a concrete mathematical admissibility criterion instead of a purely narrative one.

4. Probe reduction with quantitative search radius

For the probe functional

$$\Gamma_{\text{probe}}(\Psi_1, \Psi_2; Q) = \Gamma(\Psi_1) + \Gamma(\Psi_2) + \rho\,\Xi(\Psi_1, \Psi_2; Q),$$

define the admissible search radius

$$\mathcal{R}_\varepsilon(Q) := \{(\Psi_1, \Psi_2) : \Gamma_{\text{probe}}(\Psi_1, \Psi_2; Q) \le \varepsilon\}.$$

This produces a genuinely quantitative reading of the earlier search-space language: the probe channel is restricted to a sublevel tube around the orthogonally compatible sector.

Proposition 3 (quantitative probe reduction). Assume the probe drift satisfies the analogue of the closure-ladder estimate. Then for every sufficiently small $\varepsilon > 0$ the set $\mathcal{R}_\varepsilon(Q)$ is forward invariant under the compensated probe flow, provided the initial probe state lies in $\mathcal{R}_\varepsilon(Q)$. Hence the probe search does not merely heuristically prefer a smaller space; it is mathematically confined to a controlled admissible tube.

Proof. Apply the same differential-inequality argument to

$$E_{\text{probe}}(\tau) := \Gamma_{\text{probe}}(\Psi_1(\tau), \Psi_2(\tau); Q).$$

If $\dot{E}_{\text{probe}} \le -\mu E_{\text{probe}}$ or, more generally, $\dot{E}_{\text{probe}} \le 0$ on the boundary level set $E_{\text{probe}} = \varepsilon$, trajectories cannot leave the sublevel region. □

5. Consequence for the next development step

The present deepening changes the status of the HC/closure block in two ways. First, closure is now expressed through projector splitting and coercive defect control. Second, the master and probe channels both admit explicit admissibility tests in terms of scalar defect pairings. The next natural step is therefore no longer another meta-level consolidation, but the specialization of $T_{\text{tri}}, T_\perp, T_{\text{meas}}, T_{\text{res}}$ and of the probe drift to more definite structural or physical models. That is the point at which the current structure-first proof begins to turn into a truly term-resolved mathematical derivation.

v2.4.0 Term-Resolved Channel Specialization Core

> **Normalization note.** This block is retained as a historical specialization precursor. In the normalized architecture its active content belongs primarily to the projection/translation and field/probe realization layers; it should therefore be read as a later historical expansion, not as a competing core.

The explicit task of v3.35.0 is to pass from the admissibility language of v2.3.0 to a first term-resolved structural model. The point is not yet to claim final physics. Rather, the point is to assign mathematically disciplined roles to $T_{\text{tri}}, T_\perp, T_{\text{meas}}, T_{\text{res}}$ so that later field equations, appendix tables, and probe readouts can all be tested against the same channel map. The guiding principle is that every term must be readable simultaneously in four ways: as a structural transport rule, as a closure-defect source or sink, as a probe-interface contribution, and as a future physical interpretation slot.

1. Channel factorization principle

Let the first-layer master equation be written as

$$\partial_\tau \Psi = T_{\mathrm{tri}}(\Psi) + T_\perp(\Psi) + T_{\mathrm{meas}}(\Psi) + T_{\mathrm{res}}(\Psi) - \kappa \nabla \Gamma(\Psi).$$

In v3.35.0 we impose the factorization rule

$$T_j = A_j + B_j + C_j, \qquad j \in \{\mathrm{tri}, \perp, \mathrm{meas}, \mathrm{res}\},$$

where:

- A_j is the closure-tangent component,
- B_j is the explicitly defect-producing normal component,
- C_j is a bookkeeping remainder reserved for later model refinement.

This factorization is not claimed to be unique. Its function is instead to force each channel into a common proof grammar: tangent transport, normal defect, and controlled unresolved remainder.

Definition 1 (channel-adapted splitting). Given the projector pair $(P_{\mathrm{tan}}, P_{\mathrm{nor}})$ induced by the closure manifold, define

$$A_j := P_{\mathrm{tan}} T_j, \qquad B_j := P_{\mathrm{nor}} T_j, \qquad C_j := 0$$

at the first structural level. More general splittings with nonzero C_j are allowed when one wants to isolate scale-separated or frame-shifted contributions. At the present stage, however, the projector-induced splitting is the canonical choice because it minimizes extra assumptions.

2. Canonical specialization of the four channels

The structural role of the four channels is now fixed as follows.

Triolic channel. The term T_{tri} is the intrinsic coupling transport generated by the triolen architecture itself. Its preferred structural property is near tangency:

$$P_{\mathrm{nor}} T_{\mathrm{tri}} = O(\Gamma).$$

This expresses that the intrinsic triadic coupling should preserve admissibility to leading order and create only higher-order closure defects.

Orthogonality channel. The term $T_\perp$ enforces the standing project constraint that one branch is perpendicular to the other two. Its preferred structural property is restorative:

$$\langle \nabla \Gamma, T_\perp \rangle \le -\lambda_\perp \Gamma \quad \text{for some } \lambda_\perp > 0$$

in the near-closure regime. Thus $T_\perp$ is not merely admissible; it is a coercive correction channel.

Measurement channel. The term T_{meas} encodes the phase-space shift between the measurement frame and the underlying structural dynamics. It is therefore not expected to be sign-definite. Instead it is controlled by a relative bound of the form

$$|\langle \nabla\Gamma, T_{\mathrm{meas}}\rangle| \le a_{\mathrm{meas}}\|\nabla\Gamma\|^2 + b_{\mathrm{meas}}\Gamma.$$

This formalizes the standing project rule that measurements do not access the raw structure directly, but only through a shifted interface.

Residual channel. The term T_{res} is the disciplined storage slot for all contributions not yet promoted to a stable structural or physical interpretation. In v3.35.0 it is no longer allowed to function as a dumping ground. Instead one requires

$$T_{\mathrm{res}} = \sum_{m=1}^{M} T_{\mathrm{res}}^{(m)}$$

with every summand tagged by its origin class, such as approximation residue, frame-conversion remainder, truncation error, or phenomenological placeholder. This turns residuality into an auditable notion.

3. Channel ledger and admissibility matrix

To make the specialization operational, v2.4.0 introduces the channel ledger matrix

$$\Lambda_{\mathrm{ch}}(\Psi) = \begin{pmatrix} \ell_{\mathrm{tri}} & \delta_{\mathrm{tri}} & r_{\mathrm{tri}} \\ \ell_{\perp} & \delta_{\perp} & r_{\perp} \\ \ell_{\mathrm{meas}} & \delta_{\mathrm{meas}} & r_{\mathrm{meas}} \\ \ell_{\mathrm{res}} & \delta_{\mathrm{res}} & r_{\mathrm{res}} \end{pmatrix},$$

where

$$\ell_j := \|P_{\tan}T_j\|, \qquad \delta_j := \|P_{\mathrm{nor}}T_j\|, \qquad r_j := \|C_j\|.$$

This channel ledger matrix Λ_{ch} is not introduced for numerical decoration. Its function is to give a single interface where lemma statements, appendix tables, and future simulations can all report the same structural status.

Proposition 1 (matrix admissibility criterion). Suppose there exist constants $\alpha_j, \beta_j \ge 0$ such that

$$\delta_j^2 \le \alpha_j\Gamma + \beta_j\|\nabla\Gamma\|^2 \qquad \text{for each channel } j,$$

and assume $\sum_j \beta_j < \kappa^2$. Then the channel-specialized master flow is closure-admissible in a neighbourhood of the closure sector.

Proof. The normal contributions are exactly the defect-producing pieces entering the closure balance. By Cauchy–Schwarz,

$$|\langle \nabla\Gamma, B_j\rangle| \le \|\nabla\Gamma\|\,\delta_j.$$

Applying Young's inequality channel by channel and summing yields a bound of the same type as in the v2.3.0 admissibility criterion, now expressed entirely through the ledger entries. The compensator term dominates once the cumulative normal budget stays below the compensator threshold. □

4. Probe-channel specialization

The same channel grammar must hold for the two-probe sector. Write the probe drift in the form

$$\mathcal{T}_{\text{probe}} = \mathcal{T}_{\text{tri}}^{(1)} + \mathcal{T}_{\text{tri}}^{(2)} + \mathcal{T}_{\perp}^{(Q)} + \mathcal{T}_{\text{match}} + \mathcal{T}_{\text{res}}^{\text{probe}}.$$

Here the two intrinsic channels belong to the two probe triolen, the term $\mathcal{T}_{\perp}^{(Q)}$ encodes the orthogonal search geometry selected by Q, and $\mathcal{T}_{\text{match}}$ measures the approach or mismatch between the two probe traces. The new point of v2.4.0 is that probe matching is promoted to a channel in its own right rather than being hidden implicitly inside a heuristic search narrative.

Definition 2 (probe match defect). Let

$$\Delta_{\text{match}}(\Psi_1, \Psi_2) := \|\Pi(\Psi_1) - \Pi(\Psi_2)\|^2,$$

where Π is a chosen readout or projection interface. Then the match channel is called admissible if along the compensated probe flow,

$$\frac{d}{d\tau}\Delta_{\text{match}} \leq -\eta\,\Delta_{\text{match}} + c\,\Gamma_{\text{probe}}$$

for some $\eta > 0$ and finite c. This means that probe convergence is controlled up to the ambient closure defect.

Corollary 2 (probe convergence under closure control). If the probe closure functional remains in a forward-invariant small sublevel tube and the match channel is admissible, then probe readout mismatch decays up to an error proportional to the residual closure budget. Hence two probes can be treated as a mathematically disciplined comparison device rather than only a heuristic search metaphor.

5. Consequence for appendix tools and future derivations

The specialization introduced in v2.4.0 changes the status of the appendix again. From now on, every major table, tool, and shortcut should be readable as a compressed report of one of the ledger entries or of one of the probe-channel budgets. This gives a concrete criterion for deciding whether a future tool is structural or merely expository: a genuine structural tool must reduce, estimate, or transport one of the channel quantities.

The practical consequence is decisive. Future expansions no longer need to say only that a section "supports closure" or "helps probe reduction." They can specify which channel is being sharpened, which defect budget is reduced, and which ledger entry is stabilized. This is exactly the sort of disciplined mathematical thickening that prepares the transition from structure-first architecture to a more fully unfolded proof.

v2.5.0 Channel Budget Transport and Ledger Dynamics Core

> **Normalization note (v2.17.1 local review).** This block is retained as a *historical transport and ledger precursor*. Its proof-bearing content should be read through the normalized grouped architecture and the later unfolding phases, rather than as an independent foundational core.

The specialization step of v2.4.0 assigned every contribution to a disciplined channel and recorded its tangential, normal, and compensator-facing sizes in the ledger matrix. The next structural hardening step is to turn this static picture into an evolution rule. In v2.5.0 the ledger is therefore promoted from a reporting interface to a transport object. One no longer asks only whether a channel is admissible at a fixed stage; one asks how admissibility is propagated, recovered, or degraded along the proof flow.

1. Dynamic ledger variables

For each channel $j \in \{\mathrm{tri}, \perp, \mathrm{meas}, \mathrm{res}\}$ define the budget variables

$$B_j(\tau) := \ell_j(\tau)^2 + \delta_j(\tau)^2 + r_j(\tau)^2, \qquad N_j(\tau) := \delta_j(\tau)^2, \qquad R_j(\tau) := r_j(\tau)^2.$$

Here B_j is the total structural budget carried by the channel, N_j is the normal leakage budget, and R_j is the amount already routed through the compensator-facing slot. The point is conceptual as much as technical: future lemmas may now refer to budgets rather than to isolated terms, which makes it possible to compare sections of the document without re-expanding the full microscopic decomposition each time.

Let

$$\mathfrak{B}(\tau) := \sum_j B_j(\tau), \qquad \mathfrak{N}(\tau) := \sum_j N_j(\tau), \qquad \mathfrak{R}(\tau) := \sum_j R_j(\tau).$$

The quantity $\mathfrak{B}$ is the total visible channel load, while $\mathfrak{N}$ and $\mathfrak{R}$ separate the defect-producing part from the compensator-processed part.

2. Budget transport law

The structural thesis of v2.5.0 is that channel evolution should be read as transport with controlled exchange, not as unrelated term growth. This is encoded in the schematic law

$$\frac{d}{d\tau} B_j = -\mu_j B_j + \sum_{k \neq j} \sigma_{kj} B_k + \rho_j \Gamma + \xi_j,$$

where $\mu_j \geq 0$ is the internal damping of channel j, the coefficients $\sigma_{kj} \geq 0$ measure admissible transfer from channel k into channel j, $\rho_j \Gamma$ records closure-driven forcing, and ξ_j is a structured source term that must be tagged by origin class. In particular, ξ_j is not allowed to hide unexplained growth. It must come from one of the already sanctioned origins: projection conversion, probe matching, approximation residue, or phenomenological placeholder.

Definition 3 (transport-admissible ledger). A channel ledger is called transport-admissible on an interval I if there exist nonnegative coefficients $(\mu_j, \sigma_{kj}, \rho_j)$ and tagged sources ξ_j such that the above transport law holds on I, the cumulative transfer matrix $\Sigma = (\sigma_{kj})$ has finite operator norm, and every source term satisfies

$$|\xi_j| \leq c_j^{(0)} \Gamma + c_j^{(1)} \|\nabla \Gamma\|^2 + c_j^{(2)} \Delta_{\mathrm{match}} + c_j^{(3)} \mathrm{Res}_{\mathrm{tag}},$$

where $\mathrm{Res}_{\mathrm{tag}}$ denotes a tagged residual quantity already registered in the appendix grammar.

This definition codifies the project preference for auditable derivations: channel budgets may move and mix, but they may not mutate mysteriously.

3. Global dissipative estimate

The transport law becomes useful only once it closes against the compensated closure sector.

Proposition 1 (global budget inequality). Assume the ledger is transport-admissible and suppose the transfer coefficients satisfy

$$\mu_* := \min_j \mu_j > \|\Sigma\|.$$

Then there exist constants $A, B, C \geq 0$ such that along the compensated flow,

$$\frac{d}{d\tau}\mathfrak{B} \leq -A\mathfrak{B} + B\Gamma + C\Delta_{\mathrm{match}} + C\mathrm{Res}_{\mathrm{tag}}.$$

In particular, if Γ, Δ_{match}, and the tagged residual budget remain inside a forward-invariant small tube, then the total visible channel load stays uniformly bounded and is asymptotically damped up to that tube.

Proof. Summing the channel transport laws gives

$$\frac{d}{d\tau}\mathfrak{B} = -\sum_j \mu_j B_j + \sum_j \sum_{k\neq j} \sigma_{kj} B_k + \Big(\sum_j \rho_j\Big)\Gamma + \sum_j \xi_j.$$

The transfer contribution is bounded above by $\|\Sigma\|\mathfrak{B}$ after collecting coefficients. The strict damping gap $\mu_* > \|\Sigma\|$ yields a net negative coefficient in front of $\mathfrak{B}$. The source estimate from transport-admissibility then produces the stated inequality after regrouping the closure, match, and tagged residual terms. □

The structural meaning is important. Closure-admissibility is no longer a purely local statement about one decomposition stage. It becomes a persistent budget statement that survives transfer between channels.

4. Leakage absorption and residual discipline

The next step is to distinguish visible channel mass from true defect leakage. Because only the normal parts threaten closure, one needs a mechanism that proves that leakage either decays or is absorbed into compensator-facing storage.

Definition 4 (absorption-efficient channel). A channel j is called absorption-efficient if there exist constants $\theta_j > 0$ and $d_j \geq 0$ such that

$$\frac{d}{d\tau}N_j \leq -\theta_j N_j + d_j R_j + d_j\Gamma + d_j\Delta_{\mathrm{match}}.$$

In words: normal leakage may temporarily feed on unresolved load, but it is penalized unless supported by compensator storage or ambient closure stress.

Corollary 1 (aggregate leakage absorption). If every channel is absorption-efficient and if, in addition,

$$\sum_j d_j R_j \le \lambda \mathfrak{R} \qquad \text{with} \qquad \lambda < \theta_* := \min_j \theta_j,$$

then there exist constants $E, F \ge 0$ such that

$$\frac{d}{d\tau}\mathfrak{N} \le -E\mathfrak{N} + F\Gamma + F\Delta_{\text{match}} + F\mathfrak{R}.$$

Hence the total normal leakage is controlled by closure stress, probe mismatch, and compensator-facing storage, rather than by an uncontrolled accumulation of termwise errors.

This is exactly the desired middle statement between raw microlocal decomposition and later field-equation language: the dangerous part of the proof can be compressed into a leakage budget with explicit sinks and sources.

5. Probe-budget coupling

The two-probe sector can now be written in the same grammar. Let

$$\mathfrak{B}_{\text{probe}} := B^{(1)}_{\text{tri}} + B^{(2)}_{\text{tri}} + B^{(Q)}_{\perp} + B_{\text{match}} + B^{\text{probe}}_{\text{res}}.$$

The new content of v3.35.0 is that probe comparison is no longer controlled only through the mismatch scalar Δ_{match}, but through a full budget that separates intrinsic probe cost, orthogonal search cost, match stabilization, and residual search overhead.

Proposition 2 (probe budget stability). Assume the intrinsic probe channels and the orthogonal search channel are transport-admissible, and assume the match channel satisfies

$$\frac{d}{d\tau}B_{\text{match}} \le -\eta B_{\text{match}} + c_1\Gamma_{\text{probe}} + c_2\mathfrak{N}_{\text{probe}}.$$

Then the full probe budget obeys an estimate of the form

$$\frac{d}{d\tau}\mathfrak{B}_{\text{probe}} \le -a\mathfrak{B}_{\text{probe}} + b\Gamma_{\text{probe}} + b\mathfrak{N}_{\text{probe}} + b\text{Res}^{\text{probe}}_{\text{tag}},$$

for suitable $a, b > 0$ whenever the transfer matrix in the probe sector has a strict damping gap.

The interpretation is that two probes can consume structural budget while moving through the search geometry, but this consumption is still accounted for by the same ledger grammar as the main derivation. Nothing essentially new is hidden in the probe picture; it is a structured subledger.

6. Consequence for appendix tools and proof memory

The appendix can now be sharpened one step further. A legitimate tool is no longer merely something that estimates a channel entry. In v3.35.0 a tool should, whenever possible, declare one of four roles:

(a) budget estimator,

(b) transfer bound,

(c) leakage absorber, or

(d) source tagger.

This classification is mathematically useful because it says where the tool enters an actual differential inequality rather than only where it appears heuristically.

The same holds for proof memory. Once a lemma has been translated into budget form, its content can be stored compactly as a transport or absorption rule without repeating the entire derivation chain. This is the precise sense in which the structure-first approach compresses thousands of calculations without erasing them: it stores their net structural effect as a stable ledger law.

7. v3.35.0 readiness criterion

The present section should be regarded as successful if the following can now be done systematically:

- every major future term can be assigned a channel budget rather than only a raw symbol;
- every appendix tool can be classified by its role in a budget inequality;
- probe arguments can be reported through a subledger instead of loose search language;
- residual terms can be traced by tagged sources rather than by informal comments; and
- later field equations can read these budget laws as compressed pre-equation structure.

This is the mathematical gain of v3.35.0. The document now contains not only a closure grammar and a channel grammar, but also a transport grammar that explains how structural control is preserved as the proof unfolds.

v2.6.0 Lyapunov Ledger Closure and Forward-Invariant Regimes

> **Normalization note (v2.17.1 local review).** This block is retained as a *historical closure-stability and Lyapunov precursor*. Its role is supporting and developmental; the normalized proof-bearing closure chain is carried by the unfolding phases.

The next logical step is to turn the budget grammar into a genuine closure device. As long as one only writes transport inequalities, the proof still depends on a stagewise reading. For the structure-first programme this is not yet strong enough. One wants a single scalar functional that certifies that the admissible sector is preserved under the whole transport chain. This is the role of the Lyapunov-ledger closure introduced in v3.35.0.

1. Weighted structural energy

Choose positive weights

$$\omega_j > 0, \qquad \omega_j^N > 0, \qquad \omega_R > 0, \qquad \omega_\Gamma > 0,$$

and define the weighted structural energy

$$\mathcal{L} := \sum_j \omega_j B_j + \sum_j \omega_j^N N_j + \omega_R \mathfrak{R} + \omega_\Gamma \Gamma.$$

The aim is to choose the weights so that cross-channel transfer and residual spill can be absorbed into one dissipative inequality. In this language, the ledger is no longer just a bookkeeping device; it becomes the carrier of a monotonicity law.

Definition 1 (Lyapunov-admissible weight system). A family of weights is called Lyapunov-admissible if, after inserting the transport, absorption, and source inequalities from the previous section, one obtains constants $\kappa > 0$ and $C_{\text{tag}}, C_{\text{match}} \geq 0$ such that

$$\frac{d}{d\tau}\mathcal{L} \leq -\kappa\mathcal{L} + C_{\text{tag}}\,\text{Res}_{\text{tag}} + C_{\text{match}}\,\Delta_{\text{match}}.$$

This means that every admissible leakage path is paid for either by global decay or by an explicitly tagged source of unresolved structure.

2. Weighted closure theorem

The abstract transport statements from v2.5.0 are now compressed into a single persistence theorem.

Theorem 1 (weighted ledger closure). Assume that the channel transport matrix has a strict damping gap, that every active normal channel is absorption-efficient, and that residual production is source-tagged. Then there exists a Lyapunov-admissible weight system for which

$$\frac{d}{d\tau}\mathcal{L} \leq -\frac{\kappa}{2}\mathcal{L} + C_{\text{tag}}\,\text{Res}_{\text{tag}} + C_{\text{match}}\,\Delta_{\text{match}}.$$

In particular, if the tagged residual and probe mismatch stay integrably controlled, then the whole derivation remains in a forward-invariant admissible regime.

Proof. Choose the weights recursively along the dominance order induced by the transfer matrix, starting from the most dissipative channels and moving toward the weaker ones. Because the damping gap is strict, the transfer contributions can be absorbed by enlarging downstream weights finitely many times. The absorption-efficient normal channels contribute additional negative terms for the dangerous normal mass. Residual production enters only through tagged sources and therefore cannot create an unidentified positive contribution. After summing the weighted inequalities and absorbing the remaining off-diagonal terms, one obtains the stated estimate. □

This is the first point at which the document obtains a real forward-invariance statement rather than only a local transport picture. The admissible proof sector is now a dynamically protected cone.

3. Forward-invariant regime and restart stability

One practical advantage of the Lyapunov-ledger closure is that it explains why the proof can be resumed after interruptions without re-deriving the entire local chain. If at some stage τ_0 the state satisfies

$$\mathcal{L}(\tau_0) \le M, \qquad \int_{\tau_0}^{\tau_1} \mathrm{Res}_{\mathrm{tag}}\, d\tau < \infty, \qquad \int_{\tau_0}^{\tau_1} \Delta_{\mathrm{match}}\, d\tau < \infty,$$

then Gronwall's inequality implies a bound of the form

$$\mathcal{L}(\tau) \le e^{-\kappa(\tau-\tau_0)/2} M + \int_{\tau_0}^{\tau} e^{-\kappa(\tau-s)/2} \big(C_{\mathrm{tag}}\, \mathrm{Res}_{\mathrm{tag}}(s) + C_{\mathrm{match}}\, \Delta_{\mathrm{match}}(s) \big)\, ds.$$

Hence restarting the argument from an intermediate checkpoint is legitimate as long as the checkpoint data are reported in ledger form. This gives a rigorous version of proof memory: the document needs to preserve the weighted state, not the full local micro-history.

4. Probe inheritance of invariance

The two-probe picture inherits the same closure principle. Define the weighted probe functional

$$\mathcal{L}_{\mathrm{probe}} := \alpha_1 B_{\mathrm{tri}}^{(1)} + \alpha_2 B_{\mathrm{tri}}^{(2)} + \alpha_Q B_{\perp}^{(Q)} + \alpha_M B_{\mathrm{match}} + \alpha_R B_{\mathrm{res}}^{\mathrm{probe}}.$$

If the probe transfer system has a strict damping gap and the mismatch source remains ledger-tagged, then the same weighting argument yields

$$\frac{d}{d\tau}\mathcal{L}_{\mathrm{probe}} \le -\kappa_{\mathrm{probe}} \mathcal{L}_{\mathrm{probe}} + C_{\mathrm{tag}}^{\mathrm{probe}}\, \mathrm{Res}_{\mathrm{tag}}^{\mathrm{probe}} + C_{\mathrm{match}}^{\mathrm{probe}}\, \Delta_{\mathrm{match}}.$$

Thus the probe sector is not merely compatible with the main ledger; it inherits the same forward-invariant discipline. The search geometry therefore remains structurally confined rather than opening an uncontrolled side channel.

5. Structural consequence for later field equations

The future field-equation layer can now be read more sharply. A field equation will not merely translate channel budgets into continuous variables; it must preserve the forward-invariant regime induced by $\mathcal{L}$. In other words, admissible field variables should be chosen so that the induced energy identity mirrors the weighted ledger closure. This provides a direct criterion for accepting or rejecting later field-equation ansaetze.

6. Consequence for appendix tools

The tool appendix can be refined once more. Besides the roles introduced in v2.5.0, a strong tool may now be marked as one of the following:

(a) weight constructor,

(b) off-diagonal absorber,

(c) invariant-region verifier, or

(d) restart certificate.

This is useful because it identifies precisely which lemmas contribute to the global Lyapunov closure and which only support local transport estimates.

7. v3.35.0 readiness criterion

The present step should be regarded as successful if the following can now be done:

- the entire admissible sector can be encoded in one weighted ledger functional;
- restart points can be justified by checkpoint data in ledger form;
- probe arguments inherit the same invariant-region control as the main proof;
- later field equations can be tested against a precise closure criterion; and
- appendix tools can be sorted by whether they contribute to local transport or global invariance.

This is the gain of v3.35.0. The companion document now carries not only staged closure, channel specialization, and budget transport, but also a genuine forward-invariant structural regime that stabilizes the proof under continuation, probe comparison, and later equation-level readout.

v2.7.0 Field-Energy Realization and Semigroup Readout Core

> **Normalization note (v2.17.1 local review).** This block is retained as a *historical realization and semigroup-readout precursor.* It supports the later field/probe realization layer and should not be read as a second independent primary core.

The next structural step after forward-invariant ledger closure is to show that the weighted ledger is not only a bookkeeping device but already the shadow of a genuine field-energy balance. Version 3.35.0 therefore adds a realization layer: the channel budgets are interpreted as coarse observables of an underlying evolution system, and admissibility is reformulated as dissipative semigroup compatibility. This still does not claim the final field equations, but it fixes the exact kind of continuous dynamics those equations must generate if they are to remain faithful to the structural proof architecture.

1. Abstract state space and energy readout

Let

$$X := X_{\mathrm{tri}} \oplus X_{\perp} \oplus X_{\mathrm{meas}} \oplus X_{\mathrm{res}}$$

be an abstract Banach or Hilbert state space carrying the channel-resolved structural variables. We write a state as

$$U = (u_{\mathrm{tri}}, u_{\perp}, u_{\mathrm{meas}}, u_{\mathrm{res}}).$$

The ledger variables are no longer treated as primitive numbers but as readout functionals

$$B_i(U) \geq 0, \qquad i \in \{\mathrm{tri}, \perp, \mathrm{meas}, \mathrm{res}\},$$

with the weighted energy

$$\mathcal{E}(U) := \alpha_{\mathrm{tri}} B_{\mathrm{tri}}(U) + \alpha_{\perp} B_{\perp}(U) + \alpha_{\mathrm{meas}} B_{\mathrm{meas}}(U) + \alpha_{\mathrm{res}} B_{\mathrm{res}}(U).$$

The content of this step is that $\mathcal{E}$ should reproduce the Lyapunov ledger $\mathcal{L}$ up to uniform equivalence on the admissible sector. Thus the ledger is interpreted as a compressed energy observable of the underlying state rather than an external annotation.

2. Dissipative generator Ansatz

Assume that the channel evolution is generated by an abstract equation

$$\partial_\tau U = \mathcal{A}U + \mathcal{N}(U) + \mathcal{S}_{\mathrm{tag}}(\tau),$$

where $\mathcal{A}$ is the linear transport-dissipation generator, $\mathcal{N}$ is a structured nonlinear coupling compatible with HC confinement, and $\mathcal{S}_{\mathrm{tag}}$ collects the tagged residual sources. The admissibility requirement from the earlier sections now becomes the statement that $\mathcal{A}+\mathcal{N}$ is energy-dissipative modulo tagged production and match error, namely

$$\frac{d}{d\tau}\mathcal{E}(U(\tau)) \leq -c_0\mathcal{E}(U(\tau)) + C_{\mathrm{tag}}\,\mathrm{Res}_{\mathrm{tag}}(\tau) + C_{\mathrm{match}}\,\Delta_{\mathrm{match}}(\tau)$$

for every admissible trajectory. This is the semigroup form of the weighted ledger closure derived in v2.6.0.

3. Realization principle

The structural point can now be isolated as a single principle.

Principle 1 (field-energy realization). An eventual field equation is acceptable only if its induced state evolution admits a channel readout $U \mapsto (B_i(U))_i$ and a weighted energy $\mathcal{E}$ such that:

(a) $\mathcal{E}$ is uniformly equivalent to the admissible ledger functional $\mathcal{L}$;

(b) the induced evolution preserves the forward-invariant admissible sector;

(c) all positive production terms are either dissipatively absorbed or explicitly tagged; and

(d) probe mismatch enters only through a controlled source term Δ_{match}.

Any ansatz failing one of these four points is structurally too weak, even if it looks formally plausible at equation level.

4. Semigroup consequence and proof memory

If the above dissipative estimate holds, then the evolution operator $S(\tau)$ generated by the admissible dynamics satisfies an estimate of the form

$$\mathcal{E}(S(\tau)U_0) \leq e^{-c_0\tau}\mathcal{E}(U_0) + \int_0^\tau e^{-c_0(\tau-s)}\big(C_{\text{tag}}\,\text{Res}_{\text{tag}}(s) + C_{\text{match}}\,\Delta_{\text{match}}(s)\big)\,ds.$$

Hence proof memory can now be reformulated in semigroup language: restarting from a checkpoint is valid because the future evolution depends continuously on the current energy state and the tagged source history, not on the unrecoverable microscopic path. This upgrades the earlier restart argument into an abstract continuous-time stability statement.

5. Probe-sector realization

The same realization principle applies to the two-probe geometry. If

$$U_{\text{probe}} = (U^{(1)}, U^{(2)}, U^{(Q)}, U^{(M)}, U^{(R)})$$

denotes the probe state and $\mathcal{E}_{\text{probe}}$ the corresponding weighted energy, then admissibility requires

$$\frac{d}{d\tau}\mathcal{E}_{\text{probe}} \leq -c_{\text{probe}}\mathcal{E}_{\text{probe}} + C_{\text{tag}}^{\text{probe}}\,\text{Res}_{\text{tag}}^{\text{probe}} + C_{\text{match}}^{\text{probe}}\,\Delta_{\text{match}}.$$

Thus the probe mechanism is now seen as a dissipative comparison flow living in the same semigroup class as the main derivation. This makes the search geometry more than a metaphor: it becomes an admissible evolution subsystem with inherited energy control.

6. Consequence for future field equations and appendix tools

The field-equation programme is sharpened once again. A candidate PDE or abstract operator equation must now be checked in the following order:

(1) identify the state space and channel decomposition;

(2) construct the ledger readout functionals $B_i(U)$;

(3) verify equivalence between readout energy $\mathcal{E}$ and structural ledger $\mathcal{L}$;

(4) prove dissipativity modulo tagged sources; and

(5) verify probe inheritance and restart compatibility.

Accordingly, appendix tools can receive one further classification layer. A strong tool may now be marked as a state-space constructor, energy comparator, semigroup stability lemma, or source-tagging certificate. This helps separate tools that only support symbolic rearrangement from those that certify genuine dynamic admissibility.

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7. v3.35.0 readiness criterion

The present step should be regarded as successful if the following can now be done:

- the ledger is interpretable as a uniformly equivalent field-energy observable;
- admissibility is reformulated as dissipative semigroup compatibility;
- proof memory is supported by a continuous-time stability estimate;
- the two-probe sector is embedded into the same realization framework; and
- future field equations can be accepted or rejected by an explicit realization checklist.

This is the gain of v3.35.0. The companion monolith now carries not only closure, channel specialization, transport dynamics, and forward invariance, but also a first abstract field-energy realization principle. The later field-equation layer is therefore no longer free-form: it has to generate the same dissipative proof memory, probe inheritance, and tagged-source discipline already fixed at the structural level.

v2.8.0 Spectral Admissibility and Resolvent Gatekeeping Core

> **Normalization note (v2.17.1 local review).** This block is retained as a *historical spectral-admissibility precursor*. Its gatekeeping role is preserved, but the normalized admissibility logic is now carried by the semantic/tensor and grouped-layer unfolding phases.

The semigroup realization from v2.7.0 still leaves one decisive structural question open: which generators are admissible before one even starts evolving the system? In this version the answer is sharpened by adding a spectral gatekeeping layer. The state evolution is not accepted merely because it can be written as an abstract flow; it must also possess a resolvent and spectrum compatible with the ledger discipline, HC confinement, tagged-source accounting, and probe inheritance.

1. Spectral admissibility principle

Let $\mathfrak{X}$ be the admissible state space from the previous realization step and let $\mathcal{G}$ denote the effective generator of the dynamics, including its controlled tagged-source coupling. We call $\mathcal{G}$ *spectrally admissible* if there exist $\omega_* > 0$ and $C_* > 0$ such that for every λ with $\Re\lambda \geq -\omega_*$ the resolvent exists on the admissible sector and obeys

$$\|(\lambda - \mathcal{G})^{-1}\|_{\mathfrak{X}\to\mathfrak{X}} \leq \frac{C_*}{1+|\lambda|},$$

after projection to the HC-confined channel space. This estimate expresses that the generator does not hide uncontrolled growth modes behind formal notation. The ledger control from earlier versions is therefore promoted into a spectral entrance test.

Principle 2 (spectral gatekeeping). A candidate field equation or abstract operator model is structurally admissible only if its effective generator is spectrally admissible in the above sense and the corresponding resolvent bounds are compatible with the tagged-source decomposition. The proof programme may therefore reject an operator before any detailed PDE analysis if the spectrum already violates ledger closure.

2. Resolvent–ledger compatibility

The weighted energy $\mathcal{E}$ from v2.7.0 should now be stable not only along the flow but already at the resolvent level. Concretely, admissibility requires a bound of the schematic form

$$\mathcal{E}\big((\lambda-\mathcal{G})^{-1}F\big) \le C_{\mathrm{res}}\Big(\mathcal{E}(F)+\mathrm{Tag}(F)+\mathrm{Match}(F)\Big)$$

for every admissible forcing term F and every $\Re\lambda \ge -\omega_*$. Here $\mathrm{Tag}(F)$ denotes the explicitly source-tagged part and $\mathrm{Match}(F)$ the mismatch component inherited from probe comparison. This is the spectral analogue of the semigroup energy estimate: the resolvent may amplify forcing, but only in a way that remains ledger-readable.

3. HC confinement as spectral projector

The HC mechanism is now reinterpreted one step more sharply. Earlier versions treated it as the structural compensator that confines evolution to the septinion closure. At spectral level this becomes the statement that there exists a projector Π_{HC} with

$$(\lambda-\mathcal{G})^{-1} = \Pi_{\mathrm{HC}}(\lambda-\mathcal{G})^{-1}\Pi_{\mathrm{HC}} + \mathcal{R}_{\mathrm{tag}}(\lambda),$$

where $\mathcal{R}_{\mathrm{tag}}(\lambda)$ is either negligible in the admissible regime or explicitly source-tagged. Hence uncontrolled extra channels are not merely suppressed dynamically; they are prevented from entering the admissible resolvent class except through a tagged remainder. This gives a clean spectral formulation of HC confinement.

4. Spectral gap and restart robustness

If the admissible generator satisfies the above bounds together with a positive gap between the dominant admissible spectrum and the forbidden sector, then restart stability acquires a stronger meaning. A checkpoint is not only dynamically safe but spectrally robust: small reconstruction errors remain trapped inside the same resolvent class and do not move the system into a hidden unstable branch. In this sense proof memory is upgraded from semigroup continuity to spectral persistence.

5. Probe-sector comparison spectrum

The two-probe subsystem inherits the same criterion. Let $\mathcal{G}_{\mathrm{probe}}$ be the effective probe generator acting on the probe state space. Then admissibility requires a comparison estimate

$$\|(\lambda-\mathcal{G}_{\mathrm{probe}})^{-1}\| \le C_{\mathrm{probe,res}}(1+|\lambda|)^{-1}, \qquad \Re\lambda \ge -\omega_{\mathrm{probe}},$$

with constants controlled by the same ledger hierarchy and mismatch tags. The search geometry therefore becomes spectrally testable: a probe construction is not accepted merely because it heuristically mirrors the main evolution, but because its spectrum remains subordinate to the same admissible comparison regime.

6. Consequence for future field equations and appendix tools

The field-equation admission checklist is sharpened again. A candidate model should now be tested in the following order:

(1) identify state space and admissible channel projector;

(2) construct the energy and ledger readout;

(3) verify semigroup dissipativity modulo tagged sources;

(4) prove spectral admissibility and resolvent compatibility; and

(5) verify probe-spectrum inheritance and restart robustness.

This also refines the appendix tools. Strong tools may now additionally be marked as resolvent estimator, spectral projector certificate, gap witness, or restart persistence lemma. These labels distinguish genuinely admissibility-producing tools from merely formal symbolic aids.

7. v3.35.0 readiness criterion

The present step should be regarded as successful if the following can now be done:

- admissible generators can be filtered by a spectral entrance test;
- resolvent bounds can be read in the same ledger language as the semigroup flow;
- HC confinement is expressible as a spectral projector statement;
- restart stability is strengthened to spectral persistence; and
- the probe sector inherits a controlled comparison spectrum.

This is the gain of v3.35.0. The companion monolith now carries not only closure, channel specialization, transport dynamics, forward invariance, and field-energy realization, but also a first spectral admissibility layer. Future field equations are therefore constrained twice: they must generate the correct ledger-dissipative semigroup and they must belong to the correct resolvent class before they are even admitted into the structural programme.

v2.9.0 Spectral Readout Filters and Trace-Compatible Mode Selection Core

> **Normalization note (v2.17.1 local review).** This block is retained as a *historical spectral-readout and trace-filter precursor*. It contributes proof memory and specialized readout discipline, but should be read through the later normalized readout and interface phases.

The spectral admissibility layer from v2.8.0 decides which generators may enter the programme at all. The next structural question is subtler: once a generator is admissible, which parts of its spectrum are actually readable in a way compatible with the ledger, HC confinement, probe comparison, and later trace-style summary principles? In this version the monolith therefore adds a spectral readout layer. Not every admissible mode is automatically promoted to a structural observable. A mode must also survive a filter test that distinguishes stable, tagged, and non-readable contributions.

1. Readout filter principle

Let $\mathcal{G}$ be spectrally admissible on the HC-confined state space $\mathfrak{X}$, and let $\Sigma_{\mathrm{adm}}(\mathcal{G})$ denote the admissible part of the spectrum. We introduce a family of bounded readout filters $\{\mathcal{F}_\alpha\}_{\alpha\in A}$ acting on the admissible sector, where each $\mathcal{F}_\alpha$ is designed to isolate one structural mode class: triadic, orthogonal, measurement, residual, or probe-induced. A mode contribution is called *trace-compatible* if there exists at least one filter $\mathcal{F}_\alpha$ such that

$$\mathcal{F}_\alpha \Pi_{\mathrm{HC}} u = \Pi_{\mathrm{HC}} u + r_\alpha, \qquad \|r_\alpha\|_{\mathfrak{X}} \le \varepsilon_\alpha \|u\|_{\mathfrak{X}},$$

with either small remainder r_α or explicitly tagged remainder. Thus the readout is not merely spectral isolation but structural readability modulo tagged error.

Principle 3 (filtered observability). A spectral component may be used in the proof architecture only if it is both spectrally admissible and trace-compatible under an approved readout filter family. This prevents uncontrolled spectral clutter from being mistaken for genuine structural information.

2. Trace-compatible decomposition

The admissible spectrum should now admit a decomposition of the schematic form

$$\Sigma_{\mathrm{adm}}(\mathcal{G}) = \Sigma_{\mathrm{read}} \cup \Sigma_{\mathrm{tag}} \cup \Sigma_{\mathrm{mute}},$$

where readable modes lie in Σ_{read}, tagged but non-primary modes lie in Σ_{tag}, and the remainder Σ_{mute} is admissible but not allowed to enter the principal structural trace. The point is not that mute modes vanish, but that they cannot serve as primary carriers of theorem-level readout. This sharpens the difference between existence and interpretability.

3. HC-filter compatibility

The HC projector from v2.8.0 is now required to commute with readout at least up to controlled tagged remainder. Concretely, one seeks estimates of the form

$$\mathcal{F}_\alpha \Pi_{\mathrm{HC}} = \Pi_{\mathrm{HC}} \mathcal{F}_\alpha + \mathcal{T}_\alpha,$$

where $\mathcal{T}_\alpha$ is either small in the admissible norm or belongs to a source-tagged correction class. This means HC confinement is preserved under mode selection: filtering is not allowed to re-open forbidden channels except through a ledger-visible correction.

4. Mode budgets and readable trace mass

The budget language from v2.5.0 now acquires a spectral refinement. For every approved filter $\mathcal{F}_\alpha$ define the readable trace mass

$$\mathfrak{M}_\alpha(u) := \mathcal{E}(\mathcal{F}_\alpha u)$$

with $\mathcal{E}$ the field-energy functional from v2.7.0. Structural readability requires a balance inequality of the schematic form

$$\sum_{\alpha \in A_{\mathrm{read}}} \mathfrak{M}_\alpha(u) \le C_{\mathrm{tr}} \Big(\mathcal{E}(u) + \mathrm{Tag}(u) + \mathrm{Leak}(u) \Big).$$

Hence readable spectral mass cannot exceed the ledger budget except through already visible tagged or leakage terms. This gives a quantitative bridge between spectral filtering and earlier channel transport estimates.

5. Probe-sector mode inheritance

For the two-probe subsystem the same distinction becomes a comparison principle. Let $\mathcal{G}_{\mathrm{probe}}$ be the effective probe generator and $\mathcal{F}_\beta^{\mathrm{probe}}$ its admissible filter family. Then every readable probe mode should either descend from a readable ambient mode or be marked as a probe-only tagged correction. In particular, one asks for comparison bounds of the form

$$\mathfrak{M}_\beta^{\mathrm{probe}}(v) \le C_{\mathrm{cmp}} \big(\mathfrak{M}_\alpha(u) + \mathrm{Mismatch}(u, v) \big),$$

for matched ambient/probe states (u, v). This prevents the probe sector from manufacturing fake observables that have no ambient structural parent.

6. Consequence for future field equations and appendix tools

The admission checklist for candidate field equations is now extended once more:

(1) verify semigroup dissipativity and ledger closure;

(2) verify spectral admissibility and resolvent compatibility;

(3) construct an HC-compatible readout filter family;

(4) separate readable, tagged, and mute spectral sectors; and

(5) prove probe inheritance of readable modes.

The appendix tool taxonomy is correspondingly refined. Besides resolvent and gap certificates, a strong tool may now also be labelled as readout filter constructor, trace-mass comparator, mute-sector exclusion lemma, or probe-inheritance witness. These labels indicate that the tool produces not only existence information but also structural legibility.

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7. v3.35.0 readiness criterion

The present step should be regarded as successful if the following can now be done:

- admissible spectra can be split into readable, tagged, and mute parts;
- HC confinement survives mode filtering modulo controlled tagged remainder;
- readable trace mass is bounded in the same ledger language as channel budgets;
- probe readout inherits ambient readable modes rather than inventing new ones; and
- future field equations can be screened not only for existence, but for structural legibility.

This is the gain of v3.35.0. The companion monolith now carries not only closure, channel specialization, transport dynamics, forward invariance, field-energy realization, semigroup readout, and spectral admissibility, but also a first filtered-observability layer. Future field equations are therefore constrained three times: they must generate the correct ledger-dissipative semigroup, they must belong to the correct resolvent class, and their admissible modes must pass a trace-compatible readout filter before they can count as structural content.

v2.10.0 Trace Reconstruction and Observable Extraction Core

> **Normalization note (v2.17.1 local review).** This block is retained as a *historical reconstruction and observable-extraction precursor.* Its reconstruction language remains useful, but the normalized proof-bearing architecture now lies in the unfolding phases and the grouped transport logic.

The filtered-observability layer from v2.9.0 separates admissible modes into readable, tagged, and mute sectors. The next structural task is no longer only to screen admissible modes, but to reconstruct theorem-level observables from the readable part without double counting, without reopening forbidden channels, and without losing the ledger discipline enforced by HC confinement. The present step therefore adds a trace-reconstruction layer. It answers the question: once a mode has passed admissibility and readability tests, how is it promoted to an actual observable in the companion proof architecture?

1. Reconstruction operator for readable modes

Let $\{\mathcal{F}_\alpha\}_{\alpha\in A_{\mathrm{read}}}$ be the approved readable filter family from v2.9.0. We now introduce a reconstruction operator

$$\mathcal{R}: \bigoplus_{\alpha\in A_{\mathrm{read}}} \mathfrak{X}_\alpha \to \mathfrak{O},$$

where $\mathfrak{X}_\alpha = \mathcal{F}_\alpha\mathfrak{X}$ denotes the readable mode sector and $\mathfrak{O}$ is the structural observable space. The role of $\mathcal{R}$ is not arbitrary synthesis: it is constrained reconstruction. For a state $u \in \mathfrak{X}$ one seeks a decomposition

$$\mathrm{Obs}(u) = \mathcal{R}\big((\mathcal{F}_\alpha u)_{\alpha\in A_{\mathrm{read}}}\big)$$

such that every observable component remains HC-confined, ledger-bounded, and source-tagged whenever the reconstruction is not exact. Thus observable extraction is treated as a disciplined readout map rather than an unrestricted projection.

Principle 4 (reconstructive observability). A readable mode contributes to theorem-level content only after passing through a reconstruction operator whose output is compatible with HC confinement, channel budgets, and source tags. Readability alone is therefore necessary but not yet sufficient for observable status.

2. Observable extraction at theorem level

For each admissible theorem-context Θ let $\mathfrak{O}_\Theta$ denote the class of observables allowed in that context. One then requires extraction estimates of schematic form

$$\|\mathcal{R}_\Theta((\mathcal{F}_\alpha u)_\alpha)\|_{\mathfrak{O}_\Theta} \le C_\Theta\Big(\sum_{\alpha\in A_{\mathrm{read}}} \mathfrak{M}_\alpha(u) + \mathrm{Tag}(u) + \mathrm{Leak}(u)\Big),$$

with $\mathfrak{M}_\alpha$ the readable trace masses from v2.9.0. This means an observable is only legitimate when its theorem-level norm is controlled by the same ledger quantities already used to regulate channel transport and spectral readout. The companion proof therefore acquires an explicit extraction stage between readable modes and theorem statements.

3. Overlap ledger and anti-double-counting rule

Once multiple filters contribute to one observable quantity, overlap must be made visible. Introduce an overlap ledger

$$\mathrm{OL}(u) = \sum_{\alpha\neq\beta} \omega_{\alpha\beta}\, \langle \mathcal{F}_\alpha u, \mathcal{F}_\beta u\rangle_{\mathfrak{X}},$$

where the coefficients $\omega_{\alpha\beta}$ encode admissible correlation or subtraction rules. The purpose of $\mathrm{OL}(u)$ is not to forbid overlap but to prevent silent duplication of structural mass. Observable reconstruction is required to satisfy a compensated estimate of the form

$$\|\mathrm{Obs}(u)\|_{\mathfrak{O}} \le C_{\mathrm{obs}}\Big(\sum_{\alpha\in A_{\mathrm{read}}} \mathfrak{M}_\alpha(u) - \mathrm{OL}(u) + \mathrm{Tag}(u) + \mathrm{Leak}(u)\Big),$$

whenever the overlap ledger is well defined. Thus one can read several filtered channels at once without pretending they were disjoint when they are not.

4. HC-stable observable closure

The reconstructed observable sector must itself remain HC-stable. Concretely, one requires that

$$\Pi_{\mathrm{HC}}\,\mathrm{Obs}(u) = \mathrm{Obs}(u) + \tau(u), \qquad \|\tau(u)\|_{\mathfrak{O}} \le C_{\mathrm{HC}}\big(\mathrm{Tag}(u) + \mathrm{Leak}(u)\big),$$

so that reconstruction does not reactivate forbidden directions except through ledger-visible correction terms. In this sense HC confinement becomes not only a state-space condition and not only a filter-compatibility condition, but now also an observable-closure condition. Only outputs that survive this test may be used as stable structural readings.

5. Probe-sector reconstruction inheritance

For the two-probe subsystem the same reconstruction principle must descend to the probe readout. Let $\mathcal{R}^{\mathrm{probe}}$ be the probe reconstruction operator. Then every probe observable should either be inherited from an ambient observable through the descent chain or be explicitly tagged as probe-local correction. One asks schematically for bounds of the form

$$\|\mathrm{Obs}^{\mathrm{probe}}(v)\|_{\mathfrak{O}_{\mathrm{probe}}} \leq C_{\mathrm{pr}}\big(\|\mathrm{Obs}(u)\|_{\mathfrak{O}} + \mathrm{Mismatch}(u,v)\big),$$

for matched ambient/probe states (u, v). This ensures the probe sector remains an instrument of structural descent rather than an independent source of untracked observables.

6. Consequence for field equations and appendix tools

The admission checklist for future candidate field equations is therefore extended once more:

(1) verify semigroup dissipativity and ledger closure;

(2) verify spectral admissibility and resolvent compatibility;

(3) verify readable mode separation under HC-compatible filters;

(4) construct a reconstruction operator for theorem-level observables; and

(5) bound overlap, leakage, and probe descent in a visible ledger form.

The appendix tool taxonomy is refined accordingly. Besides resolvent estimators, readout filters, and gap witnesses, a strong tool may now also be labelled as reconstruction operator, observable extractor, overlap-ledger bound, HC-stability witness, or probe-descent certificate. These labels mark tools that do not merely prove admissibility, but actually promote admissible readable content into stable observables.

7. v3.35.0 readiness criterion

The present step should be regarded as successful if the following can now be done:

- readable spectral content can be reconstructed into theorem-level observables;
- overlaps between readable channels are accounted for instead of silently duplicated;
- observable extraction remains HC-stable modulo visible tagged/leakage correction;
- probe observables descend from ambient observables rather than appearing autonomously; and
- future field equations can be screened for observable reconstruction, not merely for readable spectra.

This is the gain of v3.35.0. The companion monolith now carries not only closure, channel specialization, transport dynamics, forward invariance, field-energy realization, semigroup readout, spectral admissibility, and filtered observability, but also a first disciplined reconstruction layer. Future field equations are therefore constrained four times: they must generate the correct ledger-dissipative semigroup, belong to the correct resolvent class, pass a trace-compatible readout filter, and finally admit HC-stable observable reconstruction before they can count as theorem-level structural content.

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v2.11.0 Multi-Readout Consistency and Fusion Core

> **Normalization note (v2.17.1 local review).** This block is retained as a *historical multi-readout and fusion precursor*. It remains relevant as supporting proof memory, but no longer functions as an independent primary structural core in the normalized architecture.

The reconstruction layer from v2.10.0 promotes readable modes into theorem-level observables. The next structural question is what happens when several reconstructions are all individually admissible but are intended to describe one and the same structural content. The task of v3.35.0 is therefore not merely to extract observables, but to fuse them without creating contradictory readouts, without hiding overlap behind notation, and without allowing probe-local views to break the ambient closure architecture. This introduces a multi-readout consistency layer.

1. Readout families and consistency targets

Let $\{\mathcal{R}_j\}_{j\in J}$ be a family of admissible reconstruction maps, possibly associated with different filter systems, theorem contexts, or probe resolutions. For a given state u one obtains a readout family

$$\mathrm{Obs}_j(u) = \mathcal{R}_j\big((\mathcal{F}_\alpha^{(j)} u)_\alpha\big).$$

The purpose of the new layer is to decide when the collection $\{\mathrm{Obs}_j(u)\}_{j\in J}$ should be regarded as several compatible views of one structural object rather than several unrelated observables. Compatibility is measured against the same closure, budget, and tagging rules that already govern the single-readout setting.

Principle 5 (multi-readout consistency). A family of theorem-level observables may be fused only when their mutual discrepancies are either controlled by a visible ledger term or assigned to explicit source tags. Silent disagreement is not an allowed mode of structural multiplicity.

2. Consistency ledger and fusion defect

Introduce a fusion-defect functional

$$\mathfrak{D}_{\mathrm{fus}}(u) := \sum_{j<k} \nu_{jk} \, \|\mathrm{Obs}_j(u) - \Gamma_{jk}\mathrm{Obs}_k(u)\|_{\mathfrak{O}_*},$$

where Γ_{jk} transports the k-th observable into the comparison frame of the j-th observable and the coefficients $\nu_{jk} \geq 0$ encode structural relevance. The associated consistency ledger is then

$$\mathrm{CL}(u) := \mathfrak{D}_{\mathrm{fus}}(u) + \mathrm{Tag}(u) + \mathrm{Leak}(u).$$

A family is called fusion-admissible when $\mathrm{CL}(u)$ stays below the threshold prescribed by the theorem context. Thus disagreement is not denied; it is made visible, budgeted, and therefore manageable.

3. Fused observable and anti-inflation rule

Whenever the family is fusion-admissible, define a fused observable

$$\mathrm{Obs}_{\mathrm{fus}}(u) = \sum_{j\in J} \lambda_j\, \Gamma_j \mathrm{Obs}_j(u) - \Xi(u),$$

where the coefficients λ_j are admissible weights, Γ_j transports to a common observable frame, and $\Xi(u)$ is an anti-inflation compensator correcting for visible overlap and recorded fusion defect. The fused observable is required to satisfy an estimate of the form

$$\|\mathrm{Obs}_{\mathrm{fus}}(u)\|_{\mathfrak{O}_*} \le C_{\mathrm{fus}}\big(\sum_{j\in J} \|\mathrm{Obs}_j(u)\|_{\mathfrak{O}_j} - \mathrm{OL}_{\mathrm{fus}}(u) + \mathrm{CL}(u)\big),$$

so that fusion cannot enlarge theorem-level content by counting the same readable structure several times through nominally different channels.

4. HC-stable fusion and theorem transport

The fused observable must still respect HC confinement. One therefore asks for

$$\Pi_{\mathrm{HC}}\, \mathrm{Obs}_{\mathrm{fus}}(u) = \mathrm{Obs}_{\mathrm{fus}}(u) + \rho(u), \qquad \|\rho(u)\|_{\mathfrak{O}_*} \le C_\rho\, \mathrm{CL}(u).$$

This means that multi-readout fusion is accepted only if the residual mismatch is fully visible in the same ledger that already recorded comparison defects. In addition, theorem transport must preserve fusion status: if a fused observable is moved into a lemma, table, or appendix tool, the same fusion certificate must remain available after transfer.

5. Probe/ambient coherence under fusion

In the two-probe sector one may have ambient and probe reconstructions of the same structural feature. The new requirement is that the probe fusion defect be bounded by ambient fusion defect plus descent mismatch:

$$\mathfrak{D}_{\mathrm{fus}}^{\mathrm{probe}}(v) \le C_{\mathrm{coh}}\big(\mathfrak{D}_{\mathrm{fus}}(u) + \mathrm{Mismatch}(u,v)\big).$$

Hence probe readouts do not become a second autonomous truth layer; they inherit coherence from the ambient structure, up to visible descent error.

6. Consequence for appendix tools

The appendix tool taxonomy is refined again. A strong appendix item may now be labelled as a fusion certificate, consistency-ledger estimate, anti-inflation compensator, theorem-transport witness, or probe-coherence bound. This fits the user's structural preference that tools be collected as reusable proof interfaces rather than scattered informally through the running text.

7. v3.35.0 readiness criterion

The present step should be regarded as successful if the following can now be done:

- several admissible reconstructions can be compared in one explicit ledger;

- disagreements between readouts are tagged or budgeted instead of hidden;
- fused observables remain protected against overlap inflation;
- probe and ambient readouts satisfy a visible coherence estimate; and
- appendix transfer preserves not only observables, but also their fusion certificates.

This is the gain of v3.35.0. The companion monolith now carries not only closure, channel specialization, transport dynamics, forward invariance, field-energy realization, semigroup readout, spectral admissibility, filtered observability, and trace reconstruction, but also a first disciplined fusion layer for theorem-level observables. Future field equations are therefore constrained five times: they must generate the correct ledger-dissipative semigroup, belong to the correct resolvent class, pass a trace-compatible readout filter, admit HC-stable observable reconstruction, and finally support multi-readout fusion without hidden inconsistency.

12 Structural Tools and Regime Tests

This appendix collects the main structural tools extracted from the proof architecture. Their purpose is not to replace the main derivations, but to normalize frequently recurring arguments into reusable tests and reduction principles.

Sector and closure tools

Tool (Closure-transfer principle). Let X be any operator on the septonic probe space. Its admissible reduction is

$$X_{\mathrm{adm}} := \hat{P}_{C_7} X \hat{P}_{C_7}.$$

For admissible states $\Psi \in \mathrm{Ran}(\hat{P}_{C_7})$, the physically relevant action of X is determined by X_{adm}, while the complementary blocks measure off-sector leakage.

Tool (Equivalent closure criteria). The following are equivalent (cf. Lemma `lem:closure-forcing` in the active-core PDF):

(i) $\mathfrak{R}_C = 0$;

(ii) $[\mathcal{M}_{16}, \hat{P}_{C_7}] = 0$;

(iii) $\mathcal{M}_{16}$ preserves the admissible sector and its complement.

Hence exact closure is equivalent to block preservation, i.e. block-diagonalization of $\mathcal{M}_{16}$ with respect to $\mathcal{H}_{\mathrm{sept}} = \hat{P}_{C_7}\mathcal{H}_{\mathrm{sept}} \oplus (\mathbf{1} - \hat{P}_{C_7})\mathcal{H}_{\mathrm{sept}}$.

Tool (Closure-preservation test). For any new structural term X, first test whether $[X, \hat{P}_{C_7}] = 0$. If yes, then X respects the admissible stabilized sector directly. If not, then X either produces leakage or must be controlled by a compensator-mediated defect term (cf. Hypothesis `hyp:frame-dependent-defect-readout` in the active-core PDF).

Tool (Sectorial normal form). Every operator X decomposes as

$$X = \hat{P}_{C_7} X \hat{P}_{C_7} + (\mathbf{1} - \hat{P}_{C_7}) X \hat{P}_{C_7} + \hat{P}_{C_7} X (\mathbf{1} - \hat{P}_{C_7}) + (\mathbf{1} - \hat{P}_{C_7}) X (\mathbf{1} - \hat{P}_{C_7}),$$

isolating: (i) admissible core block, (ii) incoming leakage, (iii) outgoing leakage, (iv) complement-sector block.

Master-equation tools

Tool (Core-defect normal form). The projected structural field operator $\mathfrak{F}_{\text{struct}} = \hat{P}_{C_7}(i\Gamma^\mu \mathcal{D}_\mu - \Omega)\hat{P}_{C_7}$ splits into

$$\mathfrak{F}_{\text{struct}} = \mathbb{D}_{\text{core}} + \mathbb{D}_{\text{defect}},$$

where $\mathbb{D}_{\text{core}}$ contains autonomous admissible transport and resonance, and $\mathbb{D}_{\text{defect}}$ collects closure mismatch, compensator activity, and off-sector suppression.

Tool (Exact-closure normal form). If $\mathfrak{R}_C = 0$, $[\mathcal{M}_{16}, \hat{P}_{C_7}] = 0$, and $\mathcal{H}_C = 0$ for the minimal compensator ansatz, then $\mathfrak{F}_{\text{struct}} = \mathbb{D}_{\text{core}}$ and $\mathbb{D}_{\text{defect}} = 0$. The visible compensator field vanishes in this regime; the kernel $C = \Pi^{(\mathcal{M})}_{\text{spec}}$ (Theorem `thm:hc-spectral-projector` in the active-core PDF) remains active (Remark `rem:hc-shadow-vs-core` in the active-core PDF).

Tool (Mass-residue test). A proposed mass contribution is classified according to whether it arises from: (i) a closed-core scalar reading of the projected resonance operator; (ii) a weak closure-defect correction; (iii) a genuine shifted-clock or projector residue $\mathfrak{m}_{\text{comm}}(\psi) = \frac{\hbar}{c}|\langle\psi|[\hat{\mathcal{R}}_\beta, \hat{P}_{C_7}]|\psi\rangle|$.

Projection and regime tools

Tool (Regime-strength test). A reduction of the structural master equation is: *strong* if exact closure plus direct slot restriction suffices; *intermediate* if projected algebra closure is additionally needed; *effective* if bilinear reconstruction, coarse-graining, or emergent parameter generation are also required. These levels correspond to the three steps of Proposition `prop:structural-lift-principle` in the active-core PDF.

Tool (Dirac gate). A Dirac-type reduction is available if: (i) exact closure holds; (ii) the projected connection freezes locally to a trivial sector; (iii) the projected resonance operator reduces to a scalar inertial reading.

Tool (Lie-closure gate). A Yang–Mills-type reduction is available only if the projected connection algebra closes on a finite-dimensional Lie algebra $\mathfrak{g}$. Without that closure, the gauge interpretation is structurally premature.

Tool (Einstein pipeline gate). An Einstein-type reduction requires all of: (i) exact closure on the admissible sector; (ii) bilinear one-forms $E^a{}_\mu$ from projected probe dynamics; (iii) maximal rank of the effective coframe; (iv) metric-compatible geometric connection; (v) coarse-grained local effective action with Einstein–Hilbert leading term. If any step is missing, the Einstein block remains effective rather than first-principles (cf. Proposition `prop:expected-probe-lensing-behaviour` in the active-core PDF).

Equivalence and anti-redundancy tools

Tool (Stone–von Neumann compression rule). If two sectors are irreducible unitary realizations of the same Weyl-type algebra on separable Hilbert spaces, differing only by representation change, symmetry implementation, or frame reading, then one representative suffices (Theorem `thm:unitary-equiv` in the active-core PDF).

Tool (Stone–von Neumann scope test). Do *not* apply Stone–von Neumann as a shortcut if the passage between two sectors involves: (i) controlled projection discarding slots; (ii) coarse-graining over internal modes; (iii) bilinear geometric reconstruction; (iv) emergent effective constants; (v) loss of irreducibility. Stone–von Neumann compresses representation multiplicity, not regime multiplicity.

Tool (Two-layer equivalence test). For candidate sectors S_1, S_2: (1) *Representation layer* — test whether they are unitary realizations of the same irreducible Weyl-type kernel; if yes, SvN-compressible. (2) *Regime layer* — test whether one sector is obtained from the other by projection, coarse-graining, geometric readout, or effective reconstruction; if yes, additional structural work is required beyond Stone–von Neumann (Definition `def:readout-relative-lemma` in the active-core PDF).

Geometry-readout tools

Tool (Coframe extraction gate). Given a closed triadic probe field $\Psi_\triangle$, define $E^a{}_\mu := \mathrm{Re}(\bar{\Psi}_\triangle \, \Gamma^a \, \mathcal{D}_\mu \Psi_\triangle)$. This defines an effective coframe only if the resulting 4×4 matrix has maximal rank.

Tool (Closure-to-geometry cleanliness test). If exact closure holds, probe bilinears are computed entirely inside the invariant C_7-sector, so reconstructed metric data is not contaminated by leakage terms. This is the minimal cleanliness criterion for any geometric readout.

Tool (Geometry-reconstruction warning). Closure by itself does not yet produce Einstein geometry. It provides: (i) an autonomous admissible sector; (ii) a well-defined projected connection; (iii) a stable basis for bilinear readout. It does *not* yet provide automatically: (i) nondegeneracy of the coframe; (ii) derivation of the effective constants; (iii) explicit identification of the collective closure fields; (iv) justified coarse-graining to a local geometric action. The bidirectional probe lensing hypothesis (Section `sec:bidirectional-probe-lensing` in the active-core PDF) proposes a two-step route that may reduce this gap.

Appendix B. Symbol Normalization for the Late Readout Layers

This appendix fixes the notation used in the late Companion expansion layers so that historical version labels and structural symbols remain auditable across the v2.4.0–v2.11.0 chain.

Historical section labels. The blocks entitled v2.4.0 through v2.11.0 are to be read as historical expansion layers. Their section headings record the version in which the corresponding layer was introduced, not merely the current document version.

Ledger matrix. The channel ledger matrix is denoted by

$$\Lambda_{\mathrm{ch}}(\Psi) = \begin{pmatrix} \ell_{\mathrm{tri}} & \delta_{\mathrm{tri}} & r_{\mathrm{tri}} \\ \ell_{\perp} & \delta_{\perp} & r_{\perp} \\ \ell_{\mathrm{meas}} & \delta_{\mathrm{meas}} & r_{\mathrm{meas}} \\ \ell_{\mathrm{res}} & \delta_{\mathrm{res}} & r_{\mathrm{res}} \end{pmatrix}.$$

It is a bookkeeping object for channel-resolved tangential, normal, and compensator-facing sizes.

Lyapunov ledger functional. The weighted scalar functional introduced in v2.6.0 keeps the notation $\mathcal{L}$. It is a Lyapunov-type control quantity and is not to be confused with the matrix-valued ledger Λ_{ch}.

Budget totals. The aggregated scalar budgets remain $\mathfrak{B}$ for total visible load, $\mathfrak{N}$ for normal leakage, and $\mathfrak{R}$ for compensator-facing storage.

Readout and fusion objects. The readable-mode trace mass, reconstruction operators, observable extraction maps, and fusion-defect quantities introduced in v2.9.0–v2.11.0 are interpreted as structure-preserving descendants of the ledger/budget layer and therefore inherit the normalization fixed here.

Session-budget rule

A further practical restriction is useful at the present stage: each local mathematics-first pass should also observe a bounded session budget. The purpose of this rule is not to slow work down, but to prevent a single pass from silently expanding from a targeted closure attempt into a diffuse rewrite wave. In concrete terms, a local subversion should normally do only one of the following as its primary task: (i) stabilize definitions for one A-priority block, (ii) discharge one dependency layer for that block, (iii) perform one focused reference/notation repair required by that block, or (iv) close one clearly delimited appendix-tool support step. If more than one such task would be required, the safer action is to end the current pass, classify the result, and continue by explicit next local subversion.

The session-budget rule therefore complements the anti-parallelization and re-entry rules. A pass is considered well-scoped if its main closure claim can be summarized in one sentence and audited against one active block. A pass should be considered over budget if it begins to change multiple theorem-status boundaries, broad notation regimes, or several major sections at once. In that case, the correct response is not to continue accumulating edits, but to stop, audit, and reopen the next step as a fresh local subversion. This rule is intentionally conservative: its goal is to make every local version legible as a finite mathematical move rather than as an opaque mass of partially mixed intentions.

Archived final release-decision template for v2.27.0

At this stage the document no longer needs another diffuse local steering note, but a compact decision template that makes explicit under which reading the next public release is justified. The present template distinguishes three cases. In the first, the document is strong enough that the next action is *release now*: this applies if the A-priority blocks are structurally frozen, the gate conditions remain green, and the remaining work consists only of optional proof-density enrichment. In the second, the document should *hold locally for proof-density enrichment*: this applies if the structural spine is already in place, but one deliberately wants a denser local unfolding before exposing the next public release. In the third, the document must *hold locally for unresolved blocker*: this applies only if a genuine release-critical dependency, hypothesis-control issue, or reference-instability problem is still active.

Under the current reading, the first A-block no longer counts as such a blocker. Its local dependency spine has been extended through admissible transport, matrix-sector realization, and admissible trace/branch persistence, and it has already been reconciled with the mathematics-first gate logic. Hence the residual caution is no longer structural incompleteness, but only whether one prefers to accept the present medium proof density or to invest one further local pass in additional unfolding. Accordingly, the document is now close to the decision boundary where a direct step to v2.27.0 becomes reasonable.

Decision states.

- **Release now:** the first A-block is structurally release-supporting, no primary blocker remains, and only optional densification is deferred.
- **Hold locally for proof-density enrichment:** no structural blocker remains, but one more local pass is preferred for denser exposition.
- **Hold locally for unresolved blocker:** a genuine reference, dependency, or hypothesis-control failure is still present.

Current reading. At the present local stage, the third state no longer appears to apply to the first A-block. The second and first states are now the relevant ones: either one accepts the current medium-density closure as sufficient for the next public release, or one performs at most one more local enrichment pass before the public step.

Release consequence. The practical consequence is that any further local subversion should no longer open a new mathematical front. It should either confirm release-readiness or add one last density-improving clarification and then stop.

2J. Open-questions closure pass and chain check

The present stage is also strong enough to classify several previously open questions more explicitly. The purpose of this pass is not to pretend that every weak-density zone has disappeared, but to distinguish between (i) questions that can already be closed reliably at the current structural level, (ii) questions that remain structurally intact but only under an explicit assumption or still-thin proof layer, and (iii) questions that should remain openly deferred. This distinction is especially important in a structure-first setting, because a

locally thin argument is not automatically broken: some passages remain admissible as conditional or contradiction-style steps, provided that their status is made explicit and no illegitimate theorem-promotion occurs.

Closure classification. At the current stage the following questions may be treated as *closable now*: the dependency spine from ground condition to HC and confinement; the compatibility of closure with projector stabilization; the admissible transport continuation on the stabilized sector; the admissible-sector action of the Millennium matrix; and the persistence of admissible trace/branch continuation once the stabilized sector has been fixed. These are not yet maximally unfolded everywhere, but they are no longer merely heuristic. By contrast, the following questions are presently better read as *intact but proof-density limited*: the strongest available mathematical reading of the Millennium matrix as converter/intertwiner; the full density of trace/branch continuation lemmas; and the extent to which some still-compact structural passages should already be promoted beyond proposition level. Finally, the following remain *intentionally open*: any full universality claim for the computational strand, any finished proof-kernel or compiler reading, and any interpretation that would make the mathematics depend on the computational teaser architecture.

Chain-check statuses. To keep the proof chain honest without overcalling local weakness as failure, the document now adopts four chain-check statuses for future mathematical passes:

- **intact:** the dependency chain carries without visible status or continuity defect;
- **intact, but proof-density weak:** the chain carries, but one or more transitions still need denser local unfolding;
- **intact under explicit assumption:** the argument is acceptable only because an assumption, contradiction setup, or provisional hypothesis is openly declared;
- **blocker:** a genuine dependency break, status violation, or contradiction prevents closure.

Current reading. Under this classification, the first major dependency spine is currently best read as *intact, but proof-density weak* rather than as blocked. No primary structural contradiction is visible in the path from ground condition through HC, closure, admissible transport, Millennium-matrix realization, and admissible trace/branch persistence. The remaining caution is therefore mainly one of density and explicitness, not one of broken continuity.

Local consequence. The practical consequence is conservative but positive. Several earlier open questions may already be marked as *reliably closable at the present stage*, provided that the document does not silently promote weak-density passages to stronger theorem status than the current support allows. The next mathematics-first passes should therefore use the chain-check statuses as a standing audit device: strengthen density where useful, keep assumption-based passages openly marked, and treat only genuine continuity failure as a blocker.

2K. Tool extraction and leverage-point scan

The next mathematics-first pass should not treat every locally thin argument as something that must immediately be expanded line by line. In a structure-first document, some still-wobbly passages already contain reusable mathematical moves that can be extracted earlier as *tools*, provided that their status is made explicit. The present scan therefore distinguishes between (i) tools that are already operationally useful now, even if their surrounding proof density is still moderate, (ii) tools that remain usable only under an explicit structural assumption, and (iii) a small number of leverage points whose clarification would simplify several later proof passages at once.

Provisional tool extraction. At the current stage the following reusable tools may already be extracted from the first dependency spine. First, a *closure-transfer tool*: once the admissible sector has been fixed, closure may be transported across projector stabilization without reopening the sector boundary. Second, a *projected transport tool*: admissible transport may be read modulo the stabilized projector rather than by re-proving closure from scratch after each transport step. Third, a *matrix-sector tool*: the Millennium matrix may already be used as an admissible-sector-preserving operator on the stabilized transported sector. Fourth, a *trace/branch persistence tool*: admissible continuation in the trace/branch layer may be used without reintroducing an external admissibility source. None of these tools should yet be oversold as maximal-density theorems everywhere; however, they are already strong enough to simplify downstream arguments by replacing repeated local closure checks with a smaller number of reusable invariance moves.

Status discipline for extracted tools. To keep the extraction mathematically honest, each such tool should be marked in one of three ways:

- **operational tool:** reusable now, with the present support level;
- **assumption-backed tool:** reusable only under an explicitly declared hypothesis or contradiction setup;
- **blocked tool candidate:** not yet safe to reuse because a real dependency gap remains.

Under the current chain check, the closure-transfer, projected transport, matrix-sector, and trace/branch persistence tools are best read as *operational tools*, though some still sit in the "proof-density weak" zone rather than the fully unfolded zone.

Leverage-point scan. The present scan suggests that the most useful immediate leverage point is not a new broad theorem family, but a single stronger organizing hypothesis: namely that the admissible stabilized sector behaves as the true fixed working domain for closure, transport, matrix action, and admissible continuation. Even if one initially formulates this only as a hypothesis-backed simplifier, it has unusually high leverage: if strengthened, it compresses several later arguments by replacing repeated local admissibility restoration with one global working-domain principle. In other words, many later proofs become shorter once they are allowed to start with "assume the admissible stabilized sector has already been fixed" and then work internally on that domain.

Use of the computational layer as a secondary aid. The computational chapter may already help here, but only in a subordinate role. MML/MMA/MCP should not drive the mathematics; however, the language layer can already serve as a *notation and workflow aid* for tool extraction. In particular, compact MML-style descriptions can help identify whether a candidate move is really one reusable operation—for example `project -> transport -> trace`—or whether it still hides an unresolved mathematical gap. Thus the computational layer may already be used as a diagnostic simplifier for tool extraction, while the mathematical status of the extracted tool remains decided purely on the mathematics-first side.

Current reading and local consequence. The current reading is therefore positive. The first dependency spine is no longer merely a place where one asks whether the chain survives; it is already rich enough to yield a first family of reusable structural tools. At the same time, the highest-leverage next target is also visible: stabilize the admissible working-domain principle as far as possible, because this one point appears to simplify closure, transport, matrix action, and trace/branch continuation simultaneously. Any next local pass should therefore prefer a leverage-point clarification over broad diffuse proof growth.

Appendix C. Julia Boundary, Master Residual η, and Reversal-Probe Consolidation

This appendix records the condensed 3.3 main line. The local residual sector is organized around the single scalar

$$\eta_f := \frac{1}{3} - \beta_f^2.$$

The point of the compression is that the previously separated Hopf-gap, Julia-boundary, isotropic defect, and reversal language are different visible faces of the same transported branch quantity.

Proposition 12.1 (Master residual transport and isotropic readout)**.** *Let f range over admissible readouts of the same closure-stable kernel, and assume that on the admissible branch the visible compensated defect is isotropic,*

$$D_{\mu\nu}^{(f)} = \eta_f \, g_{\mu\nu}^{(f)}.$$

Assume moreover that admissible transport preserves the underlying defect class. Define

$$\eta_f := \frac{1}{3} - \beta_f^2.$$

Then the following are equivalent encodings of the same local residual sector:

(i) the additive β^2-gap η_f;

(ii) the continuum-limit deformation

$$\delta_{\mathrm{CL},f} := \frac{\beta_f}{1/\sqrt{3}} = \sqrt{3}\,\beta_f, \qquad \eta_f = \frac{1-\delta_{\mathrm{CL},f}^2}{3};$$

(iii) the cosmological residuals

$$\eta_f = \frac{1}{3} - \Omega_M^{(f)} = \Omega_\Lambda^{(f)} - \frac{2}{3}$$

whenever $\Omega_M^{(f)} = \beta_f^2$ *and* $\Omega_\Lambda^{(f)} = 1 - \beta_f^2$*;*

(iv) the isotropic defect tensor $D_{\mu\nu}^{(f)} = \eta_f g_{\mu\nu}^{(f)}$*;*

(v) the transported quadratic invariant

$$\mathfrak{I}_f = \eta_f^2.$$

In particular, every admissible readout carries the same local defect class through the single scalar η_f*.*

Proof. The displayed identities are algebraic. Once the visible defect is isotropic, every normalized quadratic contraction is determined by its scalar coefficient, hence by η_f. Transport preservation identifies the same defect class in different readouts, so the admissible residual sector is represented everywhere by the same one-scalar quantity and its powers. □

Remark 12.2 (Symmetric value and the 10^{-3} scale)**.** *For the symmetric matter/tachyon readout with* $\beta_f = \beta_{\text{sym}} \approx 0.5493$ *one gets*

$$\eta_{\text{sym}} = \frac{1}{3} - \beta_{\text{sym}}^2 \approx 3.1603 \times 10^{-2}, \qquad \eta_{\text{sym}}^2 \approx 9.987 \times 10^{-4}.$$

So a number near 10^{-3} *is most naturally read not as a primitive constant, but as the quadratic visible residual of the transported master defect scalar.*

Proposition 12.3 (Julia boundary and exact-core principle)**.** *Assume that admissible branching does not introduce a new primitive local scalar beyond the continuum–discrete deformation of the same transport sector. Then the exact compensated core has two equivalent local descriptions:*

$$\textit{intrinsic core: } z^- = 0, \qquad \textit{normalized visible core: } \eta_f = 0.$$

Moreover every admissible leading-order branch/Julia defect functional satisfies

$$\mathfrak{J}_f = c_f\, \eta_f + O(\eta_f^2)$$

with a nonzero normalization constant c_f*. Thus the observed Julia-boundary sensitivity is controlled, to leading order, by the same scalar* η_f *that governs the isotropic defect readout.*

Proof. The exact compensated core is representation-stable, but it is seen in a phase-shifted readout frame. Intrinsically the core is the null-kernel relation $z^- = 0$; after Hopf normalization of the visible branch it is the vanishing of the transported residual coefficient η_f. If admissible branching opens no new primitive scalar channel, then the first visible boundary defect must be a function of the already available scalar η_f, and its first nontrivial term is therefore linear in η_f. □

Theorem 12.4 (Local η-completeness and restoring normal form)**.** *Assume a branch-local admissible transport chart in which visible observables depend only on the local admissible branch parameter and no admissible transverse scalar channel reopens. Then:*

(i) every branch-local scalar observable without derivatives is a function of η;

(ii) every first-jet scalar observable is a function of η as well, because

$$\nabla_\mu \eta = A(\eta)\, U_\mu$$

for some branch tangent U and scalar amplitude A;

(iii) every second-jet branch-local scalar observable is a function of η and the longitudinal second derivative

$$Y := U^\mu U^\nu \nabla_\mu \nabla_\nu \eta.$$

If, in addition, the branch is locally η-complete, then there exists an autonomous reduced equation

$$\frac{d^2\eta}{ds^2} = B(\eta)$$

with

$$B(0) = 0.$$

If the branch realizes a closure-restoring return law, then near the exact core one has

$$\eta'' + \omega^2 \eta = O(\eta^2)$$

with $\omega^2 > 0$ on the nondegenerate branch. Hence the informal "Umkehrwalze" is mathematically realized as a local restoring dynamics for the transported master residual.

Proof. The zeroth-order claim follows from isotropy of the visible defect sector: once $D_{\mu\nu} = \eta g_{\mu\nu}$, all algebraic scalar contractions are polynomials in η. Branch-local visibility forces the first derivative of any such scalar to be tangent to the same admissible branch direction, so the first jet remains one-dimensional. At second order the only new natural scalar is the longitudinal Hessian component Y, and local η-completeness identifies it as a function of the already visible branch value. The exact core is transport-preserved, so it must be an equilibrium of the reduced equation; closure-restoring return then fixes the linearized sign and yields the stated normal form. □

Definition 12.5 (Hybrid reversal probe)**.** *Define the condensed reversal probe by*

$$\mathfrak{S}_{\mathrm{rev}} := (\mathbb{P}_{\mathrm{B}}, \mathcal{L}_{\mathrm{L}}),$$

where $\mathbb{P}_{\mathrm{B}}$ is a hard admissibility/closure projector (the "Bombe" side) and $\mathcal{L}_{\mathrm{L}}$ is a fine residual functional on the η-branch (the "Lorenz" side).

Proposition 12.6 (Reversal-probe dichotomy)**.** *Let two admissible probe trajectories on the same local residual branch have the same visible master residual value. Then locally only two outcomes remain:*

$$\Delta\theta \equiv 0 \pmod{2\pi} \qquad \text{or} \qquad \Delta\theta \equiv \pi \pmod{2\pi}.$$

Thus the two probes either converge to the same visible branch representative, or they lie on the same transported structure with nontrivial $\mathbb{Z}_2$ phase class and close only on the spinorial/Möbius double cover.

Proof. The hard projector removes inadmissible branch continuations, while the fine functional identifies the remaining residual class through the same local variable η. Equality of the visible residual therefore leaves no generic room for an independent third local phase sector. What remains is either direct coincidence in the visible branch parameterization or the unique nontrivial $\mathbb{Z}_2$ lift compatible with spinorial sign reversal. □

C.1 First 3.4 extension: closure potential and global branch law

Definition 12.7 (Closure potential and branch energy). *Assume a connected admissible branch interval $I \ni 0$ on which the reduced residual equation is exact, i.e.*

$$\eta'' = B(\eta) = -\mathcal{U}'(\eta)$$

for some $\mathcal{U} \in C^2(I)$. We call $\mathcal{U}$ a closure potential for the branch and define the associated branch energy by

$$\mathcal{E}_\eta := \frac{1}{2}(\eta')^2 + \mathcal{U}(\eta).$$

Theorem 12.8 (Global exact branch law under closure exactness). *Assume the hypotheses of Theorem 12.4, and assume in addition branch-local closure exactness in the above sense. Then:*

(i) $\mathcal{E}_\eta$ is conserved along every admissible branch trajectory;

(ii) the exact compensated core $\eta = 0$ is an equilibrium with $\mathcal{U}'(0) = 0$;

(iii) if the branch is closure-restoring and nondegenerate, then $\mathcal{U}''(0) = \omega^2 > 0$, hence

$$\mathcal{U}(\eta) = \frac{1}{2}\omega^2\eta^2 + O(\eta^3);$$

(iv) every admissible turning point of the visible residual branch satisfies

$$\mathcal{U}(\eta) = \mathcal{E}_\eta.$$

Proof. Differentiating $\mathcal{E}_\eta$ along a trajectory gives

$$\frac{d}{ds}\mathcal{E}_\eta = \eta'\eta'' + \mathcal{U}'(\eta)\eta' = \eta'(\eta'' + \mathcal{U}'(\eta)) = 0.$$

The equilibrium statement is equivalent to $B(0) = 0$, already forced by core preservation. The nondegenerate restoring sign from the local normal form gives $B(\eta) = -\omega^2\eta + O(\eta^2)$, hence $\mathcal{U}'(\eta) = \omega^2\eta + O(\eta^2)$ and therefore the stated Taylor expansion. Turning points are exactly the points where $\eta' = 0$, so the conserved-energy identity reduces to $\mathcal{U}(\eta) = \mathcal{E}_\eta$. □

Proposition 12.9 (Energy-form reversal-probe criterion). *Assume the hypotheses of Theorem 12.8. Let two admissible probe trajectories be selected by the same hard projector $\mathbb{P}_{\mathrm{B}}$ and lie on the same connected regular level set $\mathcal{E}_\eta = E$. Assume moreover that this energy component carries no monodromy higher than the spinorial $\mathbb{Z}_2$-lift. Then exactly two global outcomes remain:*

direct convergence on the visible branch

or closure only on the spinorial/Möbius double cover

Proof. The hard projector excludes inadmissible components, while the fine residual functional identifies the same visible residual class through the common potential-energy level. On a connected regular admissible energy component, the remaining ambiguity is topological rather than scalar. By hypothesis no higher monodromy is available, so the only surviving alternatives are trivial monodromy and the unique nontrivial $\mathbb{Z}_2$ lift. □

C.2 Second 3.4 extension: reversible single-well class

Theorem 12.10 (Parity and single-well restriction of the closure potential)**.** *Assume the hypotheses of Theorem 12.8. Assume moreover that the admissible residual branch is invariant under the involution $R(\eta) = -\eta$, that the reduced branch law commutes with R, and that $\eta = 0$ is the unique admissible critical core on the connected interval I with no further admissible critical point of $\mathcal{U}$ on $I \setminus \{0\}$. Normalize $\mathcal{U}(0) = 0$. Then:*

(i) $B(-\eta) = -B(\eta)$ and $\mathcal{U}(-\eta) = \mathcal{U}(\eta)$ on I;

(ii) near the exact core,

$$\mathcal{U}(\eta) = \frac{1}{2}\omega^2\eta^2 + a_4\eta^4 + O(\eta^6), \qquad B(\eta) = -\omega^2\eta - 4a_4\eta^3 + O(\eta^5);$$

(iii) $\mathcal{U}'(\eta) > 0$ for every $\eta > 0$ in I and $\mathcal{U}'(\eta) < 0$ for every $\eta < 0$ in I; in particular $\mathcal{U}(\eta) > 0$ for all $\eta \neq 0$.

Consequently $\mathcal{U}$ belongs to the symmetric single-well class on I.

Proof. If $\eta(s)$ solves $\eta'' = B(\eta)$, branch reversibility and commutation with R imply that $-\eta(s)$ is again an admissible solution of the same reduced law. Hence

$$-\eta'' = B(-\eta),$$

while applying R to the original equation gives $-\eta'' = -B(\eta)$, so $B(-\eta) = -B(\eta)$. Since $\mathcal{U}'(\eta) = -B(\eta)$, the derivative $\mathcal{U}'$ is odd. With the normalization $\mathcal{U}(0) = 0$, integration yields $\mathcal{U}(-\eta) = \mathcal{U}(\eta)$.

Theorem 12.8 already gives $\mathcal{U}''(0) = \omega^2 > 0$. Evenness forces the Taylor series of $\mathcal{U}$ at the exact core to contain only even powers, so

$$\mathcal{U}(\eta) = \frac{1}{2}\omega^2\eta^2 + a_4\eta^4 + O(\eta^6),$$

and differentiation gives the stated odd expansion of B.

Because $\mathcal{U}''(0) = \omega^2 > 0$, one has $\mathcal{U}'(\eta) > 0$ for sufficiently small $\eta > 0$. If $\mathcal{U}'$ vanished again at some positive point, continuity would force an additional admissible critical point of $\mathcal{U}$ on $I \cap (0, \infty)$, excluded by hypothesis. Hence $\mathcal{U}'(\eta) > 0$ on the whole positive half-branch. The negative half-branch follows by oddness of $\mathcal{U}'$. Therefore $\eta = 0$ is the unique well bottom and $\mathcal{U}(\eta) > 0$ for all $\eta \neq 0$. □

Proposition 12.11 (Symmetric turning-pair criterion)**.** *Under the hypotheses of Theorem 12.10, let $E > 0$ be a regular branch-energy value. Then the turning-point equation*

$$\mathcal{U}(\eta) = E$$

has on I either no solution or exactly the symmetric pair $\eta = \pm\eta_E$ with $\eta_E > 0$. In particular every regular admissible energy shell carries at most the scalar ambiguity $\eta \leftrightarrow -\eta$.

Proof. By the preceding theorem, $\mathcal{U}$ is even and strictly increasing on each half-branch as a function of $|\eta|$. Therefore the equation $\mathcal{U}(\eta) = E$ can have at most one positive and at most one negative solution, and evenness forces them to occur as a symmetric pair whenever they exist. □

C.3 Third 3.4 extension: canonical quartic family

Proposition 12.12 (Quartic reduction under first nonlinear visibility)**.** *Assume the hypotheses of Theorem 11.4. Then on a symmetric admissible neighborhood J one has*

$$\mathcal{U}(\eta) = \frac{1}{2}\omega^2\eta^2 + \frac{\lambda}{4}\eta^4 + R_6(\eta), \qquad R_6(\eta) = O(\eta^6),$$

for some $\lambda \geq 0$.

Proof. By Theorem 11.3, the potential is even, C^6, and has a strict minimum at 0, so its Taylor expansion on J has the form

$$\mathcal{U}(\eta) = \frac{1}{2}\omega^2\eta^2 + a_4\eta^4 + a_6\eta^6 + o(\eta^6).$$

The hypotheses exclude any additional visible scalar or independent local scale at first nonlinear order, so the quartic coefficient is the only admissible datum at that order. Writing $\lambda := 4a_4$ and absorbing the higher terms into R_6 gives the claimed form. Since 0 remains the unique admissible local minimum on J, shrinking J if necessary excludes the sign choice that would open a new nearby admissible critical pair; therefore $\lambda \geq 0$ on the retained neighborhood. □

Proposition 12.13 (Turning-value parameterization)**.** *In the minimal truncation*

$$\mathcal{U}_{\min}(\eta) = \frac{1}{2}\omega^2\eta^2 + \frac{\lambda}{4}\eta^4, \qquad \lambda \geq 0,$$

regular energy shells are parameterized by the unique positive turning value $A(E)$ satisfying

$$E = \frac{1}{2}\omega^2 A(E)^2 + \frac{\lambda}{4}A(E)^4.$$

If $\lambda > 0$, then

$$A(E)^2 = \frac{-\omega^2 + \sqrt{\omega^4 + 4\lambda E}}{\lambda},$$

while for $\lambda = 0$ one has $A(E) = \sqrt{2E}/\omega$.

Proof. On a regular shell, turning points are exactly the zeros of η', so they satisfy $E = \mathcal{U}_{\min}(\eta)$. By even single-well monotonicity there is exactly one positive solution $A(E)$ and one negative solution $-A(E)$. Solving the quadratic equation in $x = A(E)^2$ gives the explicit formula when $\lambda > 0$, and the harmonic formula when $\lambda = 0$. □

Proposition 12.14 (Appendix C.4: quartic closure coefficient as visible datum)**.** *Under the hypotheses of Theorem 11.5, the visible quartic branch coefficient λ is not an independent nonlinear parameter. It is exactly the transported quartic closure coefficient μc_4, and every regular probe shell may therefore be parameterized either by its conserved energy E or by its unique positive turning value $A(E)$ without introducing an additional local scalar.*

Proof. Theorem 11.5 gives $\lambda = \mu c_4$ with $\mu > 0$, so the quartic correction is fixed by the same closure geometry that already determines the quadratic restoring core. Proposition 12.13 identifies the regular shell by either E or $A(E)$, and Theorem 11.4 excludes any further visible local scalar at quartic order. Hence the quartic visible datum is exactly the transported closure coefficient and does not enlarge the local scalar algebra. □

Proposition 12.15 (Appendix C.5: quartic timing drift as probe readout)**.** *Under the hypotheses of Theorem 11.6, the first nonlinear timing drift of a regular periodic shell is already fixed by the transported quartic closure coefficient. More precisely,*

$$T(E) = \frac{2\pi}{\omega}\Big(1 - \frac{3c_4}{4\mu c_2^2}E + O(E^2)\Big),$$

so the branch is locally isochronous at first nonlinear order if and only if $c_4 = 0$, while $c_4 > 0$ forces a strictly decreasing period for all sufficiently small regular energies.

Proof. Theorem 11.6 gives the small-energy period expansion together with the closure rewrite $\omega^2 = \mu c_2$ and $\lambda = \mu c_4$. Since $\mu > 0$ and $c_2 > 0$, the first nonlinear period correction vanishes exactly when $c_4 = 0$ and has negative sign whenever $c_4 > 0$. Thus the timing drift is a direct probe readout of the transported quartic closure coefficient and does not introduce a second nonlinear scalar. □

Proposition 12.16 (Appendix C.6: higher even hierarchy and delayed timing defect)**.** *Under the hypotheses of Theorem 11.7, the first nonzero higher nonlinear timing defect is governed by the first nonvanishing higher even closure coefficient. More precisely, if*

$$c_4 = c_6 = \cdots = c_{2m-2} = 0, \qquad c_{2m} \neq 0, \qquad m \geq 3,$$

then

$$T(E) = \frac{2\pi}{\omega}\Big(1 - \kappa_m(\omega)\,\mu c_{2m}\,E^{m-1} + O(E^m)\Big), \qquad \kappa_m(\omega) > 0.$$

Hence the branch is isochronous through all lower nonlinear orders, and the first visible higher timing drift appears exactly at order E^{m-1} with sign determined by c_{2m}.

Proof. Theorem 11.7 identifies the first visible higher nonlinear coefficient as $\alpha_{2m} = \mu c_{2m}$ and gives the corresponding small-energy period law. Because the intermediate even coefficients vanish, no lower-order contribution to the period expansion is available. Since $\kappa_m(\omega) > 0$ and $\mu > 0$, the sign of the first nonzero higher timing defect is exactly the sign of c_{2m}. □

Proposition 12.17 (Appendix C.7: analytic generator form and coefficient transport)**.** *Under the hypotheses of Theorem 11.8, the local visible branch dynamics is canonically encoded by a single even generator Φ. More precisely,*

$$\mathcal{U}(\eta) = \Phi(\eta^2), \qquad B(\eta) = -2\eta\,\Phi'(\eta^2),$$

with

$$\Phi(s) = \frac{1}{2}\omega^2 s + \frac{\lambda}{4}s^2 + \sum_{m\geq 3}\frac{\alpha_{2m}}{2m}s^m,$$

and coefficient transport

$$\omega^2 = \mu c_2, \qquad \lambda = \mu c_4, \qquad \alpha_{2m} = \mu c_{2m} \quad (m \geq 3).$$

Hence the entire local nonlinear shell hierarchy is fixed by the single generator germ Φ, and every delayed timing defect is a readout of its coefficients rather than a new independent scalar.

Proof. Theorem 11.8 gives the analytic factorization $\mathcal{U}(\eta) = \Phi(\eta^2)$ together with the coefficient identifications. Expanding Φ near $s = 0$ yields the displayed formula, while differentiating gives the canonical branch law $B(\eta) = -2\eta\Phi'(\eta^2)$. The higher-order timing and turning-value readouts were already tied to the visible coefficients in Theorems 11.6 and 11.7, so the generator form introduces no additional scalar data. □

Proposition 12.18 (Appendix C.8: generator-shell classifier)**.** *Under the hypotheses of Theorem 11.9, every sufficiently small regular admissible shell is completely determined by its generator radius $s(E) = \Phi^{-1}(E)$, and the visible shell ambiguity is exactly the transported sign pair*

$$\eta = \pm\sqrt{s(E)}.$$

Hence the hybrid reversal probe has on such a shell only two local transport classes: sign-preserving transport (direct visible convergence) and sign-reversing transport (closure only after the unique spinorial/Möbius lift).

Proof. Strict monotonicity of Φ on the small-shell interval gives the inverse shell map $s(E) = \Phi^{-1}(E)$. Since the visible potential is $\mathcal{U}(\eta) = \Phi(\eta^2)$, the shell equation $\mathcal{U}(\eta) = E$ is equivalent to $\eta^2 = s(E)$, so the shell consists exactly of the two visible representatives $\pm\sqrt{s(E)}$. Any admissible transport restricted to this shell acts on that pair by either preserving sign or reversing it. By the standing no-higher-monodromy hypothesis, these two sign actions are the only local shell transport classes. □

Proposition 12.19 (Appendix C.9: shell parity and diagnostic Möbius-reflected render)**.** *Under the hypotheses of Theorem 11.10, the local probe outcome on every sufficiently small regular shell is completely determined by the shell parity $\pi(E) \in \{+1, -1\}$. If an admissible shell cycle decomposes into regular segments and transverse reflector crossings, then*

$$\pi(E) = (-1)^{N_E},$$

where N_E is the number of transverse reflector crossings in the cycle. Accordingly, even crossing parity yields direct visible convergence, while odd crossing parity yields closure only after the spinorial/Möbius lift.

Proof. Theorem 11.10 already shows that the shell transport acts on the visible turning pair by a unique sign parity $\pi(E)$. Under the stated decomposition, regular segments preserve sign and every transverse reflector crossing contributes the involution $\eta \mapsto -\eta$. Multiplying these contributions gives $\pi(E) = (-1)^{N_E}$, from which the direct-convergence versus lifted-closure criterion follows immediately. □

Placeholder figure not bundled in this release:
textttjulia_probe_residual_scales_v3841.png

Figure 1: **Diagnostic Möbius-reflected quadratic probe render at the current residual scales.** Comparison of the iterated map $z_{n+1} = 1/\overline{(z_n^2 + c_\gamma)}$ for the baseline $\gamma = 0$ and for the three currently relevant OP 1 residual scales $\Delta_\beta \approx 7.50 \times 10^{-5}$, $\Delta_\theta \approx 1.79 \times 10^{-4}$, and $\Delta\mu \approx 3.38 \times 10^{-4}$. For the present diagnostic render, the parameter family is instantiated by $c_\gamma = -1 + i\gamma$, with the unit circle drawn as the Möbius horizon. The global morphology remains stable across all four renders, while fine boundary filaments and local transition zones respond smoothly under the residual-scale shifts. The figure is included as a diagnostic visualization of structural robustness versus boundary-level response; the earlier marker value 0.0009 is retained elsewhere only as a historical stress-test scale, not as the current main comparison value.

Figure 1 is included as a diagnostic probe image rather than as a first-principles derivation. Its role is to visualize a separation between global morphological stability and boundary-level sensitivity: the large-scale visible body remains stable, while the finest branch and transition regions react under the current OP 1 residual scales. The asset is now generated reproducibly by the local Python renderer `generate_julia_probe_residual_scales.py`, which reads the displayed values of Δ_β, Δ_θ, and $\Delta\mu$ from the TeX source and produces both a PDF and a PNG version; the document includes the PDF render when available so that the figure can be rebuilt directly from the project material.

Placeholder figure not bundled in this release: texttgamma0_julia_probe_v3841.png
$\gamma = 0$

Placeholder figure not bundled in this release: textdelta_beta_julia_probe_v3841.png
$\gamma = \Delta_\beta$

Placeholder figure not bundled in this release: textdelta_theta_julia_probe_v3841.png
$\gamma = \Delta_\theta$

Placeholder figure not bundled in this release: textdelta_mu_julia_probe_v3841.png
$\gamma = \Delta\mu$

Figure 2: **Individual diagnostic Möbius-reflected probe panels.** The four residual-scale renders from Figure 1 shown again as separate panels, so that the boundary-level response at $\gamma = 0$, Δ_β, Δ_θ, and $\Delta\mu$ can be inspected without the surrounding comparison frame.

C.9 Tenth 3.4 extension: tunnel observables and quantitative readout interface

Proposition 12.20 (Appendix support for Proposition 11.11)**.** *Under the hypotheses of Proposition 11.11, any admissible quantitative comparison between the diagnostic Julia/probe render and an external readout image may be organized through the tunnel-observable package*

$$\mathfrak{O}_{\mathrm{diag}}(\gamma) = \big(w_{\mathrm{tun}}(\gamma;\tau),\ I_{\mathrm{edge}}(\gamma),\ \Theta_{\mathrm{open}}(\gamma),\ P_{\mathbb{Z}_2}(E)\big),$$

with no additional local scalar required beyond the generator-shell radius and parity sector. In particular, if two rendering protocols produce the same $s(E)$ and the same parity law on the same thresholded shell family, then any discrepancy among the first three observables is attributable to rendering normalization rather than to a new local structural channel.

Proof. This is a direct restatement of Proposition 11.11 in appendix language. The tunnel, edge, and opening observables are post-processing functionals of the same thresholded shell geometry, and the parity response is already intrinsic to the shell classifier. Hence matching generator/parity data leaves only normalization choices, not an extra structural scalar. □

C.10 Eleventh 3.4 extension: inverse readout and shell-consistency defect

Theorem 12.21 (Appendix support for Theorem 11.12)**.** *Assume the calibrated tunnel-width law*

$$w_{\mathrm{tun}}(\gamma;\tau) = W_\tau(s(E))$$

with strictly monotone W_τ on a small-shell interval. Then the diagnostic shell data is locally equivalent to the pair

$$\big(w_{\mathrm{tun}}(\gamma;\tau), P_{\mathbb{Z}_2}(E)\big),$$

with inverse reconstruction

$$s(E) = W_\tau^{-1}(w_{\mathrm{tun}}(\gamma;\tau)), \qquad \pi(E) = P_{\mathbb{Z}_2}(E).$$

Any further shell metric extracted from the same thresholded render therefore factors through these two calibrated quantities.

Proof. Monotonicity of W_τ gives local invertibility. The shell-parity response already records the transported $\mathbb{Z}_2$ class. Since Proposition 11.11 shows that all remaining tunnel/edge/opening metrics are readout-functionals of the same shell geometry, they factor through the recovered pair. □

C.11 Twelfth 3.4 extension: inverse identifiability and stability support

Theorem 12.22 (Appendix support for Theorem 11.14)**.** *Assume the calibrated inverse-readout hypotheses of Theorem 11.12 and fix a shell point s_0 with $W'_\tau(s_0) \neq 0$. Then on a sufficiently small shell neighborhood the calibration law is bi-Lipschitz,*

$$m_0|s_1 - s_2| \leq |W_\tau(s_1) - W_\tau(s_2)| \leq M_0|s_1 - s_2|,$$

for suitable constants $m_0, M_0 > 0$. Hence the calibrated tunnel width determines the shell radius uniquely on each fixed parity branch and does so with inverse condition number bounded by $1/m_0$.

Proof. Since $W'_\tau(s_0) \neq 0$, continuity of the derivative yields a neighborhood on which W'_τ stays bounded away from zero and from infinity. The mean-value theorem gives the displayed estimate, and the inverse condition-number statement is just the corresponding inverse Lipschitz bound. □

C.13 Fourteenth 3.4 extension: obstruction-shell transition support

Theorem 12.23 (Appendix support for Theorem 11.19)**.** *Let s_* be an isolated obstruction shell. If s_* is a nondegenerate calibration critical shell, then the calibrated tunnel law has a local fold normal form with a two-sheet inverse exchange. If s_* is instead a pure parity shell, then the shell radius continues uniquely and only the transported $\mathbb{Z}_2$ class jumps. In the mixed case the local transition is the direct product of these two involutions.*

Proof. A nondegenerate critical shell of a one-dimensional smooth map is locally a fold, so the inverse consists of exactly two sheets exchanged by the local deck involution. A pure parity shell leaves the scalar calibration regular and therefore does not affect the shell coordinate, but it does flip the transported sign class by definition. Since one operation acts on the shell coordinate and the other on parity, the mixed case is their commuting composition. □

3.4.14 working interpretation. The obstruction shells are therefore organized rather than chaotic. Each isolated obstruction contributes only one of two primitive involutions—fold exchange or parity flip—and every local continuation across the inverse-readout atlas is built from these same $\mathbb{Z}_2$ pieces. This makes the spinorial/Möbius branch no longer just a visual metaphor at the image layer, but the explicit parity part of the local obstruction monodromy.

3.4.12 working interpretation. The inverse-readout block now has a local stability test. Once the calibration slope on a shell interval is known to stay away from zero, tunnel width ceases to be a merely qualitative marker and becomes a controlled coordinate for the shell radius. The remaining local obstruction is discrete—the parity branch—together with the obvious continuous failure mode at critical points of the calibration law.

C.12 Thirteenth 3.4 extension: shell atlas and obstruction support

Theorem 12.24 (Appendix support for Theorem 11.16)**.** *Let I be a shell interval for the calibrated tunnel-width law W_τ, and let*

$$\Sigma_{\mathrm{crit}} = \{W'_\tau = 0\}, \qquad \Sigma_{\mathrm{par}} = \{\textit{parity switch shells}\}.$$

Then every connected component of $I \setminus (\Sigma_{\mathrm{crit}} \cup \Sigma_{\mathrm{par}})$ is a parity-preserving inverse chart on which W_τ is one-to-one. On compact subintervals of such a chart, the inverse is bi-Lipschitz and therefore the shell-consistency defect is a well-defined internal diagnostic of model quality rather than of inverse failure.

Proof. Outside Σ_{par} the parity class is fixed. Outside Σ_{crit}, the derivative of W_τ never vanishes and therefore keeps a constant sign on each connected component, so the calibration map is strictly monotone there. Compactness then upgrades the local inverse estimate to a uniform chartwise estimate. The shell-consistency defect is therefore an internal observable on such a chart, and only the two obstruction sets can destroy the chart itself. □

3.4.13 working interpretation. The inverse detector-style layer is now stratified. Stable readout happens chart by chart, with sharp obstruction shells where either the calibration slope collapses or the transported parity changes. This gives a clean separation between chart-internal mismatch and genuine breakdown of inverse control.

3.4.11 working interpretation. The diagnostic image can therefore be treated as a small inverse instrument. After threshold calibration, tunnel width and shell parity recover the local generator shell and predict the remaining edge/opening readouts. The natural residual mismatch is then measured by the shell-consistency defect Δ_{diag}, which quantifies how well the image closes back to the same shell/parity model.

3.4.10 working interpretation. The image layer is therefore now coupled back into the mathematics in a controlled way: the diagnostic render may be discussed through tunnel width, edge intensification, and opening angle without pretending that visual similarity alone proves a detector interpretation. This keeps the observational hypothesis testable while preserving the structure-first discipline.

3.4.8 working interpretation. The shell classifier can therefore be read as a parity law. Once the generator radius $s(E)$ is fixed, the only remaining local ambiguity is whether the admissible shell cycle returns with the same sign or with reversed sign. Under a shell decomposition into regular segments and transverse reflector crossings, this becomes an explicit even/odd criterion. The accompanying Möbius-reflected diagnostic render is not used as a proof of the residual scalar; it is included only to make the stable-kernel / branch-frontier split visible at the level of the current probe geometry.

3.4.9 working hypothesis. The current diagnostic render also motivates a cautious readout hypothesis. Because the plot was not tuned to detector imagery, its visual similarity to some detector-like photon displays should not be dismissed as a mere decorative analogy, but it should likewise not be elevated to an identification claim. Within the standing phase-shift / frame-shift reading, the safer interpretation is that detector images may display readout-relative shadows of the same branch-sensitive residual geometry. This can be tested at the level of tunnel width, opening angle, boundary intensification, and shell-parity response, without assuming that the diagnostic probe already reproduces a full detector event.

3.4.7 working interpretation. The analytic generator now also controls the probe decision itself. On every sufficiently small regular shell, the energy first collapses to a unique generator radius $s(E)$, and the visible branch data then reduces to the transported sign pair $\pm\sqrt{s(E)}$. The direct-convergence versus spinorial/Möbius-lift dichotomy is therefore the $\mathbb{Z}_2$ shell classifier of the same generator family rather than an additional independent mechanism.

3.4.6 working interpretation. The even hierarchy is therefore no longer just a ladder of separate coefficients. Once the transported closure germ is analytic, the visible branch law is packaged by a single local generator $\Phi(\eta^2)$. The quartic shell, the delayed higher timing drifts, and the turning-value corrections then become coordinated readouts of one canonical germ rather than separate nonlinear sectors.

3.4.5 working interpretation. The higher nonlinear hierarchy is therefore already strongly constrained: once the quartic channel disappears, the next visible correction cannot re-enter arbitrarily, but only through the first surviving even closure coefficient. Operationally, the probe remains isochronous through all lower orders and only begins to drift at the exact order at which the closure hierarchy itself reopens.

3.4.4 working interpretation. The quartic correction is therefore not only structurally present; it is already operationally visible. The first failure of exact isochrony on a regular periodic shell is controlled by the same closure coefficient c_4 that generates the quartic branch correction. The probe may therefore read the same admissible shell through energy, through its unique positive turning value, or through its first nonlinear period drift, without opening a new scalar channel.

3.4.3 working interpretation. The nonlinear visible coefficient is therefore no longer free: once the exact branch potential is the transported readout of the same closure geometry, the quartic branch coefficient λ is simply the positive transport of the closure

coefficient c_4. The regular shell can then be indexed either by its energy E or by its unique positive turning value $A(E)$, while the nonlinear strength itself is already fixed by the closure side.

3.4.2 working interpretation. The first concrete nonlinear model is therefore already visible: once the admissible branch closes to a one-field exact potential and no additional scalar opens at first nonlinear order, the restoring branch law falls into the quartic reversible family. The global probe can then read the same admissible shell either as an energy level or as a unique positive turning value, while the residual ambiguity remains exhausted by the transported $\mathbb{Z}_2$ pair.

3.4 working interpretation. The first move toward 3.4 is therefore not a proliferation of new local lemmas, but a globalization principle: once the reduced branch equation is exact, the restoring branch law becomes a one-dimensional potential dynamics, and the reversal probe becomes an energy-component test for direct closure versus spinorial lift.

Condensed 3.3 status. The local 3.3 line is therefore fully captured by four linked claims: a single transported residual scalar η, a Julia-boundary readout of that same scalar, a restoring branch dynamics for the local reversal profile, and a hybrid reversal probe with the two-way outcome "direct convergence or spinorial/Möbius lift". Relative to the longer 3.3.0 working appendix, this is a structure-preserving compression rather than a deletion: the intermediate defect, jet, and probe lemmas are absorbed into the four main statements above.

Appendix D. Historical Release Notes and Version Record

This appendix gathers the immediate Companion version trail and historical release notes so that the mathematical main line can close before the documentation apparatus begins. The live bibliography itself is kept as the terminal back-matter block of the document.

Historical release notes and roadmap blocks

The following materials are retained deliberately as archival proof-memory and release genealogy. They preserve earlier release-stage notes, roadmap controls, and checkpoint explanations, but they are no longer kept in the opening sequence of the public document. Accordingly, the front of the release now moves directly from abstract and contents into the nomenclature, while the historical front-matter materials remain available here for reconstruction, audit, and dependency tracing.

Archived Reader's Guide and working roadmap from the public v3.13.0 release

The following note is preserved from the public v3.13.0 release stage. It records the release-facing reader's guide and roadmap exactly for that checkpoint; in the present release continuation it is archival proof-memory rather than a current status block. It presents the existing mathematical line in one monolithic text, preserves the stabilized diagnostic material already carried by the document, and keeps the operative rule: *do not shorten, do not overwrite, only sharpen, normalize, and preserve.*

Archived version identity for the public v3.13.0 checkpoint

- Historical version: 3.13.0
- Historical date: 2026-04-07
- Concept DOI: 10.5281/zenodo.19210728
- Historical status: public Zenodo release v3.13.0
- Historical public release DOI (v3.13.0): 10.5281/zenodo.19446872
- Historical version-specific DOI for v3.13.0: 10.5281/zenodo.19446872
- Previous Zenodo releases: v3.3.5 (10.5281/zenodo.19396143), v3.4.0 (10.5281/zenodo.19410949), v3.7.1 (10.5281/zenodo.19412478), v3.9.0 (10.5281/zenodo.19423689)
- Main-paper relation: Explicit mathematical companion to the current Sphaera master line [2]

Archived working note from v3.13.0

This public release proceeds from the public v3.11.0 Companion release in one preserved monolithic text. Content-wise it now keeps together three central gains:

(i) the stabilized residual-record interface and its associated structural tools,

(ii) the reproducible Möbius-shell diagnostic suite for the live residual scales, and

(iii) the OP 5 packet, breathing-envelope, and frame-flicker visual audit line.

Accordingly, that archived public release should be read as the stabilized public checkpoint of its then-current monolithic chain: it preserved the mathematical line of that stage while keeping the front cleaner and the live burdens more sharply routed.

Archived primary working goals recorded at v3.13.0

1. Present the current Companion mathematics in one public monolithic release.
2. Keep the stabilized generator–atlas–holonomy core and the residual-record package $\mathfrak{R}(B) = \big(\Phi, \Delta_{\mathrm{diag}}, \varepsilon_{\mathrm{par}}, \chi_{\mathrm{par}}, \widehat{\chi}_{\mathrm{par}}\big)$ intact as the faithful visible interface.
3. Preserve the Möbius-shell diagnostics as reproducible, active-PDF figures so that residual drift, continuation, sensitivity, and sector persistence remain directly auditable.
4. Preserve the OP 5 packet-wave, breathing-envelope, and frame-flicker visual line in one release-safe appendix block.
5. Preserve the monolithic document without shortening historical material.

Archived release priorities recorded at v3.13.0 toward Zenodo deposition

1. Turn the current OP 5 packet/phase/epsilon analysis into one cleaner structural branch theorem with a small admissible branch load and a non-escape criterion.
2. Keep OP 1 on the transported-branch reading and compress the remaining burden to the discrete correction channel without reopening settled continuum structure.
3. Normalize the document language so that internal patch chronology remains archival only, while the main line reads as one coherent proof route.
4. Collect the now-stabilized transport, readout, checkpoint, and epsilon-trap principles as explicit reusable tools for the next theorem blocks.
5. Use the present v3.13.0 public release as the stabilized next checkpoint because the OP transport closure architecture is now integrated more consistently across the monolithic text rather than remaining split between new theorem blocks and unreconciled historical passages.

Continuity of the v3.5.2 checkpoint in the present release

The present file preserves the former *v3.5.2 abstract-polish checkpoint* as compatible earlier material within the current release. Concretely, the 3.5.2 achievements remain part of the present monolith:

(i) the residual-record interface $\mathfrak{R}(B)$ and the reconstruction normal form remain unchanged;

(ii) the readout-relative lemma methodology, structural lift logic, and Theorem 7 / Hypotheses 7.4–7.6 remain preserved as dependencies;

(iii) the former OP 2 operator burden is no longer carried as a separate primary front, but is preserved in reclassified form inside the later OP-first status reading;

(iv) the abstract-polish framing of 3.5.2 is therefore not discarded, but sharpened and transported into the later residual-controlled and boundary-layer language.

Accordingly, the development line may be read as:

v3.5.2 interface normalization $\Longrightarrow$ v3.6 conservative frontend $\Longrightarrow$ v3.7 OP-first stabilization $\Longrightarrow$ v3.8 residual / boundary-layer sharpening.

Archived current mathematical status at the public v3.13.0 release stage

For the current release line, the reader should keep one status separation in mind before entering the preserved historical layers below:

- The stabilized generator–atlas–holonomy sector from v3.4 is already closed at the structural level and has been compressed into the admissible residual record

$$\mathfrak{R}(B) = \big(\Phi, \Delta_{\mathrm{diag}}, \varepsilon_{\mathrm{par}}, \chi_{\mathrm{par}}, \widehat{\chi}_{\mathrm{par}}\big).$$

- The current v3.5.x step is therefore no longer a search for new visible primitives, but a normalization of comparison, transport, and interpretation through that record.
- Historical OP 2 should no longer be read as a separate live operator problem; its T1/T2/T3 blocks are closed, and the former T4 frame-offset part is absorbed into the residual quantitative completion of OP 1.
- Historical OP 3 is closed at the present structural level; it may still be enriched later, but it is no longer one of the primary live fronts.
- Historical OP 5 has been reduced to the photon-frame Step (B) remainder only; Step (A) is already closed in the present text.
- The genuinely live fronts are therefore the residual continuum-to-measurement correction for β_{sym}, controlled emergence with explicit constants and bounds, the quantitative mass hierarchy, the three-generation splitting law, the compact Dixon-algebra reformulation, and the remaining Theta-sum cancellation step.

Release-stage caution. The present reading is strong enough for this release, but it is intentionally stated in stabilized form: the remaining live fronts are read as residual downstream tasks *in the present stabilized reading.* This is stronger than a loose hypothesis, because it rests on the exact representability and conservative-frontend results of Sections 3.5–3.6; but it is still a mathematically revisable reading if a later residual OP were to exhibit a genuinely missing visible degree of freedom. No such counter-indication is currently visible in the Companion chain.

How to read the preserved historical roadmap blocks. The longer roadmap sections that follow from earlier cycles are retained deliberately as proof-memory and dependency control. They should now be read as *historical support strata*: useful for reconstruction, closure auditing, and tool extraction, but no longer identical with the short list of primary live questions at the head of the present line.

What the 3.5.x line achieved, and what 3.6 now fixes first

At this point the 3.5.x line should be read as a *completed interface-normalization stage* rather than as an unfinished search for further visible structure. Its concrete achievements are:

1. the full 3.4 generator–atlas–holonomy package has been compressed into the admissible residual record
$$\mathfrak{R}(B) = \big(\Phi, \Delta_{\mathrm{diag}}, \varepsilon_{\mathrm{par}}, \chi_{\mathrm{par}}, \widehat{\chi}_{\mathrm{par}}\big);$$
2. visible reconstruction has been normalized through the record and the chartwise shell data (s, π);
3. record-level comparison and transport have been fixed through admissible morphisms and residual-record isomorphisms;
4. the faithful interpreter principle has made the stabilized visible sector exactly representable on record level;
5. the historical OP-layer has been reclassified so that closed, reduced, and genuinely open fronts are no longer mixed.

Accordingly, the next step should *not* attempt to reopen the visible interface. A mathematically clean 3.6 target is instead:

(i) prove the conservative-frontend principle explicitly, i.e. every later language layer must factor through the residual-record interpreter and may not introduce any new visible primitive;

(ii) sharpen the remaining live fronts only as downstream quantitative tasks: prime-lattice/continuum correction for β_{sym}, explicit constants and bounds for controlled emergence, mass hierarchy and three-generation splitting, compact Dixon-algebra reformulation, and the unresolved Theta-sum cancellation block;

(iii) prepare the eventual ZFM-/language-layer only after the conservative factorization theorem is in place.

In short: *3.5.x stabilized the interface; 3.6 begins the stabilization of the admissible languages above it.*

Reading discipline for the next stage

Every new subsection introduced below is to be read under four status labels that must be kept distinct:

(a) structurally forced,

(b) projectively read,

(c) interpretive overlay separated from theorem status,

(d) reserved for later proof expansion.

Archived mathematics-first completion path realized in v2.27.0

The present major release is to be read as a *mathematics-first completion cycle.* The computational chapter opened in v2.24.0–v2.26.0 is to remain in place as a visible teaser and reserved architecture, but it is no longer the primary work driver for the immediate next version. For the next cycle, the operative rule is instead:

close the mathematics first; let the proof engine wait for a sufficiently closed mathematical layer.

Accordingly, the completion path realized in v2.27.0 is organized into five mathematical work packages:

WP1: Core structural closure. Complete and normalize the remaining structural chains around ground condition, HC emergence, closure admissibility, and the Millennium matrix as global HC realization.

WP2: Proof-chain hardening. Close missing lemma-to-theorem dependencies, remove duplicated or merge-residual argument fragments, and restore explicit proof-memory where shortened passages still rely on hidden expansion.

WP3: Projector / transport mathematics. Systematize projector logic, carrier-group-readout transport, and the semantic/tensorial kernel interface so that projection steps are mathematically explicit before they are computationally compressed.

WP4: Tools as mathematical appendices. Gather structural tools, computational shortcuts, closure tricks, and trace/branch transfer rules as named mathematical instruments with scope conditions, rather than letting them remain scattered as implicit know-how.

WP5: Computational strand on hold-but-present. Retain MML, MMA, MCP, certificates, and host export as a prepared future chapter, but do not let that layer outrun the mathematical substance it is supposed to compress.

In practical terms, v2.27.0 therefore prioritizes mathematical completion over computational expansion. The computational chapter remains part of the Companion and should be preserved, cited, and gently normalized where needed, but only as a secondary architecture until the mathematical layer is sufficiently closed. The version-specific DOI for the present v2.27.0 release is 10.5281/zenodo.19266666.

Operational mathematical closure agenda

To make the mathematics-first rule operational rather than merely declarative, the next revision cycle is to be executed along the following closure agenda:

MA1: Definition pass. Re-read and normalize the basic definitions that the later chains depend on: collapse state, closure, admissibility, projector sector, compensator action, transport layer, carrier/group/readout distinction, and Millennium-matrix status.

MA2: Dependency pass. Identify theorem blocks whose proofs still rely on compressed or implicit dependencies, then make the lemma chain explicit enough that each forward step can be reconstructed without external memory.

MA3: Transport pass. Isolate the mathematics of projection, frame transport, Lorentz-mediated readout, and disjoint-sector return so that these are expressed first as mathematical maps and only later as computational metaphors.

MA4: Appendix-tools pass. Move recurrent structural tricks into named mathematical tools with hypotheses and scope, so that later proof reuse becomes mathematically transparent rather than conversationally remembered.

MA5: Stability pass. Check the normalized chapter sequence against merge-residual duplication, outdated internal cross-references, and proof steps that were historically preserved but not yet fully re-hardened.

Priority clusters for v2.27.0

The following clusters should receive mathematical priority before any further major proof-engine expansion:

PC1: Ground condition ⇒ HC ⇒ confinement. This chain should be rewritten as a clean mathematical dependency, with the Heisenberg compensator treated as emergent from the ground condition rather than as an independently floating postulate.

PC2: Projector and closure mathematics. The admissible-sector logic should be made fully explicit, including which projectors are primitive, which are derived, and how closure is checked at the mathematical rather than merely intuitive level.

PC3: Millennium matrix as mathematical object. The Millennium matrix should first be stabilized as a global mathematical realization of the HC/closure architecture before any stronger computational reading is allowed to dominate the exposition.

PC4: Trace / branch mathematics. Probe traces, branch transfer, Hopf-transition logic, and return structure should be stated as mathematical transformations with hypotheses and admissibility conditions, not only as algorithmic motifs.

PC5: Proof-memory restoration. Where earlier versions compressed long chains into structural shorthand, the next cycle should restore enough explicit mathematics that the shorthand remains auditable and publication-safe.

What is deliberately deferred beyond v2.27.0

The following lines are explicitly preserved but not treated as primary completion goals for the next release cycle:

D1: full MML language formalization,

D2: full MMA/MCP execution refinement,

D3: standalone proof-kernel extraction,

D4: stronger universality claims,

D5: operating-system style runtime architecture.

These elements remain valuable, but in the current sequencing they are to follow mathematical closure rather than compete with it.

Mathematical work matrix for the next pass

To make the mathematics-first agenda operational, the next pass is organized as a compact work matrix rather than as a loose wish list.

Block	Target state	Main dependencies	Priority
Ground condition → HC → confinement	explicit statement chain, normalized notation, and no ambiguity about emergence vs. axiomatics	core structural assumptions and septinion confinement language	A
Projector and closure mathematics	sharpened lemmas, admissibility conditions, and stable projector notation	closure operator layer, readout discipline, sector decomposition	A
Millennium matrix as mathematical object	operator-level formulation, domain/codomain clarity, and clarified relation to HC	projector layer, transport layer, compensator interpretation	A
Trace / branch / Hopf transport mathematics	clean separation of trace, branching, return, and transport rules	probe framework, disjointness, Lorentz-mediated transport	B
Dependency chain hardening	theorem/lemma dependency map cleaned and checked for merge artifacts	current theorem chain, proposition and lemma references	A
Structural tools appendix	collect reusable mathematical shortcuts without shortening proofs	established proof moves, normalized notation	B
Proof-memory restoration / archival recovery	recover or reintegrate historically displaced proof material where needed	older Companion states, recovered large basis, appendix chronology	B
Computational teaser retention	keep chapter visible, but frozen except for compatibility edits	mathematical closure progress	C

A-priority blocks should drive the next version boundary. B-priority blocks should be developed in service of those A-blocks. C-priority blocks should be touched only when required for compatibility with the mathematical reorganization.

Dependency map of the mathematical core

To prevent the next pass from drifting into local edits without structural order, the mathematical core is also organized as a dependency map. The intended reading is simple: if an upstream block remains unstable, all downstream blocks that consume it can only be provisionally edited.

Core block	Direct prerequisite	Main downstream consumers
Ground condition and collapse picture	none; foundational starting point	HC emergence, confinement thesis, collapse-state language, admissibility reading
HC emergence and confinement chain	ground condition, collapse picture	projector logic, trace/branch admissibility, Millennium-matrix interpretation
Projector / closure mathematics	HC chain, admissibility normalization	branch filtering, Boolean readout, proof templates, certificate layer
Transport / Lorentz layer	collapse picture, projector sector discipline	return-map reading, branch transport, reflector-like structural reversals
Millennium matrix as mathematical object	HC chain, projector/closure layer, transport layer	computational teaser chapter, co-processor view, host-export discipline
Trace / branch mathematics	projector logic, transport layer, admissibility conventions	Hopf branching examples, trace reconciliation, branch templates
Tool and appendix lemmas	all stabilized upstream definitions	proof hardening, auditability, future proof-engine extraction

In operational terms, this means that the next mathematical pass should proceed roughly in the order

$$\begin{aligned}&(\text{ground condition} \Rightarrow \text{HC} \Rightarrow \text{confinement}) \prec (\text{projector/closure})\\ &\prec (\text{transport}) \prec (\text{Millennium matrix, trace, branch})\\ &\prec (\text{appendix tools and supporting lemmas}).\end{aligned}$$

Any local repair that violates this order may still be useful as a note, but it should not be mistaken for structural closure.

Definition-and-notation pass for the next mathematical stage. Before the next major release, the Companion should also undergo a focused definition-and-notation pass. The purpose of this pass is not to shorten the document, but to reduce merge drift, ambiguous symbol reuse, and status confusion between theorem, lemma, tool, hypothesis, and teaser language.

Mathematically fixed for the current line. For the current Companion line, the following should be treated as fixed structural anchors unless explicitly revised in a later release: the collapse-state viewpoint as the local readout picture; the phase-shifted measurement-frame perspective; the ground-condition ⇒ Heisenberg-compensator ⇒ confinement chain; the projector/closure discipline for admissible sectors; the standing tri-

olen architecture with one orthogonal component; and the imaginary convention already adopted for the relevant octonionic/triadic units.

Notation to normalize before v2.27.0. Notation should be normalized especially in the following zones: projector symbols and their admissibility role; closure notation versus stabilization notation; transport/Lorentz notation across layers; Millennium-matrix notation when used as a mathematical object rather than as a computational teaser; trace versus probe-trace notation; branch/Hopf notation; and the distinction between theorem-level mathematical statements and later computational analogies. In particular, the same symbol should not silently alternate between mathematical object, physical readout, and computational metaphor.

Items that must remain hypotheses. Some parts of the Companion should remain explicitly flagged as hypotheses, interpretive proposals, or deferred claims rather than being silently upgraded into theorems. This applies in particular to the Higgs/coupling interpretation, to stronger black-hole or event-horizon extrapolations, to stronger computational-universality language, and to any claim that would identify the computational chapter with a finished proof engine rather than a prepared architecture. The next mathematical pass should therefore favor closure and definition-hardening over conceptual inflation.

Operational rule for the next pass. For the next revision cycle, each touched block should be tagged mentally by the question: *definition, theorem, lemma, tool, hypothesis, or teaser?* The next major mathematical gain is expected to come more from making these statuses explicit and stable across the Companion than from adding further new conceptual branches.

Target-state matrix for v2.27.0

Core block	Current status	Target state for v2.27.0	Minimal completion condition
Ground condition / collapse picture	Structurally central, but still partly distributed across explanatory passages.	Gathered into a visibly controlling mathematical block.	The collapse-state picture is stated once in stable mathematical language and used consistently as the local readout frame.
HC emergence / confinement chain	Present as a guiding structural principle, but not yet fully normalized as a dependency spine.	Explicitly treated as a mathematical chain: ground condition ⇒ HC ⇒ confinement.	The chain appears in one normalized theorem/lemma sequence and is referenced consistently downstream.

Core block	Current status	Target state for v2.27.0	Minimal completion condition
Projector / closure mathematics	Strongly present, but notation and admissibility wording still need tightening.	Treated as the first closed admissibility layer of the framework.	Projector symbols, closure symbols, and admissibility language are normalized and no longer drift between sections.
Transport / Lorentz layer	Conceptually active, but still mixed between mathematical transport and physical interpretation.	Clarified as the transport layer relating frames without overloading its physical extrapolations.	A stable mathematical transport notation is fixed and separated from speculative physical remarks.
Millennium matrix as mathematical object	Partly present beside its computational reading.	Brought back under mathematical control before any stronger proof-engine use.	The matrix is stated first as a mathematical object/operator, with the computational reading clearly subordinate.
Trace / branch mathematics	Available in structural language, but still partly entangled with later computational analogies.	Recast as a mathematical trace/branch layer before proof-engine expansion resumes.	Trace, probe-trace, branch, and Hopf language are normalized and tied to mathematical dependence rather than metaphor.
Tool / appendix lemmas	Growing, but not yet fully separated from main-line dependencies.	Reorganized as supporting tools rather than semi-primary arguments.	Each tool is clearly marked as tool/appendix support and no longer silently carries theorem-level load without declaration.

Use of the matrix. This target-state matrix is intended as an operational filter for the next revision cycle. A block should only be counted as *mathematically advanced* if its current status has moved measurably toward the declared target state and if its minimal completion condition has been checked explicitly.

Proof-dependency checklist for v2.27.0

To keep the next mathematical pass referentially stable, the following dependency checklist should be treated as a minimal proof-control layer. The point is not to produce exhaustive dependency graphs for every sentence, but to ensure that the main mathematical blocks are supported by a sufficiently explicit chain of definitions, lemmas, transport statements, and appendix tools.

- **Ground condition / collapse picture.** Must rest on stable base definitions, a clear distinction between structural state and measurement readout, and an explicit statement of which consequences are structural and which remain interpretive.

- **HC emergence / confinement chain.** Must depend only on the ground condition and the confinement logic, not on later computational analogies. For v2.27.0 the chain ground condition ⇒ HC ⇒ confinement should be readable without hidden appeals to deferred sections.
- **Projector / closure mathematics.** Requires normalized projector notation, at least one closure lemma family, and stable cross-references into the appendix tools where routine projector identities are collected.
- **Transport / Lorentz layer.** Must state exactly which structural transformations are taken as transport rules, which are theorem-level, and which remain hypothesis-level interpretations. Dependencies on projector compatibility should be explicit.
- **Millennium matrix as mathematical object.** Should depend on the already stabilized projector/closure and transport layer, rather than carrying foundational burden by itself. For v2.27.0 the matrix should be mathematically positioned after the base closure apparatus, not before it.
- **Trace / branch mathematics.** Requires a small stable family of supporting lemmas on admissible continuation, branch exhaustion, and trace comparison. The next pass should avoid leaving branch claims supported only by prose.
- **Appendix tools and supporting lemmas.** These should be treated as genuine support infrastructure: enough to stabilize the main line, but not so invasive that the main thread becomes unreadable. The key requirement is reference stability and not merely accumulation.

For the purpose of the next release threshold, a proof block should count as *referentially stable* only if its direct supporting definitions and lemmas can be named without ambiguity, found in the current Companion without version drift, and read in an order compatible with the mathematical dependency map already given above.

Reference-stability pass for v2.27.0

For the mathematics-first pass toward v2.27.0, internal references are themselves treated as part of structural closure. The next public version should not only add arguments, but also stabilize the minimal web of labels, theorem references, and section anchors on which the A-priority mathematical blocks depend.

- **Definition stability.** The core labels for the ground condition, Heisenberg-compensator emergence, confinement statement, projector/closure layer, transport layer, and Millennium-matrix mathematical section should remain fixed under the next hardening pass unless an explicit renumbering note is added.
- **Dependency stability.** Every A-priority theorem or proposition should point only to labels that are themselves already fixed at the level of definition, lemma, theorem, or supporting tool. Soft roadmap anchors should not be used as silent proof dependencies.
- **Appendix-tool stability.** If a proof uses an appendix tool, the tool must be referenced by a stable label and its status must remain visibly marked as tool, lemma, or hypothesis.
- **Hypothesis transparency.** Any block that still depends on a hypothesis label must keep that hypothesis status explicit in the surrounding prose and may not inherit theorem-strength merely by cross-reference.

- **Public-release threshold.** For v2.27.0, a block counts as reference-stable only if its internal labels, backward references, and appendix-tool dependencies survive a clean compilation pass without ad hoc renaming.

Operationally, the next mathematical stage should therefore include a short explicit reference-stability check before public release: (i) labels on A-priority definitions and theorems are frozen, (ii) proof dependencies resolve only to mathematically declared objects, (iii) appendix tools referenced in the main line remain stable and typed, and (iv) theorem-to-hypothesis drift is not introduced by accidental cross-linking.

Block-by-block closure checklist for the next pass

For each A-priority block, the next pass should explicitly separate three statuses: what remains to be closed, what may count as done for v2.27.0, and what may remain open without blocking the release threshold.

- **Ground condition / collapse picture.** Missing: final normalized wording and explicit distinction between structural and observational language. Done for v2.27.0: stable definition and one clear theorem-level consequence. May remain open: broader interpretive extensions.
- **HC emergence / confinement chain.** Missing: explicit derivational chain and at least one supporting lemma that fixes the confinement reading. Done for v2.27.0: readable chain ground condition $\Rightarrow$ HC $\Rightarrow$ confinement. May remain open: stronger physical extrapolations.
- **Projector / closure mathematics.** Missing: normalized notation and a compact closure lemma family. Done for v2.27.0: a stable projector/closure core with appendix support. May remain open: larger auxiliary projector calculus.
- **Transport / Lorentz layer.** Missing: clean split between structural transport statements and hypothesis-level interpretive claims. Done for v2.27.0: one stable transport layer attached to the projector/closure block. May remain open: more speculative extensions.
- **Millennium matrix as mathematical object.** Missing: definitive positioning relative to HC, projector structure, and transport. Done for v2.27.0: mathematically explicit role after closure/transport stabilization. May remain open: the computational reading beyond teaser level.
- **Trace / branch mathematics.** Missing: compact lemma support for admissible continuation, exhaustion, and trace comparison. Done for v2.27.0: one stable branch/trace family sufficient for the main line. May remain open: extended proof-program analogies.
- **Appendix tools / supporting lemmas.** Missing: sharper separation between indispensable support and purely archival material. Done for v2.27.0: tool appendix that stabilizes references without overgrowing the main line. May remain open: secondary tool families.

Explicit subversion discipline

From this point onward, intermediate work states are to be recorded explicitly as local subversions whenever substantial planning, normalization, recovery, or integration work has been performed without yet justifying a new Zenodo release. The intended pattern is:

```
public release v2.27.0 -> local subversion v2.27.1 -> further local subversions
as needed -> next public milestone
```

This rule is not cosmetic. It is meant to prevent silent rollback, silent overwrite, and accidental compression of the project memory inside a single file name.

Concrete chapter scaffold to be unfolded in the next revision cycle

Part I — Structural Core

1. **Introduction**
   - 1.1. Motivation
   - 1.2. Scope of this version
   - 1.3. Relation to the main paper
   - 1.4. Structural strategy: structure first, closure before projection
2. **Structural Preliminaries**
   - 2.1. Basic structural units
   - 2.2. Orthogonality constraint
   - 2.3. Complex and imaginary structural setting
   - 2.4. Frame shift of measurement
   - 2.5. Structural closure intuition
3. **Ground Condition and Emergent Closure**
   - 3.1. Ground condition
   - 3.2. From ground condition to HC
   - 3.3. HC as compensating constraint
   - 3.4. From HC to closure
   - 3.5. Protected invariants
   - 3.6. Proposition block: Ground Condition $\Rightarrow$ HC $\Rightarrow$ Closure
4. **Septinion / Seption Structural Embedding**
   - 4.1. From triadic units to higher structural coupling
   - 4.2. Octonionic and higher assembly logic
   - 4.3. Septinion / seption confinement
   - 4.4. Structural consequences
5. **Structural Master Equation — First Layer**
   - 5.1. Why a master equation emerges
   - 5.2. Pure structural form
   - 5.3. Term-by-term structural origin
   - 5.4. Structural vs. projective terms
   - 5.5. First consistency remarks

Part II — Probe Sector

6. **Probe Geometry**

    6.1. Two probes moving toward each other
    6.2. Each probe as a triolic structure
    6.3. Orthogonal third group
    6.4. Definition of the search space

7. **Probe Dynamics and Field Structure**

    7.1. Structural interaction regime
    7.2. Closure in the probe sector
    7.3. Probe sector equation
    7.4. Search, trace, and structural read-off

8. **Probe Tools and Guided Search**

    8.1. Frequency and pattern detection
    8.2. Routing and candidate compression
    8.3. Transition guidance
    8.4. Convergence tools
    8.5. Optional statistical layer
    8.6. Interpretation as structural toolkit

Part III — Technical Normalization

9. **Structural Tools and Computational Shortcuts**

    9.1. Why tool normalization is needed
    9.2. Closure-tools
    9.3. Frame-shift-tools
    9.4. Projection-tools
    9.5. Probe-tools
    9.6. Equivalence-tools
    9.7. Read-off / trace-tools
    9.8. Standard form for every tool

Part IV — Interpretive Bridge

10. **Projection Logic and Interpretive Discipline**

    10.1. From structure to projection
    10.2. What is forced, what is read, what is hypothesized
    10.3. Dirac / QFT / Einstein-type readings
    10.4. Higgs as coupling instance or mean coupling point

Part V — Consolidation

11. **Consistency Layer**
    - 11.1. Internal structural consistency
    - 11.2. Probe/main compatibility
    - 11.3. Closure robustness
    - 11.4. Remaining soft spots
12. **Outlook to v1.2.0**
    - 12.1. What comes next
    - 12.2. Planned lemma expansion
    - 12.3. Publication path

Appendix scaffold

A. Symbol table

B. Structural tools catalog

C. Technical lemmas

D. Computational shortcuts

E. Projection tables

F. Probe sector schematics

Minimum formal architecture reserved for v3.35.0

The next expansion pass should lock at least the following formal blocks:

- Proposition: Ground Condition implies emergence of HC.
- Proposition: HC implies structural closure.
- Proposition: Closure confines the admissible dynamics to the septinion/seption regime.
- Proposition: The probe sector inherits the same closure discipline under orthogonal coupling.
- Proposition: The first-layer master equation is structurally forced by the closure architecture.
- Auxiliary lemmata: orthogonality, frame shift, confinement, probe compatibility, read-off consistency.

Practical build order

For the actual writing sequence of the next cycle:

1. Ground Condition / HC / Closure
2. Probe sector
3. Master equation (first layer)

4. Tool normalization
5. Interpretive bridge
6. Consistency layer
7. Appendix fill-in

v2.27.5 Proof-Chain Roadmap: Identified Gaps Closure Targets

Status of this section. Version 2.27.4 completed a full structural scan of the companion. This section records the identified proof-chain gaps in order of priority, and defines the closure targets for the next local subversion sequence. It is a working roadmap, not a release statement. The DOI and version references inside this preserved v2.27.x roadmap belong to the original historical checkpoint and are retained only as archival proof-memory. They are not current release-status statements for the present Companion line.

Proof-chain inventory (v2.27.4 scan)

The structural scan of v2.27.4 identified the following main proof chain:

$$\text{Axiom} \Rightarrow \underbrace{\text{Semantic Kernel}}_{\text{def}} \Rightarrow \underbrace{\text{HC Forcing}}_{\text{thm}} \Rightarrow \underbrace{\text{Closure Stable}}_{\text{lem}}$$

$$\Rightarrow \underbrace{\text{Diagonal Traversal}}_{\text{lem}} \Rightarrow \underbrace{C = \Pi_{\text{spec}}^{(\mathcal{M})}}_{\text{thm}} \Rightarrow \underbrace{\text{SM Realization}}_{\text{cor}}$$

The following table rates each link on four axes: *proof type* (Full / Short / Sketch / Structural), *explicit refs* (references cited in proof body), *dependency gap* (missing intermediate step), and *priority* for the next closure pass.

Chain link	Type	Refs	Gap / note	Prio
`def:semantic-structural-kernel`	Short (9L)	0	Definitions; no gap.	–
`prop:semantic-forcing-chain`	Short	0	Structural; no proof.	B
`thm:hc-forcing-theorem`	Short (14L)	2	Cites `lem:re-identification-lemma`; sound.	B
`lem:closure-forcing-lemma`	Short (8L)	0	**No refs; pure prose; not auditable.**	**A**
`prop:tensorial-intertwining-m`	Short (8L)	0	**Necessity argument only; effectively a hypothesis.**	**A**
`lem:diagonal-traversal-scalar-action`	Full (24L)	3	Well-founded; Jordan-block argument solid.	–
`thm:hc-spectral-projector`	Full (30L)	5	Depends on `prop:tensorial-intertwining-m` hypothesis.	B
`cor:M-HC-fixpoint`	Short	3	Follows cleanly from theorem.	–
`cor:SM-spectral-realization`	Short	4	Depends on `thm:SM`; sound.	–
`thm:koide-triolic` (OP 3)	Short (7L)	0	**θ_K formula not derived; $K = 2/3$ by C–S only.**	**A**
`thm:bsym-hopf` (OP 1)	Short (11L)	0	**5.1% gap to $1/\sqrt{3}$ unaddressed.**	**A**

Continued on next page

Chain link	Type	Refs	Gap / note	Prio
`thm:unity` (central)	Short (9L)	5	`prop:compensator` ref resolved; label confirmed.	B
`prop:structural-lift-principle`	Short (8L)	1	Circular: uses its own assumptions as proof.	B
ClosureStable $\Leftrightarrow$ im(C)	*missing*	–	**Bijection not stated; gap in thm:hc-spectral-projector Step 3.**	**A**

Priority A: Required for structural soundness

A1. `lem:closure-forcing-lemma` hardening. The proof currently reads as structural prose with no citations. The required explicit chain is:

$$\begin{aligned} \text{Triolic}(x) &\Rightarrow \texttt{lem:triolic-minimality-lemma}, \\ \text{Anchored}(x) &\Rightarrow \texttt{lem:orthogonal-anchor-lemma}, \\ \text{HCForced}(x) &\Rightarrow \texttt{lem:hc-admissibility-lemma}. \end{aligned}$$

Combined consequence: no leakage outside the common container $\Rightarrow$ ClosureStable(x). Each implication must cite its lemma explicitly.

A2. `prop:tensorial-intertwining-m` status correction. $\mathcal{MC} = \mathcal{CM}$ is currently stated as a proposition but proved only by necessity argument. It must either be: (a) promoted to an axiom/hypothesis with explicit hypothesis-grade marker, or (b) derived from `def:tensor-structural-kernel` by explicit index calculation showing that the idempotent tensor $\mathcal{C}^\mu{}_\nu$ commutes with any structure-preserving $\mathcal{M}^\mu{}_\nu$ satisfying the admissibility conditions. Option (b) is preferred.

A3. ClosureStable$(x) \Leftrightarrow x \in$ im(C) as explicit lemma. Step 3 of `thm:hc-spectral-projector` uses `lem:closure-forcing-lemma` to identify im(C) with $\mathcal{S}_0$, but the reverse implication is still missing. The required gap may be summarized as:

- im$(C) \subseteq \mathcal{S}_0$ is already covered,
- closure-stable states must still be shown to be compensator-normalized,
- hence a dedicated closure-stable/im(C) lemma is still needed.

A4. `thm:koide-triolic` (OP 3) numerical hardening. The existing proof derives $K = 2/3$ via Cauchy-Schwarz but does not derive the angle formula $\theta_K = \pi - \sqrt{2} \cdot \arcsin(\beta_{\text{sym}})$ established in prior session work. This formula predicts the Koide angle to within 0.115% and must be incorporated as an explicit step. The residual mass-ratio discrepancy ($\sim$ 7–16%, amplification factor ~ 60 near the electron mass) must also be stated explicitly.

A5. `thm:bsym-hopf` (OP 1) gap statement. The Hopf-angle derivation gives $\beta_{\text{sym}} \approx 1/\sqrt{3}$ (continuum limit) while the measured value is ≈ 0.5493, a 5.1% gap in β_{sym} corresponding to a $\sim 61\%$ gap in γ_L due to nonlinear amplification. The theorem currently does not state this gap explicitly. A remark recording the gap and its structural interpretation (continuum S^3 vs discrete prime lattice) must be added.

Priority B: Structural hardening (next passes)

B1. `prop:semantic-forcing-chain` proof. Currently stated without proof body. The implication GroundLinked $\wedge$ FrameShifted $\Rightarrow$ HCForced needs an explicit derivation from the semantic kernel axioms (S1)–(S6).

B2. `prop:structural-lift-principle` circularity fix. The proof assumes its conclusion. A non-circular argument requires explicit reference to the admissible transport operations (for example, `lem:group-transport` and `lem:hc-admissibility-lemma`) as the mechanism by which kernel-invariant content is preserved.

B3. `thm:hc-spectral-projector` conditional status. The theorem is sound *under* the hypothesis that $\mathcal{MC} = \mathcal{CM}$ (A2). Until A2 is resolved, the theorem should carry an explicit conditional marker. A remark to this effect should be inserted immediately after the theorem statement.

Closure sequence for next subversions

v2.27.6: Close A1 (`lem:closure-forcing-lemma`), A3 (ClosureStable $\Leftrightarrow$ im(C) lemma), and B3 (conditional marker on `thm:hc-spectral-projector`).

v2.27.7: Close A2 (`prop:tensorial-intertwining-m` index calculation or hypothesis-promotion) and B2 (structural lift circularity fix).

v2.27.8: Close A4 (`thm:koide-triolic` θ_K formula, OP 3) and A5 (`thm:bsym-hopf` gap remark, OP 1).

v2.27.9: Close B1 (`prop:semantic-forcing-chain` proof). Evaluate gate readiness for next public release.

What is already sound (do not reopen)

The following blocks passed the v2.27.4 scan with no structural issues and should not be reopened unless a genuine contradiction is found:

- `lem:diagonal-traversal-scalar-action` (Jordan-block exclusion),
- `thm:fractal` (fractal self-traversal),
- `thm:symmetric-frame` (algebraically complete),
- `thm:unitary-equiv` (Stone–von Neumann, cites literature),
- `cor:M-HC-fixpoint`, `cor:SM-spectral-realization`,
- all of `sec:readout-relative-lemmas`,
- `thm:hc-readout-geometry` (conditional on A2).

v1.2.0 Consolidation Core

Normalization note. This block is retained as a *historical consolidation layer*. In the normalized grouped architecture of Version 3.35.0, its active generative content is absorbed into the normative layer $\mathfrak{G}$; the present section therefore records the earlier consolidation surface

and should no longer be read as an independent foundational core competing with the later normalized structural kernel.

This version closes the gap between the previously inserted roadmap and the already existing structural sections of the monolith. Its purpose is to state, in one contiguous place, the working discipline of the next release, the core closure chain, the probe-sector architecture, the first-layer master-equation logic, and the normalized tool catalogue. The guiding rule remains unchanged: *no shortening, no silent merging, no replacement of the leading line; only sharpening, explicit classification, and append-only completion.*

1. Structural reading discipline

Every major claim used below is assigned one of four statuses.

(a) **Structurally forced:** the statement follows from the ground condition, the triolic organization, orthogonality, and closure.

(b) **Projectively read:** the statement is not primary, but appears after passage to a chosen observational or effective sector.

(c) **Heuristically interpreted:** the structural content is stable, while its physical naming is still a disciplined reading rather than a final derivation.

(d) **Open:** the location of the claim is fixed, but its expanded proof or explicit computation is postponed to the next lemma layer.

This fourfold discipline is essential because the framework is intended to be a *structure-first* proof architecture. In particular, equations are not to be introduced as autonomous starting axioms when they can instead be read as forced compression formulas of a deeper closure mechanism.

2. Ground condition, HC, and closure

The working core of v1.2.0 is the explicit chain

$$\text{Ground Condition} \Longrightarrow \text{HC} \Longrightarrow \text{Closure}.$$

Here the ground condition is the requirement that the underlying triolic body is not free to unfold into arbitrary independent sectors. Rather, the threefold organization is constrained so that one component remains orthogonal to the other two, the admissible units are treated in the standing imaginary convention, and every physical readout is already phase-shifted with respect to the underlying structural frame.

Under this condition, the Heisenberg compensator is not an external correction patched in after the fact. It is the minimal balancing mechanism that becomes necessary once a common body must be represented consistently in dual or phase-shifted ways without allowing the structure to drift apart. In the logic of the present paper, HC therefore carries three simultaneous tasks: it re-identifies dual or shifted representatives, it suppresses unstable escape modes, and it transports the system toward a light/collapse-compatible output.

Closure is then the global consequence of this balancing action. Once the compensator has become active, admissible dynamics are no longer free to leave the confined regime; instead they remain trapped within the higher coupled architecture represented in the paper by the septinion/seption perspective. This is the sense in which HC is emergent, while closure is enforced.

Proposition A (ground condition induces HC). Given the triolic structural core, the orthogonality constraint, and the phase-shifted measurement principle, a compensating identification mechanism is forced whenever the same body is to admit more than one consistent appearance. That mechanism is the Heisenberg compensator.

Proof sketch. Without compensation, dual or shifted presentations of the same body would produce uncontrolled representation drift. Because the framework demands one common object across all sectors, a balancing map is required. This balancing map is precisely the HC in structural form.

Proposition B (HC induces closure). Whenever HC acts as the balancing constraint of the triolic system, the admissible dynamics are confined to a closure-stable regime. In particular, sectors may transform, couple, or be read out differently, but they cannot escape the common structural container without violating the ground condition.

Proof sketch. HC identifies and neutralizes precisely those deviations that would split the common body into incompatible local descriptions. The remaining degrees of freedom therefore assemble into a closed class. This class is the closure regime read throughout the monolith.

Protected invariants. At the level of v1.2.0, the most important protected invariants are: (i) the persistence of one common underlying body, (ii) orthogonality of the third triolic direction relative to the first two, (iii) frame-shift discipline between structure and observer description, and (iv) confinement of admissible dynamics to the closure-compatible sector. These invariants are not yet the full spectrum-level output, but they are the stable backbone needed for all later projections.

2A. Consolidated dependency spine for v2.27.0

For the next mathematical pass, the previous informal chain

$$extGroundCondition \Longrightarrow extHC \Longrightarrow extClosure \Longrightarrow extConfinement$$

should be treated as a single dependency spine rather than as four loosely adjacent slogans. The point of the present local pass is not to over-claim a finished theorem package, but to normalize what the monolith already uses as its operative order of dependence.

Lemma C (dependency spine lemma). Assume the triolic core, the standing orthogonality condition, and the phase-shifted readout discipline. Then every admissible multi-representative reading of one and the same body requires a compensating identification map; this yields HC. Once HC is active, admissible state evolution is restricted to those sectors that remain compatible with the common body; this yields closure. Whenever closure is read at the level of the higher coupled container, the resulting admissible class is a confinement class.

Proof sketch. The first implication is the content of Proposition A: without compensation, multiple shifted or dual descriptions of one body drift apart. The second is Proposition B: once HC acts as balancing constraint, the admissible dynamics can no longer leave the common structural container. The last implication is a change of reading level rather than a new mechanism: closure in the higher coupled setting is exactly the confinement statement used downstream in the septinion/seption language.

Operational reading. The practical value of this dependency spine is that later sections may appeal to one normalized arrow instead of reintroducing the same chain piecemeal. In particular, the later projector, transport, and Millennium-matrix sections should be read as refinements of this spine, not as independent foundations. This is also the correct mathematics-first reading for that archived public release: first the spine is stabilized, then the surrounding operator language is hardened.

Local closure target. For the purposes of the v2.26.x local sequence, this block counts as improved only if all later uses of HC emergence, closure, and confinement can be traced back to this normalized dependency spine without silent appeals to the computational teaser architecture.

2B. Projector/closure anchoring of the dependency spine

The next hardening step is to make explicit that the closure arrow in the normalized spine is not merely verbal. It is read downstream through projector language. In other words, the chain

$$\text{Ground Condition} \Longrightarrow \text{HC} \Longrightarrow \text{Closure}$$

should already be understood as carrying an operator-level consequence: once HC has been introduced as the compensating identification map, admissible readings must be projectively stabilized inside the common structural sector.

Lemma D (projector anchoring lemma). Assume the dependency spine of Lemma C. Then every admissible downstream use of closure may be rewritten as a projector-compatible statement: there exists a sector projector P_{adm} such that the HC-balanced reading is admissible only through the image of P_{adm}, while noncompatible continuations lie outside this image.

Proof sketch. If closure is the restriction to the common structural container, then a projector description is simply the operator-level reading of that restriction. The role of P_{adm} is not to invent a new mechanism, but to express in normalized language that the HC-balanced state is retained only in the closure-compatible sector. Thus projector logic is not parallel to closure logic; it is the stabilized formal reading of closure once the argument moves into the operator layer.

Downstream consequence. This anchoring is what allows the later projector/closure and transport sections to be treated as genuine mathematical refinements of the dependency spine. Ground Condition $\Rightarrow$ HC $\Rightarrow$ Closure no longer terminates in a slogan, but feeds directly into the admissible projector sector on which the later transport and Millennium-matrix mathematics are allowed to act.

Local closure target. For the v2.26.x sequence, this block counts as improved only if later projector uses can be traced back to the dependency-spine-plus-projector reading, rather than being invoked as an independent auxiliary formalism.

2C. Transport/Lorentz continuation of the dependency spine

The next hardening step is to show that the normalized spine does not stop at closure and projector admissibility. Once the admissible sector has been fixed, the later transport layer must be read as a constrained continuation of the same chain. In particular, Lorentz-transport language is not allowed to appear as an independent dynamical decoration; it must preserve the HC-balanced, projector-stabilized structural identity already enforced upstream.

Lemma E (transport continuation lemma). Assume the dependency spine of Lemma C together with the projector anchoring of Lemma D. Then every admissible transport between downstream readout frames must preserve the HC-balanced closure sector. Equivalently, Lorentz-type transport is admissible only as a continuation on the image of the stabilized projector, not as a map into an unconstrained external sector.

Proof sketch. By Lemma D, admissibility is already restricted to the projector-compatible closure image. A transport map acting on downstream representatives is therefore allowed only if it respects that image. Otherwise the map would reintroduce a break between closure logic and readout logic, contradicting the role of HC as the compensating identification mechanism. Hence transport does not weaken the spine; it inherits and preserves it.

Downstream consequence. This continuation is what permits the later Lorentz layer to be read in a mathematics-first way: not as a second origin of structure, but as the frame-translation discipline internal to the already stabilized closure sector. It also prepares the next pass in which the Millennium matrix is treated as a mathematical object acting only on admissibly transported states.

Local closure target. For the v2.26.x sequence, this block counts as improved only if later transport claims can be referred back to the dependency spine through the combined chain Ground Condition ⇒ HC ⇒ Closure ⇒ Projector Stabilization ⇒ Admissible Transport.

2D. Millennium-matrix realization of the dependency spine

The next closure step is to prevent the Millennium matrix from floating as a separate computational or architectural metaphor. If the mathematics-first sequence is to hold, then the matrix must be read downstream from the already stabilized chain of Ground Condition, HC emergence, closure, projector stabilization, and admissible transport. In that sense the Millennium matrix is not an independent origin of admissibility; it is the global matrix-level realization of the closure-preserving compensator architecture already fixed by the earlier lemmas.

Lemma F (Millennium-matrix realization lemma). Assume the dependency spine of Lemma C, the projector anchoring of Lemma D, and the transport continuation of Lemma E. Then the Millennium matrix may be introduced as a mathematically admissible operator only on the HC-balanced, closure-compatible, projector-stabilized, and transport-preserving state space. Equivalently, its admissible role is not to generate a new unconstrained sector, but to realize at matrix level the already normalized confinement-preserving structural action.

Proof sketch. By Lemma C the structural dependence runs from ground condition to HC and from HC to closure. By Lemma D this closure is stabilized as an admissible projector image, and by Lemma E every downstream transport must preserve that image. Hence a matrix-level realization can be admitted only if it acts on that same stabilized transported image. If it acted outside this chain, the Millennium matrix would become an additional primitive rather than a realization of the existing architecture, breaking the mathematics-first normalization. Therefore its mathematically acceptable status is that of a global operator reading of the already enforced confinement structure.

Proposition I (admissible matrix-sector invariance). Let $\mathcal{M}$ denote the Millennium matrix restricted to the HC-balanced, closure-compatible, projector-stabilized, and transport-

preserving sector identified by Lemmas C–F. Then $\mathcal{M}$ preserves that admissible sector; in particular,

$$\mathcal{M}\,P_{\mathrm{adm}} = P_{\mathrm{adm}}\,\mathcal{M} \qquad extonthestabilizedtransportedimage.$$

Hence the Millennium-matrix realization is mathematically acceptable only as a sector-preserving global operator, not as a generator of a fresh unconstrained layer.

Proof sketch. By Lemma F the admissible introduction of $\mathcal{M}$ is already downstream from closure, projector stabilization, and admissible transport. Proposition H states that the transport layer commutes with the admissibility projector on the stabilized sector. Therefore the only mathematically consistent matrix-level realization is one that acts within that same sector and preserves its image. If $\mathcal{M}$ failed to preserve P_{adm}, the matrix would reopen a non-admissible branch and would no longer realize the normalized confinement architecture. The commutation relation above is thus the matrix-level form of admissible sector preservation.

Mathematical significance. This proposition upgrades the matrix discussion from a merely allowed operator reading to a first invariance statement. The Millennium matrix is not only introduced downstream from the dependency spine; it is constrained to preserve the same admissible image that closure, projector stabilization, and transport have already fixed. This gives the matrix chapter a clearer mathematical status inside the normalization program.

Downstream consequence. This is the step that allows the later Millennium-matrix chapter to be retained without overtaking the mathematics. The matrix can now be cited as a downstream operator realization of the dependency spine, rather than as a free-standing meta-object. It also prepares the next A-block pass, where trace and branch mathematics can be read as acting under the matrix-level realization of the same closure discipline.

Local closure target. For the v2.26.x sequence, this block counts as improved only if later uses of the Millennium matrix can be referred back to the combined chain Ground Condition $\Rightarrow$ HC $\Rightarrow$ Closure $\Rightarrow$ Projector Stabilization $\Rightarrow$ Admissible Transport $\Rightarrow$ Millennium-Matrix Realization.

2E. Trace/branch continuation of the dependency spine

The final step in this first A-block pass is to prevent trace and branch mathematics from reopening the stabilized chain as if branching introduced a new primitive source of admissibility. In the mathematics-first sequence, trace formation and branch resolution must be read downstream from the already fixed matrix-level realization. Thus branching is not a separate beginning of structure; it is the controlled readout of differentiated admissible paths inside the same confinement-preserving spine.

Lemma G (trace/branch continuation lemma). Assume the dependency spine of Lemma C, the projector anchoring of Lemma D, the transport continuation of Lemma E, and the Millennium-matrix realization of Lemma F. Then trace formation and branch resolution are mathematically admissible only as downstream differentiations of the already HC-balanced, closure-compatible, projector-stabilized, transport-preserving, and matrix-realized state space. Equivalently, admissible trace/branch mathematics does not create a fresh unconstrained dynamics; it resolves path structure inside the same confinement-preserving architecture.

Proof sketch. By Lemma F the Millennium matrix is admitted only as an operator realization of the already stabilized dependency spine. Therefore any trace extracted from the resulting evolution must be a trace of that same admissible operator action, not a record of an independent unnormalized sector. Likewise, branch resolution can be accepted only if it separates compatible downstream continuations of the same confined structure. If branching were allowed to act as a new primitive, then the chain from ground condition through HC, closure, projector stabilization, transport, and matrix realization would no longer control the later readout layer. Hence trace and branch mathematics remain admissible only as terminal differentiations internal to the same spine.

Downstream consequence. This closes the first mathematics-first pass across the initial A-block group. The trace/branch layer can now be cited without drifting into an autonomous search or computational metaphor: it is anchored in the same structural chain that begins at the ground condition and ends in admissible differentiated readout. This is the point at which the first local freeze judgement for the A-block group becomes meaningful.

Local closure target. For the v2.26.x sequence, this block counts as improved only if later trace and branch statements can be referred back to the combined chain Ground Condition ⇒ HC ⇒ Closure ⇒ Projector Stabilization ⇒ Admissible Transport ⇒ Millennium-Matrix Realization ⇒ Trace/Branch Continuation.

Proposition J (admissible trace/branch persistence). Let Γ_{adm} denote an admissible trace family generated inside the stabilized admissible sector, and let B_{adm} denote a branch selector that only separates continuations already induced by the same admissible matrix-sector action. Then admissible trace continuation and admissible branch separation preserve the same normalized admissible sector; in particular they do not generate a new independent admissibility source outside the projector/transport/matrix spine.

$$B_{\mathrm{adm}}\,\Gamma_{\mathrm{adm}} \subseteq P_{\mathrm{adm}}\mathcal{H}_{\mathrm{conf}}.$$

Proof sketch. By Proposition H, admissible transport acts inside the stabilized projector image, and by Proposition I the Millennium-matrix realization preserves that same admissible sector. Hence every admissible trace produced downstream is already a differentiated record of motion internal to the projector-stabilized, transport-compatible, matrix-preserving sector. If a branch operation is admissible at all, it cannot introduce a fresh primitive sector; it can only separate compatible continuations already contained in that same image. Therefore both trace continuation and admissible branching persist inside the normalized admissible sector.

Mathematical significance. This upgrades the final step of the first A-block from a narrative endpoint to a persistence statement: the trace/branch layer is no longer merely attached downstream, but mathematically constrained not to reopen an external admissibility source. In that sense the first A-block now carries a full preservation chain from the ground condition through HC, closure, projector stabilization, transport, matrix realization, and finally trace/branch continuation.

Local closure target. For the v2.26.x sequence, this block counts as strengthened only if later trace/branch arguments can explicitly cite Proposition J as the persistence rule preventing downstream drift outside the admissible spine.

2F. Local freeze check for the projector/transport core

The first A-block is now strong enough that its middle core — projector, closure, and transport — can be evaluated on its own rather than only as a passing stage inside the longer dependency spine. This local freeze check asks whether that middle layer has reached a condition in which later work may treat it as stable for the next release cycle without pretending that every possible auxiliary lemma has already been unfolded to maximal density.

Local freeze status. At the present stage, the projector/closure plus transport/Lorentz core is best read as *locally freeze-ready for v2.27.0*, provided that no later pass re-opens the admissible image as an unconstrained external sector. The closure logic now fixes the admissible image; the projector logic stabilizes that image; and the transport continuation restricts downstream frame changes to that already fixed sector. In that sense, the central middle layer of the first A-block is no longer merely planned architecture: it has acquired a coherent mathematical dependency order.

Residual blocker. What still limits the block is not the absence of an admissibility spine, but rather the remaining density gap between the local lemmas and a fully expanded proof presentation. In other words, the structure is already in place, but the companion could still benefit from additional explicit micro-lemmas later if a future release wants a denser audit trail.

Release consequence. For the purposes of the mathematics-first roadmap, the projector/-transport core may therefore count as *closed for now at medium proof density*. This is enough to support the next local passes and to contribute positively to the v2.27.0 gate, but not yet enough to justify claiming that every supporting auxiliary statement has already reached final archival density.

Local closure target. The block counts as satisfactorily frozen for the next public release if later passes preserve the chain Ground Condition ⇒ HC ⇒ Closure ⇒ Projector Stabilization ⇒ Admissible Transport without reintroducing a competing admissibility source outside that normalized middle layer.

2G. Consolidated freeze and gate pass for the first A-block

The first A-block has now been hardened not only by the dependency-spine lemmas, but also by the three preservation propositions on transport, matrix action, and trace/branch continuation. It is therefore possible to perform a consolidated freeze and gate pass that asks whether the block, taken as a whole, has reached a level suitable for counting positively toward the v2.27.0 release threshold.

Consolidated freeze status. Taken together, Lemma C through Lemma G and Proposition H through Proposition J justify reading the first A-block as *structurally freeze-ready for v2.27.0 at medium proof density*. The block now carries a continuous admissible spine from ground condition through HC, closure, projector stabilization, transport, Millennium-matrix realization, and admissible trace/branch persistence. No later local pass should need to reopen the block at the level of its basic dependency order unless a genuine contradiction is found.

Gate consequence. For the global v2.27.0 gate, this means that the first A-block can now count as a *positive readiness contributor*. It does not by itself open the public release, because the document-wide gate still depends on the remaining A-blocks, reference stability, and hypothesis-control discipline. But it does remove the first major blocker from the mathematics-first release logic.

Residual blocker. The remaining limitation is not structural incompleteness, but proof density and audit granularity. Later archival passes may still unfold more micro-lemmas, but this is no longer the same as saying that the block is mathematically open. The correct current reading is: *closed in structure, not maximally dense in presentation.*

Local closure target. For the v2.26.x sequence, this pass counts as successful only if later global readiness summaries can cite the first A-block as freeze-ready without again collapsing its seven-step admissible spine into vague architectural language.

3. Higher embedding and confinement logic

The local triolic organization is not the terminal description. Its load-bearing role is to provide the minimal unit from which higher coupling becomes possible. The octonionic step is useful because it already carries nontrivial coupled imaginary directions, but within the present project logic it is not yet the final confinement container. The septinion/seption reading is introduced precisely to express that the entire compensated dynamics lives inside a more rigid higher assembly in which uncontrolled sector break-out is blocked.

This point should be read carefully: the higher embedding is not decorative algebraic escalation. It is the formal language by which the paper states that closure is a global architectural fact rather than a local numerical accident. In this sense, the passage

$$\text{local triolic unit} \to \text{compensated coupling} \to \text{higher confinement}$$

is one of the main structural arrows of the version.

4. First-layer master-equation logic

The monolith already contains the structural core section and the ToE-style master-equation section. v1.2.0 now fixes how these are to be read. The first-layer master equation is not yet the maximal explicit final law of the framework. It is the minimal compressed form that must simultaneously carry geometry, triolic sector dynamics, compensator action, closure residue, and observer-frame phase shift.

Accordingly, the five structural blocks used later in the paper should be classified as follows.

1. The geometric block is *projectively privileged*: it is forced as a large-scale closure reading, but not as the primitive starting point.
2. The triolic block is *structurally forced*: without it there is no load-bearing internal organization.
3. The HC block is *structurally forced*: without it, no stable identification of dual/shifted representatives exists.
4. The closure block is *structurally forced*: without it, the theory would have no admissibility criterion.
5. The observer-frame block is *projectively forced*: once the measurement frame is shifted, a readout term is unavoidable.

The upshot is that the master equation should always be read from left to right as a compression chain,

$$\text{structure} \to \text{compensation} \to \text{closure} \to \text{projection}.$$

Any later physical naming — Einstein-like, Dirac-like, or QFT-like — comes only after this structural order has been respected.

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Proposition C (first-layer master form is forced). Once triolic organization, HC, closure, and frame shift are all imposed, the dynamics can no longer be represented by a purely geometric or purely sectorial equation. A combined master form is forced at first layer.

Proof sketch. Each omitted block destroys one already fixed architectural demand: omitting triolicity erases the internal organization; omitting HC erases compensation; omitting closure erases admissibility; omitting the observer term erases the readout discipline. Hence only a combined structural form remains compatible.

5. Probe sector as second access path

The probe sector is not a side remark but the second major access path of v1.2.0. Two probes are sent toward one another, and each probe itself is to be treated as a triolic structure. The third group stands orthogonal to both probes and thereby defines a constrained search region rather than an arbitrary ambient space. This orthogonal third group is crucial because it prevents the probe architecture from degenerating into a mere two-body scan.

The role of the probe picture is twofold. First, it gives a structural way to read interaction and search without abandoning the closure discipline. Second, it provides an explicit place where the framework's search tools can be attached. In the present version, the probe geometry is therefore a controlled laboratory for structural read-off.

Proposition D (probe sector inherits closure). If each probe carries triolic organization and the third group remains orthogonal to both probes, then the same HC/closure discipline that governs the main sector also governs the admissible probe dynamics.

Proof sketch. The probe architecture does not introduce a new ontology. It duplicates the same structural ingredients in a constrained interaction geometry. Since the closure mechanism depends on these ingredients rather than on the label "background" or "probe", the probe sector inherits the same admissibility logic.

Search space. The search space is therefore not an unrestricted continuum of candidates. It is the region left after imposing triolic self-organization of each probe, orthogonality of the third group, compensator compatibility, and closure. Everything outside that region is structurally irrelevant for the probe readout.

6. Normalized tool apparatus

A central goal of v1.2.0 is to stop treating rechen-tricks as scattered intuition and instead store them as a reusable apparatus. The following classes are fixed for this version.

Closure-tools.
Tools that test, restore, or compare closure-compatible configurations. Their purpose is to decide admissibility and leakage.

Frame-shift-tools.
Tools that translate between the underlying structure and the observer's projected readout.

Projection-tools.
Tools that extract effective sector equations or recognizable physical forms from the common structural body.

Probe-tools.
Tools that organize the constrained search in the probe sector.

Equivalence-tools.
Tools that identify when two differently written states belong to the same structural class.

Read-off / trace-tools.
Tools that recover results from the structure without expanding the entire calculation tree every time.

Every individual tool should be documented under the same six labels: *name, purpose, prerequisite, short formal scheme, typical use, boundary of validity.* This normalization rule is itself one of the deliverables of v1.2.0.

Canonical examples in the present project. Kasisky detection is stored as a frequency/pattern detector rather than as a cryptographic side story. Hash and rainbow methods are stored as routing and candidate-compression devices. Markov organization is stored as a transition controller over admissible search paths. Newton is stored as a convergence accelerator once a structural branch has already been selected. GARCH, where used at all, remains optional and secondary.

7. Projection discipline and physics reading

The structural reading of matter, gauge, mass, and Higgs later in the monolith is to be understood under a strict hierarchy.

1. Matter as closure-stable bound sector is a structural reading.
2. Gauge symmetry as invariance of closure-compatible transport is a projective reading.
3. Mass as closure/frame residue is a projective reading with strong structural motivation.
4. Higgs as coupling instance or mean coupling point is currently a hypothesis block and must remain explicitly marked as such.

This hierarchy matters because it prevents later criticism from collapsing the whole framework into one category. The paper does not claim that every named physical object is already fully derived at the same level. It claims instead that the structural origin is common and that different physical sectors appear at different distances from that origin.

8. Consistency status of v1.2.0

The present version should therefore be regarded as *finished in working form* whenever the following points are simultaneously true:

1. the DOI, version stamp, and roadmap are fixed;
2. the chain Ground Condition $\Rightarrow$ HC $\Rightarrow$ Closure is explicitly stated;
3. the probe sector is presented as a genuine structural sector;
4. the master equation is classified as a first-layer forced compression;
5. the tool catalogue is normalized;
6. open points are named rather than hidden.

These conditions are now met in the present monolith together with the later sections *Structural Core and Closure Principles*, *Toward a Structural ToE Master Equation*, and *Structural Reading of Matter, Gauge, Mass, and Higgs*. What remains for v1.2.0 is not the recovery of a missing architecture, but the next proof-theoretic deepening: more explicit lemma chains, stronger sector-by-sector projections, and a larger appendix of normalized tools and shortcuts.

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v1.4.0 Derivation and Appendix Integration Core

Normalization note. In Version 3.35.0, this block is read as a *mixed historical derivation layer*: its closure-ladder and micro-to-macro passages belong genealogically to $\mathfrak{G}$ and $\mathfrak{P}$, while its probe-reduction and appendix-coupling material remains operational support for the normalized architecture. It is therefore preserved for proof-memory and appendix linkage, but not as a second normative origin.

The specific task of v1.4.0 is to turn the earlier consolidation and lemma-development layers into a more tightly chained proof surface. This means that the argument is no longer presented only as a list of strategic claims. Instead, the intermediate dependencies are written so that later appendix items can be cited as reusable proof modules. The rule of this stage remains simple: *no shortening, no creative overwrite, only denser local chaining and clearer main-text/appendix interlock.*

1. Closure ladder as a reusable derivation chain

The closure architecture is now fixed as a ladder rather than a single jump. The intended reading is

$$\begin{aligned}&\text{local triolic unit} \Longrightarrow \text{orthogonal stabilization}\\ &\Longrightarrow \text{compensated identification} \Longrightarrow \text{closure class}\\ &\Longrightarrow \text{effective projection.}\end{aligned}$$

The advantage of this ladder is methodological. It prevents the later sections from collapsing several distinct proof moves into one rhetorical transition. Each arrow can now be isolated, tested, and, where needed, exported into the appendix as a technical lemma or shortcut.

Lemma 12.25 (Orthogonal stabilization lemma)**.** *Whenever one member of the triolic organization remains orthogonal to the other two, the local structural body is prevented from degenerating into a freely rotating two-sector ambiguity. The third direction therefore acts as a stabilizing anchor for all later compensator and closure statements.*

Proof. Without the orthogonal anchor, the first two directions would admit a family of relabelings that preserve local appearance while destroying global structural identity. The orthogonal third component blocks this ambiguity by fixing a non-coincident direction that all admissible couplings must respect. □

Lemma 12.26 (Compensated identification lemma)**.** *If the same structural body is observed through shifted or dual frames, then a compensating map is forced in order to preserve common identity across those frames. In the present framework this map is read as the Heisenberg compensator.*

Proof. Frame-shifted representatives are not independent ontological copies. They are multiple presentations of one common body. A balancing operation is therefore necessary to identify them as one object rather than a divergent family. That operation is exactly what HC contributes at the structural level. □

Lemma 12.27 (Closure-class lemma)**.** *Once orthogonal stabilization and compensated identification are both active, the admissible states assemble into a closure class: transition is allowed, but leakage into an incompatible external regime is not.*

Proof. The orthogonal anchor limits admissible couplings, while HC neutralizes those frame-induced mismatches that would otherwise split the common body. The surviving states therefore form a self-consistent class under the allowed moves of the theory. □

2. Micro-to-macro chain tightened

The earlier versions already stated that the macroscopic projection must be read from the microscopic structural unit through HC and closure. In v1.4.0, this chain is stated more sharply:

$$\text{micro structural unit} \Longrightarrow \text{local compensated coupling}$$
$$\Longrightarrow \text{sector stability} \Longrightarrow \text{macroscopic effective law.}$$

No macroscopic term is treated as primitive if it can be located lower in this chain. This applies in particular to geometric large-scale terms, matter/gauge labels, and all observer-frame residues.

Proposition 12.28 (Micro-to-macro closure principle)**.** *Every effective macroscopic equation used in the Sphaera-Cascade line must be read as a closure-compatible projection of the local triolic body rather than as an independent starting law.*

Proof. The ground condition, orthogonality requirement, and HC mechanism already fix what counts as an admissible local evolution. A macroscopic law that failed to arise as a projection of that admissible class would describe a sector not licensed by the common body. Hence effective macroscopic forms are subordinate projections of the closure class. □

Interpretive consequence. This principle strengthens the earlier interpretive discipline. Einstein-like, Dirac-like, and QFT-like forms remain useful projective readings, but their status is now tethered to a single chain of legitimacy: they are admissible only insofar as they compress the closure-compatible microstructure.

3. Probe-reduction and guided search principle

The probe sector is now equipped with a more explicit reduction principle. Two probes approaching one another do not explore an arbitrary ambient continuum; they reduce the candidate space by repeated closure tests.

Lemma 12.29 (Probe-reduction lemma)**.** *Let each probe carry its own triolic organization, and let the third group be orthogonal to both. Then the admissible interaction region is the subset of the naive search space that survives orthogonality, compensator compatibility, and closure tests simultaneously.*

Proof. Each probe contributes an internal triolic admissibility condition. The shared orthogonal third group cuts away probe pairings that cannot be embedded in one common constrained geometry. HC removes residual representational drift, and closure excludes the remaining leakage channels. What survives is exactly the reduced search region relevant to structural read-off. □

Tool interlock. This is the point where the normalized tools become operational rather than merely classificatory. Kasisky is attached to frequency-pattern selection. Hash/rainbow devices compress routing across candidate branches. Markov control organizes transitions inside the still-admissible region, and Newton accelerates convergence once a branch has already been selected. The appendix can therefore store these devices as modular search operators without severing them from the main proof line.

4. Appendix coupling rules

The appendix is no longer a passive storage area. In v1.4.0 it is given a precise function: each appendix block must either (i) justify a compressed step in the main text, (ii) codify a reusable tool, or (iii) collect a repeated technical transformation that would otherwise bloat the main line.

The intended appendix map is therefore:

1. **Symbol and sector tables** collect notation and prevent local restatement overhead.
2. **Structural tool records** store normalized rechen-tricks in the six-field format already fixed earlier.
3. **Technical lemmas** receive derivations whose content is load-bearing but too local or repetitive for the main line.
4. **Projection tables and read-off schemes** record the disciplined passage from structural objects to effective sector forms.

Proposition 12.30 (Appendix transfer principle)**.** *A step may be moved from the main text to the appendix only if its logical role is preserved through an explicit citation path. Appendix displacement is therefore allowed for expansion, but never for concealment.*

Proof. The project rule forbids silent loss of proof substance. Therefore, every step outsourced to the appendix must remain recoverable as a named lemma, tool, or transformation block. Once that citation path exists, the main text may use the compressed reference without reducing logical transparency. □

5. v1.4.0 readiness criteria

This release should be regarded as complete in working form when the following conditions hold simultaneously:

1. the version stamp and DOI are updated to v1.4.0;
2. the closure ladder is stated as an explicit reusable chain;
3. at least one new family of intermediate lemmas is present;
4. the micro-to-macro principle is written as a formal proposition;
5. the probe sector is tied directly to the normalized tool apparatus;
6. the appendix is assigned explicit transfer rules instead of being a passive dump area.

These conditions are now satisfied in the present monolith. Relative to v1.3.0, v1.4.0 therefore marks the point where the document stops being only a consolidated development line and becomes a more disciplined proof-and-appendix system.

v1.5.0 Projection and Appendix Table Integration Core

Normalization note. In Version 3.35.0, this block is retained as a $\mathfrak{P}$-*dominant transport layer*. It introduces no independent generative axioms; rather, it records how closure-licensed structural objects are transported into projection tables, appendix targets, and effective readout interfaces.

The task of v1.5.0 is to convert the earlier derivation-and-appendix interlock into a more explicit transport layer between the main line and the technical tables. Earlier versions had already introduced the closure ladder, the micro-to-macro principle, and the appendix transfer rule. The present stage adds a stricter rule: every repeated projection, reduction, or read-off step should be capturable either as a named lemma or as a named table interface with a declared upstream structural source and a declared downstream effective use. The point is not stylistic expansion but recoverability. A compressed move must remain reconstructible.

1. Projection tables as proof interfaces

Projection tables are not merely descriptive summaries. In v1.5.0 they receive a precise logical role: a projection table records how a structural object, already licensed by the closure architecture, may be read as an effective sector form without pretending that the effective form is primary. The direction of legitimacy is therefore always

$$\text{structural source} \Longrightarrow \text{closure-compatible projection} \Longrightarrow \text{effective sector reading.}$$

This prevents the later document from treating a Dirac-like, Einstein-like, or QFT-like formula as an autonomous axiom. The table is only valid if each row preserves that direction.

Definition 12.31 (Projection-table interface)**.** *A projection-table interface is a named record that assigns to a structural source object its admissible effective readings together with the conditions under which the passage is allowed, restricted, or still open.*

Lemma 12.32 (Projection-interface lemma)**.** *Whenever a projection step occurs repeatedly in the main line, the step may be compressed into a projection-table interface provided that the source object, closure condition, and interpretation status are all explicitly recorded.*

Proof. A repeated projection is proof-relevant only because it preserves a controlled relation between source and effective reading. If that relation is named once in a table together with its conditions, the main line may cite the named interface without losing substance. Compression is legitimate precisely because the table stores the missing local detail in recoverable form. □

2. Interpretation status discipline

The earlier reading discipline distinguished between what is structurally forced, projectively read, interpretively overlaid, and reserved for later proof expansion. In v1.5.0 this becomes a transport rule for all summary tables and appendix catalogues.

Proposition 12.33 (Interpretation-status transport principle)**.** *Every entry transferred from the main line into a table, catalogue, or appendix record must preserve its interpretation status. A structurally forced item may not be downgraded into merely interpretive language, and an interpretive or open reading may not be silently upgraded into a forced statement through tabulation.*

Proof. Tables compress syntax, not logical rank. If the interpretation status were allowed to drift during compression, then the appendix would alter rather than preserve the proof surface. The transfer must therefore carry both content and status labels together. □

Practical consequence. Projection tables must visibly mark at least three columns: structural source, allowed effective reading, and status label. Optional columns may record frame conditions, probe restrictions, or unresolved hypotheses, but the status label is mandatory.

3. Appendix-citation lattice

The appendix now receives a more explicit citation lattice. Instead of treating the appendix as a linear storage area, v1.5.0 organizes it as a small network of recoverable proof targets:

1. **lemma targets** for local derivation moves;
2. **tool targets** for normalized rechen-tricks;
3. **table targets** for repeated projection or reduction patterns;
4. **schematic targets** for probe geometry and search-space layouts.

A compressed sentence in the main text is acceptable only when it points to one of these four target types.

Lemma 12.34 (Appendix-citation lattice lemma)**.** *A compressed main-text step is proof-disciplined if and only if it can be expanded along at least one explicit appendix citation path ending in a lemma, tool, table, or schematic target.*

Proof. The project rule forbids concealment by relocation. Therefore every compressed step must have a recoverable destination. The four target types above exhaust the relevant forms of local expansion used in the present monolith. If no such path exists, the compression is not yet justified. □

4. Projection robustness across frames and probes

A repeated source of ambiguity is the temptation to read the same effective term as though it had identical status in all frames and in the probe sector. The present release therefore states a robustness rule.

Proposition 12.35 (Projection-robustness principle)**.** *An effective reading is robust only if it survives both frame-shift transport and probe-sector restriction without leaving the closure-compatible class.*

Proof. A term read in one frame but destroyed by admissible frame transport cannot be a stable structural projection. Likewise, a reading that collapses once the probe sector imposes orthogonality and closure tests is not globally licensed. Robust readings are exactly those that persist under both operations. □

Use in later versions. This principle is meant to control later expansions of matter, gauge, mass, and Higgs readings. The question is no longer only whether a reading is suggestive, but whether it survives transport across the disciplined structural operations already fixed by the framework.

5. v1.5.0 readiness criteria

This release should be regarded as complete in working form when the following conditions hold simultaneously:

1. the version stamp and DOI are updated to v1.5.0;
2. projection tables are assigned an explicit proof-interface role;
3. interpretation status is required to survive table and appendix transfer;
4. the appendix-citation lattice is stated as a reusable compression rule;
5. projection robustness is tied to both frame transport and probe tests;
6. the new material is appended without shortening the inherited monolith.

These conditions are now satisfied in the present monolith. Relative to v1.4.0, v1.5.0 marks the point where appendix tables and projection summaries are integrated into the proof discipline itself rather than serving only as support material.

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Computational completeness layer: QAM / MML / MMA

The present block records the first explicit machine reading of the post-v3.31.0 line. In the current v3.34.0-r09 working candidate it should be read together with the local HC/OP interface registry of Definition `def:r07-local-language-version-registry` in the active-core PDF, the compression map of Definition `def:compression-map-from-inherited-text-to-mml-mma-operations` in the active-core PDF, and the reusable tool normal forms of Definition `def:first-reusable-hcop-tool-norma` in the active-core PDF. That registry gives the active sublanguage versions actually used by the present OP-entry chain, the compression map gives the active audit reading for inherited HC/OP text, and the r09 tool layer gives the controlled invocation form:

$$\mathrm{QAM}^{\mathrm{loc}}_{\mathrm{HCOP}} \to \mathrm{MML}^{\mathrm{loc}}_{\mathrm{HCOP}} \to \mathrm{MMA}^{\mathrm{loc}}_{\mathrm{HCOP}}.$$

The broader computational chapter remains a reserve architecture; the r07 interface is the narrow executable slice used for HC-routed OP-entry.

Its purpose is to separate three layers which were already implicit in the project architecture:

1. QAM as the finite tagged object language;
2. MML as the finite machine / instruction language;
3. MMA as the matrix-realized execution layer.

The block proceeds in three steps. First, it isolates finite machine completeness and finite universality (the Z3-style regime). Second, it exhibits the unbounded extension mechanism through extensible address towers and fresh-address allocation. Third, it formulates stage-stable universality and the resulting architectural Turing-style completeness reading.

This computational layer is to be read *subordinate* to the earlier strength hierarchy

$$\mathbf{S}_1 \to \mathbf{S}_2 \to \mathbf{S}_3(\mathcal{R}) \to \mathbf{S}_4(\mathcal{R}) \to \mathbf{S}_5$$

from Sections 19U–19Z and the subsequent machine-status decoder. Accordingly, the present section does not by itself upgrade the unconditional public status of the line beyond $\mathbf{S}_1$. Rather, it records an *internal machine-reading package*: finite machine completeness and finite universality sharpen the internal route toward $\mathbf{S}_2$, while explicit fresh-address extension and stage-stable universality formulate the internal architectural ingredients that would be needed later on any conditional route toward $\mathbf{S}_4(\mathcal{R})$ and $\mathbf{S}_5$.

Definition 12.36 (Finite machine store)**.** *A* finite machine store *is a finite tagged family*

$$\mathfrak{S}_{\mathrm{fin}} = \{(a_i, x_i)\}_{i=1}^{N}$$

where each a_i is an address tag and each x_i is a stored QAM-object.

Definition 12.37 (Finite machine state)**.** *A* finite machine state *is a tuple*

$$\Sigma = \big(\mathfrak{S}_{\mathrm{fin}}, p, q, r\big),$$

where:

1. $\mathfrak{S}_{\mathrm{fin}}$ *is a finite machine store;*
2. p *is the current program position;*
3. q *is the current control state;*

4. *r is the current route-faithful residual/control register.*

Definition 12.38 (QAM-object layer)**.** *The* QAM-object layer *is the finite tagged object language in which the machine store contents x_i are represented.*

A QAM-object may be:

1. *a scalar-like tagged object;*
2. *a packet-like tagged object;*
3. *a tuple/block object;*
4. *a routed or normalized residual object;*
5. *any finite object built from the current QAM grammar by the allowed formation rules.*

Remark 12.39 (QAM as object language)**.** *QAM is not yet used here as a new infinite set theory. At this stage it is used as the finite object language in which machine-addressable project data are encoded.*

Definition 12.40 (MML instruction alphabet)**.** *The* MML instruction alphabet *is a finite set*

$$\mathcal{I}_{\mathrm{MML}}$$

of primitive machine instructions, divided into the following classes:

1. ***read/write instructions:***

$$\mathrm{READ}(a), \qquad \mathrm{WRITE}(a, \xi);$$

2. ***routing/transfer instructions:***

$$\mathrm{ROUTE}(\xi), \qquad \mathrm{PROJECT}(\xi), \qquad \mathrm{NORMALIZE}(\xi);$$

3. ***comparison/test instructions:***

$$\mathrm{TEST}_{=}, \qquad \mathrm{TEST}_{<}, \qquad \mathrm{TEST}_{\mathrm{adm}};$$

4. ***branching/control instructions:***

$$\mathrm{SELECT}(b_1, \ldots, b_k), \qquad \mathrm{BRANCH}(j), \qquad \mathrm{RETURN};$$

5. ***cycle/termination instructions:***

$$\mathrm{STEP}, \qquad \mathrm{LOOP}, \qquad \mathrm{HALT}.$$

Remark 12.41 (MML as finite machine language)**.** *MML is the first finite machine language layer of the project. Its purpose is not yet unrestricted computation, but finite programmable symbolic control on route-faithful QAM-data.*

Definition 12.42 (Finite MML program)**.** *A* finite MML program *is a finite word*

$$P = (I_1, \ldots, I_m), \qquad I_j \in \mathcal{I}_{\mathrm{MML}}.$$

Definition 12.43 (MMA execution layer). *The* MMA execution layer *is the matrix-realized execution semantics of MML on finite machine states.*

Concretely, an MMA-step is a map

$$\mathcal{E}_{\mathrm{MMA}} : (\Sigma, I) \longmapsto \Sigma^{+},$$

sending a finite machine state Σ and one MML instruction I to the updated state Σ^{+}, in a way compatible with the current Millennium-matrix / Heisenberg-compensator realization of the route-faithful control structure.

Remark 12.44 (Why MMA is the right execution layer). *MML gives the symbolic instruction form. MMA gives the executable matrix-side realization. Thus QAM, MML, and MMA correspond respectively to object language, machine language, and execution dynamics.*

Definition 12.45 (Finite branch emulation). *A* finite branch emulation rule *is any route-faithful realization of conditional continuation by means of:*

1. *a test instruction;*
2. *a finite selector family;*
3. *a branch-resolution instruction;*
4. *an updated program position and control state.*

Remark 12.46 (Why explicit jump is not required). *For the finite completeness level pursued here, it is not necessary to require a primitive classical jump instruction. It is sufficient that the machine language can emulate conditional continuation by a finite branch-resolution mechanism.*

Definition 12.47 (Finite machine completeness). *The triple*

$$(\mathrm{QAM}, \mathrm{MML}, \mathrm{MMA})$$

is called finitely machine-complete *if:*

1. *every finite project-relevant symbolic state can be encoded as a finite QAM-store;*
2. *every finite admissible symbolic transformation on such states can be represented by a finite MML program;*
3. *every such finite MML program is executable in the MMA layer;*
4. *conditional continuation is realizable by finite branch emulation;*
5. *halting and finite return are representable on the same machine level.*

Definition 12.48 (Finite address space). *A* finite address space *is a finite nonempty set*

$$\mathfrak{A}_{\mathrm{fin}} = \{a_1, \ldots, a_N\}$$

of address tags.

Definition 12.49 (Finite machine store as address map). *A* finite machine store *on the address space*

$$\mathfrak{A}_{\mathrm{fin}}$$

is a map

$$\mathsf{Mem} : \mathfrak{A}_{\mathrm{fin}} \longrightarrow \mathsf{QAM}_{\mathrm{fin}},$$

where $\mathsf{QAM}_{\mathrm{fin}}$ denotes the class of finite QAM-objects admissible at machine level.

Definition 12.50 (Store read operation). *For $a \in \mathfrak{A}_{\text{fin}}$, the* store read operation *is*

$$\mathsf{READ}_{\mathsf{Mem}}(a) := \mathsf{Mem}(a).$$

Definition 12.51 (Store write operation). *For $a \in \mathfrak{A}_{\text{fin}}$ and $\xi \in \mathsf{QAM}_{\text{fin}}$, the* store write operation *is the updated store*

$$\mathsf{WRITE}_{\mathsf{Mem}}(a, \xi) = \mathsf{Mem}^{+},$$

defined by

$$\mathsf{Mem}^{+}(a) = \xi, \qquad \mathsf{Mem}^{+}(b) = \mathsf{Mem}(b) \text{ for all } b \neq a.$$

Definition 12.52 (Address register). *An* address register *is a distinguished element $\alpha \in \mathfrak{A}_{\text{fin}}$ serving as the currently selected address for machine-level read/write access.*

Definition 12.53 (Control register). *A* control register *is a finite tuple*

$$\mathcal{R}_{\text{ctrl}} = (p, q, r, \alpha),$$

where p is the current program position, q the current control state, r the current route-faithful residual/control datum, and α the current address register.

Definition 12.54 (Finite machine configuration). *A* finite machine configuration *is a pair*

$$\mathcal{C} = (\mathsf{Mem}, \mathcal{R}_{\text{ctrl}})$$

consisting of a finite machine store and a control register.

Definition 12.55 (Finite universality). *The stack*

$$(\text{QAM}, \text{MML}, \text{MMA})$$

is called finitely universal *if every finite admissible symbolic task on finite machine configurations admits a finite MML realization executable in MMA.*

Definition 12.56 (Extensible address tower). *An* extensible address tower *is a nested family*

$$\mathfrak{A}_1 \subset \mathfrak{A}_2 \subset \mathfrak{A}_3 \subset \cdots$$

of finite address spaces such that each inclusion preserves the previously existing address tags and every finite set of address demands is realized at some stage.

Definition 12.57 (Store extension map). *Given two stages $\mathfrak{A}_n \subset \mathfrak{A}_{n+1}$, a* store extension map *is a map*

$$\text{Ext}_n^{n+1} : \mathsf{Mem}_n \longmapsto \mathsf{Mem}_{n+1}$$

such that old address contents are preserved and every new address is initialized by a fixed admissible default QAM-object.

Definition 12.58 (Fresh tag). *Let $\mathfrak{A}_{\text{fin}}$ be the currently active finite address/tag carrier set. A* fresh tag *is a tag*

$$a_{\text{new}} \notin \mathfrak{A}_{\text{fin}}.$$

Definition 12.59 (Fresh-address realization rule). *A* fresh-address realization rule *assigns to every fresh tag a_{new} an extended finite address space*

$$\mathfrak{A}_{\text{fin}}^{+} = \mathfrak{A}_{\text{fin}} \cup \{a_{\text{new}}\}.$$

Definition 12.60 (Default initialization object)**.** *A* default initialization object *is a distinguished finite QAM-object*

$$\mathbf{0}_{\mathrm{QAM}} \in \mathsf{QAM}_{\mathrm{fin}}$$

used to initialize newly created addresses before any further write/update takes place.

Definition 12.61 (Fresh-address store extension)**.** *Let*

$$\mathsf{Mem} : \mathfrak{A}_{\mathrm{fin}} \to \mathsf{QAM}_{\mathrm{fin}}$$

be the current finite machine store, and let $a_{\mathrm{new}} \notin \mathfrak{A}_{\mathrm{fin}}$ *be a fresh tag.*

The fresh-address store extension *is the extended store*

$$\mathsf{Mem}^{+} : \mathfrak{A}_{\mathrm{fin}}^{+} \to \mathsf{QAM}_{\mathrm{fin}}$$

defined by

$$\mathsf{Mem}^{+}(a) = \mathsf{Mem}(a) \quad \textit{for all } a \in \mathfrak{A}_{\mathrm{fin}},$$

and

$$\mathsf{Mem}^{+}(a_{\mathrm{new}}) = \mathbf{0}_{\mathrm{QAM}}.$$

Definition 12.62 (Old-address semantic preservation)**.** *A fresh-address extension satisfies* old-address semantic preservation *if every admissible instruction that only refers to addresses in the old domain has exactly the same effect on those old addresses before and after extension.*

Definition 12.63 (Stage-stable universality)**.** *The stack*

$$(\mathrm{QAM}, \mathrm{MML}, \mathrm{MMA})$$

is called stage-stably universal *if every extension-compatible stage-stable task family admits a stable unbounded realization by one stage-stable finite program.*

Remark 12.64 (Computational status reading)**.** *At the present stage, the project computational architecture should be read in three levels:*

1. *finite machine completeness;*
2. *finite universality (Z3-style regime);*
3. *stage-stable universality together with fresh-address extension, yielding an architectural Turing-style regime.*

Theorem 12.65 (First finite machine completeness reading theorem)**.** *Assume:*

1. *finite machine states are encoded in the QAM-object layer;*
2. *finite admissible symbolic transformations are representable in MML;*
3. *the MMA execution layer executes finite MML programs route-faithfully;*
4. *finite branch emulation is available.*

Then the triple

$$(\mathrm{QAM}, \mathrm{MML}, \mathrm{MMA})$$

supports a finitely machine-complete internal reading.

Proof. Finite tagged QAM stores supply the data layer, finite MML programs supply the symbolic control layer, MMA supplies executable realization, and branch emulation supplies finite conditional continuation. Hence the machine architecture is complete at the finite programmable level. □

Theorem 12.66 (Finite address/store layer theorem). *Assume:*

1. *the QAM-object layer encodes finite machine stores;*
2. *the MML instruction alphabet contains finite read/write, test, branch, loop, and halt instructions;*
3. *the MMA execution layer executes address-valid MML instructions on finite machine configurations;*
4. *finite branch emulation is available.*

Then the project architecture supports an explicit complete finite address/store machine layer.

Proof. Finite stores are explicit address maps, read/write closure holds on the same finite address domain, finite control is preserved under address-valid execution, and finite conditional control is realizable without leaving the finite regime. Thus the address/store layer is complete. □

Theorem 12.67 (First finite universality reading theorem). *Assume:*

1. *the strengthened finite machine completeness theorem holds;*
2. *every finite symbolic task decomposes into finitely many canonical finite programming primitives;*
3. *finite branch emulation is available.*

Then the stack

$$(\mathrm{QAM}, \mathrm{MML}, \mathrm{MMA})$$

supports a finite-universality reading on the internal machine layer.

Proof. Every finite symbolic task can be decomposed into finite read/write, routing, projection, normalization, test, selection, branch, loop, and halt primitives; these are representable in MML and executable in MMA. Hence every finite admissible symbolic task admits a finite realization. □

Theorem 12.68 (Fresh tag = fresh address reading theorem). *Assume:*

1. *the architecture already contains an implicit extensible address-tower pattern;*
2. *fresh tags are admissibly generable;*
3. *fresh-address realization holds;*
4. *default initialization is available;*
5. *old-address semantic preservation holds.*

Then the current QAM/MML/MMA architecture supports an explicit extensible-address reading.

Proof. Fresh-address realization turns new tags into genuine new addresses, default initialization turns those addresses into usable machine cells, and old-address semantic preservation makes the extension conservative on the previous stage. Iterating this process makes the address tower explicit. □

Theorem 12.69 (First stage-stable universality reading theorem). *Assume:*

1. *the first finite universality theorem holds;*

2. *the fresh tag = fresh address theorem holds;*
3. *uniform instruction persistence holds;*
4. *uniform execution persistence holds;*
5. *finite branch emulation persists along stage extension.*

Then the stack

$$(\mathrm{QAM}, \mathrm{MML}, \mathrm{MMA})$$

supports a stage-stably universal machine reading.

Proof. Finite universality gives programmable realization on each finite stage. Fresh-address extension makes store growth explicit and conservative. Uniform instruction and execution persistence keep the same finite program semantics valid on larger stages, and persistent branch emulation stabilizes conditional control. Therefore extension-compatible stage-stable task families admit stable unbounded realizations. □

Theorem 12.70 (Conditional architectural Turing-style completeness reading theorem)**.** *Assume the first stage-stable universality reading theorem. Then the stack*

$$(\mathrm{QAM}, \mathrm{MML}, \mathrm{MMA})$$

supports a conditional architectural Turing-style completeness reading along the explicit stage-stable route singled out by the present computational layer.

Proof. By stage-stable universality, the architecture supports programmable finite symbolic control, explicit unbounded store growth, stable finite program semantics across arbitrarily large stages, and persistent conditional control. These are exactly the architectural ingredients isolated for a Turing-style machine reading. Because the older Sections 19U–19Z still distinguish unconditional status $\mathbf{S}_1$ from stronger conditional route-dependent upgrades toward $\mathbf{S}_4(\mathcal{R})$ and $\mathbf{S}_5$, the present conclusion is recorded only as a conditional architectural reading rather than as a new unconditional global machine-status theorem. □

Corollary 12.71 (Computational status embedding rule)**.** *The project computational status should now be embedded into the older strength hierarchy in the following disciplined way:*

1. *unconditionally, the public machine-status reading of the line remains at* $\mathbf{S}_1$ *unless the external route certificates of Sections 19U–19Z are invoked;*
2. *internally, the present block sharpens the machine reading from a bare* $\mathbf{S}_1$ *interpretation toward a finitely machine-complete / finitely universal (Z3-style) package, i.e. toward the primitive-core side of* $\mathbf{S}_2$*;*
3. *explicitly extensible fresh-address and stage-stable universality readings isolate the architectural ingredients that would later be needed on a conditional route toward* $\mathbf{S}_4(\mathcal{R})$ *and* $\mathbf{S}_5$*, but they do not by themselves replace the earlier route-dependent certification ladder.*

Proof. Immediate from the preceding completeness theorems. □

Remark 12.72 (No parallel machine-status hierarchy)**.** *This computational block should therefore not be read as introducing a second independent machine-status ladder beside Sections 19U–19Z. Its role is narrower and more local: it explicates the QAM/MML/MMA machine package that underlies the already existing hierarchy and sharpens the internal architectural route toward stronger conditional computational claims, while leaving the unconditional public status decoder unchanged.*

Version and DOI record

The table below records the currently preserved Companion chain as a normalized release and working-state register. Zenodo releases are listed together with a publication-date column; earlier intermediate records and local working witnesses are retained so that the release chronology remains reconstructible even where the register groups recent working states before older archive rows.

The following chronology block is the active normalized release register for the present v3.35.0 release. The concept DOI remains the stable title-page pointer; version-specific DOI claims for public releases and pre-release freeze witnesses are recorded explicitly in this chronology.

Version / record	**Status**	**Publication date**	**DOI / note**
Concept DOI	concept record	ongoing pointer	10.5281/zenodo.19210728
v3.28.0	public Zenodo release	2026-04-20	10.5281/zenodo.19648743 — earlier public release; completes the deferred tool/QAM/decoder layer and fixes the route-control infrastructure that the present release inherits as its public baseline
v3.28.1-r39	pre-release working continuation	2026-04-20	local working state above the public v3.28.0 release; assembled the certificate-aware extension/routing/probe layer, local retained-corridor closure of the live OP 1 and OP 5 Step (B) kernels, QAM/decoder shadow reattachment, certificate-aware Monte-Carlo discipline, and restored full historical preservation; carried forward here as the final pre-release working witness for v3.29.0
v3.29.0	public Zenodo release	2026-04-20	10.5281/zenodo.19670829 — prior public release; promotes the certificate-aware extension/routing/probe layer, finite decision/action/word routing, matrix/crystal/field readout tags, admissibility and routing discipline, QAM/decoder shadow reattachment, certificate-aware Monte-Carlo discipline, and local retained-corridor closure of the live OP 1 and OP 5 Step (B) kernels while preserving the full historical witness in Part II

Version / record	Status	Publication date	DOI / note
v3.29.1-r32	release-freeze working continuation	2026-04-21	local freeze witness above the public v3.29.0 release; completes the conservative upgrade-plus-probe integration line through probe-compatible packets, work queues, executable conservative programs, compact readout/output/intake surfaces, OP-facing request/fulfillment/aggregation/dispatch/program/cycle/return logic, a completed second downstream loop, and an explicit release-freeze stabilization point; no version-specific DOI assigned
v3.30.0	public Zenodo release	2026-04-21	10.5281/zenodo.19682951 — prior public release; promotes the conservative upgrade-plus-probe integration line above the public v3.29.0 release, including probe-compatible conservative upgrade packets, finite probe-ranked work queues, queue-extracted working programs, route-faithful lead tasks, executable conservative packets, compact conservative readout/output/intake surfaces, OP-facing request/fulfillment/aggregation/dispatch/program/cycle/return logic, a completed second downstream loop, and an explicit release-freeze stabilization point while preserving the full historical witness in Part II
v3.31.0	public Zenodo release	2026-04-24	10.5281/zenodo.19699827 — prior public release; freezes the groundwork-complete pre-OP frontier synthesis above the public v3.30.0 baseline and publishes the full monolith with the integrated coupled prime-lattice trigger correction and the QAM/MML/MMA machine-realization layer while preserving the full historical witness in Part II
v3.32.0	public Zenodo release	2026-04-24	10.5281/zenodo.19704489 — prior public release; current release final built above the public v3.31.0 baseline; integrates the first machine-fused coupled OP-entry transition line, the coupled prime-lattice machine realization block, and the strengthened finite-to-stage-stable universality reading while preserving the full historical witness in Part II

Version / record	**Status**	**Publication date**	**DOI / note**
v3.33.0	public Zenodo release	2026-04-24	10.5281/zenodo.19716926 — prior public release; built above the public v3.32.0 baseline; carries the HC-captured local continuation line through lock-corridor capture, minimal HC-lock step, HC menu scan, plausible HC candidate, micro-lock update, and first local HC-capture chain while preserving the full historical witness in Part II
v3.34.0	public Zenodo release	2026-04-25	10.5281/zenodo.19746821 — prior public split-bundle release; promotes the locally audit-clean v3.34.0-r27 line into a reader-facing active core plus preserved historical witness bundle, adds the QAM/MM-L/MMA store-array layer, frontmatter reader architecture, and Sphaera array mirror/stripe/parity-residual infrastructure while preserving the non-closure status of OP 1/4 and OP 5(B)
v3.35.0	public Zenodo release	2026-04-25	10.5281/zenodo.19750674 — current public pre-v4 QAM/tensor preparation release; adds the short/extended abstract architecture, Hilbert-hotel QAM reindexing plan for later v4, guarded mixed QAM readouts, optional binary mask registry, conservative two-sided structural tensor grammar, role-erasure and mixed-readout normal forms, Boolean selector grammar, and release-final DOI packaging while preserving the non-closure status of OP 1/4 and OP 5(B)
v3.34.0-r17	local working candidate	2026-04-24	no Zenodo DOI assigned — executes the first dispatched handoff-guard packet $\mathsf{W}^{\mathrm{guard}}_{\mathrm{handoff}}$ after a stable lock/capture return, installs the handoff-guard defect register $\Delta^{\mathrm{guard}}_{\mathrm{handoff}}$, and returns exactly one of handoff-ready source, bounded handoff-guard repair source, or smaller frontier source without manufacturing an OP closure certificate

Version / record	Status	Publication date	DOI / note
v3.34.0-r18	local working candidate	2026-04-24	no Zenodo DOI assigned — executes the first dispatched OP-normalization/type packet $\mathsf{W}^{\mathrm{type}}_{\mathrm{norm}}$ after a handoff-ready return, installs the normalization/type-coherence defect register $\Delta^{\mathrm{type}}_{\mathrm{norm}}$, and returns exactly one of route/rank-ready typed OP input, bounded normalization/type repair source, or smaller frontier source without manufacturing an OP closure certificate
v3.34.0-r19	local working candidate	2026-04-24	no Zenodo DOI assigned — executes the first dispatched route/rank packet $\mathsf{W}^{\mathrm{rank}}_{\mathrm{route}}$ after a normalization-ready return, installs the route/rank defect register $\Delta^{\mathrm{rank}}_{\mathrm{route}}$, and returns exactly one of comparison-ready routed OP task, bounded route/rank repair source, or smaller frontier source without manufacturing an OP closure certificate
v3.34.0-r20	local working candidate	2026-04-24	no Zenodo DOI assigned — executes the first dispatched comparison/defect packet $\mathsf{W}^{\Delta}_{\mathrm{cmp}}$ after a route/rank-ready return, installs the comparison/defect register $\Delta^{\Delta}_{\mathrm{cmp}}$, and returns exactly one of guarded bind candidate, guarded split candidate, or smaller comparison frontier without manufacturing an OP closure certificate
v3.34.0-r21	local working candidate	2026-04-24	no Zenodo DOI assigned — executes the first dispatched resolve/return packet $\mathsf{W}^{\mathrm{next}}_{\mathrm{resolve}}$ after a guarded comparison return, installs the resolve/return defect register $\Delta^{\mathrm{next}}_{\mathrm{resolve}}$, and returns exactly one of guarded bind-work continuation, guarded split-work continuation, or smaller comparison-frontier descent without manufacturing an OP closure certificate
v3.34.0-r22	local working candidate	2026-04-24	no Zenodo DOI assigned — executes the first continuation-queue packet after resolve/return, installs the continuation-queue defect register $\Delta^{\mathrm{cont}}_{\mathrm{queue}}$, and returns exactly one of a guarded common-origin bind queue, guarded ranked split queue, or protected frontier-descent queue without manufacturing an OP closure certificate

Version / record	Status	Publication date	DOI / note
v3.34.0-r23	local working candidate	2026-04-24	no Zenodo DOI assigned — executes the first queue-head selection packet after continuation queue, installs the queue-head selection defect register $\Delta^{\mathrm{select}}_{\mathrm{head}}$, and returns exactly one of a guarded common-origin-check head, guarded branch-rank-check head, or guarded frontier-descent-check head while retaining the finite tail and without manufacturing an OP closure certificate
v3.34.0-r24	local working candidate	2026-04-24	no Zenodo DOI assigned — executes the first selected queue-head execution packet, installs the selected queue-head execution defect register $\Delta^{\mathrm{exec}}_{\mathrm{head}}$, and returns exactly one of an executed common-origin-check surface, executed branch-rank-check surface, or executed frontier-descent-check surface while retaining the finite queue tail and without manufacturing an OP closure certificate
v3.34.0-r25	local working candidate	2026-04-24	no Zenodo DOI assigned — executes the first executed-head tail-entry packet $\mathsf{W}^{\mathrm{entry}}_{\mathrm{tail}}$, installs the tail-entry defect $\Delta^{\mathrm{entry}}_{\mathrm{tail}}$, and returns exactly one of a guarded bind-tail, split-tail, or frontier-tail entry without manufacturing an OP closure certificate
v3.34.0-r26	local audit-freeze candidate	2026-04-24	no Zenodo DOI assigned — installs the first audit-freeze ledger $F^{\mathrm{audit}}_{\mathrm{HCOP}}$ with audit-freeze defect $\Delta^{\mathrm{freeze}}_{\mathrm{audit}}$, recording baseline, DOI state, local spine, tail-entry output, preservation obligations, reference/compilation audit data, terminal bibliography, historical witness, and the no-false-certificate marker
v3.34.0-r27	local audit-freeze candidate	2026-04-24	no Zenodo DOI assigned — installs the audit-freeze stabilization object $F^{\mathrm{audit}}_{\mathrm{freeze}}$, permitting only audit, mechanical repair, metadata repair, release packaging, and Zenodo-description preparation before further theorem expansion

Version / record	Status	Publication date	DOI / note
v3.34.0-r15	local working candidate	2026-04-24	no Zenodo DOI assigned — turns the r14 tail-normalized HC/OP proof skeleton into the first controlled skeleton-to-workstep dispatch object $\mathsf{D}^{\mathrm{skel}}_{\mathrm{HCOP}}$, assigning finite admissible next work packets to lock/capture, handoff, normalization, routing, comparison, and resolution nodes while preserving dependency order, guards, witnesses, route tags, theorematic status, and the no-false-certificate rule
v3.34.0-r16	local working candidate	2026-04-24	no Zenodo DOI assigned — executes the first dispatched lock/capture stabilization packet $\mathsf{W}^{\mathrm{stab}}_{\mathrm{lock}}$, installs the lock-stability defect register $\Delta^{\mathrm{stab}}_{\mathrm{lock}}$, and returns exactly one of handoff-ready stable source, bounded local repair source, or smaller frontier source without manufacturing an OP closure certificate
v3.34.0-r14	local working candidate	2026-04-24	no Zenodo DOI assigned — extracts the first tail-normalized HC/OP proof skeleton $\mathsf{S}^{\mathrm{tail}}_{\mathrm{HCOP}}$, organizing the active r01–r04 spine as LOCK/CAPTURE, HANDOFF, NORMALIZE, ROUTE, COMPARE, and RESOLVE source nodes plus later guarded citation tails while preserving dependency order and preventing false OP-closure certificates
v3.34.0-r13	local working candidate	2026-04-24	no Zenodo DOI assigned — turns the r12 ledger into the first guarded-reduction pilot normal form $\mathrm{GRCite}_{\mathrm{HCOP}}(\ell; G, W, S)$, allowing only later repeated HC/OP tails to be rewritten as guarded tool citations while first occurrences, local guards, witnesses, certificate tests, proof-critical inequalities, theorematic status, and historical witness material remain explicit

Version / record	Status	Publication date	DOI / note
v3.34.0-r12	local working candidate	2026-04-24	no Zenodo DOI assigned — applies the r11 guarded-reduction protocol for the first time by installing the finite ledger $\mathcal{L}^{(1)}_{\mathrm{HCOP}}$ over the active HC/OP proof spine, allowing later repeated lock/capture, handoff, normalization, route/frontier, paired-comparison, probe-diagnostic, and archive-to-tool tails to be cited in guarded normal form while keeping first occurrences, local guards, certificate tests, proof-critical inequalities, theorematic status, and historical witness material explicit
v3.34.0-r11	local working candidate	2026-04-24	no Zenodo DOI assigned — keeps the r10 tool-contract application layer and adds the first tool-guarded reduction protocol $\mathcal{R}^{(1)}_{\mathrm{HCOP}}$, defining reducible and non-reducible HC/OP occurrences, guarded citation normal form, and the partial simplification map $\mathsf{Red}^{(1)}_{\mathrm{HCOP}}$ while preserving proof content, first definitions, local guards, certificate tests, theorematic status, and the full historical witness in Part II
v3.34.0-r10	local working candidate	2026-04-24	no Zenodo DOI assigned — applies the r09 reusable tool contracts back to the active r01–r04 HC/OP proof spine as a conservative cross-reference layer $\mathcal{P}^{\mathrm{tool}}_{\mathrm{HCOP}}$, allowing repeated route, frontier, and no-false-certificate safety prose to be replaced by typed tool-contract citations only where the original hypotheses and guards remain visible, while preserving proof content and the full historical witness in Part II
v3.34.0-r09	local working candidate	2026-04-24	no Zenodo DOI assigned — keeps the r08 compression map unchanged and extracts the first reusable HC/OP tool normal forms $\mathbb{T}^{(1)}_{\mathrm{HCOP}}$, giving typed contracts for lock/capture, handoff, normalization, routing, comparison, resolution, probe ranking, certificate tests, frontier descent, and archive-to-tool extraction, so later text can cite tool contracts instead of repeating safety prose while preserving proof content and the full historical witness in Part II

Version / record	Status	Publication date	DOI / note
v3.34.0-r08	local working candidate	2026-04-24	no Zenodo DOI assigned — keeps the r07 QAM/MML/MMA interface unchanged and adds $\mathfrak{C}_{\mathrm{HCOP}}$, a conservative MML/MMA compression map for inherited HC/OP text, so repeated capture, handoff, normalization, routing, comparison, probe, certificate-test, route-preservation, and historical-tool paragraphs can be reread as typed operations without deleting proof content or manufacturing OP closure certificates, while preserving the full historical witness in Part II
v3.34.0-r07	local working candidate	2026-04-24	no Zenodo DOI assigned — keeps the r04 paired OP-entry comparison execution and r06 bibliography hardening unchanged, assigns local interface versions $\mathrm{QAM}^{\mathrm{loc}}_{\mathrm{HCOP}}$, $\mathrm{MML}^{\mathrm{loc}}_{\mathrm{HCOP}}$, and $\mathrm{MMA}^{\mathrm{loc}}_{\mathrm{HCOP}}$, and proves that the r01–r04 HC-routed OP-entry chain is executable as one finite typed MML/MMA program without manufacturing an OP closure certificate, while preserving the full historical witness in Part II
v3.34.0-r06	local working candidate	2026-04-24	no Zenodo DOI assigned — keeps the r04/r05 mathematical and bibliography-placement state unchanged, expands the terminal bibliography after a whole-document literature scan, adds compact front/back literature anchor notes for Teichmuller/quasiconformal geometry, Riemann–GUE/RMT, Hopf/fibre-bundle topology, division algebras, gauge/QFT landmarks, CKM/PMNS mixing, Julia dynamics, set-theoretic foundations, and probe/cryptographic/computational diagnostics, while preserving the full historical witness in Part II
v3.34.0-r05	local working candidate	2026-04-24	no Zenodo DOI assigned — keeps the r04 paired OP 1/4–OP 5(B) comparison execution layer unchanged and normalizes the bibliography/back-matter placement by moving the live bibliography to the terminal Final Bibliography block while preserving the full historical witness in Part II

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v3.34.0-r04	local working candidate	2026-04-24	no Zenodo DOI assigned — executes the paired OP 1/4–OP 5(B) comparison branch above the r03 task-routing layer; compares joint versus separated HC-compatible readouts and returns either a bound common-origin state, a split ranked single-branch state, or a smaller frontier-certified comparison state while preserving route tags and the full historical witness in Part II
v3.34.0-r03	local working candidate	2026-04-24	no Zenodo DOI assigned — extracts the first finite OP-entry task-routing layer from the HC-derived normal form; maps the selected live residual branch into OP 1/4 continuum-limit work, OP 5(B) routed Θ–R balance work, anchor/admissibility refinement, or paired OP 1/4–OP 5(B) comparison while preserving route tags and the full historical witness in Part II
v3.34.0-r02	local working candidate	2026-04-24	no Zenodo DOI assigned — extracts the first downstream-readable OP-entry normal form from the transport-stable HC-to-OP handoff packet; routes the terminal live residual triple into OP 1/4, OP 5(B), and anchor/admissibility request families while preserving route tags and the full historical witness in Part II
v3.34.0-r01	local working candidate	2026-04-24	no Zenodo DOI assigned — begins the transport-stable HC-capture and OP-entry handoff line above the public v3.33.0 release; packages local HC capture into a route-faithful transport-stability test and an OP-facing handoff packet while preserving the full historical witness in Part II
v3.25.0	public Zenodo release	2026-04-16	10.5281/zenodo.19615952 — earlier public release; consolidates the optical-comparison and machine-theoretic overlay line, including compressed optical diagnostic control, bundled proof-use compression, conditional reference-core closure, and the capped machine-theoretic overlay line up to the present equivalence surface

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v3.18.0	public Zenodo release	2026-04-13	10.5281/zenodo.19548590 — earlier public release; carries the public v3.17.0 tensor/operator baseline forward into a first public OP 3/QAM closure package with shell certificates, shell-defect localization, shell-stable readout factorization, and a bundled OP 3→OP 5/global interface while keeping stronger kernel-first activation internal
v3.17.0	public Zenodo release	2026-04-10	10.5281/zenodo.19522352 — earlier public release; promotes the third public-shape QAM tensor/operator line above the public v3.16.0 release and keeps translation/round-trip statements in front-facing public form while leaving support-selection and wider compression internal
v3.16.0	public Zenodo release	2026-04-10	10.5281/zenodo.19489324 — earlier public release; promotes the second public-shape QAM branch/shell/routed layer while leaving tensor/operator and support-selection expansion internal, and records a deferred kernel-first compression path with activation criteria, staged stress tests, and migration roadmap
v3.15.0	public Zenodo release	2026-04-08	10.5281/zenodo.19463389 — earlier public release; freezes the first compressed public-shape QAM core on top of the public v3.14.0 branch/shell baseline
v3.14.0	public Zenodo release	2026-04-07	10.5281/zenodo.19460589 — earlier public release; extends the public v3.13.0 line by freezing the first post-closure branch/shell hardening layer
v3.3.0	local working state	2026-04-03	condensed Appendix-C integration of Julia-boundary, master-residual, Umkehrwalze, and reversal-probe sector
v3.3.1	local working state	2026-04-03	clean condensed Appendix-C line; 16-page block reduced to a 3-page support appendix
v3.3.2	local working state	2026-04-03	clean-mainline promotion: compact 3.3 synthesis theorem moved into main text; Appendix-C support retained
v3.3.3	local final-polish state	2026-04-03	removed embedded editorial fragments in the new 3.3 synthesis line and normalized notation in the compressed summary

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v3.3.4	local editorial-clean state	2026-04-03	global editorial cleanup: removed remaining visible drafting fragments, normalized compressed remarks, and cleaned heuristic table notes
v3.3.5	Zenodo release	2026-04-03	10.5281/zenodo.19396143 — strict wording cleanup: aligned version metadata, removed residual status-weakening phrasing in the 3.3 line, normalized the current-version record, and finalized the 3.3.5 release package
v3.4.0	local working extension	2026-04-03	first 3.4 pass: added closure-potential globalization of the residual branch law, conserved branch energy, and an energy-form reading of the reversal-probe dichotomy
v3.4.1	local working extension	2026-04-03	constrained the exact branch potential to the reversible single-well class and reduced the global scalar ambiguity to the transported $\mathbb{Z}_2$ pair
v3.4.2	local working extension	2026-04-03	extracted the canonical quartic one-field family for the visible residual branch law and added turning-value parameterization of regular probe shells
v3.4.3	local working extension	2026-04-03	tied the visible quartic branch coefficient to the transported quartic closure coefficient and replaced the technical turning-amplitude label by turning value
v3.4.4	local working extension	2026-04-03	added the first nonlinear period-drift readout, showing that the timing defect on regular periodic shells is already fixed by the same quartic closure coefficient
v3.4.5	local working extension	2026-04-03	extended the quartic readout to the full even hierarchy, proving that the first higher timing defect appears exactly at the order of the first surviving even closure coefficient
v3.4.6	local working extension	2026-04-03	repackaged the transported even hierarchy into a single analytic generator $\Phi(\eta^2)$ for the visible branch potential and its nonlinear shell readouts
v3.4.7	local working extension	2026-04-03	generator-shell probe classifier: small regular shells reduced to the generator radius $s(E) = \Phi^{-1}(E)$ and the transported sign pair $\pm\sqrt{s(E)}$

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v3.4.8	local working extension	2026-04-03	shell-parity criterion: direct convergence versus spinorial/Möbius lift determined by the even/odd reflector-crossing parity on sufficiently small regular shells; added first diagnostic Julia-type appendix render
v3.4.9	local working extension	2026-04-03	added cautious frame-shifted detector-readout hypothesis for the diagnostic Julia/probe geometry, together with explicit candidate observables for later comparison work
v3.4.10	local working extension	2026-04-03	converted the detector-readout hypothesis into a quantitative tunnel-observable interface: tunnel width, edge intensification, opening angle, and shell parity are now organized as a minimal diagnostic comparison package
v3.4.11	local working extension	2026-04-03	added a calibrated inverse-readout theorem: tunnel width plus shell parity now recover the local shell radius, while edge intensification and opening angle become shell-consistency checks measured by Δ_{diag}
v3.4.12	local working extension	2026-04-03	added a local inverse identifiability/stability theorem: the calibrated tunnel-width law now carries a shell-wise stability margin and inverse condition number, so shell reconstruction is controlled whenever the calibration slope stays away from zero
v3.4.13	local working extension	2026-04-03	organized the inverse readout globally into a shell atlas stratified by calibration-critical shells and parity-switch shells, so stable inversion now proceeds chart by chart away from the obstruction set
v3.4.14	local working extension	2026-04-03	added an obstruction-shell transition law: isolated obstructions now carry only fold exchange, parity flip, or their composition, giving the inverse-readout layer a small $\mathbb{Z}_2$ transition algebra and local obstruction monodromy
v3.4.15	local working extension	2026-04-03	organized the obstruction transitions into a global Klein-four transition cocycle on the shell atlas and showed that, after fold resolution, the remaining monodromy reduces to the parity local system underlying the spinorial/Möbius lift

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v3.4.16	local working extension	2026-04-03	promoted the parity local system to a global double-cover criterion: direct visible convergence is globally available exactly when the parity class trivializes, while every nontrivial case becomes globally single-valued on the canonical parity double cover
v3.4.17	local working extension	2026-04-03	recast the residual parity sector as a genuine cohomological obstruction class in $H^1(B, \mathbb{Z}_2)$, identifying direct visible convergence with the vanishing class and the canonical parity double cover with its associated classifying two-sheeted cover
v3.4.18	local working extension	2026-04-03	promoted the parity obstruction to an operational global holonomy/readout character: admissible closed shell paths now evaluate the class $\varepsilon_{\mathrm{par}}$ directly, so direct visible return versus spinorial/Möbius lift is read as a $\mathbb{Z}_2$ loop outcome factoring through mod-two homology
v3.4.19	local working extension	2026-04-03	turned the parity holonomy character into an empirical loop-readout estimator: chartwise parity-jump counting across admissible obstruction shells now recovers the cohomological $\mathbb{Z}_2$ holonomy, giving a detector-side loop test for direct visible return versus spinorial/Möbius lift
v3.4.20	local closing state	2026-04-03	clean 3.4 closure: compressed the full extension line into a single generator-atlas-holonomy synthesis theorem, identified the minimal handoff package $(\Phi, \Delta_{\mathrm{diag}}, \varepsilon_{\mathrm{par}}, \chi_{\mathrm{par}}, \widehat{\chi}_{\mathrm{par}})$, and marked the end of the 3.4 build-up before the next-stage simplification work
v3.5.0	local working extension	2026-04-03	first compression-stage theorem: introduced the residual record $\mathfrak{R}(B)$ and proved the reconstruction normal form for the stabilized visible sector
v3.5.1	local working extension	2026-04-03	record-level transport step: defined admissible residual-record morphisms and isomorphisms, and formalized the faithful translation principle on the stabilized visible sector

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v3.5.2	local abstract-polish state	2026-04-03	sharpened the abstract and the compression-stage framing: clarified the residual-record interface, the conservative-frontend requirement, and the split between structurally closed versus still-open global identification problems
v3.5.3	local status-cleanup state	2026-04-04	reclassified the open-questions layer by actual status, absorbed OP 2(T4) into OP 1, marked OP 3 as no longer a primary open problem, and reduced OP 5 to Step (B) only
v3.5.4	local chronology-sync state	2026-04-04	synchronized the document head with the current 3.5.x line: added the missing v3.5.0–v3.5.4 chronology entries and replaced outdated roadmap goals by the residual-record/compression-stage agenda
v3.5.5	local reader-guide alignment state	2026-04-04	aligned the Reader's Guide with the actual 3.5.x status: separated live fronts from preserved historical roadmap layers and made the OP 1/2/3/5 status split explicit at the document head
v3.5.6	local main-text status-gloss state	2026-04-04	propagated the 3.5.x status split into the main compression-stage remarks and appendix support transitions: clarified that remaining fronts are downstream residual tasks and not invitations to reopen the visible interface
v3.5.7	local stage-summary state	2026-04-04	added an explicit 3.5.x-versus-3.6 orientation block: recorded what the compression line has already achieved and fixed the next target as conservative language stabilization above the residual-record interface
v3.5.8	local figure-integration state	2026-04-04	rebuilt the missing Julia/probe comparison render, integrated it into the text with a cautious diagnostic caption, and restored a clean PDF build with the image in place
v3.6.0	local working extension	2026-04-04	introduced the conservative-frontend principle: any later language layer must factor through the residual-record interpreter and may not introduce new visible primitives

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v3.6.1	local frontend-hardening state	2026-04-04	hardened the conservative frontend line by adding closure under composition and a standalone primitive non-drift criterion; clarified that Section 3.6 fixes admissibility constraints for later language layers rather than introducing a second visible interface
v3.6.2	local frontend-finalization state	2026-04-04	promoted the closure/non-drift hardening step into the main line and stabilized the 3.6 block as a completed admissibility framework for later language chains
v3.7.0	local OP-first consolidation state	2026-04-04	formalized the current OP-first reading: the remaining live fronts are organized as residual downstream tasks over the fixed record interface, with direct language growth deliberately deferred
v3.7.1	Zenodo release	2026-04-04	10.5281/zenodo.19412478 — public release of the OP-first 3.7 line; froze the present stabilized-reading claim and fixed 3.8 as an explicit OP-closure phase while keeping QAM deferred
v3.8.0	local two-core-program state	2026-04-04	promoted 3.8 from a planning note to a working OP-closure stage: reduced the foundational historical burden to the OP 1 correction core and the OP 5 step (B) Theta-balance core, with all other live fronts treated as secondary unless dependency is shown
v3.8.1	local residual-emergence state	2026-04-04	recast the field-equation / controlled-emergence question into a residual-controlled physical interface form, concentrating the unresolved burden into the explicit remainder families $\mathcal{E}_1$ and $\mathcal{E}_5$ instead of introducing a new visible primitive basis
v3.8.2	local OP 1 reduction state	2026-04-04	reduced the remaining OP 1 burden to a single scalar residual channel $\mathcal{R}_1(\mu)$, making OP 1 a one-channel closure problem rather than a diffuse correction family
v3.8.3	local OP 1 root-isolation state	2026-04-04	proved that the OP 1 scalar residual channel has a unique root in the physical window $1.85 \leq \mu \leq 1.90$, numerically $\mu_* \approx 1.86366225$, thereby turning the remaining OP 1 burden from a broad numerical search into an exact structural-identification problem

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v3.8.4	local OP 1 frame-shift diagnostic state	2026-04-04	showed that the residual $\gamma_L - \mu_*$ shrinks on the matter-frame offset scale, behaves like a local Newton/readout correction, and does not trigger a parity/horizon jump in the current Julia/probe family at the tested residual scales
v3.8.5	local minimal neutrino-inflow test state	2026-04-04	converted the archived $k(\nu) = 4$ inflow hypothesis into a minimal OP 1 downstream correction ansatz $\mathcal{R}_{1,\nu}(\mu;\eta_\nu) = \mathcal{R}_1(\mu) + \eta_\nu/\mu^4$, with calibrated $\eta_\nu^* \approx -1.10 \times 10^{-3}$ and no reopening of the visible primitive basis
v3.8.6	local semi-structural inflow estimate state	2026-04-04	showed that the calibrated coefficient satisfies $\eta_\nu^* \approx -\varepsilon_\nu \gamma_L^4 \beta_{\mathrm{sym}}^4$ with $\varepsilon_\nu \approx 1.0012 \times 10^{-3}$, so the fitted inflow law reduces to a $k = 4$ matter-share factor times a per-mille fermionic leakage efficiency
v3.8.7	local defect-squared inflow state	2026-04-04	identified the per-mille leakage efficiency with the already existing quadratic defect invariant $\eta_{\mathrm{sym}}^2 = (\frac{1}{3} - \beta_{\mathrm{sym}}^2)^2$, yielding the sharper law $\eta_\nu^* \approx -\eta_{\mathrm{sym}}^2 \gamma_L^4 \beta_{\mathrm{sym}}^4$
v3.8.8	local second-order fermionic leakage state	2026-04-04	used the J-antisymmetric fermionic sector H_-^+ to justify why the first visible inflow scalar should be quadratic in the transported defect, leading to $\eta_\nu \approx -\lambda_\nu \eta_{\mathrm{sym}}^2 \gamma_L^4 \beta_{\mathrm{sym}}^4$ with $\lambda_\nu \approx 0.9990$
v3.8.9	local cosmological-offset link state	2026-04-04	showed that any cosmological contribution to the OP 1 inflow remainder is naturally mediated by the permanent frame-offset term β_{sym}^2 already used in the dark-offset theorem, not by a direct Hubble-rate coupling, yielding $\eta_\nu^* \approx -(\frac{1}{3} - \beta_{\mathrm{sym}}^2)^2 \gamma_L^4 \beta_{\mathrm{sym}}^4$ as the natural weak expansion reading
v3.8.10	local one-background collapse state	2026-04-04	showed that the frame/readout, fermionic-leakage, and weak cosmological-offset readings of the smooth OP 1 remainder collapse numerically to the same background law $-\eta_{\mathrm{sym}}^2 \gamma_L^4 \beta_{\mathrm{sym}}^4$, leaving the discrete prime-lattice mechanism as the only genuinely independent live burden

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v3.8.11	local readout-point cancellation state	2026-04-04	showed that the scalar prime/continuum residual $\mathcal{R}_1(\gamma_L) \approx 9.1216 \times 10^{-5}$ and the collapsed smooth background term $-\eta_{\mathrm{sym}}^2\beta_{\mathrm{sym}}^4 \approx -9.1208 \times 10^{-5}$ cancel to within $\Delta_{\mathrm{can}} \approx 8.68 \times 10^{-9}$, shifting the live OP 1 burden from finding a new correction term to explaining the near-identity itself
v3.8.12	local background-corrected bridge state	2026-04-04	defined the bridge function $\Phi_1(\mu) = \mathcal{R}_1(\mu) - \eta_{\mathrm{sym}}(\mu)^2\beta(\mu)^4$, proved that it has a unique root $\gamma_{\mathrm{br}} \approx 1.86399996799643$ on the physical window, and showed that the Companion matter-frame value 1.864 already coincides with this bridge root to about 3.2×10^{-8}
v3.8.13	local bridge-shape forcing state	2026-04-04	showed that the current Companion architecture forces any minimal admissible visible OP 1 bridge term to have the shape $\lambda_{\mathrm{br}}\eta_{\mathrm{sym}}(\mu)^2\beta(\mu)^4$, leaving only the canonical normalization $\lambda_{\mathrm{br}} \approx 1.00010$ as the live remaining freedom
v3.8.14	local multiplicative bridge-renormalization state	2026-04-04	rewrote the forced bridge equation as the dressed readout law $\beta(\mu) = \beta_{\mathrm{CL}}(\mu)\exp(\eta_{\mathrm{sym}}(\mu)^2\beta(\mu)^4)$, showing that the apparent non-canonicity of λ_{br} at 1.864 is only the tiny 8.68×10^{-9} displacement from the exact bridge root rather than a missing normalization mechanism
v3.8.15	local transport-balanced exponential state	2026-04-04	derived the exponential self-background dressing as the finite form of a repeated compensator-balanced transport law on the positive readout amplitude, reducing the remaining OP 1 burden to the path-constancy/normalization of the bridge source rather than to the appearance of the exponential itself
v3.8.16	local bridge-contraction state	2026-04-04	recast the bridge law as the fixed-point equation $T_\mu(\beta) = \beta$ and showed that the bridge map is strongly contractive at the distinguished root, with $\lvert\partial_b T_{\gamma_{\mathrm{br}}}(\beta_{\mathrm{br}})\rvert \approx 0.00311$, thereby turning the source-freezing step into a local stability statement rather than a free assumption

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v3.8.17	local OP 1 closure state	2026-04-04	packaged the 3.8.2–3.8.16 line into a local closure theorem at the distinguished matter-frame point $\gamma_{\mathrm{loc}} = \gamma_{\mathrm{br}} \approx 1.86399996799643$, relocating the remaining burden from local OP 1 readout to the question of global extension versus transition to OP 5 Step (B)
v3.8.18	final local OP 1 closure release state	2026-04-04	finalized the 3.8 release with version DOI 10.5281/zenodo.19414221; the release message is local closure of OP 1 at the distinguished matter-frame point, with OP 5 Step (B) promoted to the next primary live front
v3.8.19	local OP 5 one-point reduction state	2026-04-04	used local OP 1 closure to fix the matter-frame and photon-frame background data, reducing historical OP 5 Step (B) to the single scalar finite-cutoff identity $\Omega_{5,\mathrm{loc}} = 0$ at $T_{\mathrm{loc}} = \gamma_{\mathrm{loc}}^{-3/2} \approx 0.3929444410$ with target value $\mathcal{T}(T_{\mathrm{loc}}) \approx 0.893255$
v3.8.20	local OP 5 exact-tail state	2026-04-04	rewrote historical OP 5 Step (B) as the exact tail identity $\mathcal{E}_N(T_{\mathrm{loc}}) = 0.893255 - \mathcal{T}_N(T_{\mathrm{loc}})$ above an arbitrary truncation level N; in particular, using the archived $N = 50$ diagnostic $\mathcal{T}_{50}(T_{\mathrm{loc}}) \approx -0.146$, the required tail contribution becomes $\mathcal{E}_{50}(T_{\mathrm{loc}}) \approx 1.039255$
v3.8.21	local OP 5 counting-tail state	2026-04-04	rewrote the exact OP 5 tail as a Stieltjes integral against the photon-frame zero counting tail $N_{\mathrm{ph}}^{>N}(u)$ and, after integration by parts, as the oscillatory counting-function integral with kernel $K_{T_{\mathrm{loc}}}(u) = T_{\mathrm{loc}} \cos(T_{\mathrm{loc}} u)/u - \sin(T_{\mathrm{loc}} u)/u^2$, thereby isolating the exact analytic interaction that remains to be controlled
v3.8.22	local OP 5 smooth-density split state	2026-04-04	decomposed the exact OP 5 tail into a smooth counting-density contribution $\mathcal{E}_N^{\mathrm{sm}}(T_{\mathrm{loc}})$ plus a genuine fluctuation term $\mathcal{E}_N^{\mathrm{fl}}(T_{\mathrm{loc}})$, turning Step (B) into the precise question whether the distinguished-cutoff tail is density-dominated or fluctuation-dominated

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v3.8.23	local OP 5 leading-density test state	2026-04-04	inserted the leading transported photon-frame Riemann–von Mangoldt density, finding $\mathcal{E}^{\mathrm{sm}}_{50,\mathrm{lead}}(T_{\mathrm{loc}}) \approx 0.047867069$ versus the exact required tail $\mathcal{E}_{50}(T_{\mathrm{loc}}) \approx 1.039202153$; thus only about 4.6% of the tail is smooth and Step (B) is decisively fluctuation-dominated
v3.8.24	local OP 5 pure-fluctuation state	2026-04-04	subtracted the leading smooth counting density completely and rewrote Step (B) as the pure fluctuation-tail identity $-2\int_0^\infty S^{>N}_{\mathrm{ph}}(u)K_{T_{\mathrm{loc}}}(u)\,du = (0.893255 - \mathcal{T}_N(T_{\mathrm{loc}})) - \mathcal{E}^{\mathrm{sm}}_{N,\mathrm{lead}}(T_{\mathrm{loc}})$; in particular, at $N = 50$ the fluctuation tail must contribute about 0.991335084
v3.8.25	local OP 5 pair-block state	2026-04-04	reorganized the pure fluctuation tail into adjacent photon-frame pair blocks $P_{N,j}(T_{\mathrm{loc}})$, yielding the exact pair formula controlled by the pair means $x_{N,j}$ and half-gaps $d_{N,j}$; this turns the next OP 5 attack into a mesoscopic resonance test rather than another smooth-density refinement
v3.8.26	local OP 5 quadratic-gap state	2026-04-04	showed that the exact adjacent pair-block profile is even in the half-gap d, so the linear gap term vanishes identically and the first nontrivial local-gap correction is quadratic; consequently the first mesoscopic OP 5 layer is mean-phase dominated rather than gap-noise dominated
v3.8.27	local OP 5 pair-mean quasi-lattice state	2026-04-04	showed that the adjacent pair means are well described by a slowly drifting density-controlled quasi-lattice, with observed step ratios near the prediction $2/\rho_{\mathrm{ph}}(x_{N,j})$ having mean 0.997634; thus the first mesoscopic OP 5 burden is slow phase drift rather than linear gap disorder
v3.8.28	local OP 5 four-block symmetry state	2026-04-04	grouped adjacent pair blocks into four-blocks and showed that the resulting super-gap dependence is even, so the linear pair-mean drift term vanishes identically; thus OP 5 now exhibits a two-stage mesoscopic suppression in which both the linear local-gap term and the linear pair-mean drift term are removed

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v3.8.29	local OP 5 super-mean quasi-lattice state	2026-04-04	showed that the four-block super-means themselves form a second-stage slowly drifting density-controlled quasi-lattice, with observed step ratios near the prediction $4/\rho_{\mathrm{ph}}(X_{N,k})$ having mean 0.996050; thus the current OP 5 burden is best read on a hierarchy of quasi-lattices rather than as a raw oscillatory tail
v3.8.30	local OP 5 quarter-turn super-mean state	2026-04-04	showed that the super-mean phases at the benchmark cutoff cycle through the four quarter-turn sectors in strict order for the first 16 super-means, while drifting on average by about -0.0652531 relative to the exact $\pi/2$ increment; thus the current OP 5 super-mean layer exhibits weak quarter-turn locking with slow slip
v3.8.31	local OP 5 quarter-cycle packet state	2026-04-04	showed that one full quarter-turn cycle of the super-mean layer, i.e. a 16-zero resonance packet, already exhibits strong cancellation: the first two packets leave only about 2.21% and 3.99% of their ℓ^1-mass, so the mesoscopic OP 5 burden is best read in terms of packet drift and inter-packet residue rather than isolated block errors
v3.8.32	local OP 5 packet-defect state	2026-04-04	derived the first-order packet-defect law, showing that once quarter-cycle cancellation is taken as the zeroth-order picture, the surviving OP 5 residue is carried only by slow packet amplitude drift and quarter-turn phase-slip, not by the ideal packet itself
v3.8.33	local OP 5 packet-envelope state	2026-04-04	compressed the first-order packet-defect variables into a single scalar packet-envelope $\mathfrak{D}_{N,m}$, proving that the packet residue is controlled by $\lvert R_{N,m}(T_{\mathrm{loc}})\rvert \leq \sqrt{2}\,\mathfrak{D}_{N,m} + O(\delta\varepsilon + \varepsilon^2)$; thus OP 5 is now a packet-envelope problem at first order
v3.8.34	local OP 5 observable-envelope state	2026-04-04	introduced the observable lower packet-envelope proxy $\underline{\mathfrak{D}}_{N,m} = \lvert R_{N,m}(T_{\mathrm{loc}})\rvert/\sqrt{2}$, giving benchmark values $\underline{\mathfrak{D}}_{50,1} \approx 0.00273964$, $\underline{\mathfrak{D}}_{50,2} \approx 0.00434351$, and $\underline{\mathfrak{D}}_{50,3} \approx 0.00921439$; this moves Step (B) onto a directly observable higher-scale scalar defect sequence

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v3.8.50	local merge-reconciliation checkpoint	2026-04-04	reconciled the former v3.5.2 abstract-polish checkpoint with the current v3.8.49 boundary-layer line; declared the 3.5.2 interface-normalization results fully absorbed into the active 3.8 working monolith and recorded the merge explicitly at the document head and in the main text
v3.8.51	local reproducible-figure state	2026-04-04	regenerated the missing Julia/probe comparison figure from a local Python renderer, embedded the resulting PDF asset as the primary document figure with PNG fallback, and recorded the rebuild path directly in the appendix discussion
v3.8.52	local current-scale figure state	2026-04-04	replaced the earlier standalone comparison marker by the current residual-scale diagnostic family $\gamma \in \{0, \Delta_\beta, \Delta_\theta, \Delta\mu\}$ and updated the appendix figure/caption accordingly
v3.8.53	local caption/spacing fix state	2026-04-04	repaired caption spacing for the current residual-scale probe figure and stabilized the appendix-page layout for the diagnostic render
v3.8.55	local deep-boundary diagnostic extension	2026-04-04	extended the Appendix-C diagnostics with a focused $\Delta\mu$ deep-zoom comparison, localized difference ladders, and edge-skeleton drift overlays to test whether the strongest current residual remains boundary-local under further magnification
v3.8.56	local continuation diagnostic extension	2026-04-04	added a six-step continuation test from 0 to $\Delta\mu$, including a boundary-belt strip, cumulative motion envelope, and horizon-profile continuation to check whether the active shell deforms continuously across the whole live residual interval
v3.8.57	local sensitivity-localization extension	2026-04-04	added finite-difference sensitivity maps, regional/radial sensitivity profiles, and a horizon-angle heatmap to show that the live residual response stays concentrated on the same horizon-adjacent boundary shell rather than spreading into the core

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v3.8.58	local horizon-sector persistence extension	2026-04-04	added coarse angular-sector persistence maps and dominant-peak tracking on the Möbius horizon, showing that the active response remains locked to a small conjugate sector family instead of diffusing around the full circle
v3.8.59	local OP 5 packet-wave diagnostics extension	2026-04-05	added visual diagnostics for the quarter-turn super-mean cycle, packet-level cancellation hierarchy, signed packet-defect wave fits, and the rotating packet-mode phase portrait; these figures make the current OP 5 reading visible as packet resonance plus slow signed wave rather than as an unstructured tail
v3.8.60	local OP 5 breathing-envelope visual extension	2026-04-05	added visual diagnostics for the transverse envelope trough, the slow secondary envelope wave, the breathing-helix interpretation of the packet mode, and the carrier/envelope ratio test; these figures make the current OP 5 reading visible as an axially dominated carrier with a slower breathing boundary-layer correction rather than as a rigid fixed-radius helix
v3.8.61	local OP 5 frame-flicker visual integration	2026-04-05	added live Appendix-C frame-flicker fit, decomposition, and geometric-readout figures and moved the new visual block into the actually compiled PDF path so that the moving-frame hypothesis becomes directly auditable in the active monolith
v3.9.0	Zenodo release	2026-04-05	10.5281/zenodo.19423689 — public release preserving the stabilized residual-record interface, the reproducible Möbius-shell diagnostic suite, and the OP 5 packet / breathing-envelope / frame-flicker visual line in one monolithic text
v3.10.0	Zenodo release	2026-04-06	10.5281/zenodo.19434163 — public release based on the v3.9.0 monolith, with version and DOI record updated for the present release and front-matter release language cleaned for Zenodo
v3.10.1	local working continuation	2026-04-06	front-cleaned working continuation of the public v3.10.0 release; historical roadmap and release-note blocks moved behind the nomenclature into an archival appendix position

Version / record	Status	Publication date	DOI / note
v3.10.2	local working tool-extraction state	2026-04-06	first explicit working tool layer for v3.11 preparation: transport-first, readout-after-transport, checkpoint-absorption, and epsilon-trap tools recorded together with a synchronized diagnosis of the live OP 1 and OP 5 burdens
v3.10.3	local working OP 1 epsilon-bridge state	2026-04-06	reframed the live OP 1 burden as epsilon-trapped bridge normalization: canonical closure is now read as entry into a contractive bridge tube around the distinguished fixed point rather than as a second branch search
v3.10.4	local working OP 1 theorem-promotion state	2026-04-06	promoted the OP 1 epsilon-bridge tool to theorem level by proving a residual-to-fixed-point estimate and a contractive non-escape theorem for the canonical bridge normalization inside the local bridge tube
v3.10.5	local working OP 1 rounded-entry state	2026-04-06	proved that the standard rounded matter-frame readout already enters the local contractive bridge tube via the tiny readout-point displacement $\lvert\gamma_L - \gamma_{\mathrm{br}}\rvert \approx 3.2\times10^{-8}$, reducing the live OP 1 burden to the structural identification $b_{\mathrm{can}} = b_{\mathrm{MF}}$ rather than to a second local bridge search
v3.10.6	local working OP 1 uniqueness promotion state	2026-04-06	promoted admissible-readout uniqueness to theorem level, removing the final structural ambiguity between canonical normalization and the standard matter-frame readout and reducing OP 1 to a single explicit local bridge residue
v3.10.7	local working shared-matrix diagnostic state	2026-04-06	added shared HC/MM/J leakage diagnostics and a structure-first translation of linear, differential, and linear-differential cryptanalysis for attacking the common OP 1/OP 5 matrix remainder
v3.10.8	local working shared-phase coupling diagnostic state	2026-04-06	added a shared-carrier / quadrature diagnostic for testing whether the remaining OP 1 and OP 5 residues are one in-phase mode or two phase-shifted readouts of the same hidden leakage carrier
v3.10.9	local working shared-leakage probe-family state	2026-04-06	replaced the abstract phase-coupling parameter by a concrete HC/MM/J cross-sector probe family $\mathcal{M}_\lambda = \mathcal{M} + \lambda\Delta_{\mathrm{shared}}$ and isolated its first-order effect on the visible OP 1/OP 5 channels

Version / record	Status	Publication date	DOI / note
v3.10.10	local working probe-invariant classification state	2026-04-06	introduced the normalized determinant and dot-product invariants of the HC/MM/J probe response matrix, turning the shared-carrier versus quadrature question into a decision-grade first-order diagnostic
v3.10.11	local working canonical low-rank leakage state	2026-04-06	fixed a canonical rank-two HC/MM/J leakage class and proved that its first-order response is necessarily one-mode rather than quadrature, so any later sinus/cosinus picture would have to be emergent rather than primitive
v3.10.12	local working bidirectional leakage-plane state	2026-04-06	added the minimal paired rank-four cross-sector leakage plane and proved that it is the first structurally honest location where a quadrature/sinus-cosinus picture can arise, whereas primitive rank-two leakage remains one-mode only
v3.10.13	local working visible-ellipse state	2026-04-06	coupled the minimal paired leakage plane to the OP 1/OP 5 mask layer and proved that the visible response is generically an ellipse, collapsing to a line in rank one and to exact circle/quadrature only under conformal mask geometry on the leakage plane
v3.10.14	local working shared-mask decomposition state	2026-04-06	decomposed the visible OP 1/OP 5 mask layer into a shared channel plus separation defect, proved singular-value bounds in the near-shared regime, and showed that exact quadrature cannot be primitive when the two visible masks differ only by a small downstream correction
v3.10.17	local working observer-split control state	2026-04-06	showed that the visible OP 1/OP 5 separation defect is exactly the restriction of the observer-side mask difference to the active leakage plane, proving that near-shared one-mode behavior is the default whenever the two visible masks differ only by a small downstream readout split

Version / record	Status	Publication date	DOI / note
v3.10.18	local working shared-residual dichotomy state	2026-04-06	decomposed the remaining primitive OP 1/OP 5 visible residual into a shared matrix-leakage carrier plus a downstream observer-shell split and proved that exact grouped no-drift forces any surviving primitive residual on the active leakage plane to be common-mode rather than primitive quadrature
v3.10.19	local working active matrix-fit state	2026-04-06	pushed the remaining primitive common-mode carrier back to the active HC/MM/J fit and proved that its size is controlled exactly by the active commutator $[\mathcal{M}, J]\|_U$ on the admissible branch
v3.10.20	local working exact-transport dichotomy state	2026-04-06	used the grouped-layer admissible-transport architecture to show that exact grouped Millennium transport kills the active commutator on the admissible branch, yielding a clean dichotomy between exact primitive closure and a purely downstream observer-shell residual
v3.10.22	local working canonical observer-shell residual state	2026-04-06	promoted the remaining visible OP 1/OP 5 term, when primitive matrix leakage is absent, to a canonical downstream observer-shell residual carried by a common shell dressing mode rather than by a primitive HC/MM/J leakage mode
v3.10.23	local working canonical shell-law modeling state	2026-04-06	modeled the surviving shared OP 1/OP 5 shell mode after exact transport as a first-order phase-shifted / expansion-dressed observer-shell law, reducing the visible target to exact vanishing versus canonical post-transport shell identification
v3.10.24	local working shell-dominance classification state	2026-04-06	compressed the remaining visible post-transport question to a comparison of first-order shell loads, classifying the shared shell mode as phase-dominant, expansion-dominant, or genuinely mixed at first order
v3.10.25	local working shell-cancellation corridor state	2026-04-06	reduced the shared shell mode to a signed first-order shell-balance mismatch, isolating the cancellation corridor for exact visible closure and showing that same-sign shell loading forces a canonical nonzero shell remainder unless the linear shell load itself vanishes

Version / record	Status	Publication date	DOI / note
v3.10.26	local working normalized shell-defect state	2026-04-06	compressed the shared observer-shell problem to a single normalized shell-cancellation defect, extracted an exact cancellation-depth invariant, and proved that exact visible closure requires the normalized first-order shell mismatch to be no larger than the normalized higher-order shell remainder
v3.10.28	local working sign-corridor shell state	2026-04-06	converted the normalized shell verdict into a sharp sign/corridor dichotomy, proved that sign-coherent first-order shell loads force canonical nonzero visible residual whenever the normalized higher-order rescue budget is below one, and extracted the exact anti-phase corridor required for possible visible closure
v3.10.29	local working shell-conjugacy gain state	2026-04-06	pushed the induced phase and expansion gains back to a shell-conjugate observer geometry, introduced the downstream shell involution and transport-balance defect, and showed that the anti-phase corridor is structurally natural whenever phase and expansion are J-descended conjugate readouts of one shell displacement with nearly balanced transport weights
v3.10.30	local working shell anti-phase budget state	2026-04-06	compressed the anti-phase corridor to a single normalized shell-conjugacy load and showed that a small total shell load forces the induced shell ratio into a natural anti-phase tube around -1
v3.10.31	local working architecture-driven shell state	2026-04-06	pushed the shell-balance driver back to a combined architecture budget given by active HC/MM/J fit plus observer-shell dressing, so that first-order visible cancellation is now read as architecture-driven rather than phenomenological
v3.10.32	local working architecture-rescue corridor state	2026-04-06	reduced the remaining visible question to a single budget comparison between the normalized combined architecture load and the normalized higher-order rescue budget, giving a final first-order corridor for exact visible closure after exact transport

Version / record	Status	Publication date	DOI / note
v3.11.0	Zenodo release	2026-04-06	10.5281/zenodo.19444128 — public release consolidating the v3.10.32 architecture-rescue corridor state, explicit tool-layer extraction, and release-normalized front matter for Zenodo
v3.11.2	local working state	2026-04-07	OP 1/OP 5 closed as a native source-return pair via closure-class preservation; branchwise refinement, admissible routed aggregation, and the final visible-support mismatch decomposition added as the compact main-line hardening beyond the public v3.11.0 release
v3.12.0	Zenodo release	2026-04-07	10.5281/zenodo.19446264 — public release promoting the OP 1/OP 5 closure line into the public-facing main text, with branchwise refinement and routed-readout consequences integrated into the monolithic Companion
v0.1.0	Zenodo release	2026-03-24	10.5281/zenodo.19210729
v0.3.0	Zenodo release	2026-03-24	10.5281/zenodo.19211785
v0.4.0	Zenodo release	2026-03-24	10.5281/zenodo.19211798
v0.5.0	Zenodo release	2026-03-24	10.5281/zenodo.19211877
v0.6.0	Zenodo release	2026-03-24	10.5281/zenodo.19212073
v0.7.0	Zenodo release	2026-03-24	10.5281/zenodo.19212212
v0.8.0	Zenodo release	2026-03-24	10.5281/zenodo.19212324
v0.9.0	Zenodo release	2026-03-24	10.5281/zenodo.19220003
v1.0.0	Zenodo release	2026-03-24	10.5281/zenodo.19220118
v1.6.0	Zenodo release	2026-03-25	10.5281/zenodo.19221335
v1.7.0	Zenodo release	2026-03-25	10.5281/zenodo.19221372
v1.8.0	Zenodo release	2026-03-25	10.5281/zenodo.19221431
v1.9.0	Zenodo release	2026-03-25	10.5281/zenodo.19221473
v2.0.0	Zenodo release	2026-03-25	10.5281/zenodo.19221527
v2.1.0	Zenodo release	2026-03-25	10.5281/zenodo.19221618
v2.3.0	Zenodo release	2026-03-25	10.5281/zenodo.19221864
v2.4.0	Zenodo release	2026-03-25	10.5281/zenodo.19221954
v2.9.0	Zenodo release	2026-03-25	10.5281/zenodo.19222638
v2.10.0	Zenodo release	2026-03-25	10.5281/zenodo.19222894
v2.11.0	Zenodo release	2026-03-25	10.5281/zenodo.19223084
v2.11.1	local-only repair state	2026-03-25	non-Zenodo version-fix state between v2.11.0 and v2.11.2
v2.11.2	Zenodo release	2026-03-25	10.5281/zenodo.19223365
v2.11.3	Zenodo release	2026-03-25	10.5281/zenodo.19223515
v2.12.0	Zenodo release	2026-03-25	10.5281/zenodo.19223782
v2.13.0	Zenodo release	2026-03-25	10.5281/zenodo.19223968

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v2.14.0	Zenodo release	2026-03-25	10.5281/zenodo.19225191
v2.15.0	Zenodo release	2026-03-25	10.5281/zenodo.19226408
v2.16.0	Zenodo release	2026-03-25	10.5281/zenodo.19226445
v2.17.0	Zenodo release	2026-03-25	10.5281/zenodo.19226497
v2.18.0	Zenodo release	2026-03-25	10.5281/zenodo.19226549
v2.19.0	Zenodo release	2026-03-25	10.5281/zenodo.19226773
v2.20.0	Zenodo release	2026-03-26	10.5281/zenodo.19227889
v2.21.0	Zenodo release	2026-03-26	10.5281/zenodo.19227940
v2.22.0	Zenodo release	2026-03-26	10.5281/zenodo.19228061
v2.23.0	Zenodo release	2026-03-26	10.5281/zenodo.19239228
v2.23.1	Zenodo patch release	2026-03-26	10.5281/zenodo.19239967
v2.24.0	Zenodo release	2026-03-27	10.5281/zenodo.19258774
v2.25.0	Zenodo release	2026-03-27	10.5281/zenodo.19259425
v2.25.1	Zenodo patch release	2026-03-27	10.5281/zenodo.19262890
v2.27.0	Zenodo release	2026-03-27	10.5281/zenodo.19266666
v2.27.1	local-only repair state	2026-03-28	baseline-preserving post-release working state
v2.27.2	local-only repair state	2026-03-28	normalization and proof-chain hardening
v2.27.3	local-only repair state	2026-03-28	structural closure pass; last state before v2.27.4 integration
v2.27.4	local working state	2026-03-28	spectral-projector layer ($C = \Pi_{\mathrm{spec}}^{(\mathcal{M})}$), readout-relative lemma methodology, Theorem 7 (HC-induced readout geometry), Hypotheses 7.4–7.6, bidirectional probe lensing, SM spectral realization, regime-test appendix
v2.27.5	Zenodo release	2026-03-28	10.5281/zenodo.19297422 — full proof-chain audit; closure roadmap; 35 missing labels; five Priority-A gaps identified
v2.27.8	local working state	2026-03-28	B1+B2 closed: `prop:semantic-forcing-chain` explicit 3-part proof from axioms (S1)–(S8); `prop:structural-lift-principle` non-circular via `lem:group-transport` + `lem:hc-admissibility-lemma`; OP 2 formalized; all 7 Roadmap items closed

Version / record	**Status**	**Publication date**	**DOI / note**
v2.27.7	local working state	2026-03-28	A4+A5 closed: `thm:koide-triolic` with θ_K formula; 0.111% match and 115× amplification remark; `thm:bsym-hopf` with proof and Hopf-gap remark; 4.9%/6.8%, OP1/OP2 coupling; open: B1, B2
v2.28.0	Zenodo release	2026-03-28	10.5281/zenodo.19298586; consolidation of v2.27.4–v2.27.8; spectral-projector unconditional; bijection lemma; SM realization; readout-relative methodology; Thm 7; Hyp 7.4–7.6; OP 2 stub
v2.29.0	Zenodo release	2026-03-28	10.5281/zenodo.19299245 — OP 5 step (A) closed; step (B) quantified; renormalization explicit
v2.29.1	local working state	2026-03-28	GOE/GUE split (`prop:goe-fermionic`); γ_L = GOE/GUE ratio; $\gamma_L = 1.197$ corrected to 1.864; χ^2 improvement 75.8%; δ_{CL} exponent $\gamma_L^2 \to \gamma_L^5$; `thm:dual-origin` corrected
v2.29.2	Zenodo release	2026-03-28	10.5281/zenodo.19300351 — Einstein blunder reframed; $\Omega_M = \beta^2_{\mathrm{sym}}$ determination; three-gap convergence
v2.29.3	Zenodo release	2026-03-28	10.5281/zenodo.19300813 — OP 2 (T1)(T2) closed; $D_- = L_0\big\vert_{H^+_-}$ (corrected candidate); GOE essential self-adjointness; T4=OP1 equivalence; **Tribonacci theorem** $\gamma_L \to \mathbb{T}$; cascade normalization $1/\mathbb{T} + 1/\mathbb{T}^2 + 1/\mathbb{T}^3 = 1$

Note. The sequence above records the currently preserved release trail. The day-level publication dates are normalized from the project publication log so that Zenodo releases and local repair states remain chronologically comparable inside a single appendix table.

v2.26.0 computational consolidation note

Version **2.26.0** executes the consolidation roadmap that had been reserved in v2.25.1. The computational strand is retained but internally reorganized into a more coherent progression: historical perspective, structural machine reading, Triolen Algebra / Boolean readout, MML–MMA–MCP stratification, proof templates, certificate-bearing execution, normalization, addressing, scheduling, host export, and extraction outlook. The present version still remains exploratory in the sense of future formalization, but the material is no longer presented only as a loose teaser; it is now collected as a first internally coherent computational chapter of the Companion.

Hypothesis-control pass for v2.27.0

Before the next public milestone, the mathematical layer should distinguish more sharply between statements that are expected to become theorem-grade in v2.27.0 and statements that should remain explicitly hypothesis-grade even after the next hardening cycle. The practical rule is conservative: only statements with stabilized definitions, closed dependency chains, and reference-stable support should move upward from hypothesis status into theorem, proposition, or lemma status. Statements that continue to depend on interpretive transport, physical reading, or still-open structural identifications should remain explicitly marked as hypotheses or outlook items.

For the next pass, the intended control logic is therefore blockwise. In the ground-condition / HC / confinement chain, the target is theorem-readiness whenever the dependence on the collapse picture and confinement mechanism can be formulated without appealing to deferred computational scaffolding. In projector / closure mathematics, the target is theorem or lemma status whenever the relevant admissibility and sector-preservation claims are already internal to the mathematical layer. In the transport / Lorentz layer, statements that are still partly interpretive should remain hypothesis-grade unless they can be rewritten as strictly mathematical transport statements. For the Millennium-matrix and trace/branch layers, the default stance remains more cautious: structural mathematical claims may harden, but computational or machine-reading interpretations remain secondary and should not silently inherit theorem status from the mathematical core.

The resulting editorial discipline for v2.27.0 is simple: theorem promotion requires closure, dependency control, and stable notation; everything else remains visibly hypothesis, deferred claim, or teaser architecture. This avoids theorem-to-hypothesis drift in reverse as well: once a statement is kept at hypothesis level, later references should not read it as already mathematically closed merely because it is computationally suggestive.

v2.26.33 local mathematics-first note

Version **2.27.0** does not yet claim a new public release beyond v2.26.0. Instead, it records the explicit shift of near-term priority back to mathematical completion. The computational chapter remains preserved as a secondary but already visible reserve architecture. The immediate next target is v2.27.0 as a mathematics-first hardening cycle: close the mathematical layer, collect tools as mathematical appendices, and only then resume the proof-engine line as a compression language for already stabilized structure.

Theorem-promotion checklist for v2.27.0

The next public milestone should not promote statements merely because they have become familiar in the internal workflow. Each promotion from hypothesis-language into lemma-, proposition-, or theorem-language must therefore be explicit, local, and reference-stable. For the mathematical closure pass leading to v2.27.0, the promotion discipline is:

- statements about the ground condition to HC to confinement chain may be promoted only when the chain has been written in one direction without hidden interpretive jumps;
- projector/closure claims may be promoted only when the admissibility conditions and the closure domain are fixed in stable notation;
- transport/Lorentz statements may be promoted only when they are phrased as mathematical transport claims rather than as physical metaphor;

- Millennium-matrix statements may be promoted only when the matrix is treated first as a mathematical object with domain, codomain, and structural role, not merely as a computational image;
- trace/branch claims may be promoted only when the branch criteria and trace-compatibility conditions are stated in theorem-capable form.

In parallel, several statement families are to remain explicitly outside theorem-promotion for v2.27.0 unless a full closure pass is actually completed. These include broad computational-universality claims, strong proof-kernel claims, speculative host-language claims, and any physically interpretive reading that still rests on heuristic transport language rather than closed mathematical formulation.

A practical promotion checklist for the next pass is therefore:

1. identify candidate statements that are currently written as heuristic summary sentences;
2. decide whether each candidate should become a Definition, Lemma, Proposition, or remain a Hypothesis;
3. verify that every promoted statement has stable notation, explicit assumptions, and stable backward references;
4. verify that no promoted statement depends silently on teaser/computational language;
5. record all non-promoted but still important claims explicitly as open, deferred, or interpretive.

This checklist is intended to keep the next public release mathematically harder without forcing premature promotion of material that should still remain hypothesis-level.

Mathematical freeze criterion for v2.27.0

For the next public release, a mathematical core block should count as *frozen for release* only when it is not merely well phrased, but also stable under the local closure discipline introduced in the present subversion sequence. In particular, a block is to be treated as mathematically release-ready for v2.27.0 only if all of the following are satisfied:

1. the principal objects of the block are defined in stable notation and do not drift between theorem-language and hypothesis-language;
2. the block depends only on explicitly cited prior definitions, lemmas, propositions, or tools, rather than on narrative intuition or roadmap placeholders;
3. the key admissibility conditions are written inside the statement or proof scaffold itself;
4. all required cross-references are stable and do not point to provisional teaser/computational material;
5. the block can be read both locally and in the dependency order recorded for the mathematics-first pass without hidden backward jumps;
6. any remaining interpretive or physical reading is clearly separated from the mathematical closure claim.

A block should *not* be counted as frozen merely because its wording is polished, because it already appears in summary form elsewhere, or because it has an attractive structural interpretation. For v2.27.0, the intended release criterion is therefore conservative: closure requires definitional stability, dependency stability, reference stability, and status stability at the same time.

A practical freeze test for each A-priority block is:

- can the block be pointed to by a stable label and theorem-status marker?
- can its assumptions be listed in one compact place?
- can its proof or justification be read without consulting deferred computational scaffolds?
- can its use in later blocks be stated without introducing new notation drift?

If the answer to any of these questions remains negative, the block should remain marked as active work rather than as mathematically frozen for public release. This criterion is meant to prevent premature hardening of material that is already suggestive but not yet sufficiently closed.

Release-readiness matrix for v2.27.0

To prevent the next public release from being driven by impression rather than closure, the mathematics-first pass should end in a compact readiness matrix for the A-priority blocks. The matrix is not meant to replace proofs; it is meant to summarize, in one release-facing place, whether a core block is genuinely ready to be frozen, or whether a single missing condition still blocks public hardening.

Core block	Current state	Freeze-ready?	Missing closure condition	Primary blocker for v2.27.0
Ground condition / collapse picture	structurally central, still partly summary-driven	no	one stable definitional pass with fixed assumptions	notation and statement drift
HC emergence / confinement chain	strongly anchored, not yet written in one compact theorem-capable line	partial	explicit one-directional closure chain	hidden interpretive compression
Projector / closure mathematics	partially stabilized and reusable	partial	admissibility conditions and closure domain fixed in one notation regime	local definition spread
Transport / Lorentz layer	present but mixed with interpretive language	no	transport claims restated as mathematical transport statements	metaphor drift
Millennium matrix as mathematical object	increasingly structured, still mixed with computational imagery	no	fixed domain/codomain/operator role	object-status not yet fully hardened
Trace / branch mathematics	structurally rich, still partly template-style	partial	stable branch criterion and trace compatibility form	theorem-status not yet fixed
Tool / appendix lemmas	abundant support layer	partial	priority ordering and stable dependence tags	appendix support not yet fully normalized

For the next public milestone, the conservative reading of this matrix is intentional: *partial* does not yet count as release-ready. A block should only be treated as ready for v2.27.0 if its missing closure condition has been discharged explicitly and its blocker is no longer active. In this sense, the matrix is a release-control device rather than a rhetorical summary: it translates the mathematics-first agenda into a blockwise freeze decision surface.

v2.27.0 gate checklist

To prevent the next public release from being triggered by momentum alone, the mathematics-first pass should culminate in a compact global gate checklist. This checklist is stricter than the blockwise readiness matrix: even if several A-priority blocks look individually advanced, the public jump to v2.27.0 should only be made once the release conditions hold *together* at the document level.

- the A-priority mathematical blocks admit stable definitions, stable dependence order, and stable theorem-status markers;
- no public-facing core claim in the targeted v2.27.0 range still depends on deferred computational scaffolds for its mathematical reading;

- the blockwise freeze criteria and the release-readiness matrix no longer contain an active blocker in the A-priority line;
- the proof-dependency checklist and the reference-stability pass agree on a single release-safe label structure;
- hypothesis-control and theorem-promotion discipline are aligned, so that no statement is promoted merely because its interpretation is attractive;
- the transport / projector / Millennium-matrix line can be cited without notation drift or mixed theorem-versus-hypothesis status;
- appendix tools used by the A-priority line are stable enough to serve as support, but not so overloaded that they silently replace missing core proofs;
- the computational / MML / MMA / MCP chapter remains visibly present as teaser architecture, but no release-critical mathematical closure claim depends on it;
- the version record, roadmap, local subversion discipline, and closure-control blocks all point to the same public target without internal contradiction.

A practical release rule for the next public jump is therefore conservative: *v2.27.0 should only be cut once the A-line is simultaneously blockwise closure-ready and globally gate-clean.* If one of these conditions still fails, the correct response is another local subversion rather than a premature public release. In this sense, the gate checklist converts the mathematics-first roadmap into a final version-control criterion.

Final pre-v2.27.0 control summary

As the mathematics-first pass approaches a public release decision, the Companion should end with a compact control summary that can be read in one glance. The purpose of this summary is not to replace the detailed work matrix, target-state matrix, or release-readiness table, but to compress them into a final release-facing snapshot. In particular, the summary should answer five questions simultaneously: (i) which A-priority blocks are currently the decisive release drivers, (ii) whether each of them is already freeze-ready, (iii) what the leading blocker still is, (iv) whether the global gate checklist is currently satisfied, and (v) whether the recommended action is *continue mathematical closure* or *prepare public release.*

A-priority block	Freeze-ready?	Main blocker	Gate status impact	Release recommendation
Ground condition / collapse picture	not yet	dependence and formulation hardening	blocks full gate closure	continue mathematical closure
HC emergence / confinement chain	partial	explicit mathematical discharge	blocks full gate closure	continue mathematical closure
Projector / closure mathematics	partial	stable theorem-level closure	blocks full gate closure	continue mathematical closure
Transport / Lorentz layer	conditional	transport formalization discipline	weakens gate confidence	continue mathematical closure
Millennium matrix as mathematical object	not yet	object-level hardening before interpretive use	blocks full gate closure	continue mathematical closure
Trace / branch mathematics	partial	stable dependence on prior blocks	blocks full gate closure	continue mathematical closure

At the present stage, the control summary therefore recommends *no public jump yet beyond the current release line*: the mathematics-first agenda should remain active until the decisive A-priority blocks no longer fail the freeze criterion and the global gate checklist turns from "not yet" to "satisfied." The intended role of this final control summary is practical rather than rhetorical: it gives the next release decision a single consolidated dashboard and makes it harder to confuse local progress with document-level readiness.

One-page executive closure view

The mathematics-first steering logic introduced in the recent local subversions is now dense enough that it should also be compressible into a single executive page. The purpose of this closure view is therefore not to add another planning layer, but to place roadmap, dependency map, target-state matrix, freeze criterion, gate checklist, and pre-release control summary onto one final compact surface. In practical terms, the page should allow a reader to answer four questions immediately: (i) what the next public version is really trying to close mathematically, (ii) which A-priority blocks still drive the release boundary, (iii) whether the document is globally gate-clean, and (iv) whether the correct next action is another local subversion or a public release jump.

Control layer	Main question	Current status	Operational consequence
Roadmap / work matrix	what must be closed first?	mathematics-first line active	keep proof-engine / MML line secondary
Dependency map	what must come before what?	ordered core chain available	no closure claims out of dependency order
Definition / notation pass	are the objects and labels stable?	partially stabilized	unify notation before theorem-promotion
Target-state / closure checklist	what counts as done?	blockwise closure conditions visible	only explicit closure counts as completion
Freeze criterion / readiness matrix	is a block release-ready?	mixed, mostly partial	keep A-line blocks active until blocker-free
Gate checklist / control summary	is the document globally ready?	not yet gate-clean	continue local subversions, no public jump yet

In this condensed reading, the present Companion state remains productive but intentionally conservative: the mathematical architecture is now sufficiently organized to support disciplined closure work, but not yet sufficiently gate-clean to justify confusing local progress with public readiness. The executive closure view is therefore meant to function as a one-page release dashboard. Its decision rule is simple: as long as the A-priority mathematical blocks remain only partially frozen, and as long as the global gate still records an active blocker, the correct action is *another explicit local subversion rather than a public version jump.* Once these conditions disappear simultaneously, the dashboard should flip from *continue closure* to *prepare the next public release.*

Release-vs-local decision rule

A final practical distinction should also be made explicit before the next public jump: not every coherent local gain justifies a new Zenodo-facing version. The present Companion workflow therefore distinguishes between *local closure iterations* and *public release transitions.* A local subversion is appropriate whenever the document gains planning discipline, dependency clarity, notation stability, or blockwise closure information without yet removing the main A-line blockers. A public release transition is appropriate only when these local gains have accumulated into a document-wide state in which the mathematical core is not merely better organized, but sufficiently frozen, referentially stable, and gate-clean to support a new public boundary.

In operational terms, the decision rule may be summarized as follows. Move only by local subversion if one or more of the following remains true: (i) an A-priority block is still only partially closed, (ii) the theorem/hypothesis boundary is still under active normalization, (iii) the reference-stability pass is not yet quiet, (iv) the release-readiness matrix still lists an active primary blocker, or (v) the global gate checklist is not yet satisfied. Consider a public version jump only when all A-priority blocks satisfy their freeze criterion simultaneously, the dependency chain is stable under rereading, the hypothesis-control and theorem-promotion passes no longer produce status drift, and the release-control summary flips from *continue closure* to *prepare release.* This distinction is intentionally conservative: it protects the mathematical layer from being mistaken for release-ready merely because the planning architecture around it has become more refined.

Mathematics-first release doctrine

The present local stage adopts a simple doctrine for all remaining pre-v2.27.0 work. First, every local change is to be judged by whether it closes, stabilizes, or clarifies a mathematical dependency in the A-priority chain; additions that merely decorate the proof-engine teaser without tightening the mathematics are therefore secondary by rule. Second, no new public release is justified merely by chapter growth or architectural enthusiasm: the release threshold remains mathematical closure, reference stability, and theorem-safe status control. Third, the computational/MML/MMA/MCP layer is to remain visible and preserved, but it is not permitted to become a hidden prerequisite for mathematical claims that should stand on their own. Fourth, every local subversion should remain append-only and auditable, so that recovery from drift or accidental rollback stays possible without reconstructing the chain from memory.

In operational terms, the doctrine reads as follows:

(i) prioritize closure of ground condition → HC → confinement before expanding secondary architecture;

(ii) promote statements only when notation, dependencies, and reference anchors are already stable;

(iii) treat tools, appendices, and computational reserves as support layers unless and until they are explicitly absorbed into the mathematical core;

(iv) prefer local subversions over public version jumps whenever the work mainly sharpens internal control rather than closing a release-grade mathematical block.

This doctrine is intentionally conservative. Its role is to keep the Companion aligned with the structure-first programme: mathematics first, proof-memory preserved, computational architecture retained as a prepared secondary line, and public releases tied to genuine closure rather than mere accumulation.

Mathematics-first action queue

The next local working order is to be treated as explicit and sequential unless a later control note states otherwise. The active queue is: (1) close the ground-condition / collapse picture at the definition level; (2) harden the HC-emergence / confinement chain so that its dependency status is stable; (3) close the projector / closure layer with stable admissibility language; (4) stabilize the transport / Lorentz layer only after the closure prerequisites are fixed; (5) treat the Millennium matrix strictly as a mathematical object before any renewed computational extension; (6) move trace / branch mathematics only after the preceding transport layer is stable; and (7) keep appendix tools and the computational teaser chapter in support position unless a direct mathematical blocker requires them.

The negative rule is equally important. The next local subversions are not to prioritize new MML/MMA/MCP growth, new proof-engine macros, or broader computational rhetoric unless such additions are directly needed to preserve an already existing mathematical dependency. In this sense, the queue functions as an anti-drift device: it converts the mathematics-first doctrine into a concrete order of execution and makes explicit what should not be pulled forward in parallel.

Mathematics-first session protocol. To prevent local growth from drifting back into attractive but premature architecture work, the next internal work sessions should follow a short fixed protocol. Each session should begin by choosing exactly one A-priority mathematical block from the current action queue. The session should then record (i) the active block, (ii) the minimal closure target for that block, (iii) the exact dependencies that must already be stable,

and (iv) the specific theorem/lemma/definition/tool boundary that is being touched. Only after this boundary is explicit should any local rewriting begin.

The execution rule for such sessions is intentionally narrow. First perform a definition pass for the chosen block; then a dependency pass; then a closure pass; then a reference-stability check; and only at the end, if necessary, a supporting appendix-tool pass. If a prospective edit does not directly improve the active mathematical block or its immediate dependencies, it should be deferred. In particular, new computational language material, proof-engine rhetoric, or architecture expansion should not be introduced during a session unless the active mathematical block explicitly depends on it.

At the end of each local session, the result should be classified in one of four ways: *closed for now*, *partially hardened*, *dependency blocked*, or *still open*. This classification should then feed back into the release-vs-local decision rule, so that local subversions remain true mathematical work logs rather than merely growing source snapshots. In this way, the present protocol operationalizes the mathematics-first doctrine at the level of day-to-day document evolution.

Anti-parallelization rule

The present mathematics-first stage also requires an explicit rule against parallel drift. In the next local subversions, not more than one A-priority mathematical block should be treated as the primary closure target of a given pass. Supporting edits in neighboring blocks are allowed only insofar as they remove an immediate dependency or stabilize a reference chain for that active block. What should be avoided is simultaneous advancement of several major fronts at once—for example projector/closure mathematics, transport/Lorentz normalization, and Millennium-matrix formalization in the same pass—because this makes it too easy to confuse distributed motion with genuine closure.

The same rule applies across the mathematics/computational boundary. While the computational teaser chapter remains preserved, it should not be opened in parallel with a mathematical A-block unless a concrete blocker in that block explicitly requires it. In practical terms, the anti-parallelization rule means: one primary closure front per pass; all secondary edits must justify themselves as dependency support, reference stabilization, or notation repair for that front. This rule is intentionally restrictive, because the present problem is not lack of ideas but risk of dispersion. Its purpose is to convert the mathematics-first doctrine into a narrow execution discipline and thereby make local progress more legible, auditable, and release-relevant.

12.1 Re-entry rule after each local mathematical session

After each local mathematics-first session, the document should be re-entered through a short global control loop rather than by immediately opening a new frontier. The purpose of this rule is to prevent hidden drift, duplicated partial closure, or silent reclassification of still-open material. In practical terms, the re-entry pass should answer only four questions: (i) did the session close the targeted A-block for now, (ii) which dependencies were clarified or newly exposed, (iii) which references and labels were changed and therefore need stabilization, and (iv) whether the next session should continue the same block or move to the next dependency in the queue.

The re-entry rule therefore imposes the following order: first update the local block status (*closed for now*, *partially hardened*, *dependency blocked*, or *still open*); second reconcile the result with the target-state matrix and release-readiness matrix; third update the proof-dependency checklist if new requirements were uncovered; and only then decide whether the next local subversion should continue the same mathematical block or step forward in the mathematics-first action queue. This rule is intentionally conservative: local progress is only counted as durable progress when it is reinserted into the global control architecture without breaking reference stability, hypothesis control, or the release-vs-local decision rule.

Session-close audit rule

Each local mathematics-first session should end with a short audit pass before the next subversion is declared. This audit is intended to prevent apparent progress from being recorded merely at the level of added text. A local pass counts as a meaningful increment only if the targeted block has been re-checked against its definition status, dependency status, reference stability, and release-readiness classification.

The closing audit for a local session should therefore answer four questions explicitly: (i) was a definition actually stabilized, rather than only rephrased; (ii) was at least one dependency clarified, reduced, or made more explicit; (iii) were the relevant references and labels left in a more stable state than before; and (iv) did the session produce a genuine closure gain for the active block, rather than only a planning gain. If these questions cannot be answered positively, the session may still be useful, but it should be recorded as preparatory rather than as a closure increment.

In this sense, the session-close audit forms the final control layer of the mathematics-first protocol. It ensures that the subversion chain remains readable as a chain of actual closure work, not merely as a sequence of appended managerial notes.

Carry-forward rule between local mathematical sessions

Not every item touched in one local mathematics-first session is allowed to be carried forward as if it were already stable. The present carry-forward rule therefore distinguishes between (i) content that is eligible to be inherited by the next local subversion as a working baseline, and (ii) content that must be carried forward only as still-open material. This distinction is intended to prevent partial rewrites, provisional notation choices, or merely suggestive closure claims from silently hardening through repetition alone.

A block element may be carried forward as part of the active baseline only if all of the following are already true: its local definition is stable enough not to be rewritten immediately in the next pass; its dependency position is unchanged or clarified; its reference anchors survive rereading without ambiguity; and its status classification does not rely on a teaser, soft interpretation, or unresolved supporting note. By contrast, material should be carried forward only as *open*, *dependency blocked*, or *provisional* whenever one of these conditions still fails. In practical terms, this means that unresolved notation, unstable lemma wording, open dependency edges, or uncertain theorem-promotion status must remain visibly marked rather than being treated as already absorbed into the mathematical baseline.

The purpose of this rule is conservative continuity. It allows the next local subversion to inherit genuine closure gains, while forcing incomplete work to remain visibly incomplete. In this way, the carry-forward rule complements the session-close audit: the audit decides whether a session produced a real gain, and the carry-forward rule decides which parts of that gain are allowed to function as stable starting points for the next mathematics-first pass.

Regression-check rule for local mathematical continuity

Because the present Companion has already experienced version drift and accidental rollback in earlier phases, the mathematics-first regime also requires an explicit regression-check rule. The purpose of this rule is not to slow down productive local work, but to ensure that no apparent gain in planning, notation, or closure discipline is purchased at the cost of silently losing already integrated mathematical material. In particular, each local subversion should be checked not only for what has been added, but also for whether any previously stabilized block, appendix tool, roadmap control layer, or restored support material has shrunk, disappeared, or reverted in status.

Operationally, the regression check is conservative and document-internal. After each local session, the current source should be reread against the immediately preceding local subversion with at least four questions in mind: (i) has the active A-priority block gained real closure content without losing prior dependencies, (ii) have restored support structures such as the table of contents, version chronology, or appendix anchors remained intact, (iii) have any local control blocks been duplicated, displaced, or pushed outside the effective document body, and (iv) has any internal version or bibliographic marker fallen behind the actual subversion state. If one of these checks fails, the correct response is not to treat the current file as progressive merely because it is newer, but to repair the regression before carrying the pass forward.

In this sense, the regression-check rule complements the carry-forward rule and the session-close audit. The audit asks whether the current session produced a real local gain; the carry-forward rule asks what may legitimately be inherited as stable baseline material; and the regression-check rule asks whether that inheritance still contains everything that was already secured before. The combined effect is intentionally protective: it turns local subversions into auditable closure steps rather than a chain of potentially drifting snapshots. A local mathematics-first version should therefore be regarded as acceptable only if it is both *additively improved* and *non-regressive* relative to the immediately preceding baseline.

Baseline-lock rule for local continuation

To prevent the mathematics-first subversion chain from drifting across multiple informal starting points, the present Companion adopts an explicit baseline-lock rule. For each local pass, exactly one immediately preceding local version is treated as the binding comparison basis. All claims of progress, non-regression, and closure gain are measured against this locked baseline rather than against a vague memory of the recent editing sequence. The purpose of the rule is simple: a local file becomes meaningful as a closure step only if it can be shown to improve upon one clearly identified predecessor without sacrificing already secured material.

Operationally, a baseline may be replaced only under conservative conditions. A newly produced local subversion may become the next locked baseline if, and only if, it satisfies the regression check, preserves the restored support structures, carries forward only legitimately stable material, and moves the active mathematics block measurably closer to its stated closure target. If one of these conditions fails, the correct response is not to redefine the newer file as the new norm, but to keep the earlier baseline locked until the defect has been repaired. In particular, mere growth in page count, roadmap language, or appended managerial notes is not sufficient to justify baseline replacement.

This rule complements the anti-parallelization, re-entry, session-close, carry-forward, and regression rules by providing a stable anchor for comparison. The anti-parallelization rule limits what may be opened at once; the re-entry rule controls how local gains are fed back into the larger roadmap; the session-close audit judges whether a session produced a real gain; the carry-forward rule determines what may be inherited as stable; the regression check guards against silent loss; and the baseline-lock rule finally determines when the current subversion has become stable enough to function as the new working ground for the next mathematics-first step.

Change-scope rule for local mathematics-first sessions

The mathematics-first regime also requires an explicit rule about *scope*. In any given local session, not everything that appears editable should actually be edited. The active scope is restricted to the currently selected A-priority mathematical block, its immediate dependency edges, the notation and references required to stabilize that block, and only those appendix tools that are strictly necessary to support the same closure pass. This means that many tempting modifications—however reasonable in isolation—should remain out of scope if they do not directly

improve the currently active closure target.

The practical purpose of the change-scope rule is to protect mathematical closure from diffuse accretion. A local session may therefore adjust wording, dependencies, notation, labels, or supporting appendices only when these changes can be justified as direct support for the active block. What remains explicitly out of scope includes broad computational expansion, speculative theorem promotion outside the active dependency cone, large stylistic rewrites of unrelated sections, and opportunistic edits that merely make the document longer without bringing the selected block closer to a frozen mathematical state. In this way, the change-scope rule complements the anti-parallelization rule: anti-parallelization limits how many fronts may be open at once, while scope control limits how wide a single front is allowed to spread.

2H. Global pre-release reconciliation for v2.27.0

With the first A-block now locally frozen through Lemma C–G and Proposition H–J, the remaining question is no longer whether the dependency spine exists, but whether this local gain has been fed back consistently into the document-wide release logic. The present pre-release reconciliation therefore records the first A-block not merely as locally strengthened, but as positively reconciled with the mathematics-first gate architecture for the next public release. In particular, the block is now to be read as structurally closed enough that its remaining limitation is no longer missing architecture, but only moderate proof density and future optional unfolding.

Operationally this means three things. First, the block may already count as a positive contribution to the global v2.27.0 gate, because it now satisfies the internal chain ground condition o HC o closure o projector stabilization o admissible transport o matrix realization o trace/branch persistence. Second, no computational teaser layer is needed in order to maintain its current mathematical status. Third, any later expansion of this block should be treated as proof-density enrichment rather than as a repair of a still-broken logical spine. The correct release reading is therefore conservative but positive: the first A-block is globally reconciled with the current mathematics-first release doctrine and should no longer be counted among the primary blockers for v2.27.0.

Global release consequence. For the next release decision, the first A-block is to be classified as *mathematically release-supporting*: not yet maximally unfolded, but no longer structurally open. The remaining pre-release work should therefore shift away from re-opening this block and toward checking whether the remaining A-priority material can be brought into the same state without introducing theorem drift, reference instability, or dependence on the computational teaser architecture.

Local closure target. After this reconciliation pass, the active goal for the local subversion chain is no longer to add more managerial logic around the first A-block, but to preserve it as a frozen support block while the remaining release blockers are tested against the same gate discipline.

C.14 Fifteenth 3.4 extension: transition cocycle support

Theorem 12.73 (Appendix support for Theorem 12.74)**.** *Let $\mathfrak{U} = \{U_\alpha\}$ be the stable shell atlas and suppose each isolated obstruction shell carries one of the four local transition types from Theorem 11.19. Then these local transitions assemble into a locally constant V_4-valued cocycle on admissible overlaps. After adjoining a sheet label that resolves fold exchange, the induced cocycle factors through the quotient $V_4 \twoheadrightarrow \mathbb{Z}_2^{\mathrm{par}}$.*

Proof. Every admissible overlap arises from continuation across finitely many isolated obstruction shells. By Theorem 11.19, each such shell contributes one of the four involutive transition types.

Their compositions therefore stay inside V_4, because the fold and parity generators commute and square to the identity. This gives a locally constant overlap label.

On triple overlaps, composing continuation maps in two stages or one stage leads to the same inverse shell identification, because both are obtained by following the same continuation class through the same finite obstruction data. Hence the cocycle relation holds. Resolving fold exchange by adjoining the local inverse-sheet label kills the fold generator on the lifted data, leaving only the parity component, which is exactly the quotient to $\mathbb{Z}_2^{\mathrm{par}}$. □

3.4.15 working interpretation. The inverse-readout atlas is therefore not just stratified; it is glued by a tiny and completely explicit transition law. Before fold resolution, the global bookkeeping lives in a Klein-four transition cocycle. After fold resolution, the only surviving topological memory is the parity lift. This is the most economical global form of the spinorial/M"obius mechanism obtained so far.

3.4 Sixteenth extension: transition cocycle and parity-resolved local system

The local obstruction law of Section 3.4.14 gives the primitive transition moves, but the next structural step is to organize those moves over the whole shell atlas rather than shell by shell. The natural object here is a tiny transition cocycle. It records how inverse shell data are reidentified when one passes between adjacent stable charts and then separates the fold ambiguity from the genuine parity lift.

Theorem 12.74 (3.4 transition cocycle and parity-resolved local system)**.** *Let $\mathfrak{U} = \{U_\alpha\}$ be the shell atlas from Theorem 11.16, where each U_α is a connected component of $I \setminus \Sigma$ carrying stable inverse readout. Then the following hold.*

(i) For every nonempty admissible overlap obtained by continuation across a single isolated obstruction shell, there is a locally constant transition element

$$g_{\alpha\beta} \in V_4 := \{\mathrm{id}, T^{\mathrm{fold}}, T^{\mathrm{par}}, T^{\mathrm{par}} \circ T^{\mathrm{fold}}\}$$

such that inverse shell data satisfy

$$(s, \pi)_\beta = g_{\alpha\beta}\,(s, \pi)_\alpha.$$

(ii) On every admissible triple overlap one has the cocycle relation

$$g_{\alpha\gamma} = g_{\beta\gamma} \circ g_{\alpha\beta},$$

so the inverse-readout layer defines a V_4-valued local transition system over the shell atlas.

(iii) After passing to the fold-resolved two-sheet inverse cover, the fold generator is absorbed into the choice of sheet, and the remaining transition system reduces to a parity local system valued in

$$\mathbb{Z}_2^{\mathrm{par}} = \{\mathrm{id}, T^{\mathrm{par}}\}.$$

Proof. Statement (i) is exactly the content of Theorem 11.19: every isolated obstruction contributes only fold exchange, parity flip, or their composition. Since those operations act discretely and the obstruction type cannot vary inside a sufficiently small continuation corridor, the corresponding transition element is locally constant on the overlap.

For (ii), the continuation from U_α to U_γ through U_β is represented by successive application of the two local transition maps. Because the local transitions are honest involutive reidentifications of the same inverse shell data, their composition agrees with direct continuation, which is the cocycle relation.

For (iii), resolving the fold ambiguity means that one fixes one of the two local inverse sheets as part of the enlarged state. The fold involution then acts trivially on the lifted data, because it has been absorbed into the sheet label itself. What remains is only whether the transported parity class is preserved or flipped, hence a residual $\mathbb{Z}_2$-valued local system. □

Corollary 12.75 (3.4 parity monodromy class)**.** *Under the hypotheses of Theorem 12.74, every closed admissible continuation of inverse shell data determines a well-defined class in the parity local system $\mathbb{Z}_2^{\mathrm{par}}$. The visible direct-return case corresponds to the trivial class, while the spinorial/M"obius continuation is exactly the nontrivial parity class.*

Proof. By Theorem 12.74(iii), all fold ambiguity is removed on the lifted inverse cover, so a closed continuation can leave only the trivial or nontrivial parity action. The trivial action returns the same visible branch, whereas the nontrivial parity action returns only after the $\mathbb{Z}_2$ lift. This is precisely the spinorial/M"obius case already identified locally in Corollary 11.20. □

Interpretive consequence. The shell atlas now carries a genuine global readout organization. Stable charts are not merely patched together ad hoc: they are glued by a tiny V_4-valued transition cocycle, and after resolving the fold choice the remaining global obstruction is purely parity-theoretic. This is the cleanest form so far of the claim that the visible spinorial/M"obius branch is not decorative language, but the nontrivial global parity sector of the inverse-readout geometry.

3.4 Seventeenth extension: parity double cover and lift-triviality criterion

The parity-resolved local system of Section 3.4.15 already identifies the spinorial/M"obius sector as a genuine global class. The next structural step is to say exactly when that class disappears and, when it does not, what the canonical repair is. The natural answer is a parity double cover: it trivializes precisely the residual $\mathbb{Z}_2$ obstruction and therefore converts the parity-resolved inverse readout into an honest global section.

Theorem 12.76 (3.4 parity double cover and lift-triviality criterion)**.** *Let B be a connected admissible shell base carrying the parity local system $\mathcal{L}_{\mathrm{par}}$ from Theorem 12.74. Then the following are equivalent.*

(i) The parity local system $\mathcal{L}_{\mathrm{par}}$ is globally trivial on B.

(ii) There exists a globally continuous parity-resolved branch choice

$$\sigma : B \to \{\pm 1\}$$

such that, after fold resolution, inverse shell data admit a globally single-valued visible representative

$$(s, \sigma\sqrt{s}, \pi)$$

on B.

(iii) Every closed admissible continuation in B has trivial parity monodromy.

If these equivalent conditions fail, then there exists a canonical connected double cover

$$p : \widetilde{B} \to B$$

trivializing $\mathcal{L}_{\mathrm{par}}$. On $\widetilde{B}$, the parity-resolved inverse readout becomes globally single-valued, and every admissible closed continuation lifts to direct visible convergence.

Proof. The equivalence of (i) and (iii) is the standard characterization of a $\mathbb{Z}_2$ local system: triviality is exactly the vanishing of its monodromy on all closed admissible loops. If $\mathcal{L}_{\mathrm{par}}$ is trivial, one may choose a global parity frame, which is precisely the global sign selection in (ii). Conversely, any such global sign choice identifies the two parity fibers coherently over all of B, so the local system is trivial.

If the parity class is nontrivial, the associated principal $\mathbb{Z}_2$-bundle defines the canonical connected double cover $p : \widetilde{B} \to B$. Pulling back $\mathcal{L}_{\mathrm{par}}$ to $\widetilde{B}$ trivializes it by construction. Hence the parity-resolved inverse shell data admit a global branch choice upstairs, and every lifted closed continuation returns with trivial parity action, i.e. as direct visible convergence on the cover. □

Corollary 12.77 (3.4 direct-return versus lift criterion)**.** *On a connected admissible shell base, the direct visible-return regime is globally available if and only if the parity local system is trivial. The spinorial/M"obius regime is therefore not an extra local ambiguity, but exactly the nontrivial global parity class; it disappears on the canonical parity double cover.*

Proof. The direct visible-return regime is exactly the existence of a globally coherent parity choice after fold resolution, which is condition (ii) of Theorem 12.76. The second claim is the nontrivial case of the same theorem. □

Interpretive consequence. The inverse-readout geometry now has a clean global dichotomy. Either the parity local system is already trivial, in which case the visible branch closes globally without further enlargement, or the residual obstruction is genuinely $\mathbb{Z}_2$ and the correct global object is the canonical parity double cover. In that sense, the spinorial/M"obius sector is no longer only a language for local continuation failure; it is exactly the statement that the visible readout trivializes only after passing to the natural two-sheeted cover.

3.4 Eighteenth extension: cohomological parity obstruction class

The parity double cover already gives the correct global repair. The next and cleanest reformulation is to record the remaining parity sector as a genuine cohomological obstruction. This does not add a new mechanism; it simply packages the already extracted $\mathbb{Z}_2$ data in the natural global invariant attached to the shell atlas.

Theorem 12.78 (3.4 cohomological parity obstruction class)**.** *Let B be a connected admissible shell base equipped with the parity-resolved inverse-readout atlas from Sections 3.4.15–3.4.16. After fold resolution, let*

$$g^{\mathrm{par}}_{\alpha\beta} \in \mathbb{Z}_2$$

be the parity transition functions on an admissible shell atlas $\{U_\alpha\}$. Then:

(i) the family $\{g^{\mathrm{par}}_{\alpha\beta}\}$ is a Čech 1-cocycle with coefficients in $\mathbb{Z}_2$;

(ii) its cohomology class

(iii) one has

$$\begin{aligned} \varepsilon_{\mathrm{par}} = 0 \quad &\Longleftrightarrow \quad \textit{the parity local system is trivial} \\ &\Longleftrightarrow \quad \textit{direct visible convergence is globally available}; \end{aligned}$$

(iv) if $\varepsilon_{\mathrm{par}} \neq 0$, then the canonical parity double cover from Theorem 12.76 is the two-sheeted cover classified by $\varepsilon_{\mathrm{par}}$.

Proof. Theorem 12.74 already reduces the shell-atlas transition data to the parity sector after fold resolution. Hence the remaining transition functions take values in the constant group $\mathbb{Z}_2$, and the cocycle relation on triple overlaps is exactly the cocycle identity for a Čech 1-cochain. Changing the admissible atlas or parity frame modifies $g^{\mathrm{par}}_{\alpha\beta}$ by a Čech coboundary, so the cohomology class $\varepsilon_{\mathrm{par}}$ is well defined.

The equivalence in (iii) is the standard identification of a $\mathbb{Z}_2$ local system with its class in $H^1(B, \mathbb{Z}_2)$: vanishing means that the cocycle is cohomologous to zero, hence a global parity frame exists; this is exactly the direct visible-return regime from Theorem 12.76. If the class does not vanish, the same theorem provides the canonical parity double cover trivializing the local system, and that cover is precisely the one classified by $\varepsilon_{\mathrm{par}}$. □

Corollary 12.79 (3.4 parity obstruction as the global lift defect)**.** *The spinorial/Möbius regime is completely encoded by the single class*

$$\varepsilon_{\mathrm{par}} \in H^1(B, \mathbb{Z}_2).$$

It vanishes exactly in the direct visible-return regime and is nonzero exactly when the inverse readout trivializes only after passage to the canonical parity double cover.

Proof. This is the reformulation of Theorem 12.78(iii)–(iv) in terms of the unique cohomological obstruction carried by the parity local system. □

Interpretive consequence. The parity side of the 3.4 program is now fully compressed. Local parity flips, shell-wise monodromy, the residual $\mathbb{Z}_2$ local system, the canonical double cover, and the distinction between direct visible closure and spinorial/Möbius lift are no longer separate stories. They are all expressions of the single class

$$\varepsilon_{\mathrm{par}} \in H^1(B, \mathbb{Z}_2).$$

In that sense, the remaining lift defect is now a genuine cohomological obstruction rather than merely a descriptive continuation phenomenon.

3.4 Nineteenth extension: parity holonomy character and loop-readout criterion

The cohomological obstruction class from Section 3.4.17 is already the smallest global invariant. The next step is to make it directly readable along admissible closed shell paths. The natural object is the parity holonomy character: it evaluates the class on a loop and therefore tells us, in the smallest possible way, whether visible closure occurs already on the base or only after passage to the parity double cover.

Theorem 12.80 (3.4 parity holonomy character and loop-readout criterion)**.** *Let B be a connected admissible shell base with parity obstruction class*

$$\varepsilon_{\mathrm{par}} \in H^1(B, \mathbb{Z}_2)$$

from Theorem 12.78. Fix a base point $b_0 \in B$. Then:

(i) there is a canonical parity holonomy character

$$\chi_{\mathrm{par}} : \pi_1(B, b_0) \to \mathbb{Z}_2, \qquad \chi_{\mathrm{par}}([\gamma]) = \langle \varepsilon_{\mathrm{par}}, [\gamma]_2 \rangle,$$

where $[\gamma]_2 \in H_1(B, \mathbb{Z}_2)$ denotes the mod-two homology class of γ;

(ii) $\chi_{\mathrm{par}}([\gamma]) = 0$ *if and only if every parity-resolved inverse continuation along* γ *returns with direct visible closure on* B*;*

(iii) $\chi_{\mathrm{par}}([\gamma]) = 1$ *if and only if continuation along* γ *closes only after the nontrivial deck transformation of the canonical parity double cover from Theorem 12.76;*

(iv) the character factors through mod-two homology,

$$\pi_1(B, b_0) \twoheadrightarrow H_1(B, \mathbb{Z}_2) \xrightarrow{\langle \varepsilon_{\mathrm{par}}, - \rangle} \mathbb{Z}_2,$$

so two admissible loops with the same class in $H_1(B, \mathbb{Z}_2)$ *have the same parity readout.*

Proof. A $\mathbb{Z}_2$ local system determines a monodromy representation into $\mathbb{Z}_2$, and for the parity local system this is exactly the parity action picked up by admissible closed continuation. By Theorem 12.78, the local system is classified by the cohomology class $\varepsilon_{\mathrm{par}}$. Standard evaluation of $H^1(B, \mathbb{Z}_2)$ on mod-two 1-cycles gives the homological formula

$$\chi_{\mathrm{par}}([\gamma]) = \langle \varepsilon_{\mathrm{par}}, [\gamma]_2 \rangle.$$

The value 0 means that the parity transport around γ is trivial, which is exactly direct visible return on the base. The value 1 means that the transported branch returns with the nontrivial parity action, so visible closure occurs only after passing to the parity double cover. Because the target group is abelian, the character factors through the abelianization of $\pi_1(B, b_0)$, and with mod-two coefficients this is precisely the factorization through $H_1(B, \mathbb{Z}_2)$. □

Corollary 12.81 (3.4 loop-equivalence principle for parity readout)**.** *On a connected admissible shell base, the global direct-return versus spinorial/Möbius decision for any admissible closed continuation class depends only on the value of*

$$\langle \varepsilon_{\mathrm{par}}, [\gamma]_2 \rangle \in \mathbb{Z}_2.$$

In particular, homologous admissible loops mod 2 *are parity-indistinguishable for the inverse readout.*

Proof. This is exactly statement (iv) of Theorem 12.80: the parity readout factors through the mod-two homology class of the loop. □

Interpretive consequence. The parity obstruction is now not only a global class but an operational readout. A closed admissible shell path carries a tiny $\mathbb{Z}_2$ outcome obtained by evaluating $\varepsilon_{\mathrm{par}}$ on its mod-two class. The visible direct-return regime corresponds to holonomy value 0, while the spinorial/Möbius regime corresponds to holonomy value 1. This is the first fully global and directly readable form of the parity defect.

3.4 Twentieth extension: empirical parity-holonomy estimator and loop calibration

The parity holonomy character from Section 3.4.18 is already the smallest global obstruction. The next step is to make it measurable from the readout side itself. Since the shell atlas already resolves admissible continuations into stable chart arcs separated by obstruction-shell crossings, the only possible source of a nontrivial parity outcome is the parity-changing part of those crossings. This yields a direct empirical estimator for the global $\mathbb{Z}_2$ holonomy.

Theorem 12.82 (3.4 empirical parity-holonomy estimator)**.** *Let γ be an admissible closed shell path on a connected shell base B, decomposed into stable chart arcs whose endpoints meet obstruction shells transversely. For each encountered obstruction-shell crossing, let*

$$\delta_k \in \mathbb{Z}_2$$

be defined by

$$\delta_k = \begin{cases} 0, & \textit{for a pure fold transition}, \\ 1, & \textit{for a parity transition}, \\ 1, & \textit{for a mixed fold-plus-parity transition}, \end{cases}$$

so that δ_k records only the parity-changing component of the local transition law from Theorem 11.19. Define the empirical parity estimator

$$\widehat{\chi}_{\mathrm{par}}(\gamma) := \sum_{k=1}^{N_\gamma} \delta_k \pmod 2.$$

Then:

(i) $\widehat{\chi}_{\mathrm{par}}(\gamma)$ depends only on the parity labels of the stable chart arcs and is independent of the chosen admissible chart decomposition;

(ii) the empirical estimator agrees with the cohomological holonomy character,

$$\widehat{\chi}_{\mathrm{par}}(\gamma) = \chi_{\mathrm{par}}([\gamma]) = \langle \varepsilon_{\mathrm{par}}, [\gamma]_2 \rangle;$$

(iii) consequently, $\widehat{\chi}_{\mathrm{par}}(\gamma) = 0$ if and only if the calibrated inverse readout closes directly on the base, while $\widehat{\chi}_{\mathrm{par}}(\gamma) = 1$ if and only if closure occurs only after the nontrivial deck action on the parity double cover.

Proof. By Theorem 11.19, only the parity component of an obstruction-shell crossing contributes to the post-crossing branch label; pure folds only exchange inverse sheets and therefore do not affect the residual $\mathbb{Z}_2$ parity class. Summing the parity-changing contributions modulo 2 along the closed path therefore computes the total parity transport accumulated by the continuation. Refining the chart decomposition inserts pairs of complementary local transitions whose parity contributions cancel modulo 2, so the quantity $\widehat{\chi}_{\mathrm{par}}(\gamma)$ is well defined.

After fold resolution, the remaining transition data is exactly the parity local system from Theorem 12.74. Hence the mod-two sum of parity changes along γ is the monodromy of that local system around the loop, which by Theorem 12.80 is precisely $\chi_{\mathrm{par}}([\gamma]) = \langle \varepsilon_{\mathrm{par}}, [\gamma]_2 \rangle$. The final visible-return statements are the two cases of that holonomy value. □

Corollary 12.83 (3.4 detector-side loop test for the parity obstruction)**.** *Suppose a calibrated inverse readout assigns stable chart arcs and parity labels to a detector-side closed shell trace. Then the parity obstruction can be tested directly by the mod-two count of parity-changing crossings:*

$$\widehat{\chi}_{\mathrm{par}}(\gamma) = 0 \Longleftrightarrow \textit{detector-side direct visible return},$$

$$\widehat{\chi}_{\mathrm{par}}(\gamma) = 1 \Longleftrightarrow \textit{detector-side spinorial/Möbius lift signal.}$$

In particular, the global $\mathbb{Z}_2$ lift defect becomes empirically checkable once the shell trace has been chartwise calibrated.

Proof. This is exactly statement (iii) of Theorem 12.82, read through the calibrated inverse readout from Sections 3.4.11–3.4.13. □

Interpretive consequence. The parity side now closes the loop between geometry and readout. The cohomological class is no longer only a classification object and the holonomy character is no longer only an abstract loop functional. Once a detector-side or probe-side shell trace has been calibrated chartwise, the global lift defect is measured by a single mod-two crossing count. This is the sharpest operational form so far of the statement that the spinorial/Möbius regime is a real readout phenomenon rather than a decorative analogy.

C.15 Sixteenth 3.4 extension: parity double-cover support

Theorem 12.84 (Appendix support for Theorem 12.76)**.** *Let B be a connected admissible shell base and let $\mathcal{L}_{\mathrm{par}}$ be the parity local system obtained after fold resolution. Then the parity monodromy determines a unique connected two-sheeted cover $p : \widetilde{B} \to B$ up to isomorphism. Pulling back $\mathcal{L}_{\mathrm{par}}$ along p trivializes the parity transport, and any other connected cover with the same property factors through $\widetilde{B}$.*

Proof. The parity local system is a locally constant $\mathbb{Z}_2$-bundle over B. Its monodromy representation sends every admissible closed continuation class to either the trivial or nontrivial parity action. The associated kernel subgroup therefore defines a connected index-two cover $p : \widetilde{B} \to B$, unique up to canonical isomorphism. By construction, loops lifted to $\widetilde{B}$ lie in the kernel of the parity monodromy, so the pulled-back local system has trivial transport. If another connected cover trivializes the same monodromy, its deck group must also kill that kernel and therefore factor through the index-two quotient defining $\widetilde{B}$. □

3.4.16 working interpretation. The parity sector has now reached its cleanest possible global form. The shell atlas first produced local obstruction types, then a Klein-four transition cocycle, then a reduced parity local system, and now finally the canonical object that removes the remaining obstruction: the parity double cover. Once lifted to that cover, the inverse-readout branch is globally single-valued again. This is the sharpest mathematical version so far of the statement that the visible spinorial/M"obius branch is a double-cover phenomenon rather than a decorative metaphor.

C.16 Seventeenth 3.4 extension: cohomological obstruction support

Theorem 12.85 (Appendix support for Theorem 12.78)**.** *Let $\{U_\alpha\}$ be an admissible shell atlas on B, and let $g^{\mathrm{par}}_{\alpha\beta} \in \mathbb{Z}_2$ be the parity transition cocycle after fold resolution. Then the cohomology class*

$$\varepsilon_{\mathrm{par}} = [g^{\mathrm{par}}_{\alpha\beta}] \in H^1(B, \mathbb{Z}_2)$$

classifies the parity local system and its associated connected two-sheeted cover up to isomorphism.

Proof. Because the post-fold transition functions take values in the constant group $\mathbb{Z}_2$, they define a Čech 1-cocycle. Two admissible atlases related by refinement or change of parity frame produce cohomologous cocycles, so the resulting class in $H^1(B, \mathbb{Z}_2)$ is intrinsic. Standard classification of principal $\mathbb{Z}_2$-bundles over B identifies this cohomology class with the isomorphism class of the parity local system. The associated connected two-sheeted cover is the cover classified by the same class and is exactly the parity double cover constructed in Theorem 12.84. □

3.4.17 working interpretation. The remaining global defect has now been compressed to its sharpest invariant. What previously appeared as parity flips, spinorial lift behavior, and double-cover repair is encoded by one class in $H^1(B, \mathbb{Z}_2)$. This does not change the mechanism; it shows that the mechanism is already as small as it can be.

C.17 Eighteenth 3.4 extension: parity holonomy support

Theorem 12.86 (Appendix support for Theorem 12.80). *Let $\mathcal{L}_{\mathrm{par}}$ be the parity local system on a connected admissible shell base B, classified by*

$$\varepsilon_{\mathrm{par}} \in H^1(B, \mathbb{Z}_2).$$

Then the monodromy representation of $\mathcal{L}_{\mathrm{par}}$ is exactly the homomorphism obtained by evaluating $\varepsilon_{\mathrm{par}}$ on mod-two loop classes. Its kernel consists precisely of those admissible closed continuation classes that lift to direct visible closure on the canonical parity double cover.

Proof. A locally constant $\mathbb{Z}_2$-bundle on B has monodromy in the discrete group $\mathbb{Z}_2$, hence a homomorphism from $\pi_1(B, b_0)$ to $\mathbb{Z}_2$. By Theorem 12.85, the same local system is classified by the cocycle class $\varepsilon_{\mathrm{par}} \in H^1(B, \mathbb{Z}_2)$. Under the standard identification of $H^1(B, \mathbb{Z}_2)$ with homomorphisms out of mod-two 1-cycles, the monodromy around a loop is given by evaluation of $\varepsilon_{\mathrm{par}}$ on its mod-two homology class. The lifted double cover from Theorem 12.84 trivializes the parity transport, so the loops in the kernel are exactly those whose lifted continuation closes directly on the cover. □

3.4.18 working interpretation. The cohomological class now has a direct operational meaning. It is not only something that classifies the parity defect abstractly; it can be evaluated on a closed admissible shell path and returns a two-valued global outcome. This is the cleanest bridge so far between the atlas/cohomology side and the actual readout side.

C.18 Nineteenth 3.4 extension: empirical parity-holonomy support

Theorem 12.87 (Appendix support for Theorem 12.82). *Let γ be an admissible closed shell path decomposed into stable chart arcs meeting obstruction shells transversely. Then the mod-two count of parity-changing crossings is equal to the monodromy of the parity local system around γ, and hence equal to the evaluation of the parity obstruction class on the mod-two loop class of γ.*

Proof. Each local obstruction crossing carries one of the four transition types from Theorem 11.19. Passing to the post-fold parity local system collapses the fold component and retains only whether the crossing changes the residual parity label. Therefore the transported parity label after traversing the full loop is obtained by multiplying the local $\mathbb{Z}_2$ transition factors, which in additive notation is exactly the mod-two sum of the local parity-change indicators δ_k. This product is the monodromy of the parity local system along the loop. By Theorem 12.86, that monodromy is also the evaluation of $\varepsilon_{\mathrm{par}}$ on the mod-two homology class of γ. □

3.4.19 working interpretation. The parity obstruction can now be read directly from a calibrated shell trace. Once the trace has been decomposed into admissible chart arcs, the full global lift defect collapses to a mod-two count of parity-changing transitions. This is the first point at which the global cohomological obstruction becomes an explicit detector-side or probe-side measurement rule.

C.19 Twentieth 3.4 extension: closing synthesis support

Theorem 12.88 (Appendix support for Theorem 11.21). *The stabilized 3.4 line factors through a single visible package*

$$\left(\Phi, \Delta_{\mathrm{diag}}, \varepsilon_{\mathrm{par}}, \chi_{\mathrm{par}}, \widehat{\chi}_{\mathrm{par}}\right)$$

with branch variable η, in the sense that every local regular shell observable and every global parity-readout statement used in the mainline 3.4 section is a consequence of these data together with the shell atlas away from the obstruction set.

Proof. Theorem 11.8 reduces the visible branch dynamics to the analytic generator $\Phi(\eta^2)$, and Theorems 12.21–12.24 show that calibrated shell inversion away from the obstruction set is entirely controlled by the shell coordinate and parity label, with Δ_{diag} measuring chartwise mismatch. Theorems 12.73–12.86 then compress the remaining global ambiguity to the parity local system, its cohomological class, and its holonomy character. Finally, Theorem 12.87 identifies the empirical estimator with this same holonomy class along admissible closed traces. Hence the listed package captures the entire visible 3.4 layer. □

3.4.20 working interpretation. The 3.4 material is now closed as a single generator-atlas-holonomy block. Local residual dynamics is carried by the one-field generator, local diagnostic mismatch is measured by Δ_{diag}, and the only global leftover is the parity obstruction class with its loop readout. This is the natural handoff point for the next stage.

C.20 First 3.5 extension: residual-record support

Theorem 12.89 (Appendix support for Theorem 11.24)**.** *Let B be a connected admissible shell base satisfying the stabilized 3.4 hypotheses. Then the visible branch layer admits a faithful record description by*

$$\mathfrak{R}(B) = \big(\Phi, \Delta_{\mathrm{diag}}, \varepsilon_{\mathrm{par}}, \chi_{\mathrm{par}}, \widehat{\chi}_{\mathrm{par}}\big),$$

in the sense that every chartwise regular-shell observable and every global parity-readout statement in the stabilized 3.4 sector is recoverable from this record together with the chartwise shell coordinate and parity label.

Proof. Theorem 12.88 already proves that the full stabilized 3.4 line factors through the same five visible data together with the shell atlas away from the obstruction set. The generator Φ determines the local branch law, the inverse shell charts determine the regular-shell coordinate and parity label, Δ_{diag} measures chartwise diagnostic mismatch, and the parity package records the only remaining global ambiguity together with its operational loop estimator. Therefore the visible branch layer is faithfully encoded by the record $\mathfrak{R}(B)$, and the only extra ingredients needed for local reconstruction are the chartwise shell coordinate and parity label already present in the inverse atlas. □

3.5.0 working interpretation. The first simplification step does not yet introduce a new formal language. It only observes that the visible branch layer has already closed enough to be handled as a single admissible record. This is exactly the right level of compression before any later language-level or proof-economy layer is introduced. In the current status reading, this also marks the point after which later progress should be judged mainly by record-level sharpening, not by adding fresh visible ingredients.

C.21 Second 3.5 extension: residual-record morphism support

Theorem 12.90 (Appendix support for Theorem 11.27)**.** *Let $F : B_1 \to B_2$ be a residual-record isomorphism between connected admissible shell bases. Then the entire stabilized visible generator-atlas-holonomy sector is preserved under transport along F: the transported generator law, calibrated inverse shell data, shell-consistency diagnostics, parity obstruction class, parity holonomy, and empirical loop estimator all remain faithful to the same residual record.*

Proof. By Theorem 12.89, the stabilized visible sector factors through the admissible record together with chartwise shell coordinates. The definition of residual-record isomorphism requires preservation of exactly these ingredients at the level at which they enter the reconstruction normal form: the generator data, the shell-consistency defect, and the parity package. Therefore

every chartwise visible formula and every global parity-readout statement already proved in the stabilized sector is transported faithfully along F, and no new visible branch datum is opened by this translation. □

3.5.1 working interpretation. The first move beyond the residual record is not to enlarge the visible layer, but to declare when two residual presentations should count as the same. This makes later compression work more disciplined: any future formal front-end must preserve the record-level morphism class if it is to remain faithful to the present mathematics. Historically open blocks that survive beyond this point therefore survive as residual identification or quantitative tasks, not as unresolved questions about what the visible data are.

C.22 Third 3.5 extension: residual-record interpreter support

Theorem 12.91 (Appendix support for Theorem 11.30)**.** *Let B be a connected admissible shell base in the stabilized generator-atlas-holonomy regime. Then every visible statement in this regime is representable by a record-level formula built from*

$$\mathfrak{R}(B) = \big(\Phi, \Delta_{\mathrm{diag}}, \varepsilon_{\mathrm{par}}, \chi_{\mathrm{par}}, \widehat{\chi}_{\mathrm{par}}\big)$$

together with chartwise shell coordinate and parity label, and this representation is unique up to residual-record isomorphism.

Proof. Theorem 12.89 already establishes that the entire stabilized visible branch layer factors through the admissible residual record together with shell-chart data. Theorem 12.90 shows that these data are preserved exactly under residual-record isomorphisms. Hence every visible theorem in the stabilized sector may be rewritten as a formula over the record package and shell-chart coordinates, and any two such rewritings can differ only by a residual-record isomorphism. This proves exact representability at record level. □

3.5.2 working interpretation. The meaning of the interpreter step is simple: from this point onward, the stabilized visible theory has a fixed admissible translation interface. Any future compression layer may become more economical or more set-theoretically explicit, but it should still be judged by whether it preserves the same record-level content. This is also the right point to separate the later live fronts from the historical OP labels: the surviving questions concern residual corrections, controlled emergence, spectral identification, Dixon compression, and the Theta-sum remainder, while OP 2 as a separate operator problem and OP 3 as a primary live front are no longer active blockers.

C.23 First 3.6 extension: conservative frontend support

Theorem 12.92 (Appendix support for Theorem 11.33)**.** *Let* $(\mathrm{Fr}(B), \mathcal{F}_B, \mathcal{J}_B)$ *be a conservative frontend over the stabilized visible sector of a connected admissible shell base B. Then every frontend statement coming from a visible statement is fixed, up to unique record-level translation, by the same residual-record content as the original visible statement.*

Proof. Theorem 12.91 already identifies the exact record-level representative of every visible theorem in the stabilized sector. The defining factorization rule for a conservative frontend requires frontend presentation to pass through that same representative, while transport-compatibility and primitive non-drift forbid any deviation on the visible image. Hence the frontend contributes only a conservative presentation layer and no second visible basis. □

3.6.0 working interpretation. The gain of the first 3.6 step is methodological but real: later language work is no longer a vague future promise. It now has a theorem-level admissibility test. This is precisely what keeps the planned ZFM-/frontend line subordinate to the mathematics-first strategy of the Companion.

Appendix C.124. Visual frame-flicker diagnostics for OP 5

The following figures make the Section 3.8.48 hypothesis auditable on the current benchmark window. They do not prove the frame-flicker ansatz, but they show that a single smooth packet-scale moving-frame mode can already account for most of the mismatch beyond the rigid carrier and can do so in the two correction channels predicted by the current first-order frame-flicker decomposition.

Remark 12.93 (Systems-level reading of the OP 5 visual block)**.** *In the systems language summarized in Figure* `fig:qam-vm-os-zfs` *in the active-core PDF, the present OP 5 figures should be read as visible-support views rather than as independent kernel identities. The rigid carrier, the packet breathing, and the localized readout slip are therefore shown here precisely because they belong to the auditable view layer of one transported channel. Their role is to expose where the visible readout departs from the rigid carrier picture, while the structural closure judgment itself still remains delegated to the deeper branch/shell/transport architecture.*

Remark 12.94 (Visual-to-proof workflow for the OP 5 block)**.** *The present OP 5 figures should therefore be used in the following order. First, they identify a visible defect pattern at the readout layer. Second, they motivate a packet/carrier/slip decomposition that can be audited channelwise. Third, that decomposition must be checked against the active transport and closure-preservation architecture rather than treated as a proof by itself. Only after such a check has been made may the same pattern be recorded as a later kernel-first compression candidate. In particular, the figures are admissible as* diagnostic entry points *and as* audit surfaces*, but not as autonomous closure theorems.*

Remark 12.95 (What the OP 5 figures are allowed to certify)**.** *At the present stage, the OP 5 visual block may certify only that a single packet-scale moving-frame correction remains empirically competitive as a visible-support description on the chosen benchmark window, and that the induced residual can be organized in the two currently tracked correction channels. It may not certify uniqueness of the correction mechanism, global closure of the OP 5 hypothesis, or activation of a new transport rule beyond the active theorem chain.*

Placeholder figure not bundled in this release:
textttop5_frame_flicker_fit_v3862.png

Figure 3: **OP 5 packet/readout fit on the current benchmark window.** Upper panel: the observed OP 5 packet/readout trace, the rigid carrier baseline, and the first-order packet/readout correction inside the visible packet-support window. Lower panel: the residual beyond the rigid carrier split into the breathing-envelope channel $\alpha\psi_{N,m}\sin\Theta_{N,m}$ and the readout-slip channel $\beta A_{N,m}\psi_{N,m}\cos\Theta_{N,m}$.

Placeholder figure not bundled in this release:
textttop5_frame_flicker_components_v3862.png

Figure 4: **Packet-support mode, visible channels, and incremental OP 5 build-up.** The four panels show the packet-support window $\psi_{N,m}$, the rigid carrier/readout quadratures, the two visible channels fed by one packet-scale moving-frame mode, and the stepwise build-up from rigid carrier to packet/breathing correction and then to the full OP 5 readout trace.

Placeholder figure not bundled in this release:
textttop5_frame_flicker_geometry_v3862.png

Figure 5: **Geometric OP 5 reading: carrier helix, breathing packet, and readout slip.** The panels compare the rigid carrier helix with the packet/readout-distorted helix, show the top-view deviation, display the packet breathing envelope along the helix, and isolate the localized readout phase slip. The natural reading is therefore not a second bulk spiral, but the same carrier seen through one softened visible frame.

Reading guide. Figures 3–5 should be read in order as *trace* → *channel decomposition* → *geometry.* In that sequence, the packet/readout defect moves from the visible time trace, to the two correction channels, to the helical geometric picture.

Appendix conclusion. On the present benchmark window, the frame-flicker ansatz remains hypothesis-grade, but it is no longer only verbal. A single smooth moving-frame mode reproduces most of the residual correction beyond the rigid carrier and does so in exactly the two channels predicted by the first-order decomposition. This strengthens the current reading that OP 5 is better treated as a carrier-plus-readout-defect problem than as a missing second rigid transport wave.

Remark 12.96 (Packet/flicker diagnostics as readout evidence)**.** *Within the current OP transport architecture, the packet-wave, breathing-envelope, and frame-flicker figures should be read as visible-support diagnostics rather than as primary closure criteria. Their mathematical role is to show that a large part of the observed OP 5 residue is compatible with a carrier-plus-readout-defect picture, which is exactly the kind of situation filtered later by the visible/structural mismatch separation tool.*

Bibliography placement note. The bibliography that formerly appeared at this historical appendix position has been moved to the final back matter so that the document closes with the references. The version-record and historical-release material continues below unchanged as archival proof-memory.

Working common-mode carrier is downstream from the active HC/MM/J fit

The previous subsection reduced the primitive OP 1/OP 5 obstruction to a common-mode carrier plus a downstream observer-shell split. The next structural step is to push that common-mode carrier itself back to the active HC/MM/J fit. On the active admissible branch, the shared carrier is not an independent visible object: it is simply the shared visible mask applied to the complementary output generated by the active matrix-leakage operator. In particular, if the active branch is exactly HC/MM/J-compatible, then the primitive common-mode carrier must vanish identically.

Definition 12.97 (Working active matrix-fit defect and shared carrier readout)**.** *Let* $U := \operatorname{span}\{u_1, u_2\}$ *be the active admissible plane and* $W := \operatorname{span}\{w_1, w_2\}$ *the active complementary leakage plane. Define the active matrix-fit defect by*

$$E_{\mathrm{fit}}^{(U)} := \big((\mathrm{id} - C)\mathcal{M}C\big)\big|_U : U \to W.$$

For a unit active probe state

$$x(\theta) := \cos\theta\, u_1 + \sin\theta\, u_2 \in U,$$

define the active hidden leakage response and the shared visible carrier by

$$y_{\mathrm{fit}}(\theta) := E_{\mathrm{fit}}^{(U)} x(\theta) \in W, \qquad c_{\mathrm{sh}}^{\mathrm{fit}}(\theta) := L_{\mathrm{sh}}^{(W)}\big(y_{\mathrm{fit}}(\theta)\big).$$

Proposition 12.98 (Working exact commutator identity on the admissible branch)**.** *For every $x \in U \subseteq \mathrm{im}(C)$ one has*

$$(\mathrm{id} - C)\mathcal{M}x = \frac{1}{2}\,[\mathcal{M}, J]x.$$

Equivalently,

$$E_{\mathrm{fit}}^{(U)} = \frac{1}{2}\,[\mathcal{M}, J]\big|_U.$$

In particular,

$$\|E_{\mathrm{fit}}^{(U)}\| \le \frac{1}{2}\,\|[\mathcal{M}, J]\big|_U\|.$$

Proof. Since $U \subseteq \mathrm{im}(C)$, every $x \in U$ satisfies $Cx = x$ and $Jx = x$. Therefore

$$(\mathrm{id} - C)\mathcal{M}x = \frac{1}{2}(\mathrm{id} - J)\mathcal{M}x = \frac{1}{2}\big(\mathcal{M}x - J\mathcal{M}x\big).$$

Using $Jx = x$, this becomes

$$\frac{1}{2}\big(\mathcal{M}Jx - J\mathcal{M}x\big) = \frac{1}{2}\,[\mathcal{M}, J]x.$$

Restricting to U gives the operator identity, and the norm bound is immediate. □

Theorem 12.99 (Working shared carrier is controlled by the active HC/MM/J fit)**.** *For every active probe angle θ one has*

$$c_{\mathrm{sh}}^{\mathrm{fit}}(\theta) = L_{\mathrm{sh}}^{(W)}\big(E_{\mathrm{fit}}^{(U)} x(\theta)\big) = \frac{1}{2}\,L_{\mathrm{sh}}^{(W)}\big([\mathcal{M}, J]x(\theta)\big).$$

Hence

$$|c_{\mathrm{sh}}^{\mathrm{fit}}(\theta)| \le \|L_{\mathrm{sh}}^{(W)}\|_{W^*}\,\|E_{\mathrm{fit}}^{(U)}\|\,\|x(\theta)\| \le \frac{1}{2}\,\|L_{\mathrm{sh}}^{(W)}\|_{W^*}\,\|[\mathcal{M}, J]\big|_U\|.$$

In particular, if

$$[\mathcal{M}, J]\big|_U = 0,$$

then

$$c_{\mathrm{sh}}^{\mathrm{fit}}(\theta) = 0$$

for all θ, so the primitive common-mode OP 1/OP 5 carrier vanishes on the active branch.

Proof. The first displayed identity is the definition of $c_{\mathrm{sh}}^{\mathrm{fit}}$. The second follows from the previous proposition. Since $\|x(\theta)\| = 1$ on the active unit circle in U, the norm estimate follows immediately. If the restricted commutator vanishes, then $E_{\mathrm{fit}}^{(U)} = 0$, hence $y_{\mathrm{fit}}(\theta) = 0$ and therefore $c_{\mathrm{sh}}^{\mathrm{fit}}(\theta) = 0$ for all θ. □

Corollary 12.100 (Working primitive common-mode residual is a pure matrix-fit obstruction)**.** *Assume exact grouped no-drift on the active leakage plane, so that $\Delta L^{(W)} = 0$. Then every surviving primitive OP 1/OP 5 residual on the active branch is completely controlled by the active HC/MM/J fit defect:*

$$d(\theta) = c_{\mathrm{sh}}^{\mathrm{fit}}(\theta)\begin{pmatrix}1\\1\end{pmatrix}, \qquad |c_{\mathrm{sh}}^{\mathrm{fit}}(\theta)| \le \frac{1}{2}\,\|L_{\mathrm{sh}}^{(W)}\|_{W^*}\,\|[\mathcal{M}, J]\big|_U\|.$$

In particular, under exact grouped no-drift and exact active commutation with the umkehrwalze, the primitive OP 1/OP 5 channel closes identically on the active branch.

Proof. Under exact grouped no-drift one has $c_{\mathrm{obs}} \equiv 0$ by the previous shared-residual dichotomy, so only the common-mode component remains. The displayed bound is the previous theorem. If the restricted commutator also vanishes, then the common-mode carrier vanishes identically, hence the primitive visible channel closes. □

Working interpretation. This is the clean matrix-side endpoint of the current reduction. Once the observer-shell split is removed on the active leakage plane, the remaining primitive OP 1/OP 5 carrier is not a mysterious visible residue anymore: it is exactly the shared mask applied to the active HC/MM/J commutator defect. So the primitive common-mode channel survives only to the extent that the Millennium matrix still fails to commute with the umkehrwalze on the active admissible branch.

Working architectural promotion of the active commutator defect

The active commutator defect was introduced as the primitive common-mode obstruction on the active branch. However, the older grouped-layer architecture already contains a stronger structural principle: admissible Millennium transport is only allowed when it intertwines with the involution. Therefore the active commutator should not be read as a free new defect by default, but as a quantity that must vanish whenever the transport really is the admissible grouped-layer converter claimed by the document architecture. What then remains, if anything, must be downstream dressing rather than primitive HC/MM/J leakage.

Theorem 12.101 (Working admissible grouped transport kills the active commutator)**.** *Assume the Millennium matrix $\mathcal{M}$ acts on the active admissible plane U as an admissible grouped-layer converter in the sense of Definition* `def:millennium-transport-grouped-layers` *in the active-core PDF, and assume its transport role is structure preserving with respect to the involution/umkehrwalze in the sense of Lemma* `lem:millennium-intertwining-lemma` *in the active-core PDF. Then*

$$[\mathcal{M}, J]\big|_U = 0.$$

Consequently,

$$E_{\mathrm{fit}}^{(U)} = 0, \qquad c_{\mathrm{sh}}^{\mathrm{fit}}(\theta) = 0 \quad \textit{for all } \theta.$$

Proof. By Lemma `lem:millennium-intertwining-lemma` in the active-core PDF, every Millennium transport that claims to preserve the normalized involutive kernel must satisfy

$$\mathcal{M}\mathcal{J} = \mathcal{J}\mathcal{M}.$$

On the active branch the visible involution is realized by the operator J, so the same intertwining statement on the active branch reads

$$\mathcal{M}Jx = J\mathcal{M}x \qquad (x \in U).$$

Hence

$$[\mathcal{M}, J]x = 0 \qquad (x \in U),$$

which is exactly the claimed vanishing of the restricted commutator. The identities

$$E_{\mathrm{fit}}^{(U)} = \tfrac{1}{2}[\mathcal{M}, J]\big|_U$$

and

$$c_{\mathrm{sh}}^{\mathrm{fit}}(\theta) = L_{\mathrm{sh}}^{(W)}\big(E_{\mathrm{fit}}^{(U)}x(\theta)\big)$$

were established in the previous subsection, so both vanish immediately. □

Corollary 12.102 (Working primitive OP 1/OP 5 common mode cannot survive exact admissible transport)**.** *Assume exact grouped no-drift on the active leakage plane and admissible grouped Millennium transport on the active branch. Then the primitive visible OP 1/OP 5 channel closes identically on that branch:*

$$d(\theta) = 0 \qquad \textit{for all } \theta.$$

In particular, any still visible residual must be interpreted as downstream observer-shell dressing, phase-shifted readout, or some other post-transport effect rather than as a primitive HC/MM/J leakage mode.

Proof. Exact grouped no-drift gives $c_{\mathrm{obs}}(\theta) = 0$ by the shared-residual dichotomy. The previous theorem gives $c_{\mathrm{sh}}^{\mathrm{fit}}(\theta) = 0$. Since the primitive visible channel decomposes as the sum of the common-mode part and the observer-split part, both pieces vanish and therefore $d(\theta) = 0$ for all θ. □

Remark 12.103 (Working interpretation: exact closure versus canonical observer-shell residual)**.** *The current architecture therefore supports a sharp dichotomy.*

(i) If the Millennium matrix really acts as the admissible grouped-layer converter claimed by the structural architecture, then the active commutator vanishes on the active branch and no primitive common-mode OP 1/OP 5 leakage survives there.

(ii) If a residual nevertheless remains visible, then it should no longer be hunted as a primitive HC/MM/J closure defect. It must instead be read as a canonical downstream observer-shell residual: phase-shifted readout, shell dressing, expansion dressing, or another post-transport effect living after the exact admissible transport stage.

This is the strongest structural narrowing achieved so far: primitive matrix leakage and canonical observer-shell residual are no longer being mixed. The architecture itself says that only one of them may remain once admissible grouped transport is enforced.

Canonical downstream observer-shell residual after exact admissible transport

The previous dichotomy showed that once grouped no-drift holds on the active leakage plane and admissible grouped Millennium transport is exact on the active branch, no primitive HC/MM/J leakage can remain visible. Any still visible OP 1/OP 5 remainder must therefore live strictly downstream from the exact transport stage. The next structural step is to package that remainder as a single canonical observer-shell mode rather than as two independent unfinished closures.

Definition 12.104 (Working downstream observer-shell dressing mode)**.** *Assume exact grouped no-drift on the active leakage plane W and exact admissible grouped Millennium transport on the active branch U. A downstream observer-shell dressing mode is a scalar family*

$$r_{\mathrm{obs}}(\theta)$$

and a common visible shell profile

$$e_{\mathrm{obs}} := \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

such that the residual visible channel can be written in the form

$$d(\theta) = r_{\mathrm{obs}}(\theta)\, e_{\mathrm{obs}} + d_{\mathrm{split}}(\theta),$$

where $d_{\mathrm{split}}(\theta)$ is a pure observer-side split term living after the exact transport stage.

Theorem 12.105 (Working canonical observer-shell residual after exact admissible transport)**.** *Assume exact grouped no-drift on the active leakage plane W and exact admissible grouped Millennium transport on the active branch U. Then any remaining visible OP 1/OP 5 residual admits the canonical decomposition*

$$d(\theta) = r_{\mathrm{obs}}(\theta)\begin{pmatrix} 1 \\ 1 \end{pmatrix} + c_{\mathrm{obs}}(\theta)\begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Moreover, the primitive matrix-leakage contribution vanishes identically, so the common-mode coefficient $r_{\mathrm{obs}}(\theta)$ is a purely downstream observer-shell residual rather than a primitive HC/MM/J leakage carrier.

Proof. By the shared-residual dichotomy, the primitive visible channel decomposes as a common-mode part and an observer-split part,

$$d(\theta) = c_{\mathrm{sh}}(\theta)\begin{pmatrix}1\\1\end{pmatrix} + c_{\mathrm{obs}}(\theta)\begin{pmatrix}1\\-1\end{pmatrix}.$$

The previous active-matrix-fit analysis identifies the primitive common-mode part with the active HC/MM/J fit carrier. Under exact admissible grouped Millennium transport on U, the active commutator vanishes and therefore the primitive common-mode carrier vanishes as well. Hence any surviving common-mode coefficient can only appear after the exact transport stage, i.e. as downstream observer-shell dressing. Renaming this remaining downstream common-mode coefficient by $r_{\mathrm{obs}}(\theta)$ yields the stated decomposition. □

Corollary 12.106 (Working no-drift collapses to one observer-shell mode)**.** *Under the hypotheses of the theorem, if the observer-side split also vanishes on the active leakage plane,*

$$c_{\mathrm{obs}}(\theta) = 0 \qquad \textit{for all } \theta,$$

then the full visible OP 1/OP 5 remainder collapses to a single canonical observer-shell residual:

$$d(\theta) = r_{\mathrm{obs}}(\theta)\begin{pmatrix}1\\1\end{pmatrix}.$$

In particular, OP 1 and OP 5 then do not carry two independent unfinished closures on the active branch; they are merely two reads of the same downstream shell residual.

Proof. The theorem already gives the decomposition

$$d(\theta) = r_{\mathrm{obs}}(\theta)\begin{pmatrix}1\\1\end{pmatrix} + c_{\mathrm{obs}}(\theta)\begin{pmatrix}1\\-1\end{pmatrix}.$$

If $c_{\mathrm{obs}}(\theta) = 0$, only the common shell term remains, which is exactly the stated formula. □

Remark 12.107 (Working interpretation of the surviving residual)**.** *This is the intended structural endpoint of the current OP 1/OP 5 reduction. Once exact admissible transport kills the primitive HC/MM/J leakage and grouped no-drift removes the active observer split, any remaining visible residue should not be interpreted as evidence for two separate unfinished closures. It is a single canonical downstream shell residual, shared by OP 1 and OP 5 and only seen through two observer labels.*

From two unfinished closures to one shell-mode law

The previous reduction reached a decisive structural endpoint: once exact admissible transport removes primitive HC/MM/J leakage and grouped no-drift removes the active observer split, OP 1 and OP 5 no longer describe two independent unfinished closures on the active branch. What remains is a single downstream observer-shell mode. The next step is therefore not to force two separate zero-conditions, but to state explicitly that the visible target has collapsed to one scalar shell law.

Definition 12.108 (Working shared shell-mode amplitude)**.** *Assume the hypotheses of the previous corollary, so that*

$$d(\theta) = r_{\mathrm{obs}}(\theta)\begin{pmatrix}1\\1\end{pmatrix}.$$

The scalar family

$$r_{\mathrm{obs}}(\theta)$$

is called the shared shell-mode amplitude of the visible OP 1/OP 5 residual.

Theorem 12.109 (Working one-law reduction of OP 1 and OP 5 after exact transport)**.** *Assume exact grouped no-drift on the active leakage plane W, exact admissible grouped Millennium transport on the active branch U, and vanishing active observer split on W. Then the visible OP 1/OP 5 channel is governed by a single scalar law:*

$$d_1(\theta) = d_5(\theta) = r_{\mathrm{obs}}(\theta).$$

In particular, the visible target is no longer a pair of independent closure equations, but the single shell-mode equation

$$r_{\mathrm{obs}}(\theta) = 0$$

if exact visible closure is required, or the single shell-mode identification problem

$$r_{\mathrm{obs}}(\theta) = r_{\mathrm{can}}(\theta)$$

if a canonical nonzero observer-shell residual is allowed.

Proof. Under the stated hypotheses, the previous corollary gives

$$d(\theta) = r_{\mathrm{obs}}(\theta)\begin{pmatrix}1\\1\end{pmatrix}.$$

Reading off the two components immediately yields

$$d_1(\theta) = r_{\mathrm{obs}}(\theta), \qquad d_5(\theta) = r_{\mathrm{obs}}(\theta).$$

Hence the visible OP 1/OP 5 target collapses from two residual equations to one scalar shell-mode law. If one demands exact visible closure, that law is $r_{\mathrm{obs}}(\theta) = 0$. If one instead admits a canonical downstream observer-shell remainder, the problem is to identify the canonical shell law $r_{\mathrm{can}}(\theta)$ and match the visible residual against it. □

Corollary 12.110 (Working exact closure and canonical residual are one-mode alternatives)**.** *Under the same hypotheses, one cannot simultaneously interpret the surviving visible channel as two independent unfinished closures and as a single canonical observer-shell mode. Exactly one of the following descriptions may hold:*

(i) exact visible closure, i.e.

$$r_{\mathrm{obs}}(\theta) = 0 \qquad \textit{for all } \theta;$$

(ii) canonical observer-shell remainder, i.e.

$$r_{\mathrm{obs}}(\theta) = r_{\mathrm{can}}(\theta) \not\equiv 0.$$

In either case, OP 1 and OP 5 remain locked to the same scalar mode.

Proof. The theorem shows that both visible channels are equal to the same scalar amplitude $r_{\mathrm{obs}}(\theta)$. Therefore any interpretation of the surviving visible remainder must be an interpretation of this single scalar family. It cannot simultaneously serve as two independent residual equations and as one shared shell mode. The two listed alternatives are exactly the two structurally admissible readings. □

Remark 12.111 (Working interpretation: closure target versus shell-law target)**.** *This is the point where the present architecture changes the question. The old reading asked whether OP 1 and OP 5 can both be closed independently. The new reading says: after exact admissible transport and active no-drift, there is only one visible target left. The remaining decision is whether that target should vanish or whether it should be identified with a canonical post-transport shell law, for example expansion dressing or phase-shifted observer-shell readout.*

Canonical phase-shifted / expansion-dressed shell law

The previous reduction showed that after exact admissible transport and active no-drift, the visible OP 1/OP 5 question collapses to a single scalar shell-mode amplitude

$$r_{\mathrm{obs}}(\theta).$$

The next structural step is to stop treating this quantity as a bare leftover and instead give a first explicit law for it. The guiding hypothesis is that the observer shell is not a second ontology but a small post-transport dressing of the same exact ground closure. In that case the surviving visible mode should arise from a phase-shifted and possibly expansion-dressed readout of the same underlying shell carrier.

Definition 12.112 (Working observer-shell phase and expansion defects)**.** *A post-transport observer-shell defect state is described by two scalar families*

$$\delta_{\mathrm{obs}}(\theta), \qquad \eta_{\mathrm{exp}}(\theta),$$

where

- $\delta_{\mathrm{obs}}(\theta)$ *measures the local phase shift between the exact transport shell and the visible observer shell,*
- $\eta_{\mathrm{exp}}(\theta)$ *measures the local expansion / shell-spacing drift of the visible shell relative to the exact transport shell.*

Both quantities are downstream observer-shell parameters: they vanish on the exact transport shell itself.

Definition 12.113 (Working first-order canonical shell law)**.** *A first-order canonical observer-shell law is an identity of the form*

$$r_{\mathrm{can}}(\theta) = A_{\mathrm{sh}}(\theta)\,\delta_{\mathrm{obs}}(\theta) + B_{\mathrm{sh}}(\theta)\,\eta_{\mathrm{exp}}(\theta) + R_{\geq 2}(\theta),$$

where $A_{\mathrm{sh}}(\theta)$ *and* $B_{\mathrm{sh}}(\theta)$ *are shell-response coefficients and the remainder satisfies*

$$|R_{\geq 2}(\theta)| \leq C_{\mathrm{sh}}(\theta)\big(|\delta_{\mathrm{obs}}(\theta)| + |\eta_{\mathrm{exp}}(\theta)|\big)^2.$$

In the strictly phase-shifted case one has $B_{\mathrm{sh}} \equiv 0$*, while in the pure expansion-dressing case one has* $A_{\mathrm{sh}} \equiv 0$*.*

Theorem 12.114 (Working first-order phase-shift / expansion law for the shared shell mode)**.** *Assume exact admissible grouped Millennium transport on the active branch, exact grouped no-drift on the active leakage plane, and vanishing active observer split on that plane. Assume moreover that the visible observer shell is a smooth downstream perturbation of the exact transport shell, parameterized by phase defect* $\delta_{\mathrm{obs}}(\theta)$ *and expansion defect* $\eta_{\mathrm{exp}}(\theta)$*. Then the shared visible mode admits a first-order shell law*

$$r_{\mathrm{obs}}(\theta) = A_{\mathrm{sh}}(\theta)\,\delta_{\mathrm{obs}}(\theta) + B_{\mathrm{sh}}(\theta)\,\eta_{\mathrm{exp}}(\theta) + R_{\geq 2}(\theta),$$

with a quadratic remainder in the total shell defect magnitude. In particular, exact visible closure is equivalent to the vanishing of this canonical shell law, while a nonzero visible remainder is compatible with exact transport provided it is entirely generated by downstream shell dressing.

Proof. Under the stated hypotheses the visible channel has already collapsed to a single scalar amplitude,

$$d(\theta) = r_{\mathrm{obs}}(\theta)\begin{pmatrix}1\\1\end{pmatrix}.$$

Hence every surviving visible effect must be read as a downstream perturbation of the same exact transport shell rather than as a primitive matrix-leakage channel. By the assumed smooth dependence of the visible observer shell on the two downstream shell parameters $\delta_{\mathrm{obs}}(\theta)$ and $\eta_{\mathrm{exp}}(\theta)$, the scalar amplitude $r_{\mathrm{obs}}(\theta)$ is a smooth scalar function of these two arguments. A first-order Taylor expansion around the exact shell point

$$(\delta_{\mathrm{obs}}, \eta_{\mathrm{exp}}) = (0, 0)$$

yields

$$r_{\mathrm{obs}}(\theta) = A_{\mathrm{sh}}(\theta)\,\delta_{\mathrm{obs}}(\theta) + B_{\mathrm{sh}}(\theta)\,\eta_{\mathrm{exp}}(\theta) + R_{\geq 2}(\theta),$$

where the remainder is quadratic in the total shell defect magnitude. This is exactly the stated first-order canonical shell law. □

Corollary 12.115 (Working exact closure versus canonical nonzero shell law)**.** *Under the same hypotheses, the remaining visible OP 1/OP 5 target is reduced to the dichotomy*

(i) exact visible closure:

$$A_{\mathrm{sh}}(\theta)\,\delta_{\mathrm{obs}}(\theta) + B_{\mathrm{sh}}(\theta)\,\eta_{\mathrm{exp}}(\theta) + R_{\geq 2}(\theta) = 0;$$

(ii) canonical nonzero observer-shell remainder:

$$r_{\mathrm{obs}}(\theta) = r_{\mathrm{can}}(\theta) \not\equiv 0$$

with r_{can} given by the same first-order shell law.

In both cases the visible question is one scalar shell law rather than two independent OP 1 / OP 5 closures.

Proof. The previous theorem identifies the surviving scalar mode with a first-order observer-shell law in the two downstream shell parameters. Therefore the visible target is exhausted by deciding whether that law vanishes identically or whether it should be retained as a canonical nonzero shell remainder. □

Remark 12.116 (Working interpretation of the shell-law coefficients)**.** *The coefficients $A_{\mathrm{sh}}(\theta)$ and $B_{\mathrm{sh}}(\theta)$ are the first visible response coefficients of the exact transport shell to observer-side phase shift and observer-side expansion dressing. They are not new primitive leakage parameters. Their role is to measure how a single exact transport closure is seen from a displaced observer shell. This is why OP 1 and OP 5 remain locked to the same scalar law even after exact transport has already removed the primitive HC/MM/J leakage.*

Phase-dominant, expansion-dominant, and mixed shell-law regimes

The canonical shell law now reduces the visible post-transport question to one scalar combination of two downstream shell effects. The next structural step is therefore not to search for two further closures, but to decide which of the two first-order shell loads is actually primary.

Definition 12.117 (Working shell loads and dominance ratio). *Define the first-order shell loads*

$$\Lambda_{\mathrm{ph}}(\theta) := |A_{\mathrm{sh}}(\theta)\,\delta_{\mathrm{obs}}(\theta)|, \qquad \Lambda_{\mathrm{exp}}(\theta) := |B_{\mathrm{sh}}(\theta)\,\eta_{\mathrm{exp}}(\theta)|,$$

and the total first-order shell load

$$\Lambda_{\mathrm{lin}}(\theta) := \Lambda_{\mathrm{ph}}(\theta) + \Lambda_{\mathrm{exp}}(\theta).$$

Whenever $\Lambda_{\mathrm{ph}}(\theta) \neq 0$, *define the shell-load ratio*

$$\rho_{\mathrm{sh}}(\theta) := \frac{\Lambda_{\mathrm{exp}}(\theta)}{\Lambda_{\mathrm{ph}}(\theta)}.$$

Thus $\rho_{\mathrm{sh}} \ll 1$ *indicates phase-dominant shell dressing, while* $\rho_{\mathrm{sh}} \gg 1$ *indicates expansion-dominant shell dressing.*

Definition 12.118 (Working phase-dominant, expansion-dominant, and mixed regimes). *Fix* $\varepsilon_{\mathrm{dom}}, \varepsilon_{\mathrm{rem}} \in [0,1)$ *with*

$$\varepsilon_{\mathrm{dom}} + \varepsilon_{\mathrm{rem}} < 1.$$

At a given parameter value θ*:*

(i) the shell law is called phase-dominant *if*

$$\Lambda_{\mathrm{exp}}(\theta) \leq \varepsilon_{\mathrm{dom}}\,\Lambda_{\mathrm{ph}}(\theta) \quad \textit{and} \quad |R_{\geq 2}(\theta)| \leq \varepsilon_{\mathrm{rem}}\,\Lambda_{\mathrm{ph}}(\theta);$$

(ii) it is called expansion-dominant *if*

$$\Lambda_{\mathrm{ph}}(\theta) \leq \varepsilon_{\mathrm{dom}}\,\Lambda_{\mathrm{exp}}(\theta) \quad \textit{and} \quad |R_{\geq 2}(\theta)| \leq \varepsilon_{\mathrm{rem}}\,\Lambda_{\mathrm{exp}}(\theta);$$

(iii) it is called genuinely mixed at first order *if neither of the previous two dominance conditions holds.*

Theorem 12.119 (Working shell-law dominance classification). *Assume the first-order canonical shell law of the previous subsection. Then:*

(i) in the phase-dominant regime one has

$$|r_{\mathrm{obs}}(\theta) - A_{\mathrm{sh}}(\theta)\,\delta_{\mathrm{obs}}(\theta)| \leq (\varepsilon_{\mathrm{dom}} + \varepsilon_{\mathrm{rem}})\,\Lambda_{\mathrm{ph}}(\theta),$$

so the shared shell mode is controlled by the observer-side phase-shift term up to a strictly smaller relative error;

(ii) in the expansion-dominant regime one has

$$|r_{\mathrm{obs}}(\theta) - B_{\mathrm{sh}}(\theta)\,\eta_{\mathrm{exp}}(\theta)| \leq (\varepsilon_{\mathrm{dom}} + \varepsilon_{\mathrm{rem}})\,\Lambda_{\mathrm{exp}}(\theta),$$

so the shared shell mode is controlled by the observer-shell expansion term up to a strictly smaller relative error;

(iii) in the genuinely mixed regime, the visible shell mode cannot be reduced at first order to only one of the two shell defects.

Proof. By the first-order shell law,

$$r_{\mathrm{obs}}(\theta) = A_{\mathrm{sh}}(\theta)\,\delta_{\mathrm{obs}}(\theta) + B_{\mathrm{sh}}(\theta)\,\eta_{\mathrm{exp}}(\theta) + R_{\geq 2}(\theta).$$

Hence

$$|r_{\mathrm{obs}}(\theta) - A_{\mathrm{sh}}(\theta)\,\delta_{\mathrm{obs}}(\theta)| \leq \Lambda_{\mathrm{exp}}(\theta) + |R_{\geq 2}(\theta)|,$$

and similarly

$$|r_{\mathrm{obs}}(\theta) - B_{\mathrm{sh}}(\theta)\,\eta_{\mathrm{exp}}(\theta)| \leq \Lambda_{\mathrm{ph}}(\theta) + |R_{\geq 2}(\theta)|.$$

In the phase-dominant regime this gives

$$|r_{\mathrm{obs}}(\theta) - A_{\mathrm{sh}}(\theta)\,\delta_{\mathrm{obs}}(\theta)| \leq (\varepsilon_{\mathrm{dom}} + \varepsilon_{\mathrm{rem}})\,\Lambda_{\mathrm{ph}}(\theta),$$

which is strictly smaller than the dominant phase load because $\varepsilon_{\mathrm{dom}} + \varepsilon_{\mathrm{rem}} < 1$. The expansion-dominant case is identical after swapping the two shell loads. If neither dominance condition holds, then neither first-order term controls the shell mode up to a strictly smaller remainder, so the shell mode is genuinely mixed at first order. □

Corollary 12.120 (Working next decision for the shared shell mode)**.** *After exact admissible transport, active no-drift, and collapse to a single visible shell mode, the next structural decision is reduced to the comparison of the two first-order shell loads*

$$\Lambda_{\mathrm{ph}}(\theta) \quad \textit{and} \quad \Lambda_{\mathrm{exp}}(\theta),$$

not to two independent closure equations. In particular, the visible question is now whether the common shell mode is primarily

(i) phase-shift driven,

(ii) expansion-drift driven,

(iii) or genuinely mixed at first order.

Proof. This is the direct content of the dominance classification theorem. □

Remark 12.121 (Working interpretation of the present status)**.** *The current document does not yet prove globally which regime is realized. What it does prove is that the decision has been compressed to a single shell-law comparison problem. The remaining visible burden is therefore much narrower than the earlier picture of two unfinished OP 1 / OP 5 closures: it is now a one-mode question about which downstream observer-shell load is structurally primary.*

Signed shell-balance mismatch and the cancellation corridor

The shell-load classification identifies which first-order observer-shell contribution is dominant, but exact visible closure depends on something sharper: whether the two signed first-order shell contributions actually cancel. The next structural step is therefore to replace the unsigned load comparison by a signed shell-balance mismatch.

Definition 12.122 (Working signed shell coordinates and shell-balance mismatch)**.** *Define the signed first-order shell coordinates*

$$p_{\mathrm{sh}}(\theta) := A_{\mathrm{sh}}(\theta)\,\delta_{\mathrm{obs}}(\theta), \qquad e_{\mathrm{sh}}(\theta) := B_{\mathrm{sh}}(\theta)\,\eta_{\mathrm{exp}}(\theta),$$

and the signed shell-balance mismatch

$$\chi_{\mathrm{sh}}(\theta) := p_{\mathrm{sh}}(\theta) + e_{\mathrm{sh}}(\theta).$$

Whenever $p_{\mathrm{sh}}(\theta) \neq 0$, define the signed shell-balance ratio

$$\zeta_{\mathrm{sh}}(\theta) := \frac{e_{\mathrm{sh}}(\theta)}{p_{\mathrm{sh}}(\theta)}.$$

Thus exact first-order cancellation corresponds to $\chi_{\mathrm{sh}}(\theta) = 0$, equivalently $\zeta_{\mathrm{sh}}(\theta) = -1$.

Definition 12.123 (Working shell-cancellation corridor). *Fix* $\varepsilon_{\mathrm{bal}}, \varepsilon_{\mathrm{rem}} \in [0,1)$ *with*

$$\varepsilon_{\mathrm{bal}} + \varepsilon_{\mathrm{rem}} < 1.$$

At a given parameter value θ*, the shell law is said to lie in the* cancellation corridor *if*

$$|\chi_{\mathrm{sh}}(\theta)| \le \varepsilon_{\mathrm{bal}}\, \Lambda_{\mathrm{lin}}(\theta) \qquad \textit{and} \qquad |R_{\ge 2}(\theta)| \le \varepsilon_{\mathrm{rem}}\, \Lambda_{\mathrm{lin}}(\theta).$$

Equivalently, the first-order signed mismatch and the higher-order shell remainder are both smaller than the total first-order shell load.

Theorem 12.124 (Working shell-balance reduction and exact cancellation corridor). *Assume the first-order canonical shell law. Then:*

(i) one has the exact reduction

$$r_{\mathrm{obs}}(\theta) = \chi_{\mathrm{sh}}(\theta) + R_{\ge 2}(\theta);$$

(ii) in the cancellation corridor one has

$$|r_{\mathrm{obs}}(\theta)| \le (\varepsilon_{\mathrm{bal}} + \varepsilon_{\mathrm{rem}})\, \Lambda_{\mathrm{lin}}(\theta),$$

so exact visible closure is reduced to a small signed shell-balance mismatch rather than to two separate closures;

(iii) if $p_{\mathrm{sh}}(\theta)$ *and* $e_{\mathrm{sh}}(\theta)$ *have the same sign, then*

$$|\chi_{\mathrm{sh}}(\theta)| = |p_{\mathrm{sh}}(\theta)| + |e_{\mathrm{sh}}(\theta)| = \Lambda_{\mathrm{lin}}(\theta),$$

so exact visible closure at first order is impossible unless the total first-order shell load itself vanishes.

Proof. By definition of the signed shell coordinates,

$$p_{\mathrm{sh}}(\theta) + e_{\mathrm{sh}}(\theta) = A_{\mathrm{sh}}(\theta)\, \delta_{\mathrm{obs}}(\theta) + B_{\mathrm{sh}}(\theta)\, \eta_{\mathrm{exp}}(\theta) = \chi_{\mathrm{sh}}(\theta),$$

so the first-order shell law becomes

$$r_{\mathrm{obs}}(\theta) = \chi_{\mathrm{sh}}(\theta) + R_{\ge 2}(\theta).$$

This proves (i). If the shell law lies in the cancellation corridor, then

$$|r_{\mathrm{obs}}(\theta)| \le |\chi_{\mathrm{sh}}(\theta)| + |R_{\ge 2}(\theta)| \le (\varepsilon_{\mathrm{bal}} + \varepsilon_{\mathrm{rem}})\, \Lambda_{\mathrm{lin}}(\theta),$$

which proves (ii). If $p_{\mathrm{sh}}(\theta)$ and $e_{\mathrm{sh}}(\theta)$ have the same sign, then they add without cancellation, hence

$$|\chi_{\mathrm{sh}}(\theta)| = |p_{\mathrm{sh}}(\theta) + e_{\mathrm{sh}}(\theta)| = |p_{\mathrm{sh}}(\theta)| + |e_{\mathrm{sh}}(\theta)| = \Lambda_{\mathrm{lin}}(\theta),$$

which proves (iii). □

Corollary 12.125 (Working next decision after shell-load dominance). *After the shell-load dominance classification, the remaining visible decision is whether the signed ratio* $\zeta_{\mathrm{sh}}(\theta)$ *enters a neighborhood of* -1*. In particular:*

(i) if $\zeta_{\mathrm{sh}}(\theta) \approx -1$*, the visible shell mode lies near the cancellation corridor and exact visible closure remains possible;*

(ii) if $\zeta_{\mathrm{sh}}(\theta) > 0$*, then the first-order shell law is sign-coherent and forces a canonical nonzero observer-shell remainder unless the total first-order shell load itself vanishes;*

(iii) if $\zeta_{\mathrm{sh}}(\theta)$ stays bounded away from -1, then the shared shell mode is canonically nonzero already at first order.

Proof. The previous theorem shows that the visible shell mode is controlled by the signed mismatch $\chi_{\mathrm{sh}}(\theta) = p_{\mathrm{sh}}(\theta)(1 + \zeta_{\mathrm{sh}}(\theta))$ together with the higher-order remainder. Thus entry into the cancellation corridor is equivalent to placing the signed ratio near -1, while positive ratio excludes first-order cancellation. □

Remark 12.126 (Working interpretation of the present shell-law reduction)**.** *This is the first place where the two-shell-defect picture collapses all the way to a single signed comparison. The visible question is no longer which of two unfinished closures dominates, but whether the phase-shift and expansion-drift contributions cancel each other or reinforce each other on the observer shell. If they reinforce, the visible residual is canonically nonzero; if they nearly cancel, exact visible closure remains structurally possible.*

Normalized shell-cancellation defect and the exact first-order verdict

The signed shell-balance mismatch already reduces the visible problem to a single first-order comparison. The next structural compression is to normalize that mismatch by the total first-order shell load itself. This removes the overall shell scale and leaves only the genuinely visible cancellation defect.

Definition 12.127 (Working normalized shell defect and cancellation depth)**.** *Whenever $\Lambda_{\mathrm{lin}}(\theta) > 0$, define the normalized shell-cancellation defect by*

$$\rho_{\mathrm{sh}}(\theta) := \frac{\chi_{\mathrm{sh}}(\theta)}{\Lambda_{\mathrm{lin}}(\theta)} = \frac{p_{\mathrm{sh}}(\theta) + e_{\mathrm{sh}}(\theta)}{|p_{\mathrm{sh}}(\theta)| + |e_{\mathrm{sh}}(\theta)|}.$$

Set $\rho_{\mathrm{sh}}(\theta) := 0$ when $\Lambda_{\mathrm{lin}}(\theta) = 0$. Define the cancellation depth by

$$\kappa_{\mathrm{sh}}(\theta) := 1 - |\rho_{\mathrm{sh}}(\theta)|.$$

Thus $|\rho_{\mathrm{sh}}|$ measures the normalized first-order noncancellation, while κ_{sh} measures how deeply the shell law enters the first-order cancellation corridor.

Theorem 12.128 (Working normalized shell defect, cancellation depth, and exact first-order criterion)**.** *For every parameter value θ one has:*

(i)

$$|\rho_{\mathrm{sh}}(\theta)| \le 1, \qquad 0 \le \kappa_{\mathrm{sh}}(\theta) \le 1;$$

(ii) if $p_{\mathrm{sh}}(\theta)$ and $e_{\mathrm{sh}}(\theta)$ have the same sign or one of them vanishes, then

$$|\rho_{\mathrm{sh}}(\theta)| = 1, \qquad \kappa_{\mathrm{sh}}(\theta) = 0;$$

(iii) if $p_{\mathrm{sh}}(\theta)$ and $e_{\mathrm{sh}}(\theta)$ have opposite sign, then

$$|\rho_{\mathrm{sh}}(\theta)| = \frac{\big||p_{\mathrm{sh}}(\theta)| - |e_{\mathrm{sh}}(\theta)|\big|}{|p_{\mathrm{sh}}(\theta)| + |e_{\mathrm{sh}}(\theta)|},$$

$$\kappa_{\mathrm{sh}}(\theta) = \frac{2\min\{|p_{\mathrm{sh}}(\theta)|, |e_{\mathrm{sh}}(\theta)|\}}{|p_{\mathrm{sh}}(\theta)| + |e_{\mathrm{sh}}(\theta)|};$$

(iv) exact first-order shell cancellation holds if and only if

$$\rho_{\mathrm{sh}}(\theta) = 0,$$

which is equivalent to opposite sign and equal first-order shell magnitudes,

$$|p_{\mathrm{sh}}(\theta)| = |e_{\mathrm{sh}}(\theta)|.$$

Proof. By definition,

$$|\chi_{\mathrm{sh}}(\theta)| = |p_{\mathrm{sh}}(\theta) + e_{\mathrm{sh}}(\theta)| \leq |p_{\mathrm{sh}}(\theta)| + |e_{\mathrm{sh}}(\theta)| = \Lambda_{\mathrm{lin}}(\theta),$$

so $|\rho_{\mathrm{sh}}(\theta)| \leq 1$, and the cancellation depth automatically lies in $[0, 1]$. This proves (i). If p_{sh} and e_{sh} have the same sign, or one vanishes, then

$$|p_{\mathrm{sh}} + e_{\mathrm{sh}}| = |p_{\mathrm{sh}}| + |e_{\mathrm{sh}}|,$$

so $|\rho_{\mathrm{sh}}| = 1$ and therefore $\kappa_{\mathrm{sh}} = 0$, proving (ii). If the signs are opposite, then

$$|p_{\mathrm{sh}} + e_{\mathrm{sh}}| = \big||p_{\mathrm{sh}}| - |e_{\mathrm{sh}}|\big|,$$

which yields the displayed formula for $|\rho_{\mathrm{sh}}|$ after division by $|p_{\mathrm{sh}}| + |e_{\mathrm{sh}}|$. The formula for κ_{sh} follows because

$$1 - \frac{\big||p| - |e|\big|}{|p| + |e|} = \frac{|p| + |e| - \big||p| - |e|\big|}{|p| + |e|} = \frac{2\min\{|p|, |e|\}}{|p| + |e|}.$$

This proves (iii). Finally, $\rho_{\mathrm{sh}}(\theta) = 0$ is equivalent to $\chi_{\mathrm{sh}}(\theta) = 0$, which is exactly first-order shell cancellation. Under opposite sign this is equivalent to equal magnitudes, proving (iv). □

Definition 12.129 (Working normalized higher-order shell remainder)**.** *Whenever $\Lambda_{\mathrm{lin}}(\theta) > 0$, define the normalized higher-order shell remainder by*

$$\rho_{\mathrm{rem}}(\theta) := \frac{|R_{\geq 2}(\theta)|}{\Lambda_{\mathrm{lin}}(\theta)}.$$

Set $\rho_{\mathrm{rem}}(\theta) := 0$ when $\Lambda_{\mathrm{lin}}(\theta) = 0$.

Corollary 12.130 (Working exact visible closure requires normalized first-order cancellation)**.** *Assume the first-order canonical shell law and let $\Lambda_{\mathrm{lin}}(\theta) > 0$. Then:*

(i)

$$\frac{|r_{\mathrm{obs}}(\theta)|}{\Lambda_{\mathrm{lin}}(\theta)} \leq |\rho_{\mathrm{sh}}(\theta)| + \rho_{\mathrm{rem}}(\theta);$$

(ii) a necessary condition for exact visible closure at θ is

$$|\rho_{\mathrm{sh}}(\theta)| \leq \rho_{\mathrm{rem}}(\theta);$$

(iii) if

$$|\rho_{\mathrm{sh}}(\theta)| > \rho_{\mathrm{rem}}(\theta),$$

then the shared visible shell mode is canonically nonzero at θ.

Proof. By the shell-balance reduction theorem,

$$r_{\mathrm{obs}}(\theta) = \chi_{\mathrm{sh}}(\theta) + R_{\geq 2}(\theta),$$

so after division by $\Lambda_{\mathrm{lin}}(\theta)$ one gets

$$\frac{|r_{\mathrm{obs}}(\theta)|}{\Lambda_{\mathrm{lin}}(\theta)} \leq \frac{|\chi_{\mathrm{sh}}(\theta)|}{\Lambda_{\mathrm{lin}}(\theta)} + \frac{|R_{\geq 2}(\theta)|}{\Lambda_{\mathrm{lin}}(\theta)} = |\rho_{\mathrm{sh}}(\theta)| + \rho_{\mathrm{rem}}(\theta),$$

which proves (i). If exact visible closure holds, then

$$0 = r_{\mathrm{obs}}(\theta) = \chi_{\mathrm{sh}}(\theta) + R_{\geq 2}(\theta),$$

so $|\chi_{\mathrm{sh}}(\theta)| = |R_{\geq 2}(\theta)|$. Division by $\Lambda_{\mathrm{lin}}(\theta)$ gives (ii). Statement (iii) is the contrapositive of (ii). □

Remark 12.131 (Working interpretation of the normalized shell verdict)**.** *The visible shell problem has now collapsed to a single normalized comparison. The first-order part is measured by $|\rho_{\mathrm{sh}}|$, the higher-order rescue budget by ρ_{rem}, and exact visible closure is possible only if the normalized first-order noncancellation does not exceed the normalized higher-order shell remainder. In particular, once $|\rho_{\mathrm{sh}}|$ is visibly larger than ρ_{rem}, the shared shell mode is forced to remain canonically nonzero.*

Sign-coherent obstruction and the anti-phase closure corridor

The normalized shell verdict can be compressed one step further. The visible question is not merely whether the normalized first-order mismatch is larger than the higher-order rescue budget, but also whether the two first-order shell loads are sign-coherent or anti-phase. In the sign-coherent regime, first-order cancellation is impossible; in the anti-phase regime, exact visible closure can survive only inside a sharp corridor around the ratio value -1.

Theorem 12.132 (Working sign-coherent obstruction and anti-phase corridor for visible closure)**.** *Assume the first-order canonical shell law and let $\Lambda_{\mathrm{lin}}(\theta) > 0$.*

(i) If $p_{\mathrm{sh}}(\theta)$ and $e_{\mathrm{sh}}(\theta)$ have the same sign or one of them vanishes, then

$$|\rho_{\mathrm{sh}}(\theta)| = 1.$$

Consequently, if

$$\rho_{\mathrm{rem}}(\theta) < 1,$$

then exact visible closure at θ is impossible and the shared shell mode is canonically nonzero.

(ii) Suppose $p_{\mathrm{sh}}(\theta)$ and $e_{\mathrm{sh}}(\theta)$ have opposite sign and $p_{\mathrm{sh}}(\theta) \neq 0$. Set

$$\zeta_{\mathrm{sh}}(\theta) := \frac{e_{\mathrm{sh}}(\theta)}{p_{\mathrm{sh}}(\theta)}.$$

Then necessarily $\zeta_{\mathrm{sh}}(\theta) < 0$, and one has

$$|\rho_{\mathrm{sh}}(\theta)| = \frac{|1 + \zeta_{\mathrm{sh}}(\theta)|}{1 + |\zeta_{\mathrm{sh}}(\theta)|}.$$

Hence a necessary condition for exact visible closure is

$$\frac{|1 + \zeta_{\mathrm{sh}}(\theta)|}{1 + |\zeta_{\mathrm{sh}}(\theta)|} \leq \rho_{\mathrm{rem}}(\theta),$$

that is,

$$|1 + \zeta_{\mathrm{sh}}(\theta)| \leq \rho_{\mathrm{rem}}(\theta)\big(1 + |\zeta_{\mathrm{sh}}(\theta)|\big).$$

(iii) In particular, if

$$|1+\zeta_{\mathrm{sh}}(\theta)| > \rho_{\mathrm{rem}}(\theta)\big(1+|\zeta_{\mathrm{sh}}(\theta)|\big),$$

then the shared visible shell mode is canonically nonzero at θ.

Proof. Statement (i) is immediate from the normalized shell-defect theorem, which gives $|\rho_{\mathrm{sh}}| = 1$ in the sign-coherent regime. The exact visible closure criterion then yields the necessary inequality

$$1 = |\rho_{\mathrm{sh}}(\theta)| \le \rho_{\mathrm{rem}}(\theta).$$

Therefore, if $\rho_{\mathrm{rem}}(\theta) < 1$, exact visible closure is impossible and the shell mode remains canonically nonzero.

For (ii), opposite sign implies $\zeta_{\mathrm{sh}}(\theta) < 0$. Since

$$\chi_{\mathrm{sh}}(\theta) = p_{\mathrm{sh}}(\theta) + e_{\mathrm{sh}}(\theta) = p_{\mathrm{sh}}(\theta)(1+\zeta_{\mathrm{sh}}(\theta)),$$

and

$$\Lambda_{\mathrm{lin}}(\theta) = |p_{\mathrm{sh}}(\theta)| + |e_{\mathrm{sh}}(\theta)| = |p_{\mathrm{sh}}(\theta)|\big(1+|\zeta_{\mathrm{sh}}(\theta)|\big),$$

one gets

$$|\rho_{\mathrm{sh}}(\theta)| = \frac{|\chi_{\mathrm{sh}}(\theta)|}{\Lambda_{\mathrm{lin}}(\theta)} = \frac{|1+\zeta_{\mathrm{sh}}(\theta)|}{1+|\zeta_{\mathrm{sh}}(\theta)|}.$$

The displayed corridor condition then follows from the exact visible closure criterion $|\rho_{\mathrm{sh}}| \le \rho_{\mathrm{rem}}$.

Statement (iii) is the contrapositive of that necessary condition. □

Remark 12.133 (Working interpretation of the sign/corridor shell verdict)**.** *The visible shell problem has now become extremely sharp. If the phase and expansion shell loads are sign-coherent, then first-order cancellation is impossible and higher order would have to rescue the full shell scale to produce exact visible closure. If they are anti-phase, then exact visible closure can survive only inside a narrow corridor where the signed shell ratio ζ_{sh} sits sufficiently close to -1. Outside that corridor, the shared shell mode is forced to remain canonically nonzero.*

Single-shell-coordinate reduction and naturality of the anti-phase corridor

The anti-phase corridor around $\zeta_{\mathrm{sh}} = -1$ is sharp, but it is not yet clear whether that corridor is structurally natural or only a fine-tuned accident. The next structural step is therefore to ask whether the two downstream shell defects are actually independent, or whether they are both induced by one and the same observer-shell displacement. If the latter is true, then the signed shell ratio collapses at first order to one structural gain ratio, and the question "why near -1?" becomes a question about one induced shell geometry rather than about two unrelated corrections.

Definition 12.134 (Working single-shell-coordinate observer law)**.** *A downstream observer shell is said to be controlled by a* single shell coordinate *if there exists a scalar family $\xi_{\mathrm{sh}}(\theta)$ and coefficient families $a_{\mathrm{sh}}(\theta), b_{\mathrm{sh}}(\theta)$ such that*

$$\delta_{\mathrm{obs}}(\theta) = a_{\mathrm{sh}}(\theta)\,\xi_{\mathrm{sh}}(\theta) + r_\delta(\theta), \qquad \eta_{\mathrm{exp}}(\theta) = b_{\mathrm{sh}}(\theta)\,\xi_{\mathrm{sh}}(\theta) + r_\eta(\theta),$$

with quadratic remainder bounds

$$|r_\delta(\theta)| + |r_\eta(\theta)| \le C_\xi(\theta)\,|\xi_{\mathrm{sh}}(\theta)|^2.$$

Thus both downstream shell defects are first-order images of the same observer-shell displacement variable.

Definition 12.135 (Working shell gains induced by one shell coordinate)**.** *Under the single-shell-coordinate observer law, define the induced first-order shell gains*

$$\Gamma_{\mathrm{ph}}(\theta) := A_{\mathrm{sh}}(\theta)\, a_{\mathrm{sh}}(\theta), \qquad \Gamma_{\mathrm{exp}}(\theta) := B_{\mathrm{sh}}(\theta)\, b_{\mathrm{sh}}(\theta),$$

and, whenever $\Gamma_{\mathrm{ph}}(\theta) \neq 0$*, the induced shell gain ratio*

$$\zeta_\Gamma(\theta) := \frac{\Gamma_{\mathrm{exp}}(\theta)}{\Gamma_{\mathrm{ph}}(\theta)}.$$

Theorem 12.136 (Working first-order shell ratio collapses to an induced gain ratio)**.** *Assume the first-order canonical shell law and the single-shell-coordinate observer law. Then the signed first-order shell coordinates satisfy*

$$p_{\mathrm{sh}}(\theta) = \Gamma_{\mathrm{ph}}(\theta)\, \xi_{\mathrm{sh}}(\theta) + O\big(|\xi_{\mathrm{sh}}(\theta)|^2\big)\,,$$

$$e_{\mathrm{sh}}(\theta) = \Gamma_{\mathrm{exp}}(\theta)\, \xi_{\mathrm{sh}}(\theta) + O\big(|\xi_{\mathrm{sh}}(\theta)|^2\big)\,,$$

and hence

$$\chi_{\mathrm{sh}}(\theta) = \big(\Gamma_{\mathrm{ph}}(\theta) + \Gamma_{\mathrm{exp}}(\theta)\big)\, \xi_{\mathrm{sh}}(\theta) + O\big(|\xi_{\mathrm{sh}}(\theta)|^2\big)\,.$$

If moreover $\Gamma_{\mathrm{ph}}(\theta) \neq 0$ *and* $\xi_{\mathrm{sh}}(\theta) \neq 0$*, then*

$$\zeta_{\mathrm{sh}}(\theta) = \zeta_\Gamma(\theta) + O(|\xi_{\mathrm{sh}}(\theta)|)\,.$$

In particular, the anti-phase corridor is first-order natural exactly when the induced gain ratio satisfies

$$\zeta_\Gamma(\theta) \approx -1.$$

Proof. Insert the single-shell-coordinate expansions into the definitions

$$p_{\mathrm{sh}}(\theta) = A_{\mathrm{sh}}(\theta)\, \delta_{\mathrm{obs}}(\theta), \qquad e_{\mathrm{sh}}(\theta) = B_{\mathrm{sh}}(\theta)\, \eta_{\mathrm{exp}}(\theta).$$

This gives

$$p_{\mathrm{sh}}(\theta) = A_{\mathrm{sh}}(\theta) a_{\mathrm{sh}}(\theta)\, \xi_{\mathrm{sh}}(\theta) + A_{\mathrm{sh}}(\theta) r_\delta(\theta) = \Gamma_{\mathrm{ph}}(\theta)\, \xi_{\mathrm{sh}}(\theta) + O\big(|\xi_{\mathrm{sh}}(\theta)|^2\big)\,,$$

with the analogous formula for $e_{\mathrm{sh}}(\theta)$. Summing yields the formula for $\chi_{\mathrm{sh}}(\theta)$. If $\Gamma_{\mathrm{ph}}(\theta) \neq 0$ and $\xi_{\mathrm{sh}}(\theta) \neq 0$, then division of the two first-order expansions gives

$$\zeta_{\mathrm{sh}}(\theta) = \frac{\Gamma_{\mathrm{exp}}(\theta) + O(|\xi_{\mathrm{sh}}(\theta)|)}{\Gamma_{\mathrm{ph}}(\theta) + O(|\xi_{\mathrm{sh}}(\theta)|)} = \frac{\Gamma_{\mathrm{exp}}(\theta)}{\Gamma_{\mathrm{ph}}(\theta)} + O(|\xi_{\mathrm{sh}}(\theta)|)\,,$$

which is exactly the stated collapse to the induced gain ratio. □

Corollary 12.137 (Working structural meaning of the anti-phase corridor)**.** *Under the same hypotheses, the anti-phase corridor around* $\zeta_{\mathrm{sh}}(\theta) = -1$ *is not a separate tuning problem between two independent shell defects. It is, to first order, the statement that the single observer-shell displacement variable* $\xi_{\mathrm{sh}}(\theta)$ *is seen by the phase and expansion readouts with opposite and nearly equal induced gains:*

$$\Gamma_{\mathrm{exp}}(\theta) \approx -\Gamma_{\mathrm{ph}}(\theta).$$

Thus the visible shell question reduces further to deciding whether the induced gain pair is

(i) sign-coherent,

(ii) anti-phase but unbalanced,

(iii) or anti-phase and nearly balanced.

Proof. This is exactly the content of the theorem after replacing $\zeta_{\mathrm{sh}}(\theta)$ by its first-order approximation $\zeta_{\Gamma}(\theta)$. □

Remark 12.138 (Working next decision after the shell-ratio corridor)**.** *The visible shell problem has now moved from two apparent shell defects to one observer-shell displacement and one induced gain pair. This is a genuine structural compression. It means that the relevant question is no longer whether two unrelated corrections accidentally cancel, but whether one downstream shell displacement is read with opposite and nearly equal gains by the phase and expansion channels.*

Shell-conjugate observer geometry and induced gain balance

The single-shell-coordinate reduction showed that the anti-phase corridor is natural exactly when the induced gains satisfy $\Gamma_{\mathrm{exp}} \approx -\Gamma_{\mathrm{ph}}$. The next structural step is to push that gain relation back to the observer-shell architecture itself. Rather than treating Γ_{ph} and Γ_{exp} as free coefficients, we ask whether they arise as two downstream readouts of one shell tangent under a shell involution inherited from the umkehrwalze. If so, then anti-phase is no longer a tuning accident between two unrelated gains, but the first-order signature of one shell-conjugate observer geometry.

Definition 12.139 (Working shell-conjugate observer response)**.** *Let S_{obs} be the downstream observer-shell tangent space. A* shell-conjugate observer response *consists of*

(i) an involution $R_{\mathrm{sh}} : S_{\mathrm{obs}} \to S_{\mathrm{obs}}$ with $R_{\mathrm{sh}}^2 = \mathrm{id}$, interpreted as the observer-shell descendant of the umkehrwalze,

(ii) a shell tangent family $u_{\mathrm{sh}}(\theta) \in S_{\mathrm{obs}}$,

(iii) a base shell readout functional $\ell_{\mathrm{sh}}(\theta) \in S_{\mathrm{obs}}^*$,

(iv) and a shell-conjugacy defect $\varepsilon_{\mathrm{conj}}(\theta)$,

with

$$a_{\mathrm{sh}}(\theta) = \ell_{\mathrm{sh}}(\theta)\big(u_{\mathrm{sh}}(\theta)\big),$$
$$b_{\mathrm{sh}}(\theta) = -\ell_{\mathrm{sh}}(\theta)\big(R_{\mathrm{sh}}u_{\mathrm{sh}}(\theta)\big) + \varepsilon_{\mathrm{conj}}(\theta).$$

If moreover

$$R_{\mathrm{sh}}u_{\mathrm{sh}}(\theta) = u_{\mathrm{sh}}(\theta),$$

then the phase and expansion shell readouts are said to be read from the same shell tangent with conjugate sign, up to the defect $\varepsilon_{\mathrm{conj}}(\theta)$.

Definition 12.140 (Working transport-balance defect)**.** *Define the downstream transport-balance defect by*

$$\Delta_{AB}(\theta) := B_{\mathrm{sh}}(\theta) - A_{\mathrm{sh}}(\theta).$$

Exact transport balance on the downstream shell means $\Delta_{AB}(\theta) = 0$.

Theorem 12.141 (Working shell-conjugate observer geometry makes anti-phase natural)**.** *Assume the single-shell-coordinate observer law and a shell-conjugate observer response with*

$$R_{\mathrm{sh}}u_{\mathrm{sh}}(\theta) = u_{\mathrm{sh}}(\theta).$$

Then

$$b_{\mathrm{sh}}(\theta) = -a_{\mathrm{sh}}(\theta) + \varepsilon_{\mathrm{conj}}(\theta),$$

and hence the induced gains satisfy

$$\Gamma_{\mathrm{exp}}(\theta) + \Gamma_{\mathrm{ph}}(\theta) = A_{\mathrm{sh}}(\theta)\,\varepsilon_{\mathrm{conj}}(\theta) + \Delta_{AB}(\theta)\,b_{\mathrm{sh}}(\theta).$$

In particular, if $\Gamma_{\mathrm{ph}}(\theta) \neq 0$*, then*

$$\zeta_\Gamma(\theta) + 1 = \frac{A_{\mathrm{sh}}(\theta)\,\varepsilon_{\mathrm{conj}}(\theta) + \Delta_{AB}(\theta)\,b_{\mathrm{sh}}(\theta)}{\Gamma_{\mathrm{ph}}(\theta)}.$$

Thus the anti-phase corridor $\zeta_\Gamma(\theta) \approx -1$ *is structurally natural whenever both the shell-conjugacy defect and the downstream transport-balance defect are small relative to* $\Gamma_{\mathrm{ph}}(\theta)$*.*

Proof. By shell-conjugacy,

$$b_{\mathrm{sh}}(\theta) = -\ell_{\mathrm{sh}}(\theta)\big(R_{\mathrm{sh}}u_{\mathrm{sh}}(\theta)\big) + \varepsilon_{\mathrm{conj}}(\theta) = -\ell_{\mathrm{sh}}(\theta)\big(u_{\mathrm{sh}}(\theta)\big) + \varepsilon_{\mathrm{conj}}(\theta) = -a_{\mathrm{sh}}(\theta) + \varepsilon_{\mathrm{conj}}(\theta).$$

Therefore

$$\begin{aligned}\Gamma_{\mathrm{exp}}(\theta) + \Gamma_{\mathrm{ph}}(\theta) &= B_{\mathrm{sh}}(\theta)b_{\mathrm{sh}}(\theta) + A_{\mathrm{sh}}(\theta)a_{\mathrm{sh}}(\theta)\\ &= B_{\mathrm{sh}}(\theta)\big(-a_{\mathrm{sh}}(\theta) + \varepsilon_{\mathrm{conj}}(\theta)\big) + A_{\mathrm{sh}}(\theta)a_{\mathrm{sh}}(\theta)\\ &= (A_{\mathrm{sh}}(\theta) - B_{\mathrm{sh}}(\theta))a_{\mathrm{sh}}(\theta) + B_{\mathrm{sh}}(\theta)\varepsilon_{\mathrm{conj}}(\theta).\end{aligned}$$

Since $b_{\mathrm{sh}} = -a_{\mathrm{sh}} + \varepsilon_{\mathrm{conj}}$, one may rewrite this exactly as

$$\Gamma_{\mathrm{exp}}(\theta) + \Gamma_{\mathrm{ph}}(\theta) = A_{\mathrm{sh}}(\theta)\,\varepsilon_{\mathrm{conj}}(\theta) + \Delta_{AB}(\theta)b_{\mathrm{sh}}(\theta).$$

Division by $\Gamma_{\mathrm{ph}}(\theta)$ gives the formula for $\zeta_\Gamma(\theta) + 1$. □

Corollary 12.142 (Working exact shell conjugacy and balanced transport force first-order anti-phase)**.** *Under the same hypotheses, if*

$$\varepsilon_{\mathrm{conj}}(\theta) = 0, \qquad \Delta_{AB}(\theta) = 0,$$

then

$$\Gamma_{\mathrm{exp}}(\theta) = -\Gamma_{\mathrm{ph}}(\theta), \qquad \zeta_\Gamma(\theta) = -1.$$

Hence the anti-phase corridor is then not merely admissible but exact at first order.

Proof. This is the immediate specialization of the theorem to vanishing shell-conjugacy and transport-balance defects. □

Remark 12.143 (Working structural reading of the shell-gain pair)**.** *The gain pair* $(\Gamma_{\mathrm{ph}}, \Gamma_{\mathrm{exp}})$ *is no longer best read as two unrelated coefficients. In the shell-conjugate regime it is the image of one shell tangent under one observer-shell involution and one downstream transport imbalance. Therefore the real structural question is not "why should two gains accidentally cancel?", but rather "how far is the shell-conjugate observer geometry from exact sign-conjugacy and balanced transport?"*

Shell anti-phase defect budget and epsilon-trap

The shell-conjugate observer geometry showed that anti-phase is natural whenever the shell-conjugacy and transport-balance defects are small. The next step is to compress those two structural errors and the single-shell-coordinate remainder into one normalized anti-phase budget. This turns the visible shell question into an epsilon-trap statement: once the induced shell ratio enters a sufficiently small corridor around -1, the first-order visible mode cannot escape that corridor unless the higher-order shell remainder is of comparable size.

Definition 12.144 (Working normalized shell anti-phase budget)**.** *Assume* $\Gamma_{\mathrm{ph}}(\theta) \neq 0$*. Define the normalized shell-conjugacy, transport-balance, and single-shell linearization loads by*

$$\rho_{\mathrm{conj}}(\theta) := \frac{|A_{\mathrm{sh}}(\theta)\,\varepsilon_{\mathrm{conj}}(\theta)|}{|\Gamma_{\mathrm{ph}}(\theta)|},$$

$$\rho_{AB}(\theta) := \frac{|\Delta_{AB}(\theta)\,b_{\mathrm{sh}}(\theta)|}{|\Gamma_{\mathrm{ph}}(\theta)|},$$

$$\rho_{\xi}(\theta) := \left|\frac{\zeta_{\mathrm{sh}}(\theta) - \zeta_{\Gamma}(\theta)}{1 + |\zeta_{\Gamma}(\theta)|}\right|.$$

The total normalized shell anti-phase budget is

$$\rho_{\mathrm{trap}}(\theta) := \rho_{\mathrm{conj}}(\theta) + \rho_{AB}(\theta) + \rho_{\xi}(\theta).$$

Theorem 12.145 (Working shell anti-phase epsilon-trap)**.** *Assume the single-shell-coordinate observer law and shell-conjugate observer geometry with*

$$R_{\mathrm{sh}} u_{\mathrm{sh}}(\theta) = u_{\mathrm{sh}}(\theta), \qquad \Gamma_{\mathrm{ph}}(\theta) \neq 0.$$

Then

$$|\zeta_{\Gamma}(\theta) + 1| \leq \rho_{\mathrm{conj}}(\theta) + \rho_{AB}(\theta),$$

and hence

$$|\zeta_{\mathrm{sh}}(\theta) + 1| \leq \rho_{\mathrm{trap}}(\theta)\,\big(1 + |\zeta_{\Gamma}(\theta)|\big).$$

In particular, if for some $\varepsilon_{\mathrm{trap}} > 0$ *one has*

$$\rho_{\mathrm{trap}}(\theta) \leq \frac{\varepsilon_{\mathrm{trap}}}{1 + |\zeta_{\Gamma}(\theta)|},$$

then

$$|\zeta_{\mathrm{sh}}(\theta) + 1| \leq \varepsilon_{\mathrm{trap}}.$$

Thus the shell ratio is forced into an anti-phase epsilon-tube around -1 *by one normalized structural budget.*

Remark 12.146 (Reading of the shell epsilon-trap after OP3 localization)**.** *After the OP3 shell/residual theorem, the anti-phase epsilon-trap is best read as a non-escape criterion for the visible shell ratio inside one OP3-local corridor. By itself it does not yet prove global OP closure or global branch admissibility; rather, it says that once the shell-side refinement stays inside a sufficiently small anti-phase budget, the visible first-order shell mode cannot leave that corridor without a comparably sized higher-order defect.*

Proof. The shell-conjugate observer theorem gave

$$\zeta_{\Gamma}(\theta) + 1 = \frac{A_{\mathrm{sh}}(\theta)\,\varepsilon_{\mathrm{conj}}(\theta) + \Delta_{AB}(\theta)\,b_{\mathrm{sh}}(\theta)}{\Gamma_{\mathrm{ph}}(\theta)}.$$

Taking absolute values yields

$$|\zeta_{\Gamma}(\theta) + 1| \leq \frac{|A_{\mathrm{sh}}(\theta)\,\varepsilon_{\mathrm{conj}}(\theta)|}{|\Gamma_{\mathrm{ph}}(\theta)|} + \frac{|\Delta_{AB}(\theta)\,b_{\mathrm{sh}}(\theta)|}{|\Gamma_{\mathrm{ph}}(\theta)|} = \rho_{\mathrm{conj}}(\theta) + \rho_{AB}(\theta).$$

Now

$$|\zeta_{\mathrm{sh}}(\theta) + 1| \leq |\zeta_{\mathrm{sh}}(\theta) - \zeta_{\Gamma}(\theta)| + |\zeta_{\Gamma}(\theta) + 1|.$$

By the definition of $\rho_\xi(\theta)$,

$$|\zeta_{\mathrm{sh}}(\theta) - \zeta_\Gamma(\theta)| = \rho_\xi(\theta)\,\big(1 + |\zeta_\Gamma(\theta)|\big).$$

Since $1 \le 1 + |\zeta_\Gamma(\theta)|$, the previous bound implies

$$|\zeta_\Gamma(\theta) + 1| \le \big(\rho_{\mathrm{conj}}(\theta) + \rho_{AB}(\theta)\big)\big(1 + |\zeta_\Gamma(\theta)|\big).$$

Combining the two estimates gives

$$|\zeta_{\mathrm{sh}}(\theta) + 1| \le \rho_{\mathrm{trap}}(\theta)\,\big(1 + |\zeta_\Gamma(\theta)|\big).$$

The epsilon-tube statement is immediate. □

Corollary 12.147 (Working exact visible closure reduces to one shell-budget comparison)**.** *Assume the anti-phase sign regime and $\Gamma_{\mathrm{ph}}(\theta) \neq 0$. If exact visible closure holds, then necessarily*

$$\rho_{\mathrm{trap}}(\theta) \le \frac{\rho_{\mathrm{rem}}(\theta)\big(1 + |\zeta_{\mathrm{sh}}(\theta)|\big)}{1 + |\zeta_\Gamma(\theta)|}.$$

Conversely, if

$$\rho_{\mathrm{trap}}(\theta) > \frac{\rho_{\mathrm{rem}}(\theta)\big(1 + |\zeta_{\mathrm{sh}}(\theta)|\big)}{1 + |\zeta_\Gamma(\theta)|},$$

then exact visible closure is impossible and the shared shell mode is canonically nonzero.

Proof. The anti-phase corridor theorem for visible closure gave the necessary condition

$$|1 + \zeta_{\mathrm{sh}}(\theta)| \le \rho_{\mathrm{rem}}(\theta)\,\big(1 + |\zeta_{\mathrm{sh}}(\theta)|\big).$$

The shell anti-phase epsilon-trap gives

$$|1 + \zeta_{\mathrm{sh}}(\theta)| \le \rho_{\mathrm{trap}}(\theta)\,\big(1 + |\zeta_\Gamma(\theta)|\big).$$

If exact visible closure is to be possible, the first-order shell mismatch forced by the second estimate must fit inside the higher-order rescue corridor from the first. This yields the displayed necessary inequality. Its negation rules out exact visible closure, hence forces a canonically nonzero shared shell mode. □

Remark 12.148 (Working structural reading of the shell anti-phase budget)**.** *The anti-phase corridor is no longer best read as an unexplained near-accident between two signed shell loads. After the shell-conjugate reduction, the visible first-order defect is controlled by one budget only: shell-conjugacy error, transport-balance error, and the residual error from replacing the induced shell ratio by the gain ratio. Therefore the next structural question is not "can phase and expansion accidentally cancel?", but rather "is the combined shell anti-phase budget genuinely smaller than the higher-order rescue scale?"*

3.10.31 Shell-balance driver as an architecture defect

The remaining first-order shell driver should not be treated as an independent phenomenological parameter. It is more natural to regard it as the visible shadow of a joint architecture defect involving the active HC/MM/J fit on the admissible branch together with the observer-shell dressing of the same branch.

Definition 12.149 (Working architecture-side shell driver loads)**.** *Set*

$$\lambda_{\mathrm{fit}}(\theta) := \frac{\|[(\mathcal{M}, J)]|_U\| \, \|x(\theta)\|}{|\Gamma_{\mathrm{ph}}(\theta)|}, \qquad \lambda_{\mathrm{obs}}(\theta) := \lambda_{\mathrm{sh}}(\theta).$$

Here λ_{fit} measures active architecture mismatch through the restricted commutator, while λ_{obs} is the normalized observer-shell anti-phase load already extracted above.

Theorem 12.150 (Working shell-balance driver is downstream from HC/MM/J fit plus shell dressing)**.** *Assume the active-branch setup from the previous section, with visible shared shell mode*

$$r_{\mathrm{obs}}(\theta) = \Xi_{\mathrm{bal}}(\theta)\, \xi_{\mathrm{sh}}(\theta) + O\big(|\xi_{\mathrm{sh}}(\theta)|^2\big)\,.$$

Then there is a first-order architecture estimate of the form

$$\frac{|\Xi_{\mathrm{bal}}(\theta)|}{|\Gamma_{\mathrm{ph}}(\theta)|} \le \lambda_{\mathrm{fit}}(\theta) + \lambda_{\mathrm{obs}}(\theta).$$

In particular, if both

$$[\mathcal{M}, J]\big|_U = 0 \qquad \textit{and} \qquad \lambda_{\mathrm{sh}}(\theta) = 0,$$

then

$$\Xi_{\mathrm{bal}}(\theta) = 0,$$

and the shared visible shell mode has no first-order driver.

Proof. By the exact commutator identity on the admissible branch, the primitive common-mode carrier is controlled by

$$E_{\mathrm{fit}}^{(U)} = \frac{1}{2}[\mathcal{M}, J]\big|_U.$$

Hence every first-order visible common-mode contribution inherited from the active transport branch is bounded, up to the visible readout coefficient entering Γ_{ph}, by a multiple of $\|[\mathcal{M}, J]|_U\| \, \|x(\theta)\|$. Normalizing by $|\Gamma_{\mathrm{ph}}(\theta)|$ gives the λ_{fit} term.

The remaining first-order imbalance after exact admissible transport was already compressed into the shell-conjugacy and transport-balance loads, combined in $\lambda_{\mathrm{sh}} = \lambda_{\mathrm{conj}} + \lambda_{\mathrm{bal}}$. These are exactly the observer-shell contributions entering the shell-balance driver Ξ_{bal}. Adding the architecture-side and observer-side contributions yields

$$\frac{|\Xi_{\mathrm{bal}}(\theta)|}{|\Gamma_{\mathrm{ph}}(\theta)|} \le \lambda_{\mathrm{fit}}(\theta) + \lambda_{\mathrm{sh}}(\theta).$$

If both terms vanish, then the normalized driver vanishes and therefore $\Xi_{\mathrm{bal}}(\theta) = 0$. □

Corollary 12.151 (Working first-order visible cancellation is architecture-driven)**.** *If*

$$\lambda_{\mathrm{fit}}(\theta) + \lambda_{\mathrm{sh}}(\theta) < 1,$$

then the shared shell mode is forced into a first-order small-driver regime. If

$$\lambda_{\mathrm{fit}}(\theta) + \lambda_{\mathrm{sh}}(\theta) = 0,$$

then any remaining visible residual is necessarily of strictly higher observer-shell order.

Proof. The theorem gives

$$|\Xi_{\mathrm{bal}}(\theta)| \le |\Gamma_{\mathrm{ph}}(\theta)|\big(\lambda_{\mathrm{fit}}(\theta) + \lambda_{\mathrm{sh}}(\theta)\big).$$

The first claim is the corresponding small-driver statement. If the sum vanishes, then $\Xi_{\mathrm{bal}}(\theta) = 0$, and the first-order shell law reduces to

$$r_{\mathrm{obs}}(\theta) = O\big(|\xi_{\mathrm{sh}}(\theta)|^2\big) + R_{\ge 2}(\theta),$$

so every visible remainder is of higher order. □

Remark 12.152 (Working architectural reading of the remaining shell law). *The visible common-mode residual is now best read as the downstream image of one combined architectural budget. The active HC/MM/J fit supplies the primitive common-mode leakage budget, while the shell-conjugacy and transport-balance terms supply the observer-side dressing budget. In this reading, the next real question is no longer whether OP 1 and OP 5 are separate closures, but whether the combined architecture budget vanishes exactly or leaves a canonical higher-order observer-shell remainder.*

3.10.32 Final architecture-versus-rescue budget corridor

After the shell-balance driver has been pushed back to the combined architecture budget, the remaining visible question should no longer be read as an independent shell law. It is now a final corridor comparison: can the higher-order rescue sector absorb the first-order architecture load, or does a canonically nonzero visible shell mode necessarily survive?

Definition 12.153 (Working normalized architecture budget and quadratic shell truncation). *Set*

$$\rho_{\mathrm{arch}}(\theta) := \frac{|\Gamma_{\mathrm{ph}}(\theta)\,\xi_{\mathrm{sh}}(\theta)|}{\Lambda_{\mathrm{lin}}(\theta)}\big(\lambda_{\mathrm{fit}}(\theta) + \lambda_{\mathrm{obs}}(\theta)\big),$$

and define the normalized quadratic shell truncation term by

$$\rho_{\xi^2}(\theta) := \frac{|O(|\xi_{\mathrm{sh}}(\theta)|^2)|}{\Lambda_{\mathrm{lin}}(\theta)}.$$

The first quantity measures the visible first-order architecture load inherited from the combined HC/MM/J-fit plus observer-shell budget; the second measures the normalized first neglected shell order.

Theorem 12.154 (Working final architecture-rescue corridor under sharp active loading). *Assume the sharp active-loading regime on a window where*

$$|\Gamma_{\mathrm{ph}}(\theta)\,\xi_{\mathrm{sh}}(\theta)| \geq c_0\,\Lambda_{\mathrm{lin}}(\theta)$$

for some fixed constant $c_0 > 0$. Then the normalized first-order visible shell defect satisfies

$$|\rho_{\mathrm{sh}}(\theta)| \geq c_0\,\rho_{\mathrm{arch}}(\theta) - \rho_{\xi^2}(\theta).$$

Consequently, exact visible closure can hold only if

$$c_0\,\rho_{\mathrm{arch}}(\theta) \leq \rho_{\mathrm{rem}}(\theta) + \rho_{\xi^2}(\theta).$$

Proof. By the architecture-driven shell law from the previous subsection,

$$r_{\mathrm{obs}}(\theta) = \Xi_{\mathrm{bal}}(\theta)\,\xi_{\mathrm{sh}}(\theta) + O\big(|\xi_{\mathrm{sh}}(\theta)|^2\big) + R_{\geq 2}(\theta),$$

with

$$\frac{|\Xi_{\mathrm{bal}}(\theta)|}{|\Gamma_{\mathrm{ph}}(\theta)|} \leq \lambda_{\mathrm{fit}}(\theta) + \lambda_{\mathrm{obs}}(\theta).$$

Dividing the first-order term by $\Lambda_{\mathrm{lin}}(\theta)$ yields the normalized architecture load $\rho_{\mathrm{arch}}(\theta)$. Under the sharp loading hypothesis, the visible contribution of this term is bounded below by $c_0\rho_{\mathrm{arch}}(\theta)$ up to the normalized quadratic truncation error $\rho_{\xi^2}(\theta)$. This proves

$$|\rho_{\mathrm{sh}}(\theta)| \geq c_0\,\rho_{\mathrm{arch}}(\theta) - \rho_{\xi^2}(\theta).$$

If exact visible closure is to be possible, then by the earlier normalized shell-defect theorem one must have

$$|\rho_{\mathrm{sh}}(\theta)| \leq \rho_{\mathrm{rem}}(\theta).$$

Combining both inequalities gives the displayed necessary corridor condition. □

Corollary 12.155 (Working final reduced decision after exact transport)**.** *Under exact admissible transport and sharp active loading, the visible OP 1/OP 5 question is reduced to a single final budget comparison:*

$$c_0\,\rho_{\mathrm{arch}}(\theta) \overset{?}{\leq} \rho_{\mathrm{rem}}(\theta) + \rho_{\xi^2}(\theta).$$

If the inequality fails, then the common visible shell mode is canonically nonzero. If it holds, then exact visible closure remains possible only through higher-order rescue of a first-order architecture load.

Remark 12.156 (Working interpretation of the final corridor)**.** *The remaining visible problem is now compressed as far as the current structure allows. It is no longer a pair of unfinished closures, nor even two shell loads. It is one corridor question: can the higher-order rescue sector absorb the normalized combined architecture load after exact transport, or does a canonical observer-shell remainder necessarily survive?*

12.2 QAM-over-ZF: operator families and lambda-like structural maps

We now close the first internal QAM package by adding an operator-family and lambda-like structural layer on top of the already introduced transport, branch/shell, tensor, and translation strata. The point remains conservative: no higher-order foundation is introduced. Instead, families of coded operators and lambda-like structural maps are themselves represented as set-codes over ZF and are required to preserve closure content on the admissible fragment.

Definition 12.157 (Operator-family code)**.** *A set*

$$\mathfrak{o} = \langle 9, I, F\rangle$$

is called an operator-family code *if I is an index code and F is a graph-code assigning to each $i \in I$ a coded operator o_i.*

Definition 12.158 (Admissible operator family)**.** *An operator-family code*

$$\mathfrak{o} = \langle 9, I, F\rangle$$

is called admissible *if every member operator o_i coded by F is admissible on its intended coded state domain. Equivalently, for every admissible coded source state s and every coded target state t linked by some member o_i, one has*

$$\mathrm{QAdmTransport}(o_i, s, t) \Longrightarrow \mathrm{QClEq}(s, t).$$

Remark 12.159. *Thus admissibility of an operator family means pointwise closure preservation across the family. The family layer does not weaken the closure-first discipline already imposed on single coded operators.*

Definition 12.160 (Lambda-like structural map code)**.** *A set*

$$\lambda = \langle 10, D, C, L\rangle$$

is called a lambda-like structural map code *if D codes an intended domain class of QAM-coded objects, C codes an intended codomain class, and L is a graph-code assigning to each coded object in D a coded object in C.*

Definition 12.161 (Closure-compatible lambda map)**.** *A lambda-like structural map code λ is called* closure-compatible *on a coded state domain if for all coded states s, t in that domain,*

$$\mathrm{QClEq}(s,t) \Longrightarrow \mathrm{QClEq}(\lambda(s), \lambda(t)),$$

whenever the values $\lambda(s)$ and $\lambda(t)$ are coded states.

Remark 12.162. *This is the lambda-level analogue of admissible transport. A closure-compatible lambda map need not itself be read as a transport along one fixed route; it may instead be a structural selector, reindexer, collector, or support-transforming map, provided it preserves closure class on the admissible fragment.*

Definition 12.163 (Support-selection lambda)**.** *A closure-compatible lambda-like structural map is called a* support-selection lambda *if it acts on admissible branch-shell support codes and returns a coded sub-support, preferred support, or selected routed support without leaving the admissible closure regime.*

Proposition 12.164 (Pointwise closure preservation for admissible operator families)**.** *Let*

$$\mathfrak{o} = \langle 9, I, F \rangle$$

be an admissible operator family. If o_i is a member of the family and

$$\mathrm{QAdmTransport}(o_i, s, t),$$

then

$$\mathrm{QClEq}(s,t).$$

Consequently, every closure-compatible readout identifies the closure-relevant content of s and t after factorization.

Proof. This is immediate from pointwise admissibility of the family members together with admissibility-implies-closure preservation and closure-compatible readout. □

Proposition 12.165 (Closure preservation under closure-compatible lambda maps)**.** *Let λ be a closure-compatible lambda-like structural map on an admissible coded state domain. If*

$$\mathrm{QClEq}(s,t),$$

then

$$\mathrm{QClEq}(\lambda(s), \lambda(t)),$$

whenever the values are coded states.

Proof. This is exactly the defining property of closure-compatible lambda maps. □

Theorem 12.166 (First QAM operator-family/lambda theorem)**.** *Assume:*

1. *$\mathfrak{o} = \langle 9, I, F \rangle$ is an admissible operator family;*
2. *$\lambda = \langle 10, D, C, L \rangle$ is a closure-compatible lambda-like structural map on the relevant admissible fragment;*
3. *r is a closure-compatible coded readout.*

Then the following hold.

1. ***Familywise transport preservation.*** *For every member operator o_i of $\mathfrak{o}$ and every admissible transport step*
$$\mathrm{QAdmTransport}(o_i, s, t),$$
one has
$$\mathrm{QClEq}(s, t).$$

2. ***Lambda-level closure preservation.*** *If in addition*
$$\mathrm{QClEq}(s, t),$$
then
$$\mathrm{QClEq}(\lambda(s), \lambda(t)).$$

3. ***Readout stability after family transport and lambda action.*** *Whenever the composed expressions are defined on the admissible fragment, the closure-relevant readout content of*
$$s, \quad t, \quad \lambda(s), \quad \lambda(t)$$
agrees after factorization through the associated class-target maps.

Proof. The first claim is pointwise admissibility of the operator family. The second is closure preservation under the lambda-like map. The third follows because closure-compatible readout factors through coded closure class, and all four coded states listed in the statement lie in the same closure class whenever the hypotheses apply. □

Corollary 12.167 (Operator-family/lambda lift of the admissible OP fragment)**.** *The admissible OP fragment translated into QAM-over-ZF remains stable not only under single admissible coded operators, but also under admissible operator families and closure-compatible lambda-like structural maps acting on the translated fragment.*

Remark 12.168. *This is the first point at which the QAM layer becomes more than a static coding scheme. It now supports controlled families of operations and closure-preserving structural maps on the admissible fragment while remaining conservative over ZF.*

Remark 12.169 (Transition to support selection)**.** *The preceding operator-family/lambda layer is the last part of the internal QAM package that still acts directly on admissible transport and closure data. The next layer is different in status: support selection and degeneracy handling do not generate admissibility, but only refine already admissible branch-shell support after the closure-first filters have been applied.*

12.3 QAM-over-ZF: support selection and degeneracy handling

We now add a first support-selection layer on top of the already translated admissible QAM fragment. The purpose of this layer is not to replace closure-admissibility or shell-admissibility, but to order admissible support after those primary filters have already been applied. Thus QAM selection is subordinate to closure preservation and only acts inside the admissible branch-shell sector.

Definition 12.170 (Support candidate code)**.** *A set*
$$\varsigma = \langle 11, \Omega, w \rangle$$
is called a support candidate code *if Ω codes a family of admissible branch-shell pairs and w codes a weight or preference assignment on Ω.*

Remark 12.171. *A support candidate code is the QAM analogue of a visible support sector restricted in advance to admissible branch-shell data. It is therefore a post-admissibility object rather than a primitive closure object.*

Definition 12.172 (Support preference relation)**.**

$$\begin{aligned} &\textit{closure preservation} > \textit{branch admissibility} > \textit{shell admissibility} \\ &> \textit{support preference} > \textit{visible routed weight.} \end{aligned}$$

Remark 12.173. *Thus support preference never overrides admissibility. It only compares already admissible support candidates.*

Definition 12.174 (Closure-compatible support selector)**.** *A lambda-like structural map*

$$\lambda_{\mathrm{sel}} = \langle 10, D, C, L \rangle$$

is called a closure-compatible support selector *if:*

1. *its domain consists of admissible support candidate codes;*
2. *its values are coded selected supports or coded preferred subsupports;*
3. *whenever two input support candidates encode the same closure-bearing admissible content up to support degeneracy, their images under λ_{sel} are closure-equivalent or support-equivalent at coded level.*

Definition 12.175 (Support degeneracy)**.** *Two admissible support candidate codes ς_1, ς_2 are called* support-degenerate *if they differ only by admissible re-indexing, admissible redistribution of coded visible weight, or admissible closure-equivalent regrouping, while preserving the same coded closure-bearing support content.*

Remark 12.176. *Support degeneracy is the QAM-over-ZF shadow of carrier degeneracy and readout multiplicity in the current Companion line. It records when several admissible support realizations should still be read as one structural support type.*

Definition 12.177 (Selected routed support code)**.** *If λ_{sel} is a closure-compatible support selector and ς is an admissible support candidate code, then*

$$\lambda_{\mathrm{sel}}(\varsigma)$$

is called the corresponding selected routed support code*, whenever the value is defined.*

Proposition 12.178 (Selection stays inside the admissible QAM fragment)**.** *Let λ_{sel} be a closure-compatible support selector and let ς be an admissible support candidate code. Then every selected routed support code*

$$\lambda_{\mathrm{sel}}(\varsigma)$$

remains inside the admissible QAM support fragment.

Proof. By definition, the domain of λ_{sel} consists only of admissible support candidate codes, and the selector is required to return coded selected supports or preferred subsupports without leaving the closure-compatible admissible regime. Hence the selected support remains inside the admissible fragment. □

Proposition 12.179 (Closure preservation under support selection)**.** *Let ς be an admissible support candidate code and let λ_{sel} be a closure-compatible support selector. Then the closure-relevant content of*

$$\lambda_{\mathrm{sel}}(\varsigma)$$

agrees with the closure-relevant content of ς after passage through the associated closure-factor map.

Proof. Support selection acts only after admissibility has already been established and is required to be closure-compatible. Therefore the selector cannot change the coded closure-bearing content of the admissible support candidate; it may only refine, reduce, or prefer one admissible representation over another. Closure-compatible readout therefore identifies the closure-relevant content of the input and output supports after factorization. □

Corollary 12.180 (Support selection does not create closure)**.** *A selected routed support is closure-preserving only because it is selected from an already admissible support candidate. Selection itself is not an independent source of closure.*

Remark 12.181. *This is the coded QAM analogue of the structure-first rule that preferred support must be read only after closure and admissibility have already been fixed.*

Proposition 12.182 (Degenerate supports may be identified after selection)**.** *Let ς_1, ς_2 be support-degenerate admissible support candidate codes. If λ_{sel} is a closure-compatible support selector, then*

$$\lambda_{\mathrm{sel}}(\varsigma_1) \qquad \textit{and} \qquad \lambda_{\mathrm{sel}}(\varsigma_2)$$

may be treated as equivalent selected supports at the coded structural level.

Proof. Since ς_1 and ς_2 differ only by admissible re-indexing, admissible redistribution of visible weight, or closure-equivalent regrouping, they carry the same coded closure-bearing support content. Closure-compatibility of the selector ensures that this content is preserved under selection. Therefore their selected images may be identified at the coded structural level. □

Theorem 12.183 (First QAM support-selection theorem)**.** *Assume:*

1. *ς is an admissible support candidate code;*
2. *λ_{sel} is a closure-compatible support selector;*
3. *r is a closure-compatible coded readout on the relevant support fragment.*

Then:

1. *the selected routed support code $\lambda_{\mathrm{sel}}(\varsigma)$ remains admissible;*
2. *the closure-relevant readout content of $\lambda_{\mathrm{sel}}(\varsigma)$ agrees with that of ς after factorization;*
3. *support-degenerate admissible candidates yield equivalent selected supports at coded structural level.*

Proof. The first claim is selection-stays-inside-admissible-fragment. The second is closure preservation under support selection. The third is the degeneracy proposition above. Together they show that support selection in QAM is a post-admissibility refinement layer rather than an independent closure mechanism. □

Remark 12.184. *This is the first QAM-over-ZF support-selection layer. It does not yet impose one final universal selector. It only provides the structural place where preferred support, selected support, and degeneracy handling may be coded without violating the closure-first architecture inherited from the current Companion line.*

Remark 12.185 (Current closure of the internal QAM package)**.** *At the present internal stage, the QAM-over-ZF line has reached a first connected package state: coded objects, primitive predicates, admissible transport, coded closure-class preservation, closure-compatible readout, branch/shell/routed admissibility, tensor closure rules, faithful translation of the admissible OP fragment, operator-family/lambda-like maps, and support-selection/degeneracy handling are now all present inside one conservative ZF-grounded coding line. The next natural work should therefore focus on normalization, compression, and later public extraction rather than immediate uncontrolled enlargement.*

Bibliographic hardening scan for v3.35.0

The r06 scan was conservative: it did not alter the theorem chain and did not promote any external source to an axiom. It only checked which external literatures are repeatedly invoked by terminology, analogies, or mathematical background in the monolith and then expanded the terminal bibliography accordingly. The highest-priority additions are Teichmuller/quasiconformal references, because Teichmuller, Beltrami, and quasiconformal language occurs often while the r05 bibliography had no dedicated Teichmuller anchor. The second-priority additions are the Riemann–GUE/RMT and operator/spectral references, because the document repeatedly uses GUE pacing, Riemann-zero language, and self-adjoint/spectral framing. The third-priority additions are structural background references for Hopf/fibre bundles, division algebras, gauge/QFT landmarks, CKM/PMNS mixing, Julia dynamics, ZF/set theory, and the probe/cryptographic/-computational diagnostic vocabulary.

The present bibliography is therefore still not meant to be exhaustive. In particular, a later release could still add a dedicated literature group for Teichmuller geodesic flow, measured foliations, moduli-space dynamics, geometric quantization, modern G_2 model building, neutrino phenomenology, and numerical zeta-zero datasets. Those are useful follow-on enrichments, but they are not required for the r06 hardening pass.

Final Bibliography

Note. The bibliography below mixes (i) the current Companion release, (ii) earlier project releases, and (iii) external references. Only the Companion concept DOI 10.5281/zenodo.19210728 belongs on the title line; version-specific Companion DOIs belong in the version record or the dedicated Companion bibliography entry, while other DOIs below refer to distinct publications.

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